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Data Communication CS601 Lecture No 20

Lecture Prepared By

Muhammad Abid Ali

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CS601 Lecture No 20 Topic 100 to 113

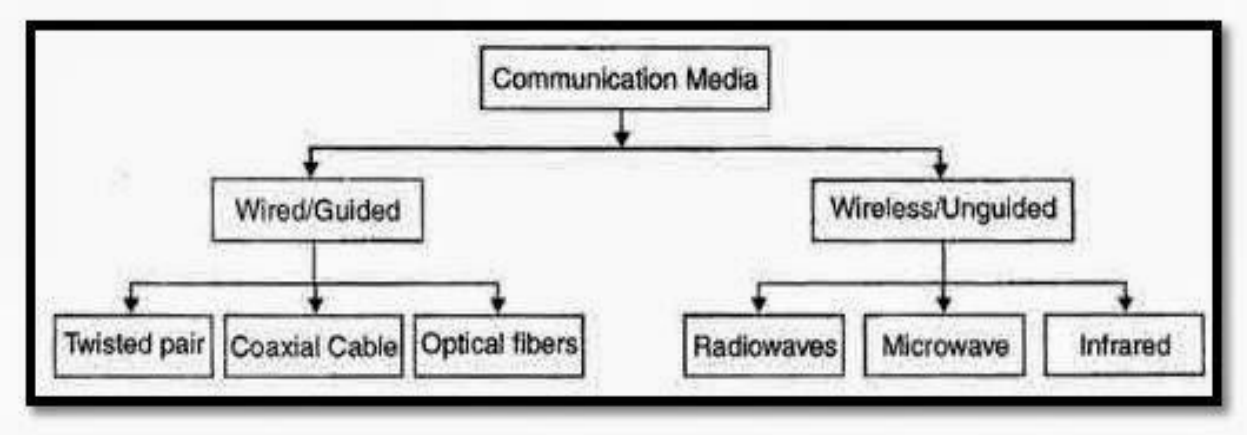
What do you mean by transmission medium?

The transmission medium can be defined as **a pathway that can transmit information from a sender to a receiver**. Transmission media are located below the physical layer and are controlled by the physical layer. Transmission media are also called communication channels.

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In transmission media is the way the systems are connected to route data signals in a network. Unguided media (wireless).

...

Differences.

Guided media	Unguided media
It is called wired communication or bounded transmission media.	It is called wireless communication or unbounded transmission media.

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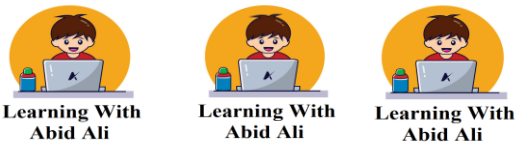
BASIS FOR COMPARISON	GUIDED MEDIA	UNGUIDED MEDIA
Basic	The signal requires a physical path for transmission.	The signal is broadcasted through air or sometimes water.
Alternative name	It is called wired communication or bounded transmission media.	It is called wireless communication or unbounded transmission media.
Direction	It provides direction to signal for travelling.	It does not provide any direction.
Types	Twisted pair cable, coaxial cable and fibre optic cable.	Radio wave, microwave and infrared.

Definition of Guided Media

Guided transmission media are more commonly known as the **wired communication** or **bounded transmission media.** The electromagnetic signals travel between the communicating devices through a physical medium/conductor. As the medium

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for transmission is a physical conductor, it also provides **direction** to the signal. But there are physical limitations of the conductor in the guided media. Like the length of the conductor, its installation cost, its maintenance, etc.

The guided media is categorized further into three categories that are **twisted-pair cable, coaxial cable** and **fiber-optic cable**. The twisted pair cable has two conductors' wires wounded around each other and each surrounded by an insulating material. The twisted pair cable is flexible and easy to install. But it has *low bandwidth* and provide *less protection* from *interference*. Twisted pair cable is also of two types **shielded and unshielded twisted pair cable**.

The **coaxial cable** has a central core conductor (usually copper) enclosed in an insulating sheath, which is further encased in an outer metallic braid, it serves as both protection against noise and as a second conductor which completes the circuit. Now, the outer metallic covering is also covered by an insulating sheath. The coaxial cable carries signals of *higher frequency* than the twisted pair cable.

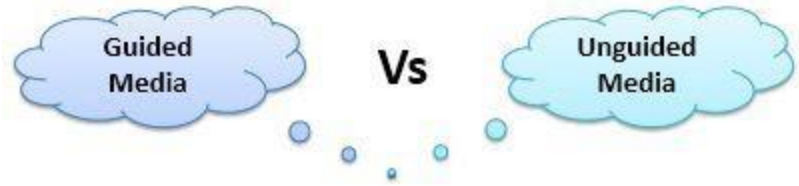
The third category is the **optical fiber** which is made of glass or plastic, and it transmits signals in the form of light. The optical fiber is *noise resistance, has less signal attenuation and has a higher bandwidth* in comparison to twisted pair cable and coaxial cable. But it also has some drawbacks like; it is *very expensive; i*t requires a lot of *installation and maintenance charge* as any defect in the cable can diffuse light and alter the signals. As the optical fiber is made of glass, it is very *fragile*.

Difference Between Guided and Unguided Media

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The computer and other communicating devices represent data in the form of signals. The signals transmit between communicating devices in the form of electromagnetic energy, and hence the signals are called **electromagnetic signals**. Electromagnetic signals are the combination of electric and magnetic fields that vibrates in relation to each other. The electromagnetic signals can travel through various transmission media. The transmission media is broadly classified into two categories that are **guided** and **unguided media**.

The basic difference between guided and unguided media is that in the **guided media**, the signal travels through a physical medium whereas, in **unguided media,** the signal travel through the air. There are some more differences between guided and unguided media which I have discussed with the help of comparison chart shown below.

Content: Guided Vs Unguided Media

1. Comparison Chart
2. Definition
3. Key Differences
4. Conclusion

Comparison Chart

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Definition of Guided Media

Guided transmission media are more commonly known as the **wired communication** or **bounded transmission media.** The electromagnetic signals travel between the communicating devices through a physical medium/conductor. As the medium for transmission is a physical conductor, it also provides **direction** to the signal. But there are physical limitations of the conductor in the guided media. Like the length of the conductor, its installation cost, its maintenance, etc.

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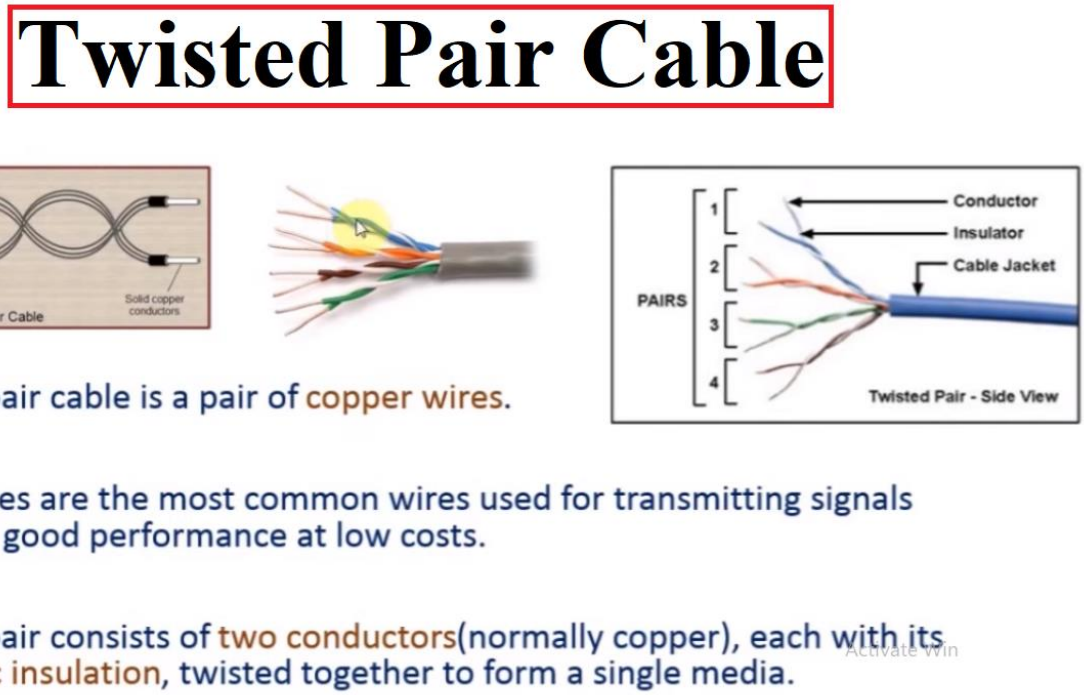
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The guided media is categorized further into three categories that are **twisted-pair cable, coaxial cable** and **fiber-optic cable**.

Twisted Pair Cable

The twisted pair cable has two conductors' wires wounded around each other and each surrounded by an insulating material. The twisted pair cable is flexible and easy to install. But it has low bandwidth and provide less protection from interference. Twisted pair cable is also of two types **shielded and unshielded twisted pair cable**.

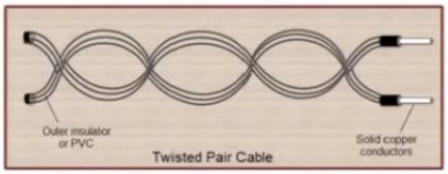


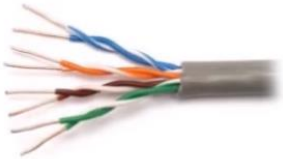
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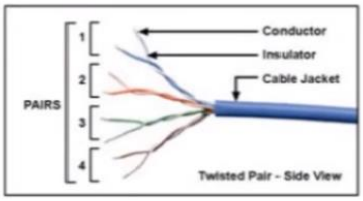
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Twisted Pair Cable







- Out of these two wires, only one carries actual signal and another is used for ground reference.
- To identify every cable, these cables are colour coated.
- The twists between wires are helpful in reducing noise (electro-magnetic interference) and crosstalk.
- This type of cable is used in telephone lines to provide voice and data channels.

There Are Two Main Types of Twisted Pair Cables

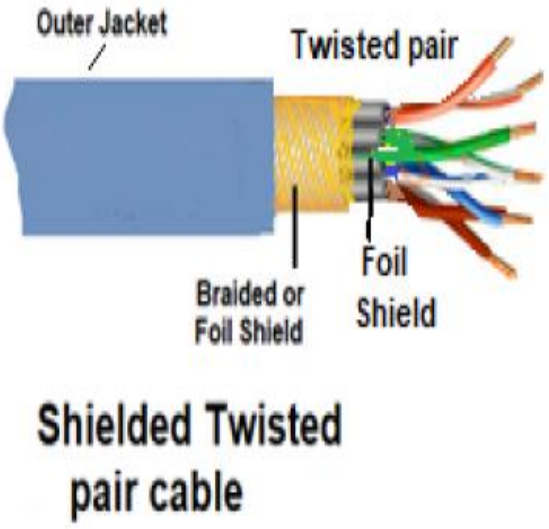
Unshielded Twisted Pair (UTP)

Shielded Twisted Pair (STP)

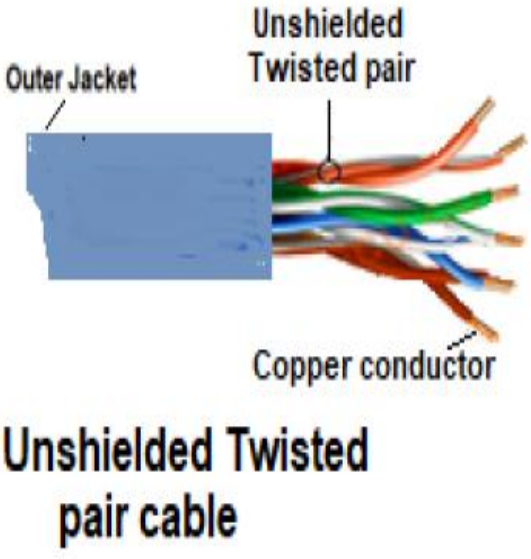
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Shielded Twisted pair cable



Unshielded Twisted pair cable

What is the difference between UTP and STP When would you use STP?

Both UTP and STP are widely used in transmitting information across distance. UTP is a cable with wires that are twisted together to reduce noise generated by an external source. On the other hand, STP is a twisted pair cable with an additional mesh shield that guards the cable against electromagnetic interference.

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Distinguish between STP and UTP:

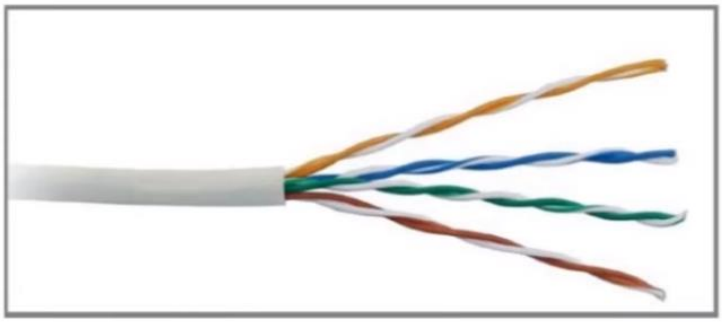
Shielded Twisted Pair	Un-shielded Twisted Pair
1. STP has a metal foil covering.	1. UTP does not have a metal foil Covering.
2. STP gives better resistance to electromagnetic interference as compared to UTP.	2. UTP does not provide better resistance to electromagnetic interference as compared to STP.
3. STP is little expensive then UTP.	3. UTP is less expensive than STP.
4. Grounding is possible.	4. Grounding is not possible.
5. Possibility of crosstalk is less as it resist to EMI.	5. Possibility of crosstalk is more as it provide less resistance to EMI.
6. Distance travelled is large and can be used in MAN.	6. Distance travelled is less and used in LAN.

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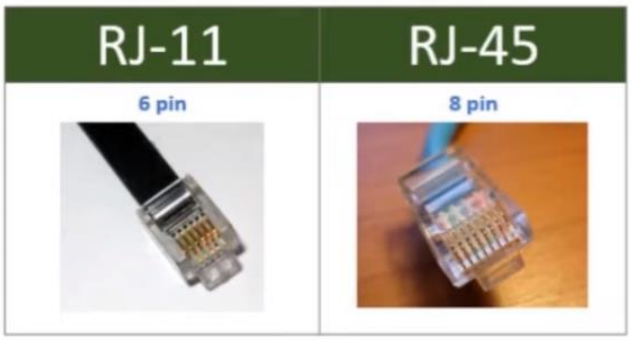


- **Unshielded twisted pair (UTP):-**
  - UTP is more common.
  - UTP cost less than STP and easily available due to its many use.
  - Due to its low cost, UTP cabling is used extensively for local-area networks (LANs) and telephone connections.
  - UTP cables consist of 2 or 4 pairs of twisted cable.
  - Cable with 2 pair use RJ-11 connector and 4 pair cable use RJ-45 connector.

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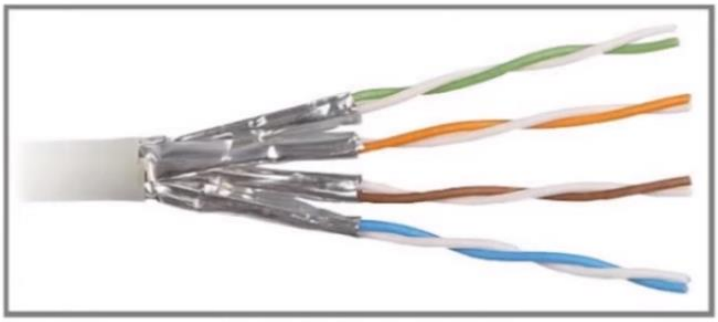


- RJ-45 connectors is used with Ethernet cables in computer networking.
- RJ-11 connectors is used in connecting telephone units.

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- **Shielded twisted pair (STP):-**
  - This type of cable has a metal foil covering which encases each pair of insulator conductors.
  - Electromagnetic noise penetration is prevented by metal casing. Shielding also eliminates crosstalk.
  - It is similar to UTP but has a mesh shielding that's protects it from EMI which allows for higher transmission rate.
  - It is more expensive than coaxial and unshielded twisted pair.

Coaxial Cable

The **coaxial cable** has a central core conductor (usually copper) enclosed in an insulating sheath, which is further encased in an outer metallic braid, it serves as both protection against noise and as a second conductor which completes the circuit. Now, the outer metallic covering is also covered by an insulating sheath. The coaxial cable carries signals of *higher frequency* than the twisted pair cable.

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Coaxial Cable



- **Coaxial cables** are copper cables with better shielding than twisted pair cables, so that transmitted signals may travel longer distances at higher speeds.
- The shield minimizes electrical and radio frequency interference.
- Coaxial cabling is the primary type of cabling used by the cable television industry and is also widely used for computer networks such as Ethernet.

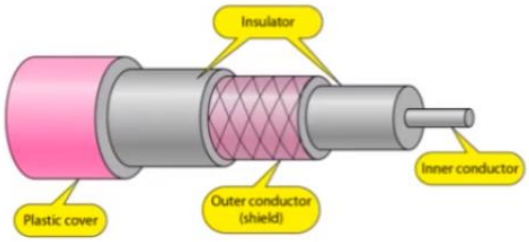
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Coaxial Cable





- Coaxial cable has two wires of copper.
- The core/inner copper wire in centre and is made of solid conductor. It is enclosed in an insulating sheath.
- The second/outer copper wire is wrapped around, and is used to protect from external electromagnetic interference(Noise).
- This all is covered by plastic cover used to protect the inner layers from physical damage such as fire or water.

Coaxial Cable Standards

- Coaxial cables are categorized by their Radio Government (RG) ratings. Each RG number denotes a unique set of physical specifications
  - 50-Ohm RG-7 or RG-11 : used with thick Ethernet.
  - 50-Ohm RG-58 : used with thin Ethernet
  - 75-Ohm RG-59 : used with cable television

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Optical Fiber

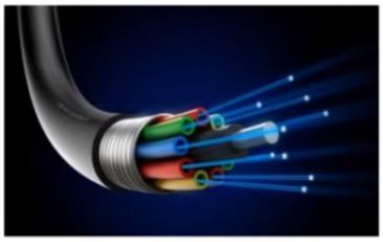
The third category is the **optical fiber** which is made of glass or plastic, and it transmits signals in the form of light. The optical fiber is *noise resistance, has less signal attenuation and has a higher bandwidth* in comparison to twisted pair cable and coaxial cable. But it also has some drawbacks like; it is very expensive; it requires a lot of installation and maintenance charge as any defect in the cable can diffuse light and alter the signals. As the optical fiber is made of glass, it is very fragile.

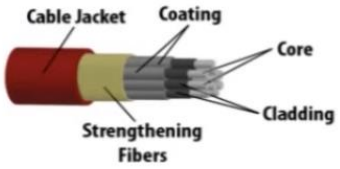
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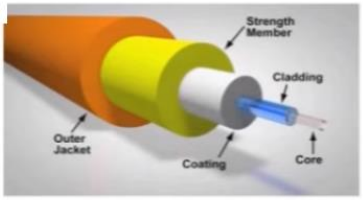
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Optical Fiber







- A fibre optic cable is made of high quality of thin glass or plastic and is used to transfer digital data signals in the form of light up to distance of thousand of miles.
- Fibre optic cables are not affected by electromagnetic interference, so noise and distortion is very less.
- Fibre optic cables carry communication signals using pulses of light generated by small lasers or light-emitting diodes (LEDs).
- The cable consists of one or more strands of glass, each only slightly thicker than a human hair. The centre of each strand is called the core, which provides the pathway for light to travel. The core is surrounded by a layer of glass called cladding that reflects light inward to avoid loss of signal and allow the light to pass through bends in the cable. No light escapes the glass core because of this reflective **cladding.**

- **Advantages of Optical Fibre:-**
  - **Fibre optic cables have a much High bandwidth than metal cables. This means that they can carry more data.**
  - **Smaller Size and Lighter weight.**
  - **low attenuation**
  - **Not affected electromagnetic interference (No EMI interference)**
  - **Signals carrying data can travel long distances without weakening**
  - **Suitable for industrial and noisy areas**

- **Disadvantages of Optical Fibre:-**
  - **Optical fibre cables are expensive**
  - **Difficult to install**
  - **Maintenance is expensive and difficult**

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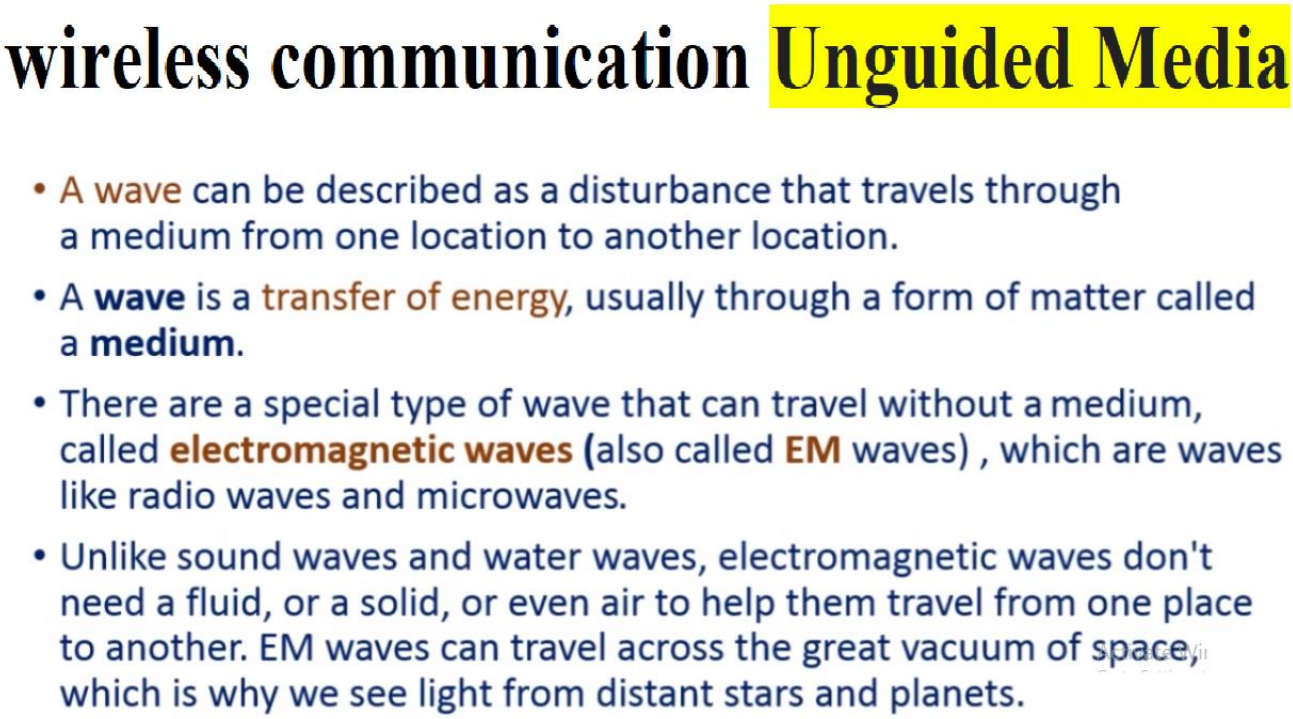
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Definition of Unguided Media

The **unguided media** is also called **wireless communication**. It does not require any physical medium to transmit electromagnetic signals. In unguided media, the electromagnetic signals are broadcasted through air to everyone. These signals are available to one who has the device capable of receiving those signals.

The unguided media is also called unbounded media as it does not have any border limitation. The unguided media allows the user to connect all the time, as the communication is wireless the user can connect himself from anywhere to the network.

The unguided media is categorized into **radio waves, microwaves and infrared waves**.



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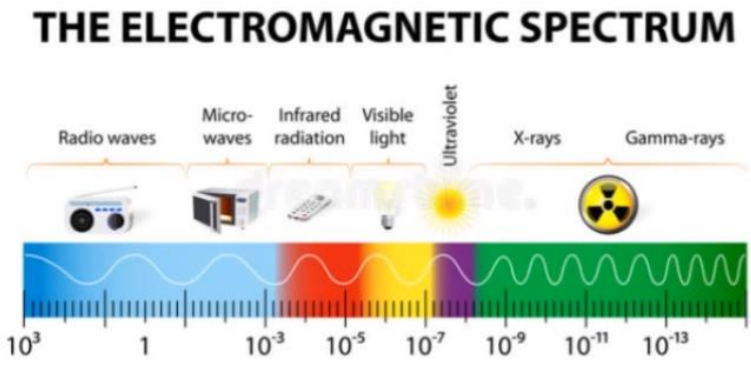
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wireless communication Unguided Media

- Electromagnetic waves are formed when an electric field comes in contact with a magnetic field. They are hence known as 'electromagnetic' waves.
- Electromagnetic (EM) radiation is a form of energy that is all around us and takes many forms, such as radio waves, microwaves, X-rays and gamma rays.
- Sunlight is also a form of EM energy. Electromagnetic energy from the sun comes to Earth in the form of radiation.
- The Electromagnetic Spectrum describes a wide range of different **electromagnetic waves**.

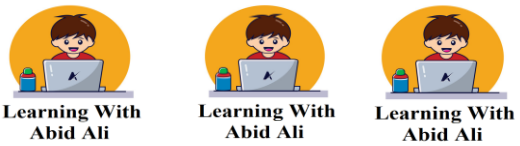
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- A little part of electromagnetic spectrum can be used for wireless transmission.

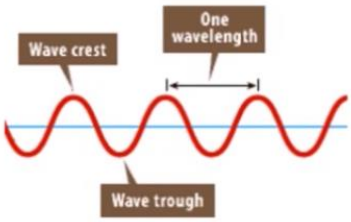


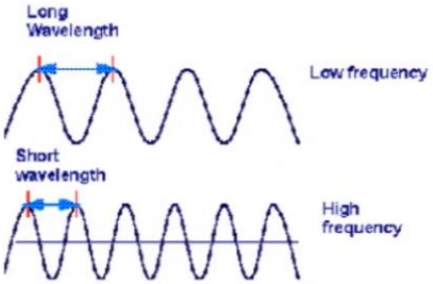
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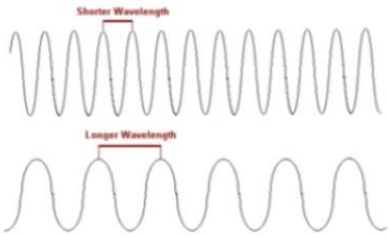
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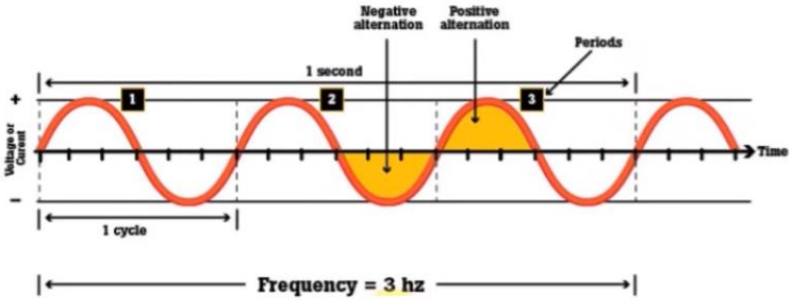


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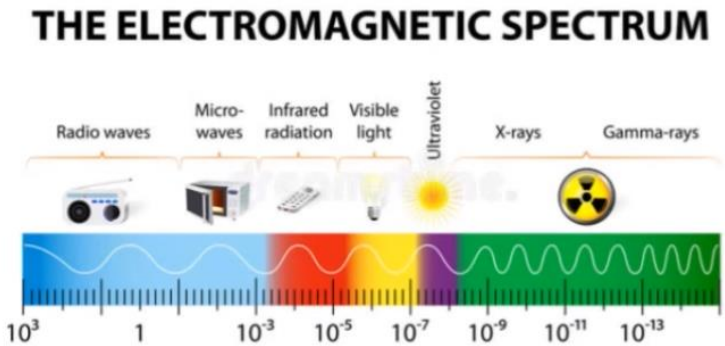


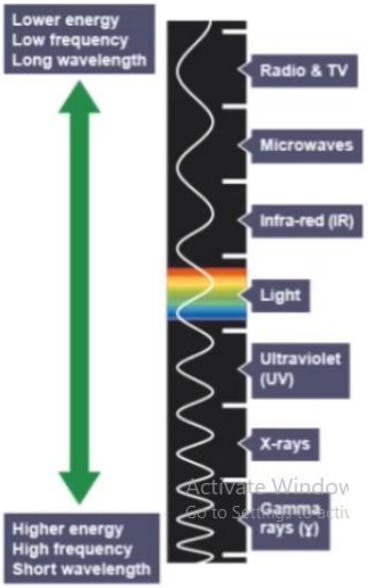






- A little part of electromagnetic spectrum can be used for wireless transmission.





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Radio Waves

The **radio waves** are generated easily; they are *low-frequency signals* and can travel a *long distance*. The radio waves can penetrate through the buildings.

Radio Waves

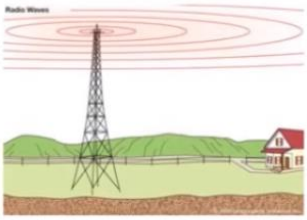


- Radio waves are EM (Electromagnetic)waves that have wavelengths between 1 millimetre and 100 kilometres (or 300 GHz and 3 kHz in frequency).
- Radio frequency is easy to generate because its has large wavelength and can travel long distance.
- Radio waves are generated by radio transmitters and received by radio receivers.
- Radio stations transmit radio waves using transmitters, which are received by the receiver installed in our devices. Both transmitters and receivers use antennas to radiate or capture radio signals
- It can penetrate walls easily, so these waves are widely used for communication both indoors and outdoors.

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Radio Waves



- Radio waves are omnidirectional means they travel in all the directions from the source.
- When an antenna transmits radio waves, they are propagated in all directions.
- A sending antenna send waves that can be received by any receiving antenna. The omnidirectional property has disadvantage, too. The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signal using the same frequency or band.
- It is Used Mobile, AM/FM radio, television

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Radio Waves



- Radio waves are EM (Electromagnetic)waves that have wavelengths between 1 millimetre and 100 kilometres (or 300 GHz and 3 kHz in frequency).
- Radio frequency is easy to generate because its has large wavelength and can travel long distance.
- Radio waves are generated by radio transmitters and received by radio receivers.
- Radio stations transmit radio waves using transmitters, which are received by the receiver installed in our devices. Both transmitters and receivers use antennas to radiate or capture radio signals
- It can penetrate walls easily, so these waves are widely used for communication both indoors and outdoors.

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Radio Spectrum

Radio Band	Frequency	Some Applications
Very Low Frequency VLF	3 KHZ to 30 KHz	Radio Navigation
Low Frequency LF	30 KHz to 300 KHz	Long Wave Radio
Medium Frequency MF	30 KHz to 3 MHz	AM Radio
High Frequency HF	3 MHz to 30 MHz	CB Radio (HAM) Point to Point Radio Search and Rescue Services
Very High Frequency VHF	30 MHz to 300 MHz	FM radio 88-108 MHz VHF Broadcast TV
Ultra High Frequency UHF	300 MHz to 3 GHz	UHF Broadcast TV Cellular Phones Microwave Links Wi-Fi in 2.4 Band Satellite Communications
Super High Frequency SHF	3 GHz to 30 GHz	Microwave Links Wi-Fi in 5 GHz Band Satellite Communications
Extra High Frequency EHF	30 GHz to 300 GHz	Microwave Links

Microwaves

The **microwaves** are transmitted in a straight line and hence require the **line-of-sight transmission**. The distance covered by the microwave signal depend on the height of the two antennas. More the taller are antennas longer is the distance covered by the signal. The microwave has a *frequency higher* than the radio waves. Microwave are used for telephone communication mobile phones, television distribution, etc.

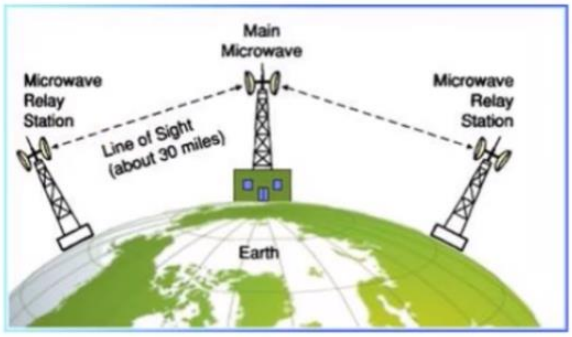
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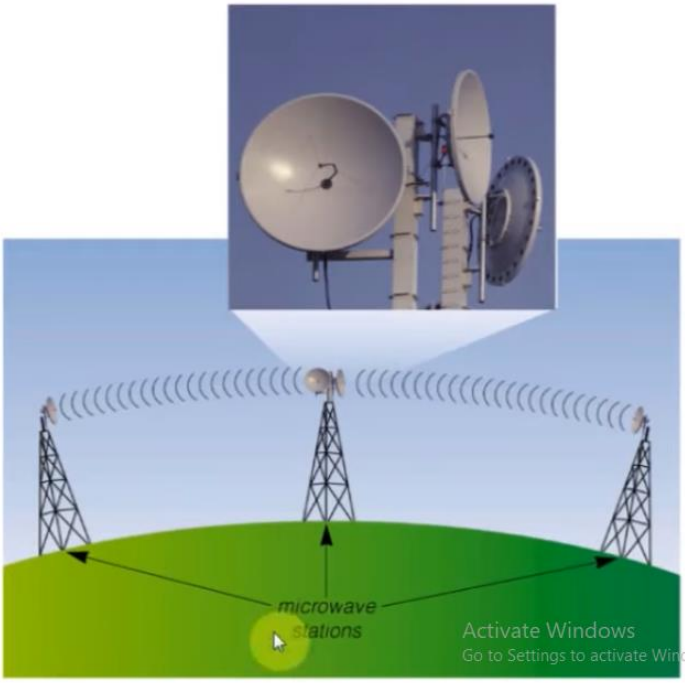
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Microwaves

- Microwaves are a type of radio waves with high frequencies. It can be classified as a subclass of radio waves. The frequency of microwaves lies in the 300 MHz to 300 GHz.
- Unlike radio waves, microwaves are unidirectional, in which the sending and receiving antennas need to be aligned.
- Microwaves are widely used for point-to-point communications because their small wavelength, which means that the signal is focused into a narrow beam. Additionally, each antenna must be within line of sight of the next antenna.
- Electromagnetic waves above 100 MHz tend to travel in a straight line and signals over them can be sent by beaming those waves towards one particular station. Because Microwaves travels in straight lines, both sender and receiver must be aligned to be strictly in line-of-sight.





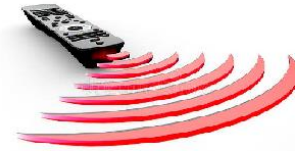
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Infrared waves are used for short range communication. Like, the remote control for televisions, VCRs, etc. uses infrared waves. It can not penetrate through obstacles. The government license is not required, to operate an infrared system as it is more secure against eavesdropping.

Infrared waves



- **Infrared signals have frequencies between 300 GHz to 400 THz. They are used for short-range communication.**
- **Infrared waves are used for very short distance communication like TV remote, wireless speakers, automatic doors, hand held devices etc.**
- **Infrared waves having high frequencies prevents interference b/w one system to another.**
- **Infrared signals have high frequencies and cannot penetrate walls. Due to its short-range communication system, the use of an infrared communication system in one room will not be affected by the use of another system in the next room. This is why using an infrared TV remote control in our home will not interfere with the use of our neighbour's infrared TV remote control.**
- ***THE DISADVANTAGES OF USING INFRARED***
  - **Infrared signals cannot be used for long distance communication. In addition, we cannot use infrared waves outside a building because sun's rays contain infrared waves that can interfere with communication.**

Activate Windows
Go to Settings to act



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Key Differences Between Guided and Unguided Media

1. The key difference between guided and unguided media is that guided media uses a **physical path or conductor** to transmit the signals whereas, the unguided media **broadcast the signal** through the air.
2. The guided media is also called **wired communication** or **bounded transmission media**. However, the unguided media is also called **wireless communication** or **unbounded transmission media**.
3. The guided media provide **direction** to the signal whereas, the unguided media **does not direct** the signal.
4. Categories of guided media are **twisted pair cable, coaxial cable** and **optical fibre**. On the other hands, the categories of unguided media are **radio wave, microwave,** and **infrared signal**.

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CS601 Lecture No 21 Topic 114 to 123

Switching

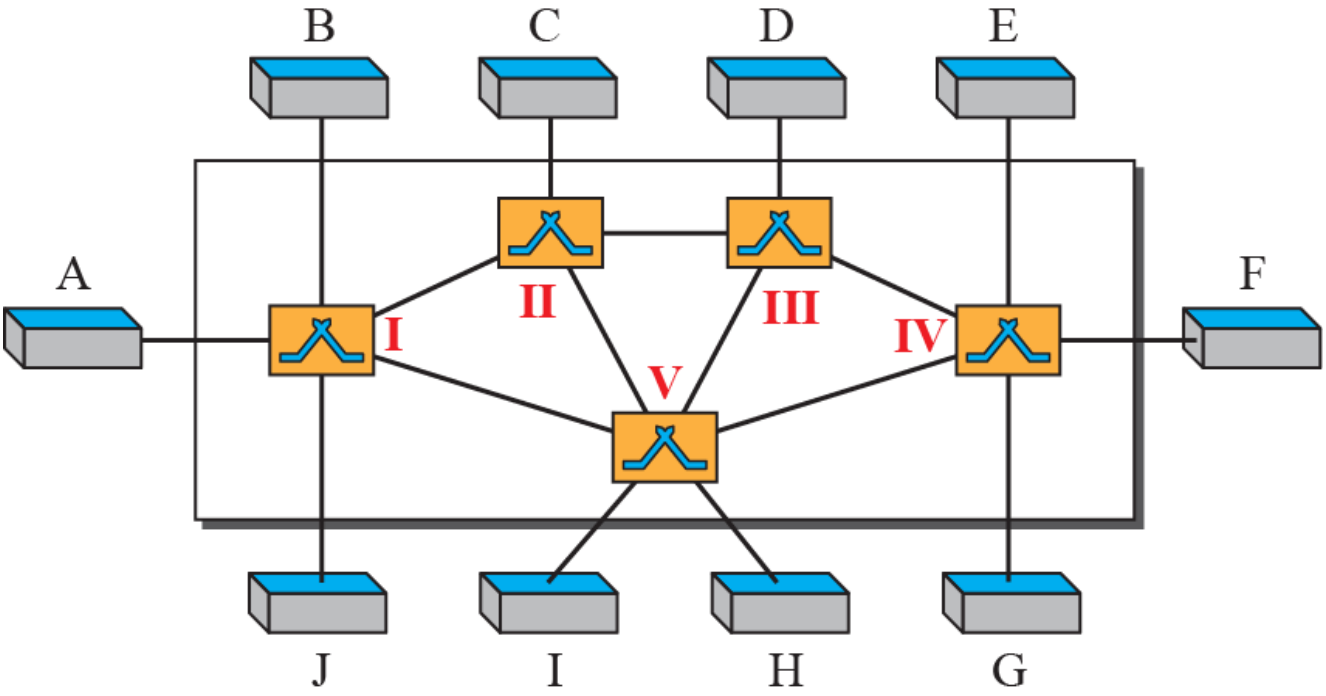
- **A network is a set of connected devices**
- **Problem of how to connect multiple devices to make one-to-one communication possible**

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- **The solution is Switching**
- **Switched network consists of a series of switches**



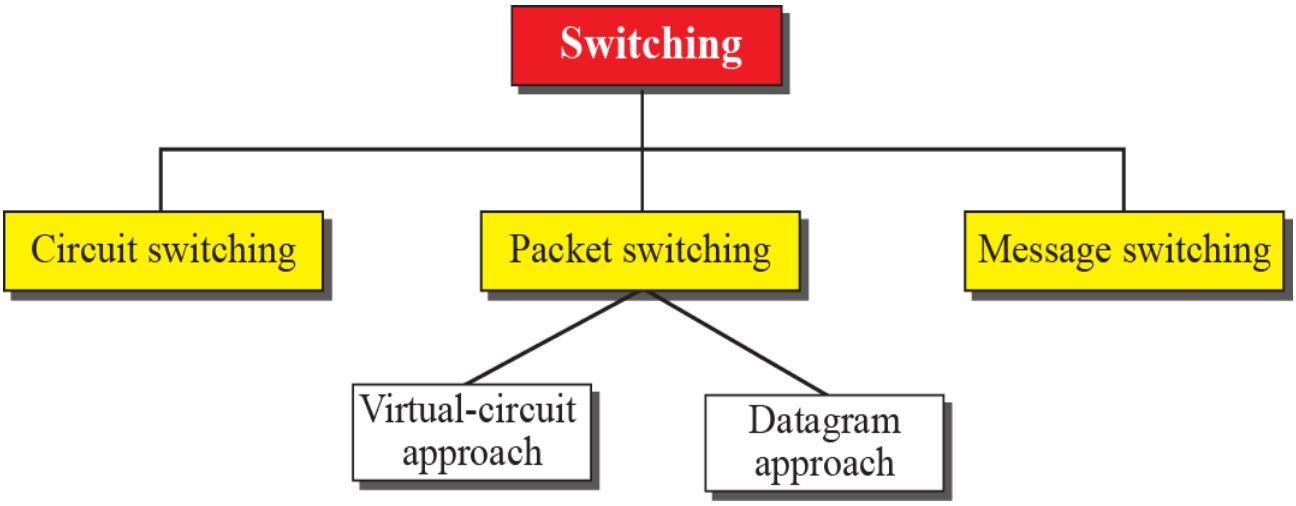
Three Methods of Switching

- **Three Methods:**
  - ✓ **Circuit Switching**
  - ✓ **Packet Switching**
  - ✓ **Message switching**

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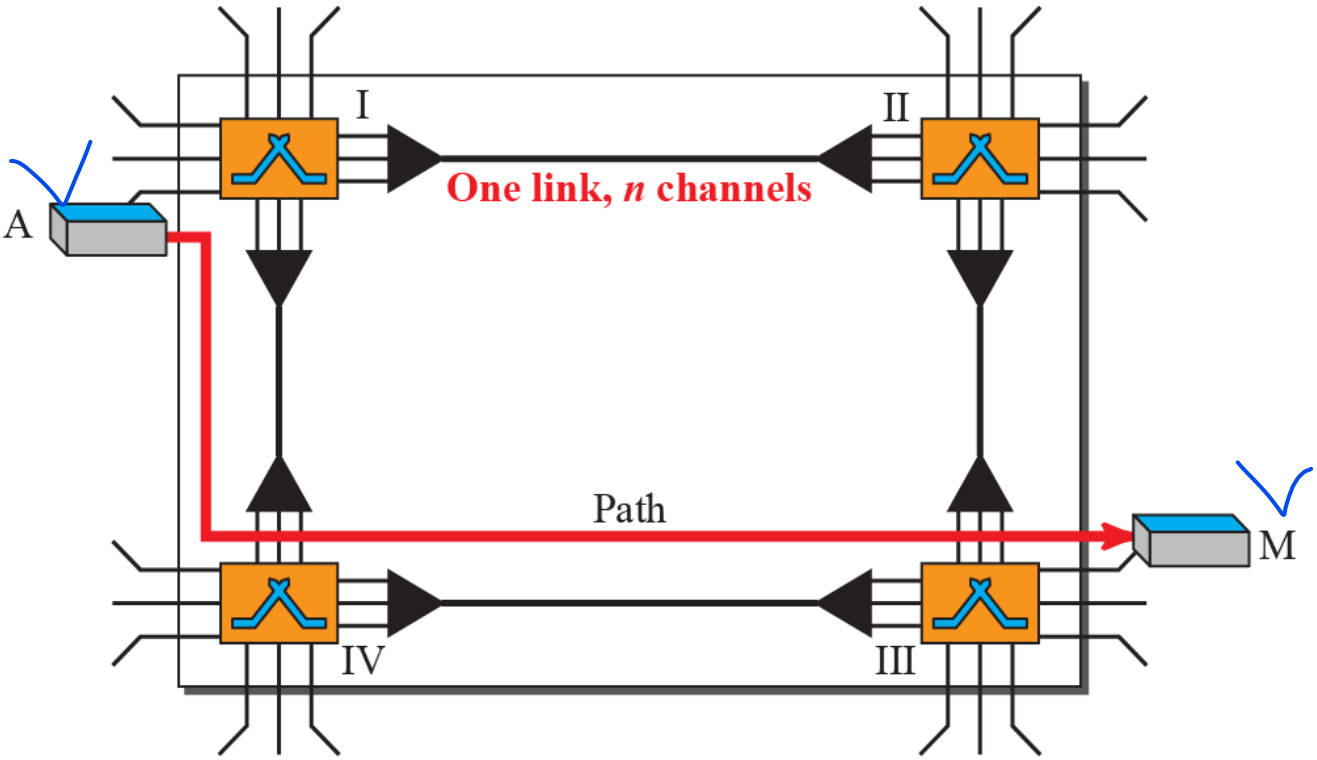
Circuit-switched Networks

- **A set of switches connected by physical links**
- **A connection between two stations is a dedicated path made of one or more links**
- **Each connection uses only one dedicated channel on each link**
- **Each link is normally divided into n channels by using FDM or TDM**

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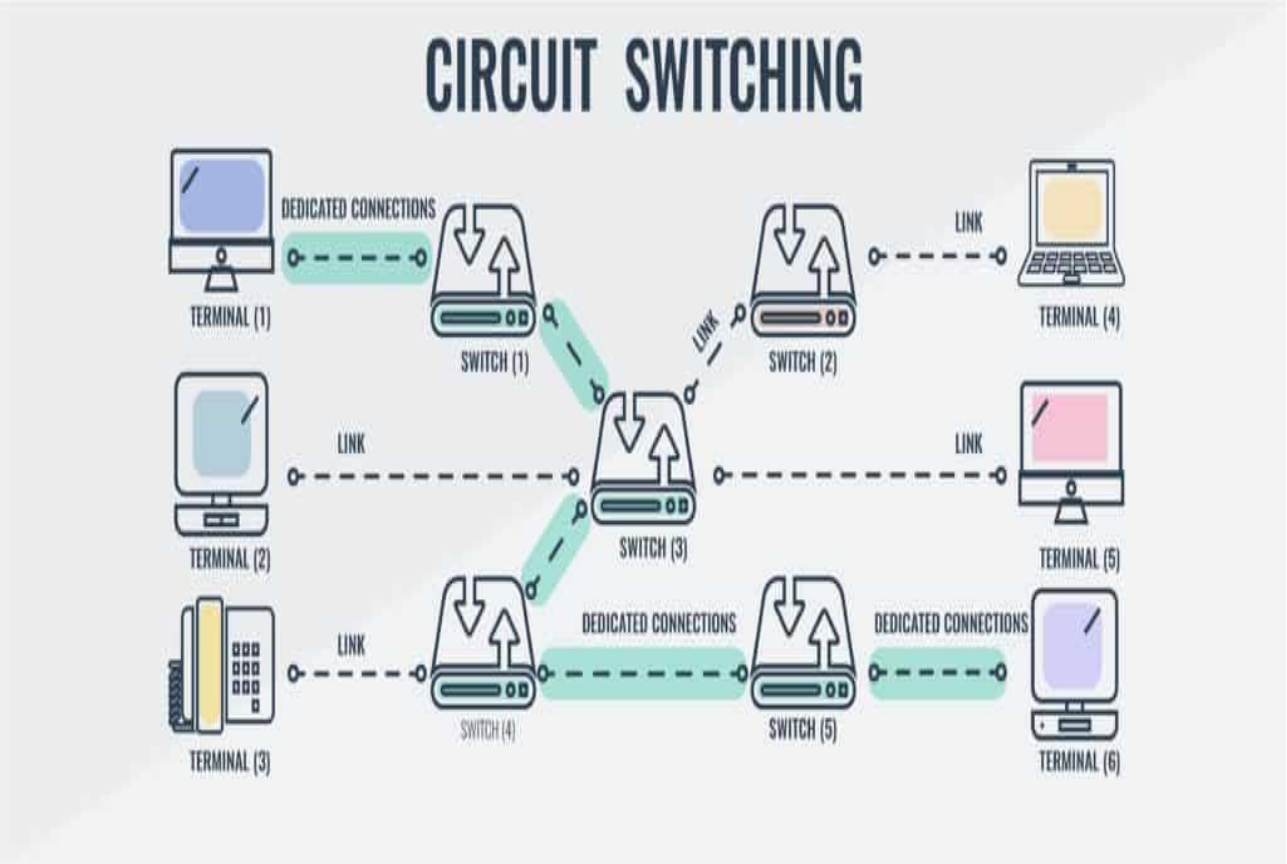
Which is a good example of a circuit-switched network?

Circuit switching is used for connections that must be continuous for long periods of time, such as long-distance communication. **Traditional telephone systems -- i.e., landlines --** are an example of a technology that uses circuit switching. Dial-up network connections.

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Three Phases in a Circuit Switched Network

- **The actual communication in a circuit-switched network requires 3 phases:**
  - ✓ **Connection Setup**
  - ✓ **Data Transfer**
  - ✓ **Connection Teardown**

Efficiency of a Circuit-Switched Network

- **Not as efficient as packet switching because resources are allocated during the entire duration of the connection and these resources are unavailable to other connections**

Efficiency of a Circuit-Switched Network

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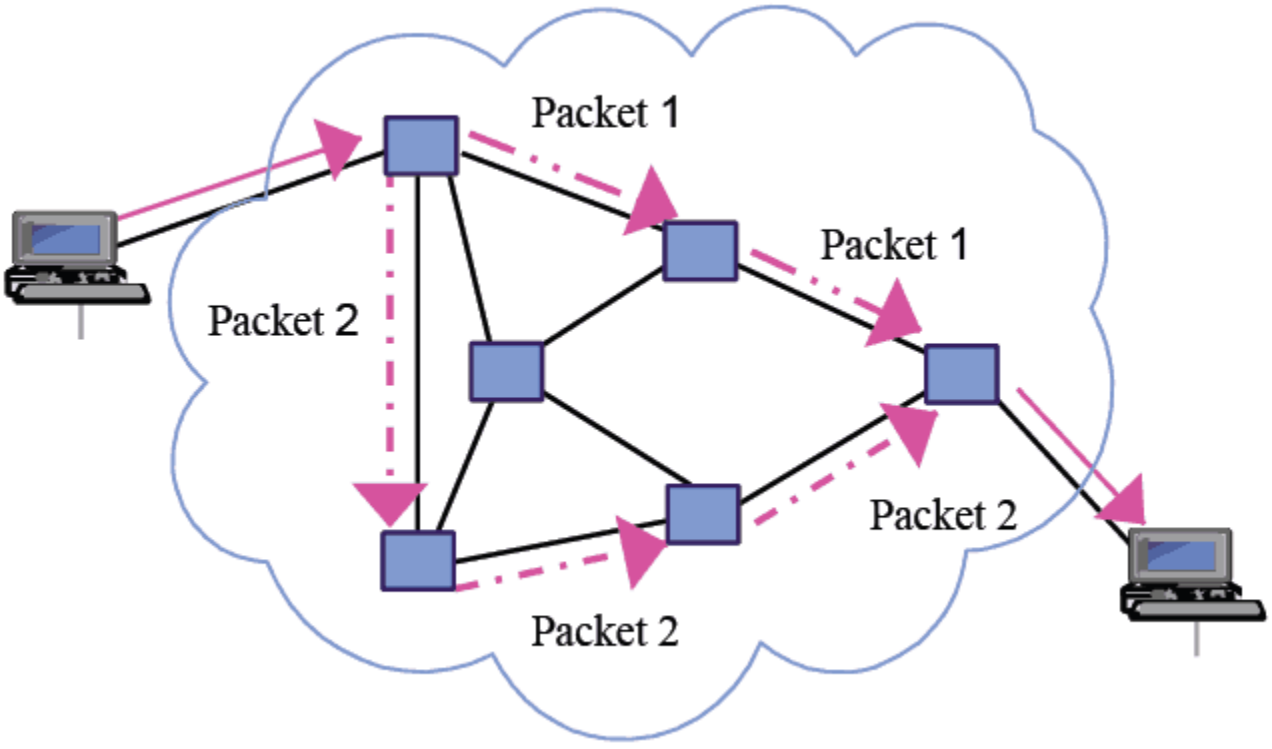
- **In a telephone network, people normally terminate the communication when they have finished their conversation**
- **Data Network is an issue**

Delay in a Circuit-Switched Network

- **Circuit switched networks have low efficiency but minimal delay**
- **Data is not delayed at each switch; the resources are allocated for the duration of the connection**

Packet Switching

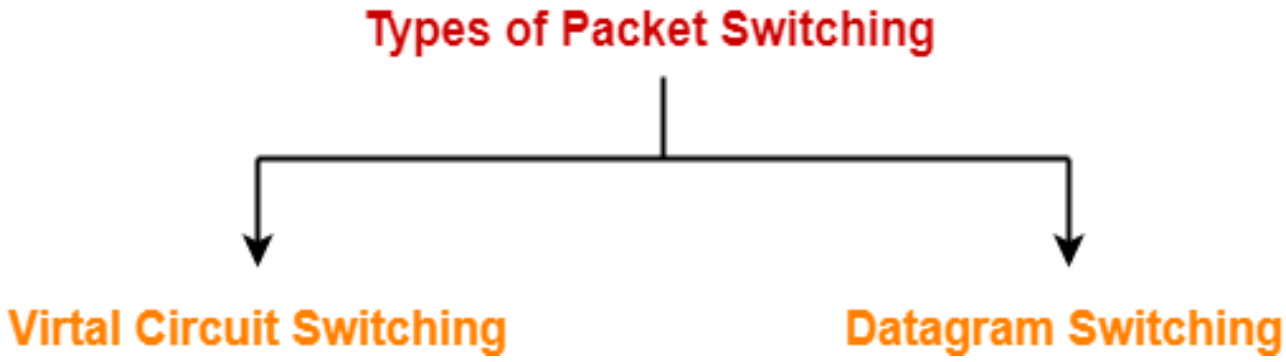
- **If the message is going to pass through a packet-switched network, it needs to be divided into packets of fixed or variable size**
- **The size of the packet is determined by the network and the governing protocol**



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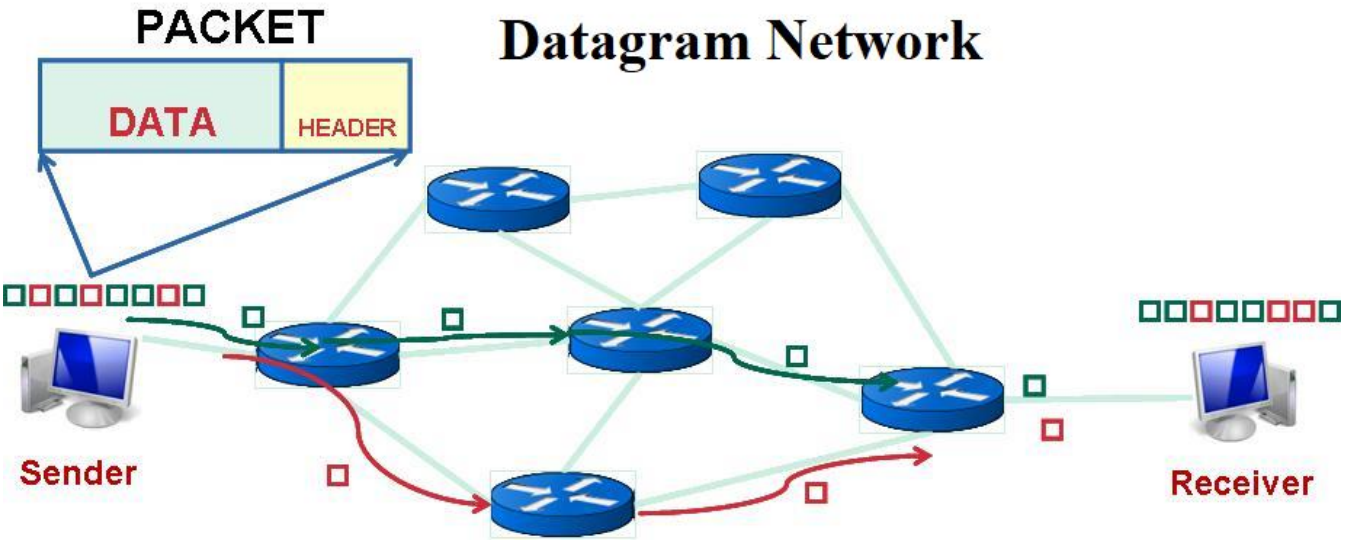
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Datagram Networks

- **Each packet is treated independently of all others.**
- **Even if a packet is part of a multi-packet transmission, the network treats it as though it existed alone**
- **Packets are referred to as datagrams**



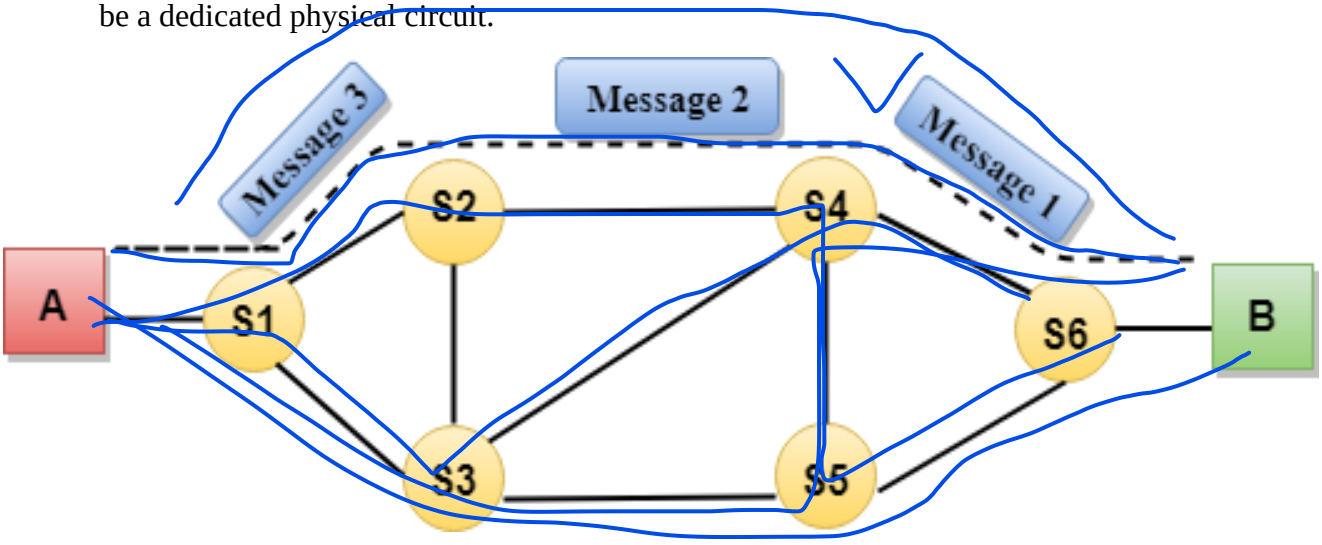
Virtual-Circuit Networks

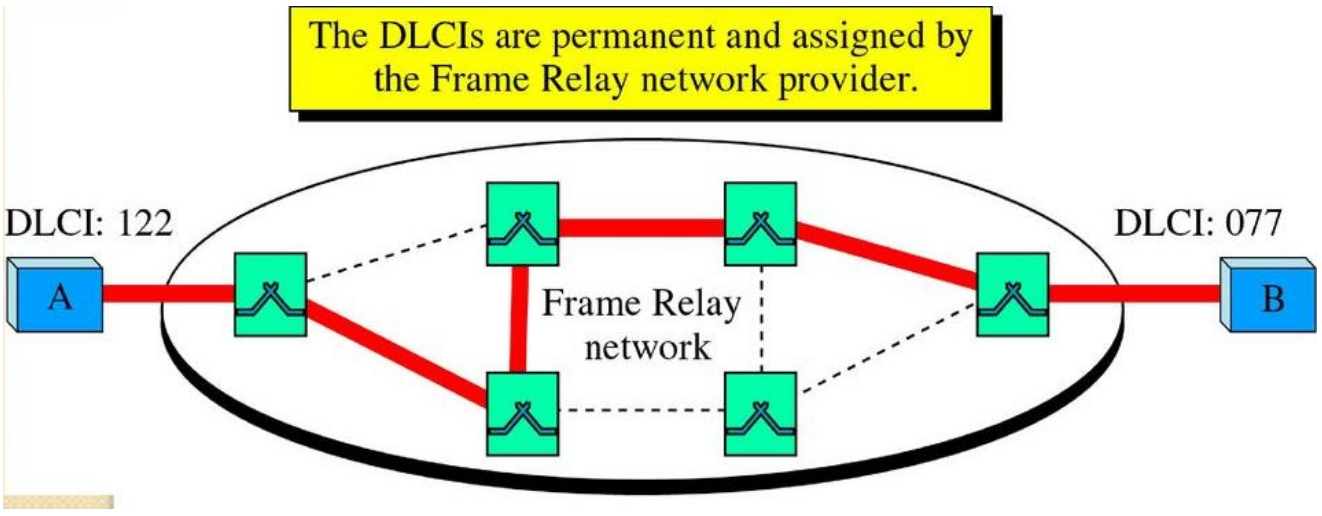
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- **A virtual-circuit network is a cross between a circuit-switched network and a datagram network**
- Virtual circuit switching is **a packet switching methodology whereby a path is established between the source and the final destination through which all the packets will be routed during a call**. This path is called a virtual circuit because to the user, the connection appears to be a dedicated physical circuit.





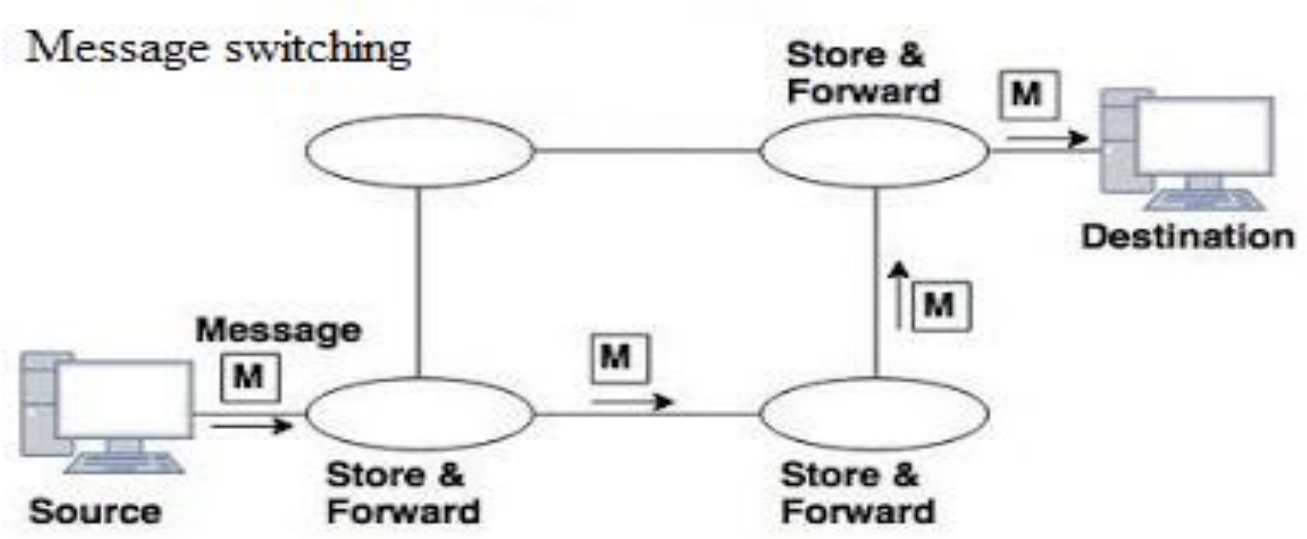
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Message switching

Message switching is **a network switching technique in which data is routed in its entirety from the source node to the destination node, one hope at a time**. During message routing, every intermediate switch in the network stores the whole message.



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CS601 Lecture No 21 Topic 124 to 128

Data Link Layer (Layer 2)

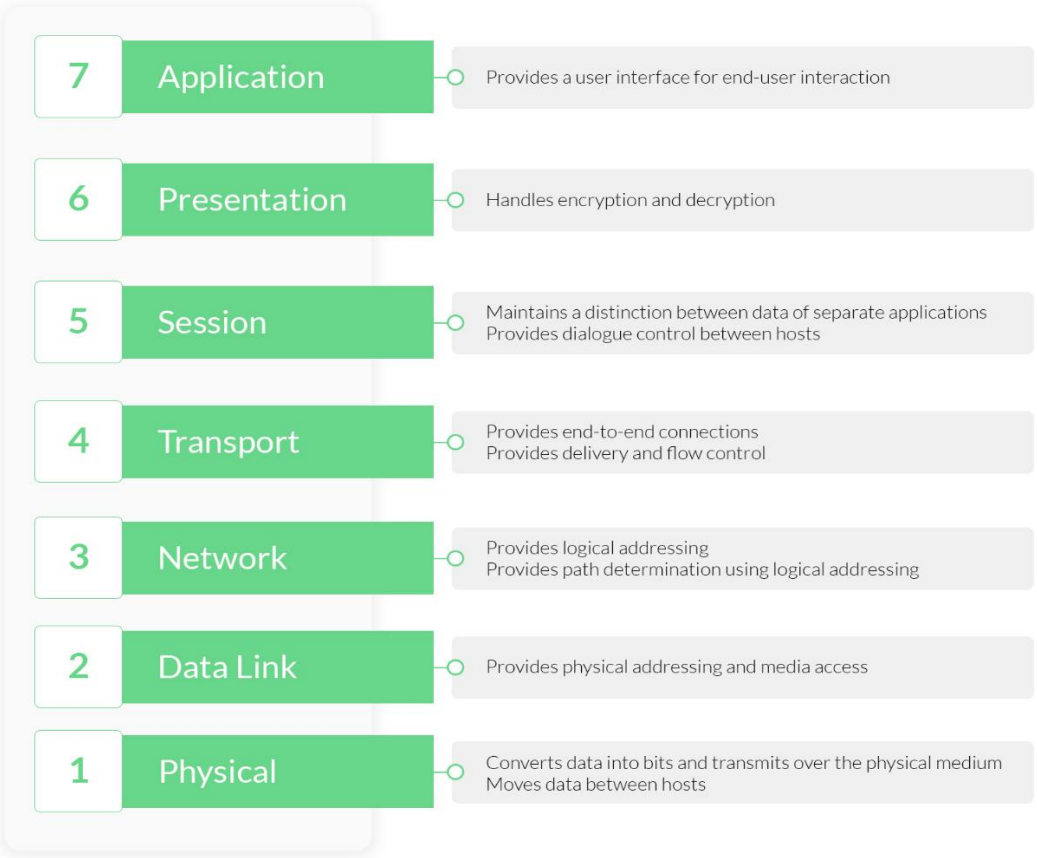
- The Internet is a combination of networks glued together by connecting devices (routers or switches)
- If a packet is to travel from a host to another host, it needs to pass through these networks

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- Data Link layer controls node-to-node communication

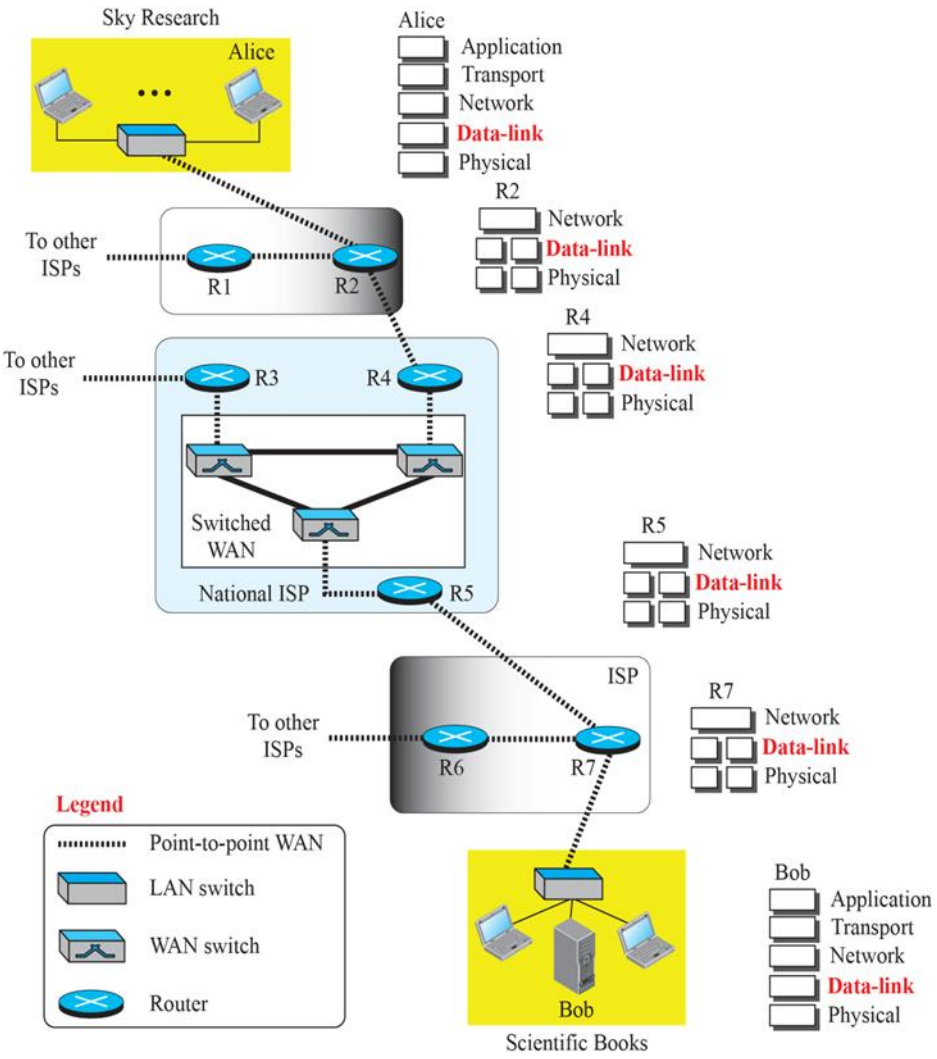


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Nodes and Links

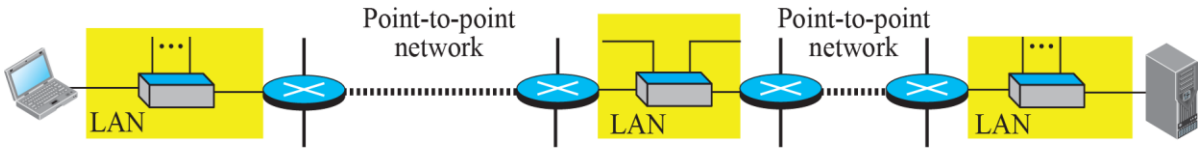
- **Communication at the data-link layer is node-to-node**
- **A data unit from one point in the Internet needs to pass through many networks (LANs and WANs) to reach another point**
- **We refer to the two end hosts and the routers as nodes and the networks in between as links**

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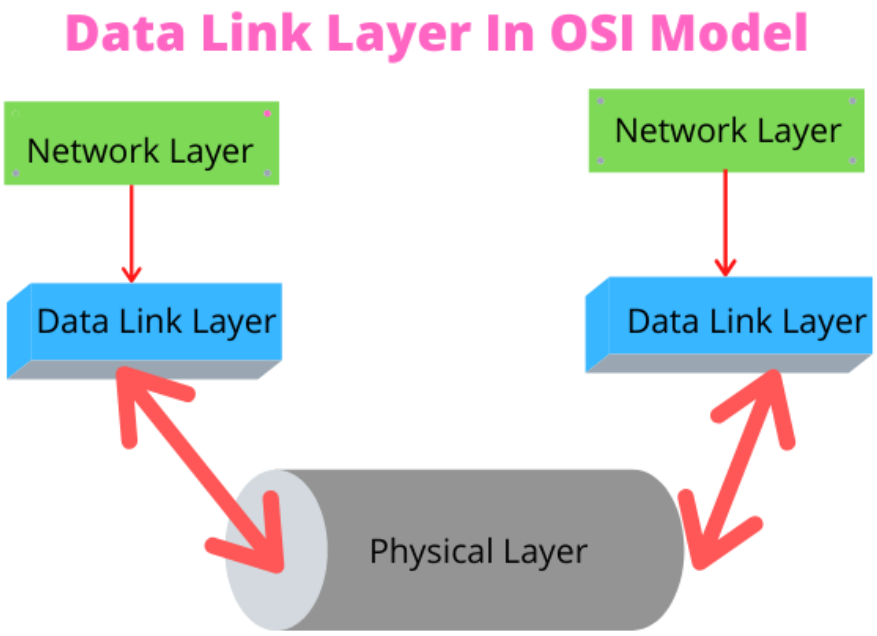
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a. A small part of the Internet



b. Nodes and links



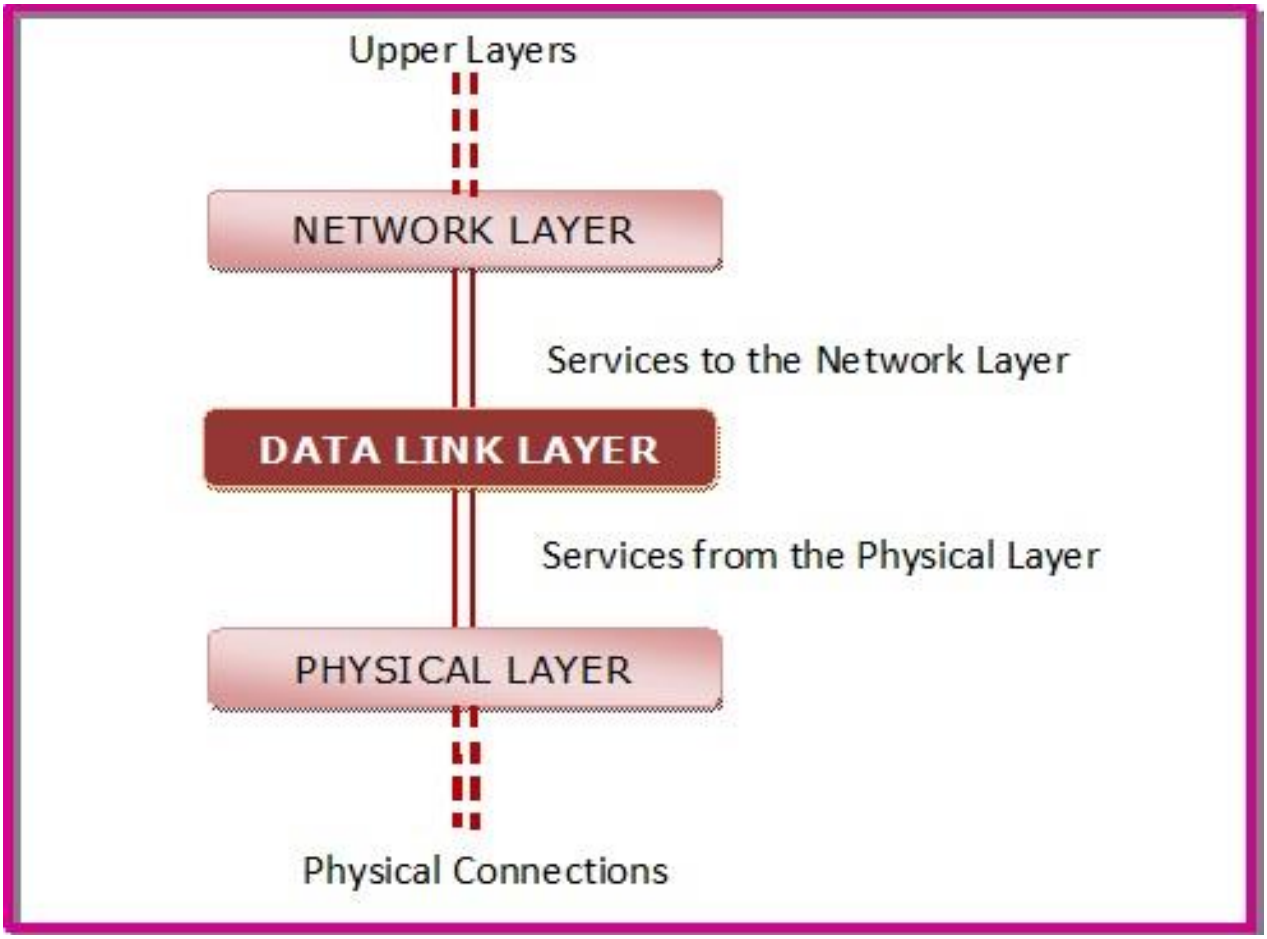
Services provided by Data-Link Layer

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- **Located between the physical and the network layers**
- **Provides services to Network Layer and receives services from Physical layer**
- **Framing**
- **Flow Control**
- **Error Control**
- **Congestion Control**



Framing

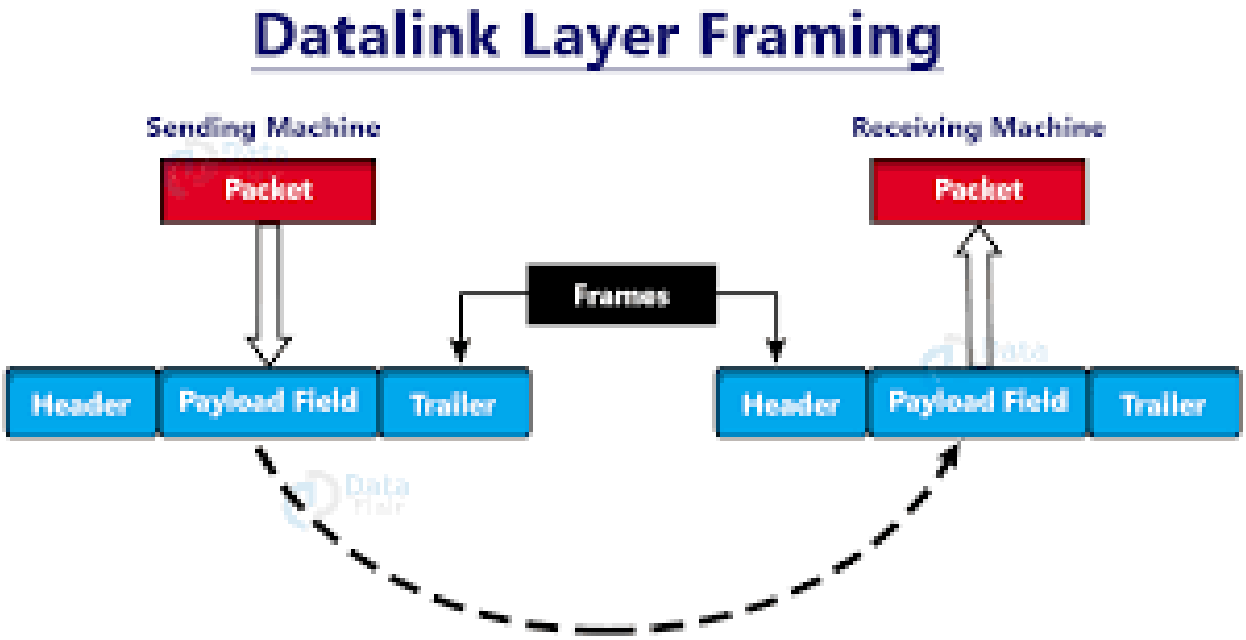
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Framing in data link layer is a point-to-point connection between the sender and receiver. The framing is the primary function of the data link layer and it provides a way to transmit data between the connected devices. Framing uses frames to send or receive data.

The data link divides the stream of bits from Network layer into manageable data units called "FRAMES". This process is known as Framing.



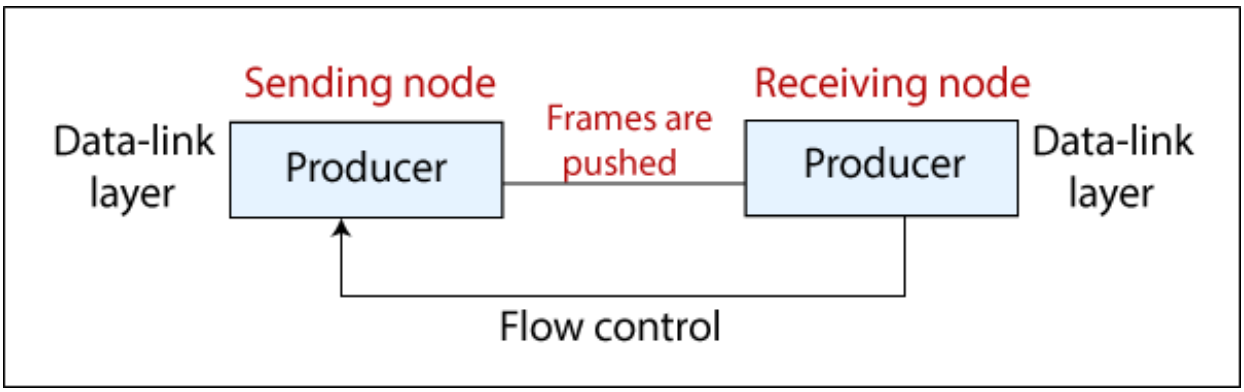
Flow Control

Flow control is **a technique that allows two stations working at different speeds to communicate with each other**. It is a set of measures taken to regulate the amount of data that a sender sends so that a fast sender does not overwhelm a slow receive

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Error Control

Error control in data link layer is the process of detecting and correcting data frames that have been corrupted or lost during transmission. In case of lost or corrupted frames, the receiver does not receive the correct data-frame and sender is ignorant about the loss.

There are different techniques for detecting errors in the transmitted data:

(a)Parity Checks: Parity check deploys single parity bit and is considered the simplest form of error detection method. It comprises of even parity and odd parity methods.

(b)Checksum: It is a simple type of redundancy check that is used to detect error. Through this method, an algorithm calculates the binary values in a packet or other block of data and stores the results with the data.

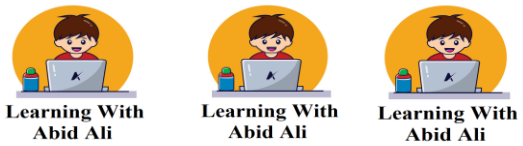
(c)Cyclic Redundancy Check: In the TCP/IP or OSI reference model, CRC is added to a packet frame at the data link layer. It is a method of checking for errors in data that have been transmitted on a communications link.

(d) Hamming code: It is an error-detecting and error-correcting binary code.

Congestion Control

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Congestion control in data link layer **occurs when a node or link carries data beyond its limit**. This often leads to the queuing of packets—and in the worst case, loss of packets—as well as a decrease in the network's Quality of Service (QoS).

Two Categories of Links

- **We can have the following two types of links:**
  - ✓ **Point-to-point link or a**
  - ✓ **Broadcast link**

Point-to-point mode of communication

Point to point communication mode is the communication over a link between a single transmitter and a receiver.

Broadcast mode of communication

In broadcast mode large number of receivers are linked to a single transmitter.

Two Sublayers of Data-Link Layer

- **We can divide the data-link layer into two sublayers:**
  - **Data Link Control (DLC) Point-to-point link**
  - ✓ **Media Access Control (MAC) Broadcast link**

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CS601 Lecture No 23 Topic 129 to 134

Why LINK-LAYER ADDRESSING?

- **IP addresses are the identifiers at the network layer**
- **In Internet we cannot make a packet reach its destination using only IP addresses**

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- **Source and destination IP addresses define the two ends but cannot define which links the packet will take**

2.2 Link Layer Addressing

- A Link layer address is also called as **Link address or physical address or MAC address.**
- In a connectionless internetwork such as the Internet we cannot make a datagram reach its destination using only IP addresses. The reason is that each datagram in the Internet, from the same source host to the same destination host, may take a different path. The source and destination IP addresses define the two ends but cannot define which links the datagram should pass through.

2.2.1 Working Operation

- When a datagram passes from the network layer to the data link layer, then the datagram will be encapsulated in a frame and two data link addresses are added to the frame header.
- These addresses are changed every time the frames moves from one link to another.
- Each frame carries the same datagram with the same source and destination IP addresses.. The figure below demonstrates the concepts in a small internet:

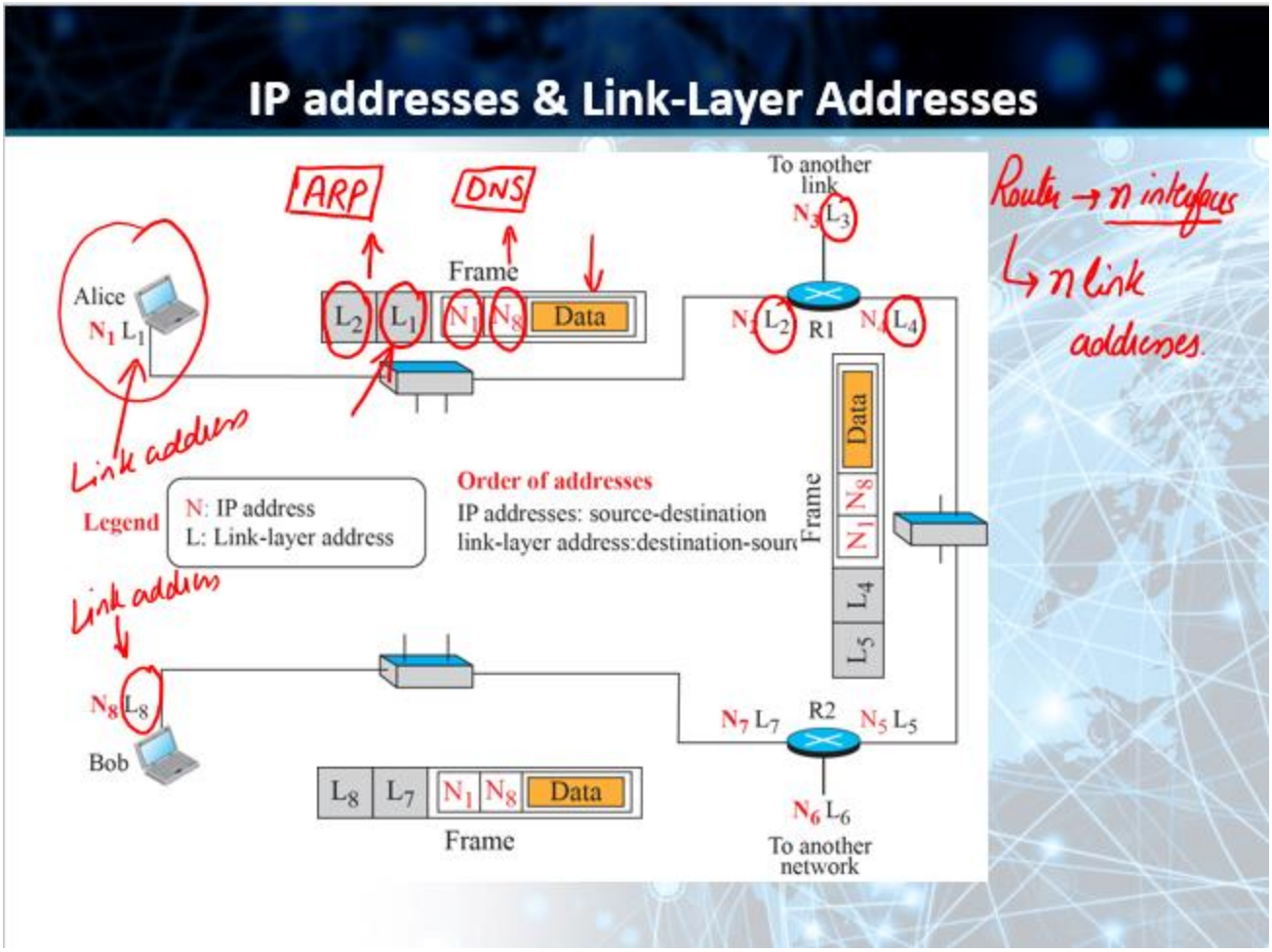
What is link layer addressing?

A link-layer address is called **a link address, called a physical address, and sometimes a MAC address**. Since a link is controlled at the data-link layer, the addresses need to belong to the data-link layer.

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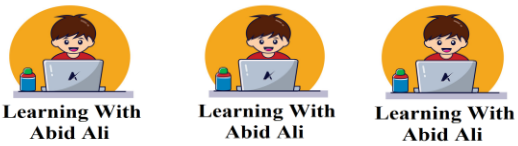


Three Types of addresses

- **Some link-layer protocols define three types of addresses:**
- **Unicast**
- **Multicast**
- **Broadcast**

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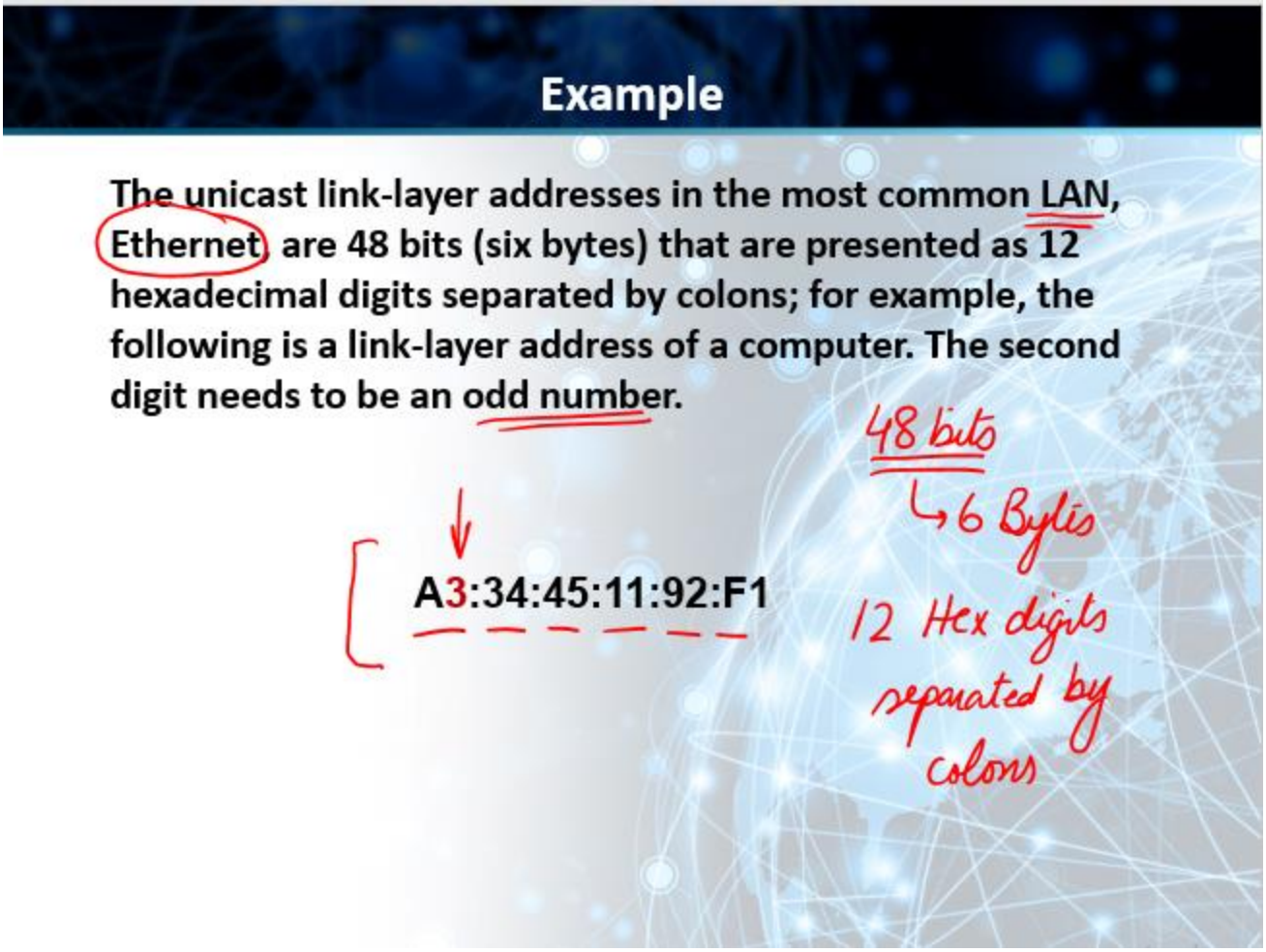
2.2.2 Types of Link layer Addresses

- Some link-layer protocols define three types of addresses: unicast, multicast, and broadcast.
- **Unicast:** Each host or each interface of a router is assigned a unicast address. Unicasting means one-to-one communication. A frame with a unicast address destination is destined only for one entity in the link.
- The unicast link-layer addresses in the most common LAN, Ethernet, are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons; for example, the following is a link-layer address of a computer. The second digit needs to be an odd number.
- A3:34:45:11:92:F1

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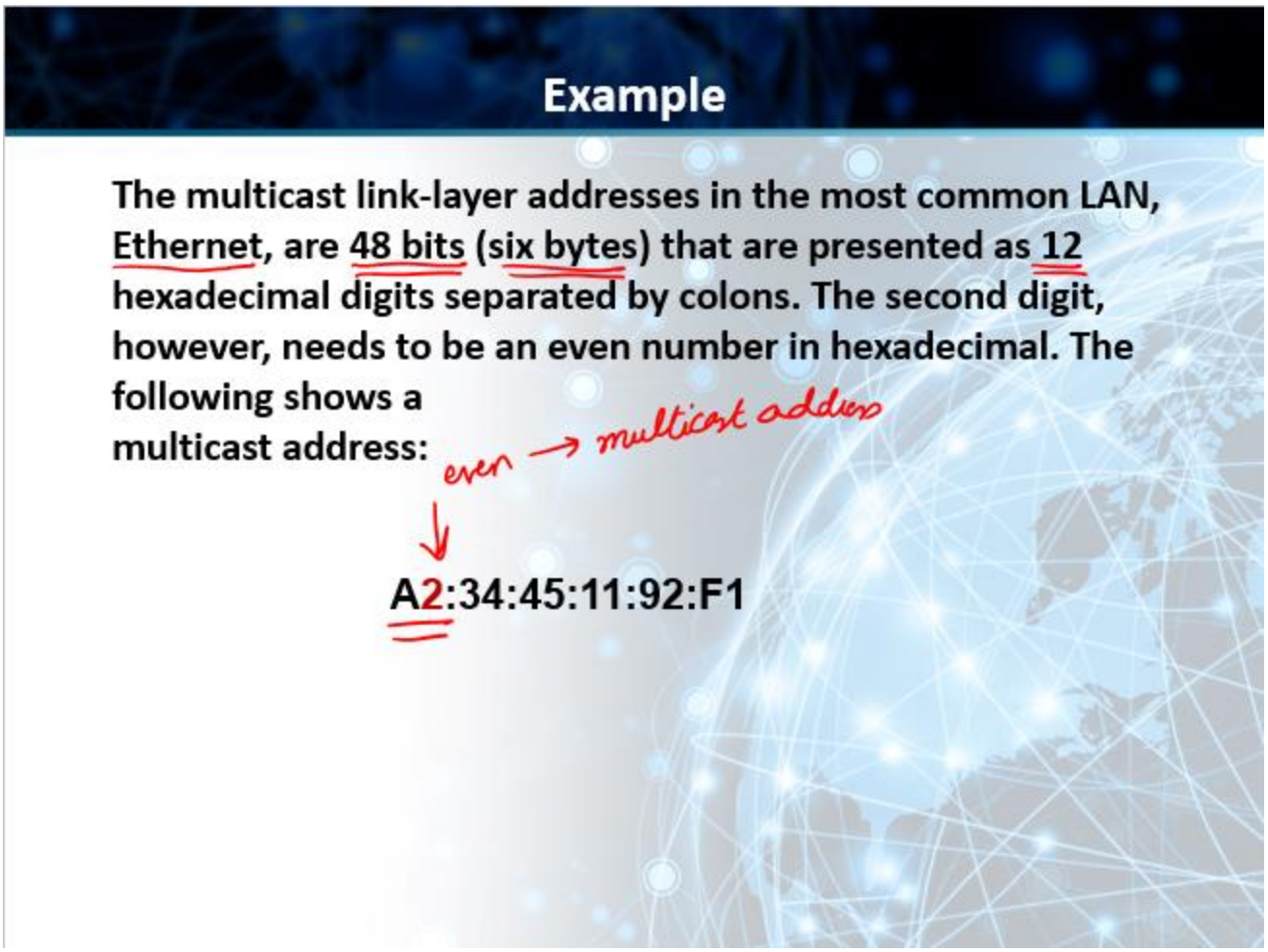
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- **Multicast**: one-to-many communication. However, the jurisdiction is local (inside the link).
- The multicast link-layer addresses in the most common LAN, Ethernet, are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons. The second digit, however, needs to be an even number in hexadecimal. The following shows a multicast address:
- A2:34:45:11:92:F1
- **Broadcast:** one-to-all communication. A frame with a destination broadcast address is sent to all entities in the link.
- The broadcast link-layer addresses in the most common LAN, Ethernet, are 48 bits, If all digits are Fs, then the address is broadcast all 1s, that are presented as 12 hexadecimal digits separated by colons. The following shows a broadcast address:
- FF:FF:FF:FF:FF:FF

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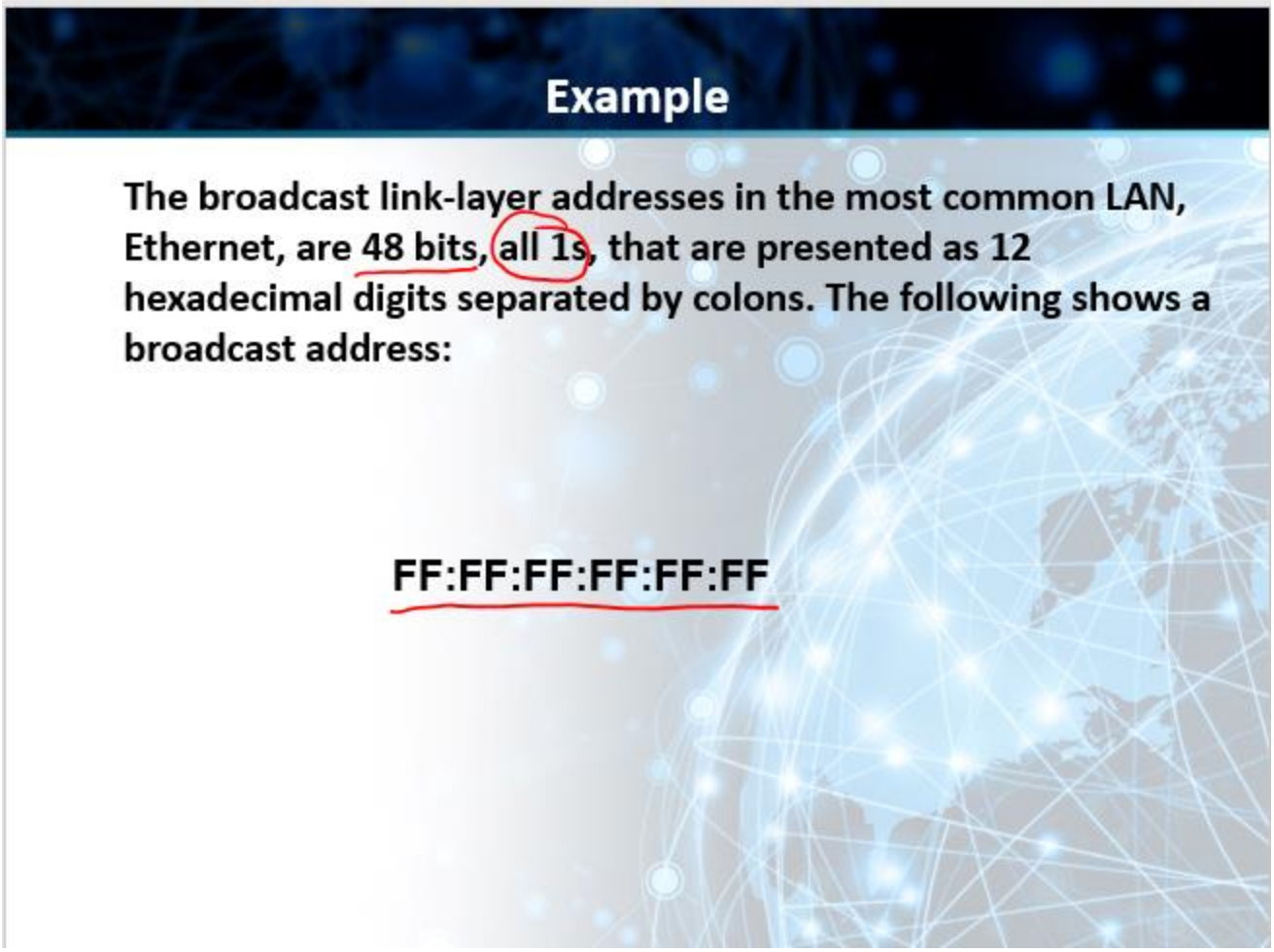
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Address Resolution Protocol (ARP)

- **Anytime a node has an IP packet to send to another node in a link, it has the IP address of the receiving node**
- **IP address of the next node is not helpful in moving a frame through a link; we need the link-layer address of the next node**

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ARP operation

- When a host or a router needs to find the link-layer address of another host or router in its network, A host or a router sends an ARP request packet that includes the link-layer and IP addresses of the sender and the IP address of the receiver. Because the sender does not know the link-layer address of the receiver, the query is broadcast over the link using the link-layer broadcast address.
- Every host or router on the network receives and processes the ARP request packet, but only the intended recipient recognizes its IP address and sends back an ARP response packet. The response packet contains the recipient's IP and link-layer addresses. The packet is unicast directly to the node that sent the request packet.

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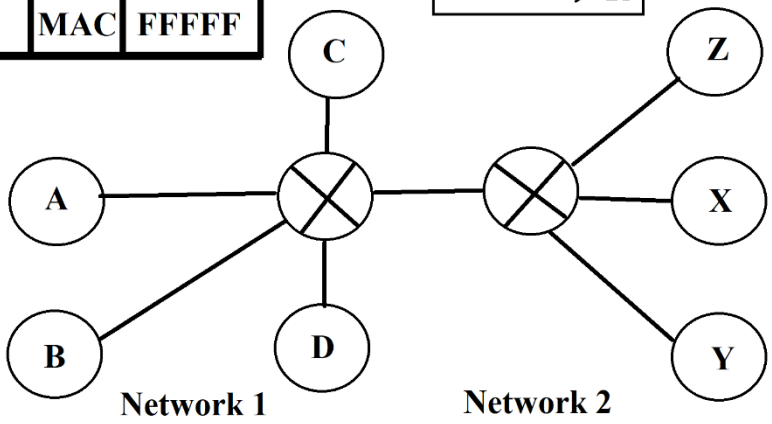
Address Resolution Protocol (ARP)

IP Adress ⟶ Logical

Mac Adress ⟶ Physical

H	H
H	R
R	H
R	R

IP	MAC	FFFFF



6	48	FFFFF OR 00000000

Hardware: LAN or WAN protocol

Protocol: Network-layer protocol

0	8	16	31
Hardware Type		Protocol Type	
Hardware length	Protocol length	Operation Request:1, Reply:2	
Source hardware address 6			
Source protocol address 48			
Destination hardware address (Empty in request) empty			
Destination protocol address			

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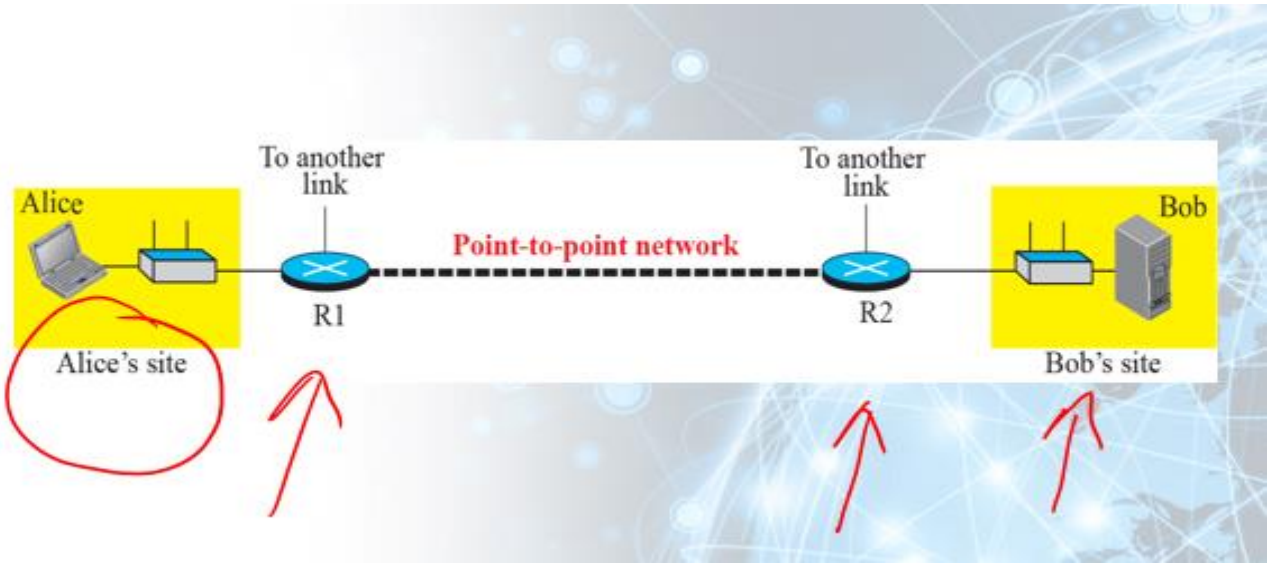
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- **ARP packet**
- The hardware type field defines the type of the link-layer protocol; Ethernet is given the type 1.
- The protocol type field defines the network-layer protocol: IPv4 protocol is $(0800)_{16}$.
- Hardware Length defines the physical address length in bytes
- Protocol Length defines the logical address length in bytes
- The source hardware and source protocol addresses are variable-length fields defining the link-layer and network-layer addresses of the sender.
- The destination hardware address and destination protocol address fields define the receiver link-layer and network-layer addresses.
- An ARP packet is encapsulated directly into a data-link frame.

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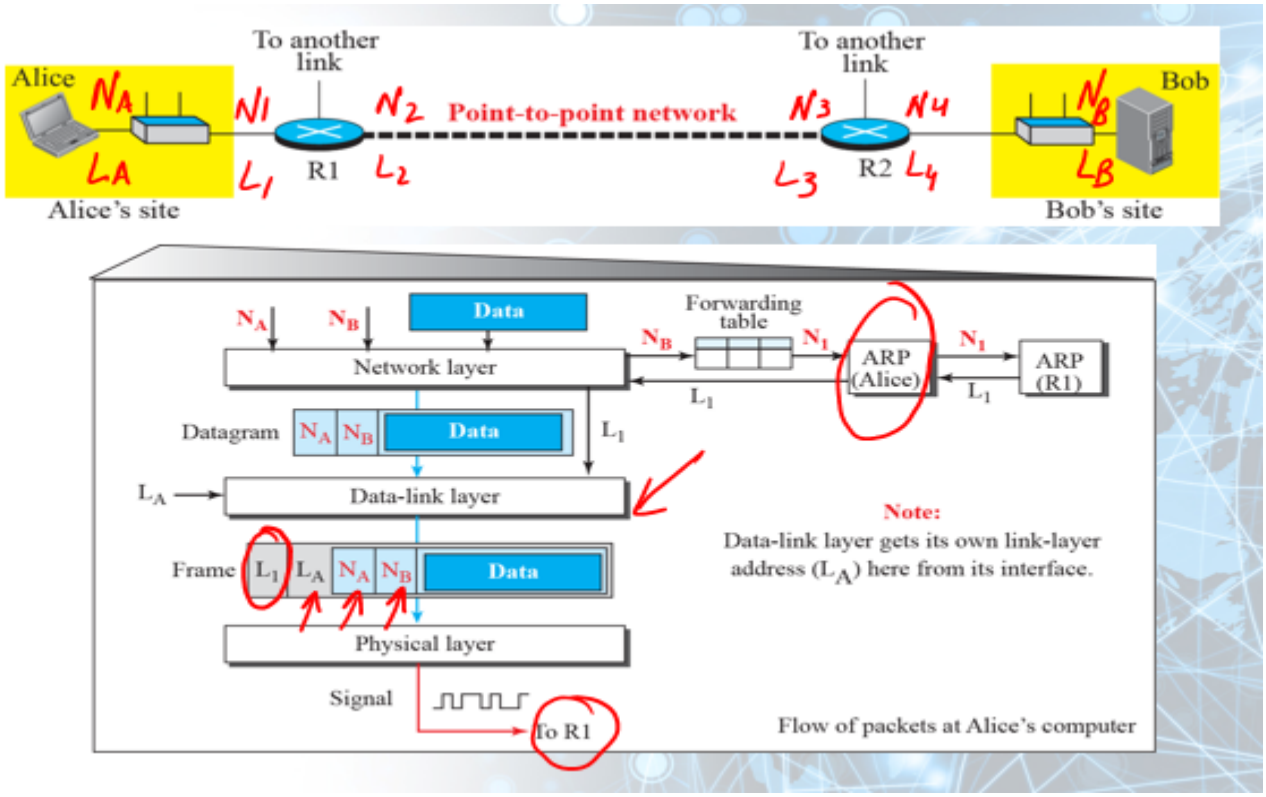


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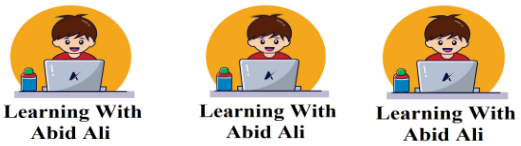
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Types of Errors

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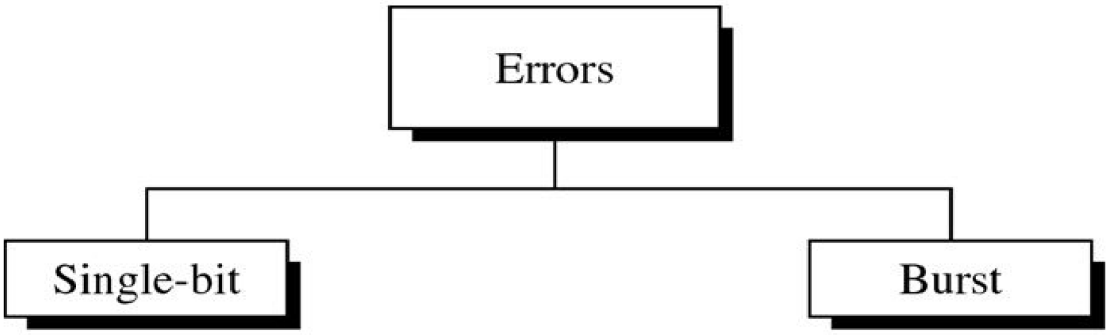
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Type of Errors

- An electromagnetic signal is subject to interference from heat, magnetism, and other forms of electricity
- Single-bit error: $0 \rightarrow 1$ or $1 \rightarrow 0$
- Burst error: 2 or more bits have changed



- **Data transmission suffers unpredictable changes because of interference**
- **The interference can change the shape of the signal**

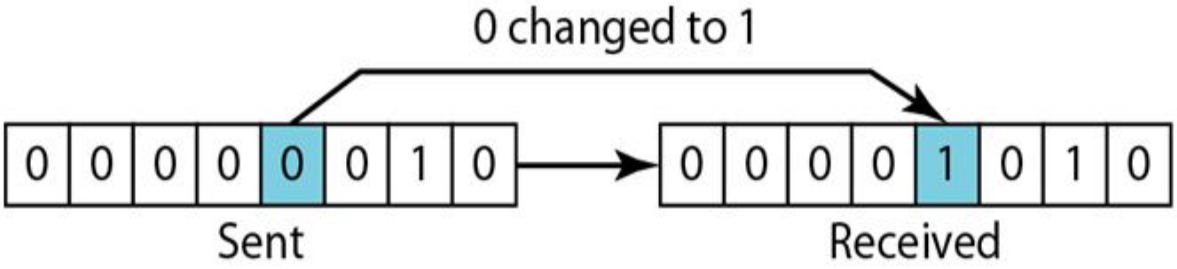
Single-bit error

Single-bit error means that only 1 bit of a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1

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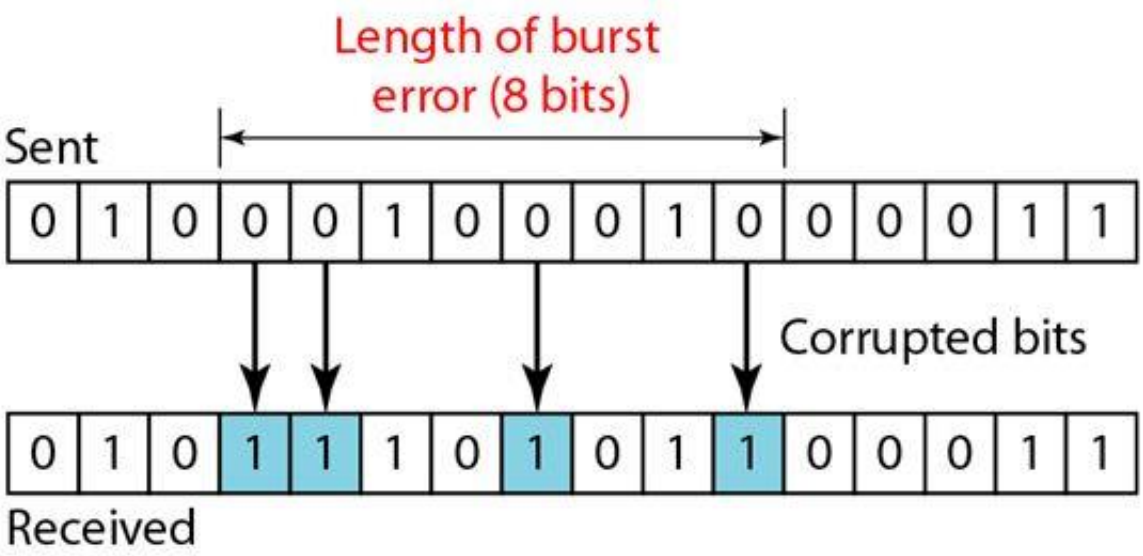
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Burst Error

Burst Error means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1



Number Of Impacted Bits Depend on Two Things

Data rate

Duration of rate

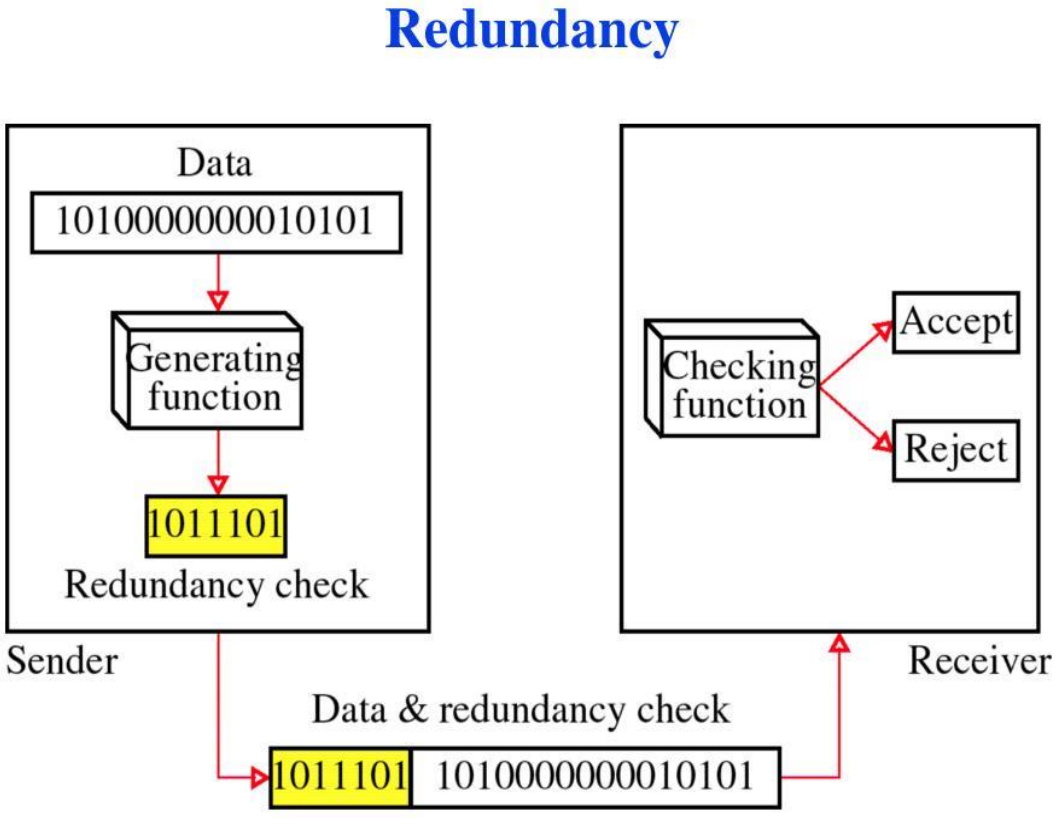
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Redundancy

- **Central concept in detecting or correcting errors is Redundancy**
- **To be able to detect or correct errors, we send some extra bits with our data**
- **The presence of these redundant bits allows the receiver to detect or correct corrupted bits**



Detection versus Correction

- **Correction is more difficult than the detection**
- **In error detection, we are only looking to see if any error has occurred (Yes or No)**

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- **We are not interested in the number of corrupted bits in Detection**
- **Single-bit error is same as a Burst error**
- In Error Correction, we need to know:
- The exact number of bits that are corrupted and,
- Their location in the message

Types of Coding Schemes

- **Coding schemes can be divided into 2 broad categories:**
  - ✓ **Block Coding**
  - ✓ **Convolution Coding**

Block Coding

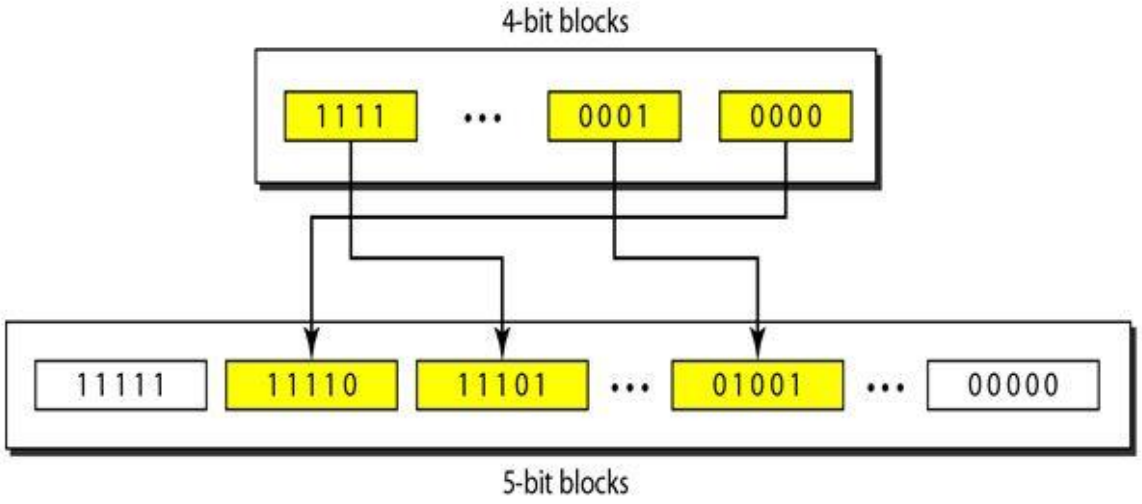
- **We divide our message into blocks, each of 'k' bits, called datawords**
- **We add 'r' redundant bits to each block to make the length 'n = k + r'**
- **The resulting 'n-bit' blocks are called codewords**

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BLOCK CODING in Error Detection

- **If the following two conditions are met, the receiver can detect a change in the original codeword:**
  - ✓ **The receiver has (or can find) a list of valid codewords**
  - ✓ **The original codeword has changed to an invalid one**

The sender creates code words out of data words by using a generator that applies the rules and procedures of encoding (discussed later). Each code word sent to the receiver may change during transmission. If the received code word is the same as one of the valid code words, the word is accepted; the corresponding data word is extracted for use.

If the received code word is not valid, it is discarded. However, if the code word is corrupted during transmission but the received word still matches a valid code word, the error remains undetected. This type of coding can detect only single errors. Two or more errors may remain undetected.

For example, consider the following table of data words and Code words:

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Datawords	*Codewords*
00	000
01	011
10	101
11	110

Assume the sender encodes the data word 01 as 011 and sends it to the receiver. Consider the following cases:

1. The receiver receives 011. It is a valid code word. The receiver extracts the data word 01 from it.

2. The code word is corrupted during transmission, and 111 is received (the leftmost bit is corrupted). This is not a valid code word and is discarded.

3. The code word is corrupted during transmission, and 000 is received (the right two bits are corrupted). This is a valid code word. The receiver incorrectly extracts the data word 00. Two corrupted bits have made the error undetectable.

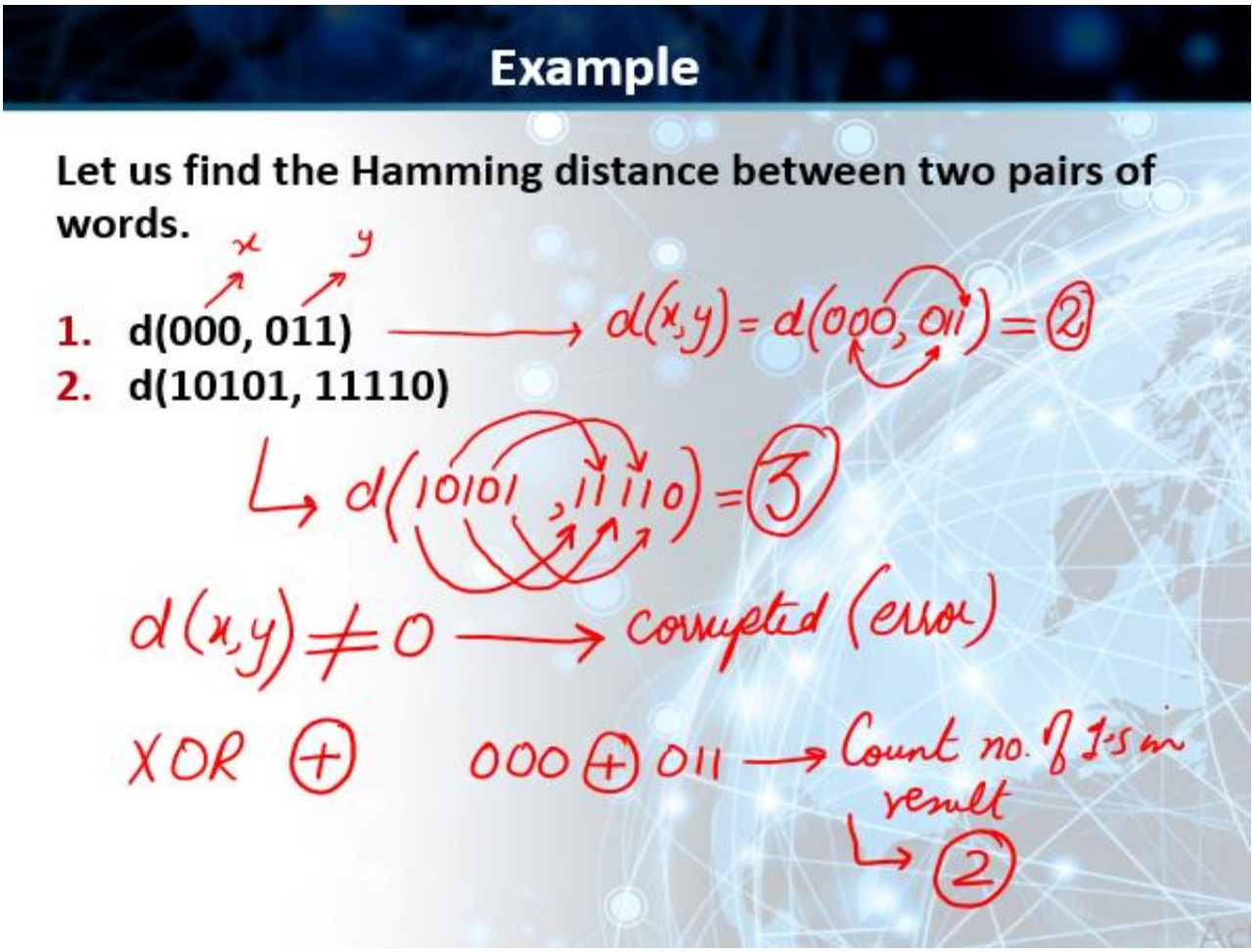
Hamming Distance

- **Hamming Distance between two words of the same size is the number of differences between the corresponding bits**
- **Hamming Distance between two words x and y is *d* (x, y)**
- **Hamming distance between received codeword and sent codeword is number of bits corrupted**

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Minimum Hamming Distance

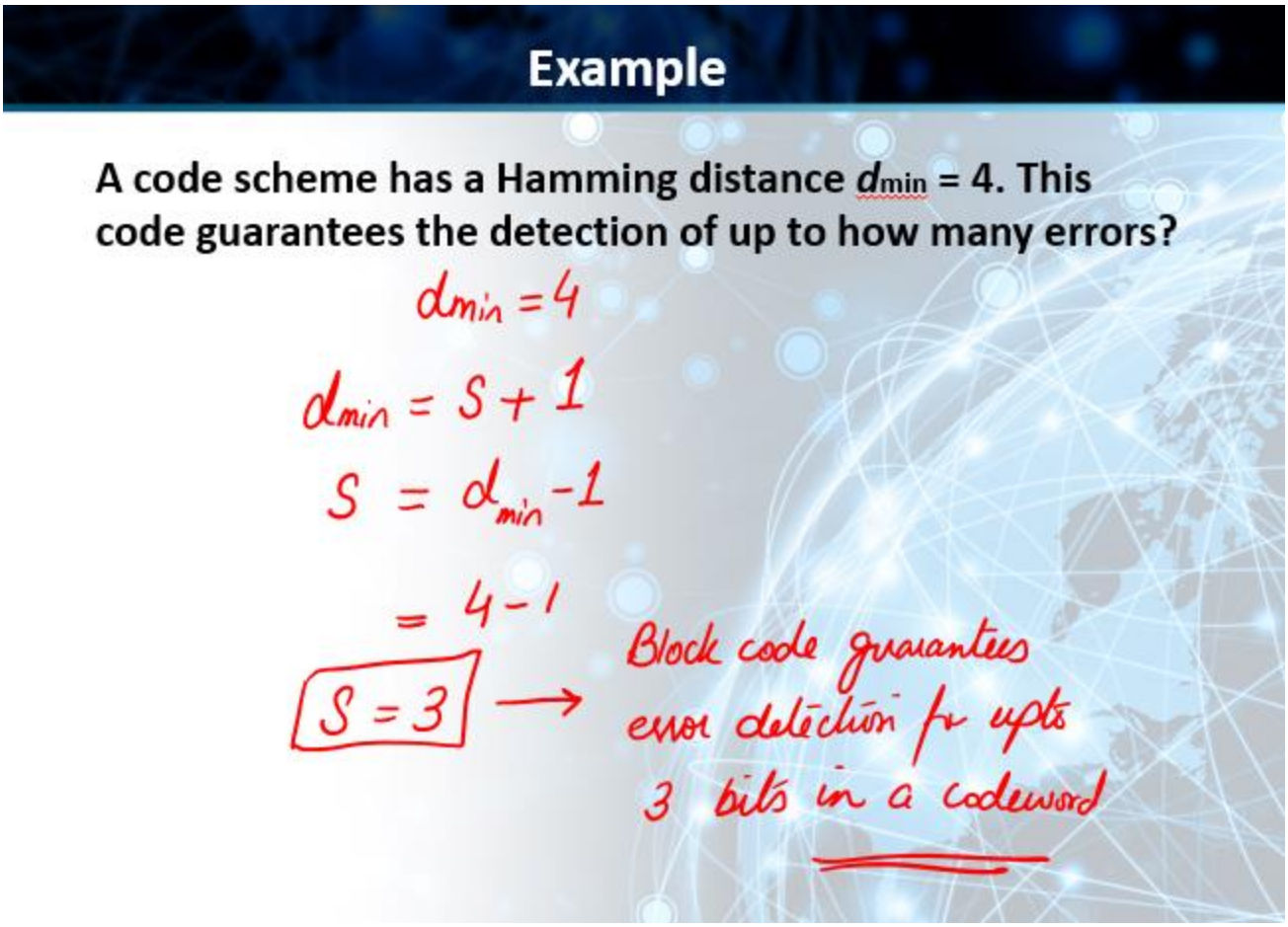
- **Minimum Hamming Distance is smallest hamming distance between all possible pairs of codewords**
- **dmin= s+1**

were, s →no. of detectable errors dmin→ minimum hamming distance

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CS601 Lecture No 25 Topic 143 to 148

Parity-Check Code

- **Most common error-detecting code**
- **Linear block code (n=k+1)**

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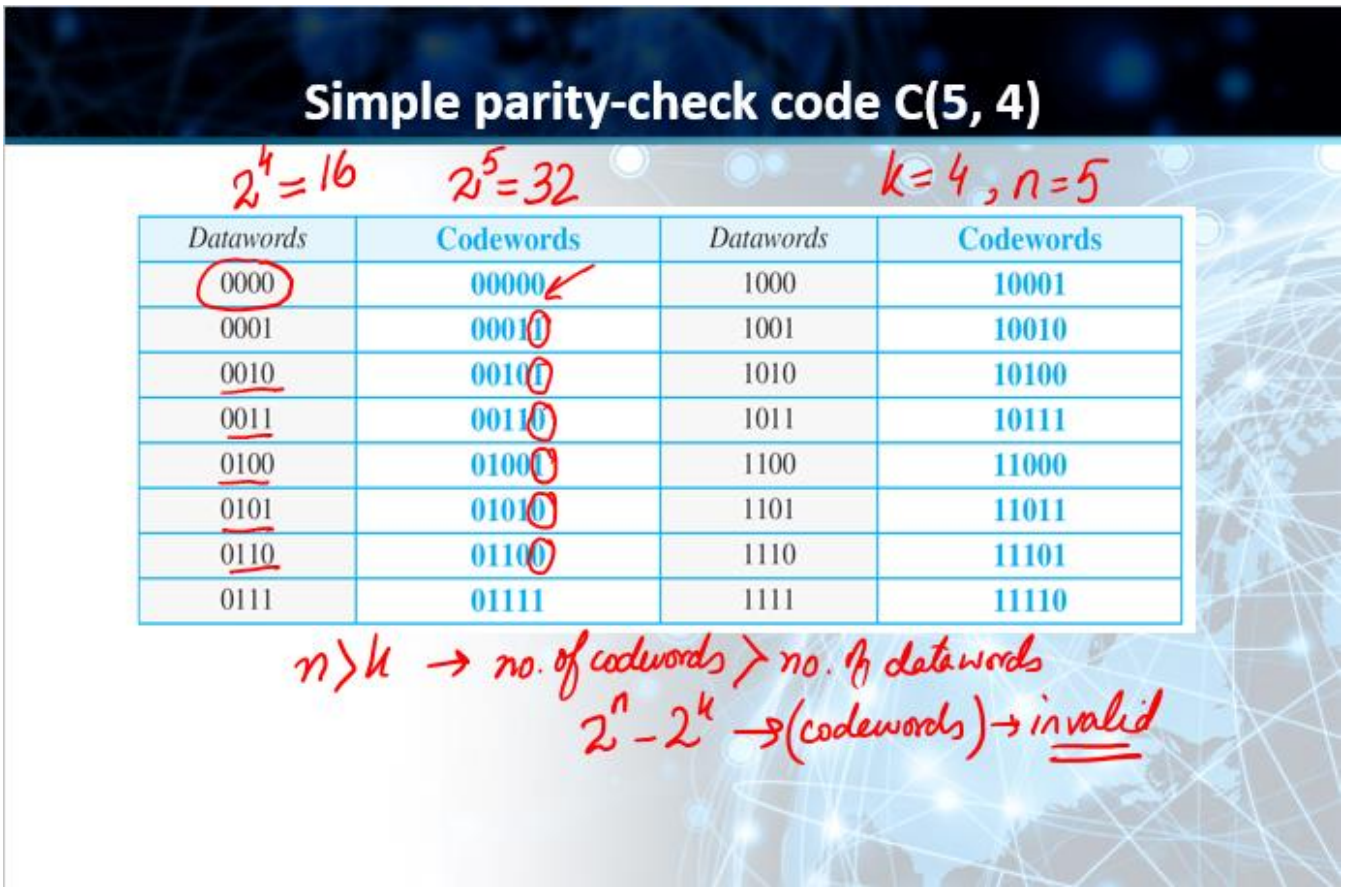
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- **The extra parity bit is selected to make total number of 1s in codeword even**

The simple parity-check code is the most familiar error-detecting code. In this code, **a k-bit data word is changed to an n-bit code word where n = k + 1**. The extra bit, called the parity bit, is selected to make the total number of 1s in the code word even.



Datawords	Codewords	Datawords	Codewords
0000	00000	1000	10001
0001	00011	1001	10010
0010	00101	1010	10100
0011	00110	1011	10111
0100	01001	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110

CYCLIC CODES

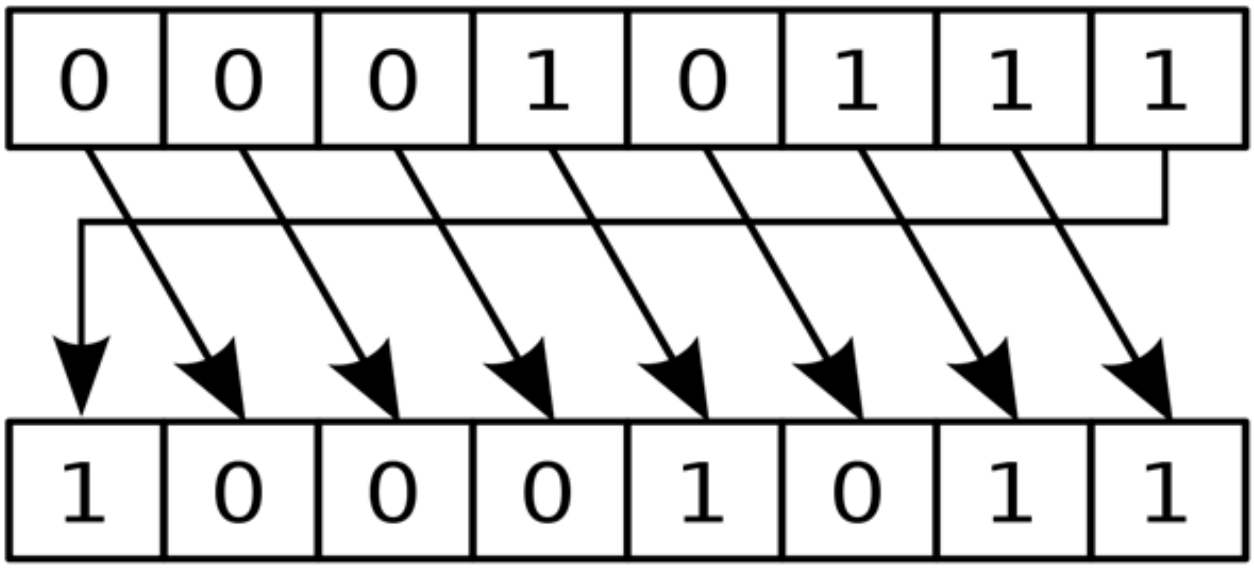
- **Special linear block codes with one extra property**
- **If a codeword is cyclically shifted (rotated), the result is another codeword**
- **If 1011000 is a codeword and we cyclically left-shift, then 0110001 is also a codeword**

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In coding theory, a cyclic code is **a block code, where the circular shifts of each codeword gives another word that belongs to the code**.



Cyclic Redundancy Check (CRC)

- **Subset of Cyclic Codes**
- **Cyclic redundancy check (CRC) is used in networks such as LANs and WANs**

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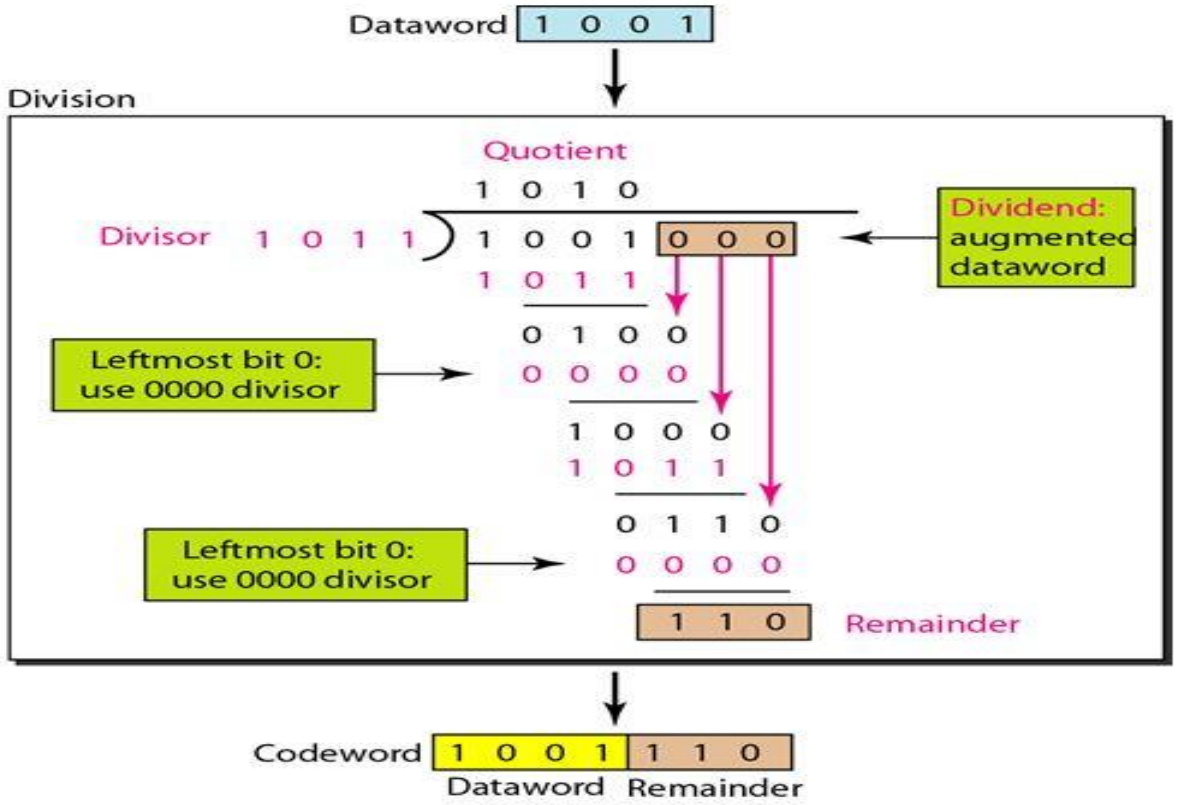
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A CRC code with C(7, 4)

Dataword	*Codeword*	*Dataword*	*Codeword*
0000	0000000	1000	1000101
0001	0001011	1001	1001110
0010	0010110	1010	1010011
0011	0011101	1011	1011000
0100	0100111	1100	1100010
0101	0101100	1101	1101001
0110	0110001	1110	1110100
0111	0111010	1111	1111111

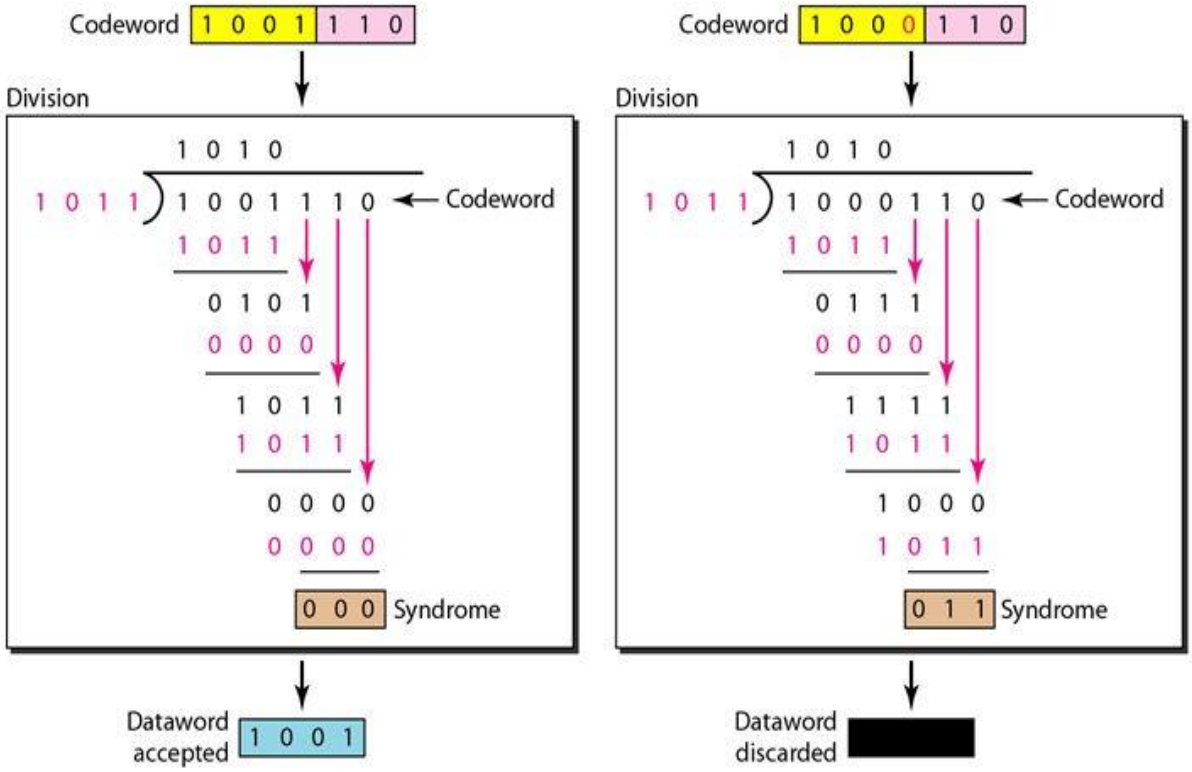


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Decoder:



Advantages of Cyclic Codes

- **Good performance in detection:**
  - **Single-bit errors**
  - **Double errors**
  - **Odd number of errors**
  - **Burst errors**
- **Easy Implementation**
- **Fast Implementation**

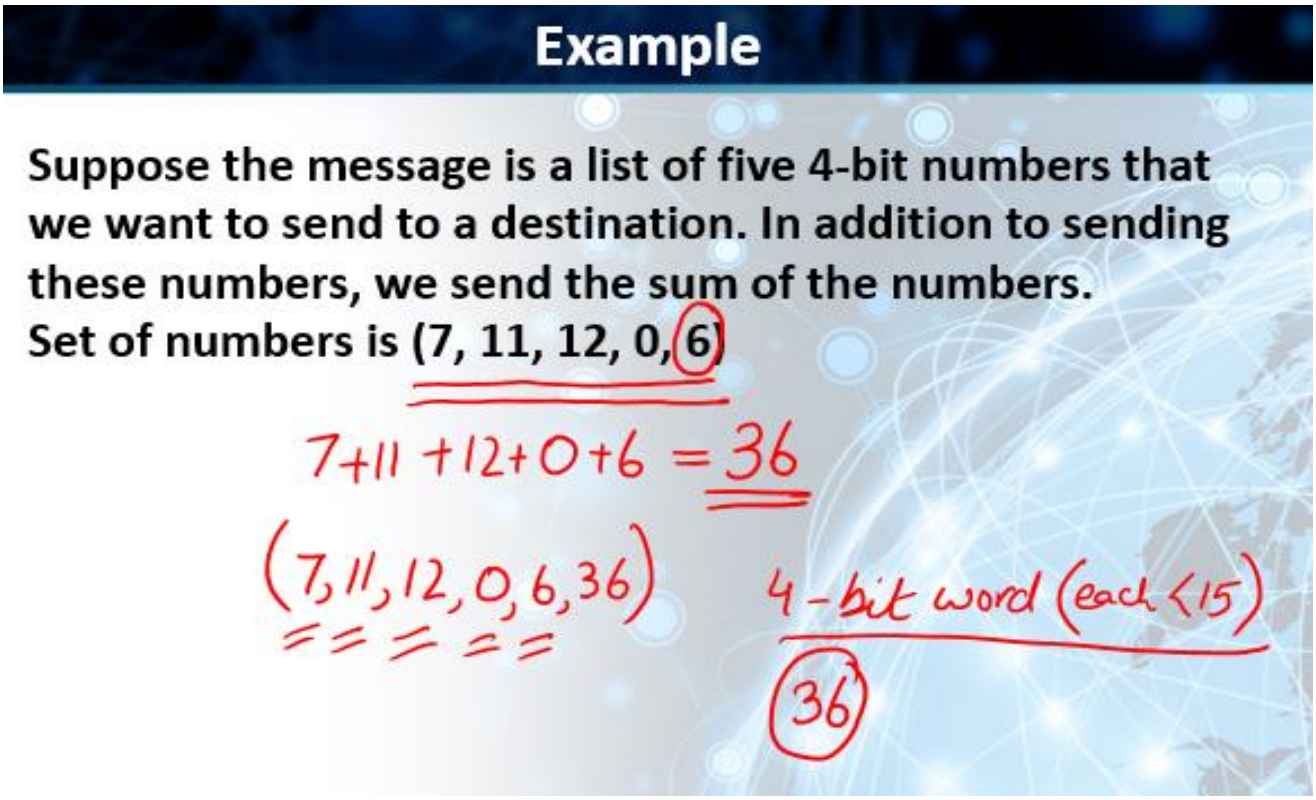
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CHECKSUM

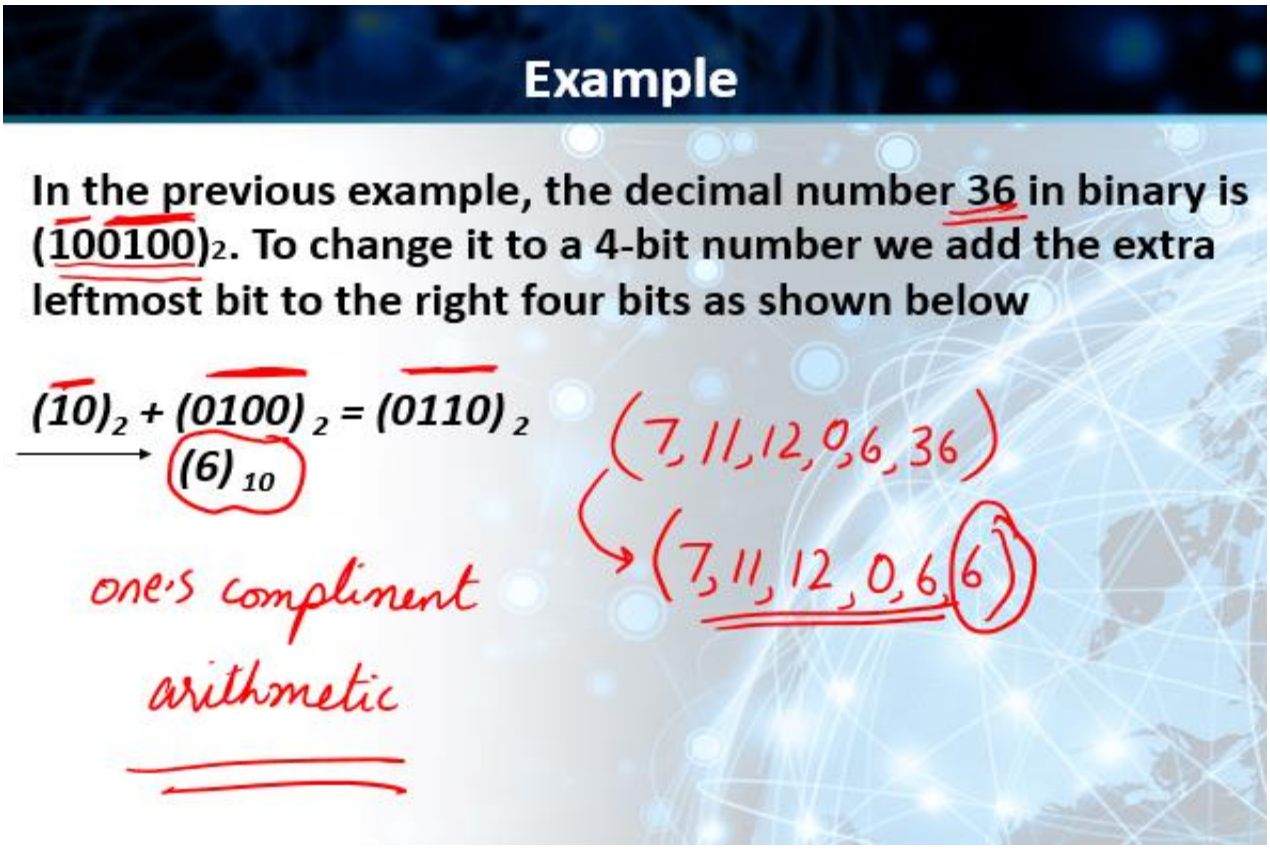
- **Error-detection technique that can be applied to a message of any length**
- **Checksum mostly used at the network and transport layer rather than the data-link layer**



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Forward Error Correction

- **Retransmission of corrupted and lost packets is not useful for real-time multimedia transmission**
- **We need to correct the error or reproduce the packet immediately**
- **Several techniques developed and are commonly called Forward Error Correction techniques**

Using Hamming Distance

- **For error detection, we definitely need more distance**
- **It can be shown that to correct 't' errors, we need to have:**

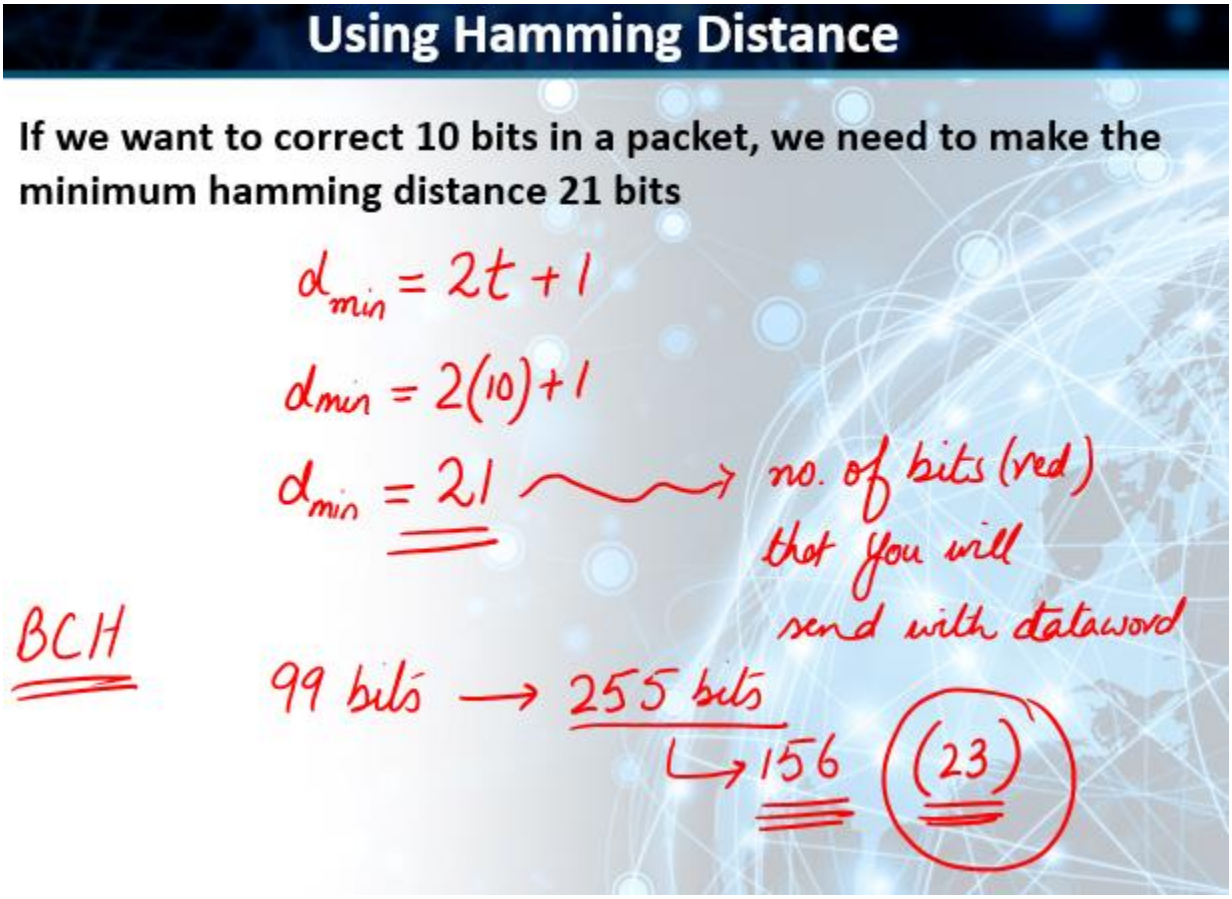
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$d_{min} = 2t + 1$

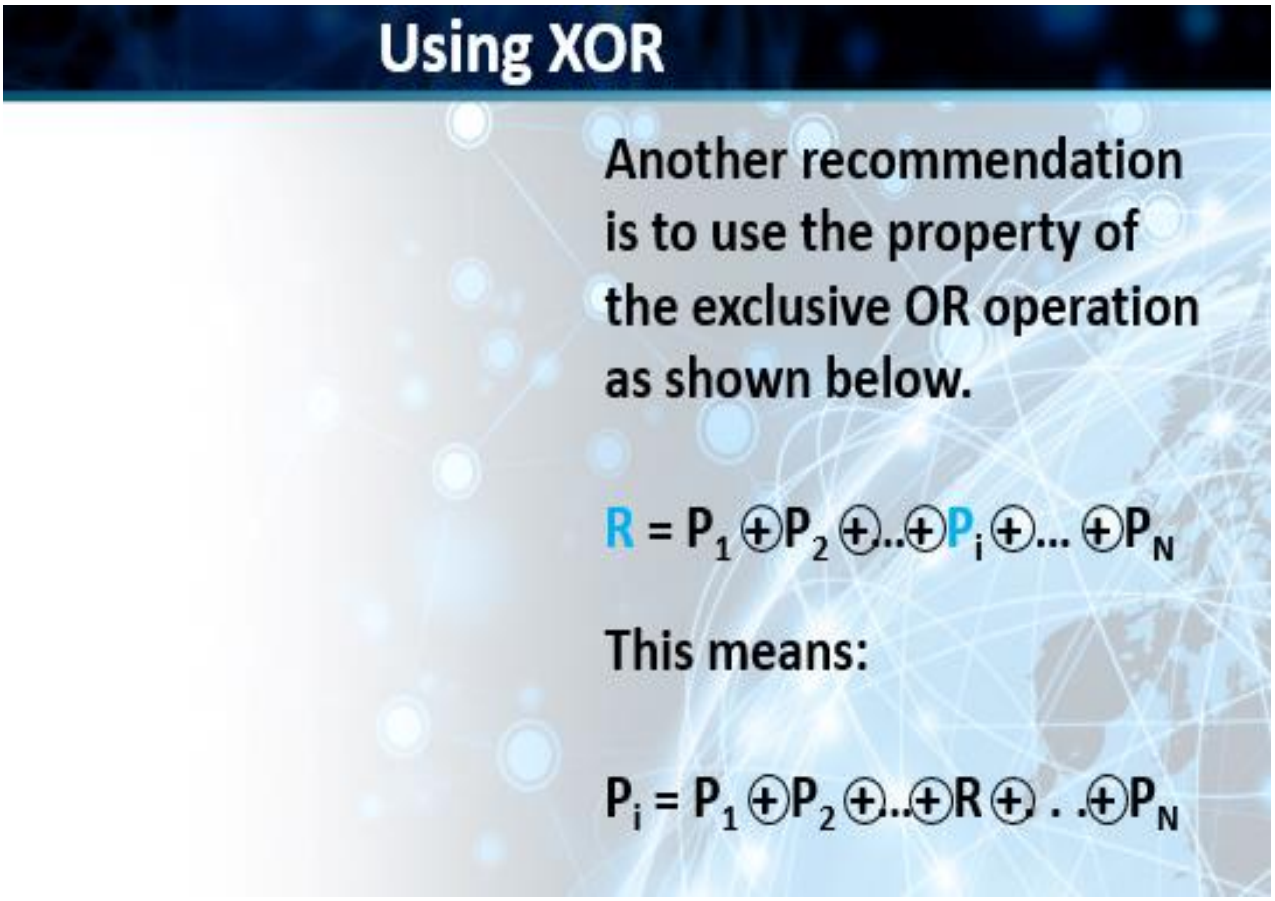
- **If we want to correct 10 bits in a packet, we need to make the minimum hamming distance 21 bits**
- **A lot of redundant bits need to be sent with the data**



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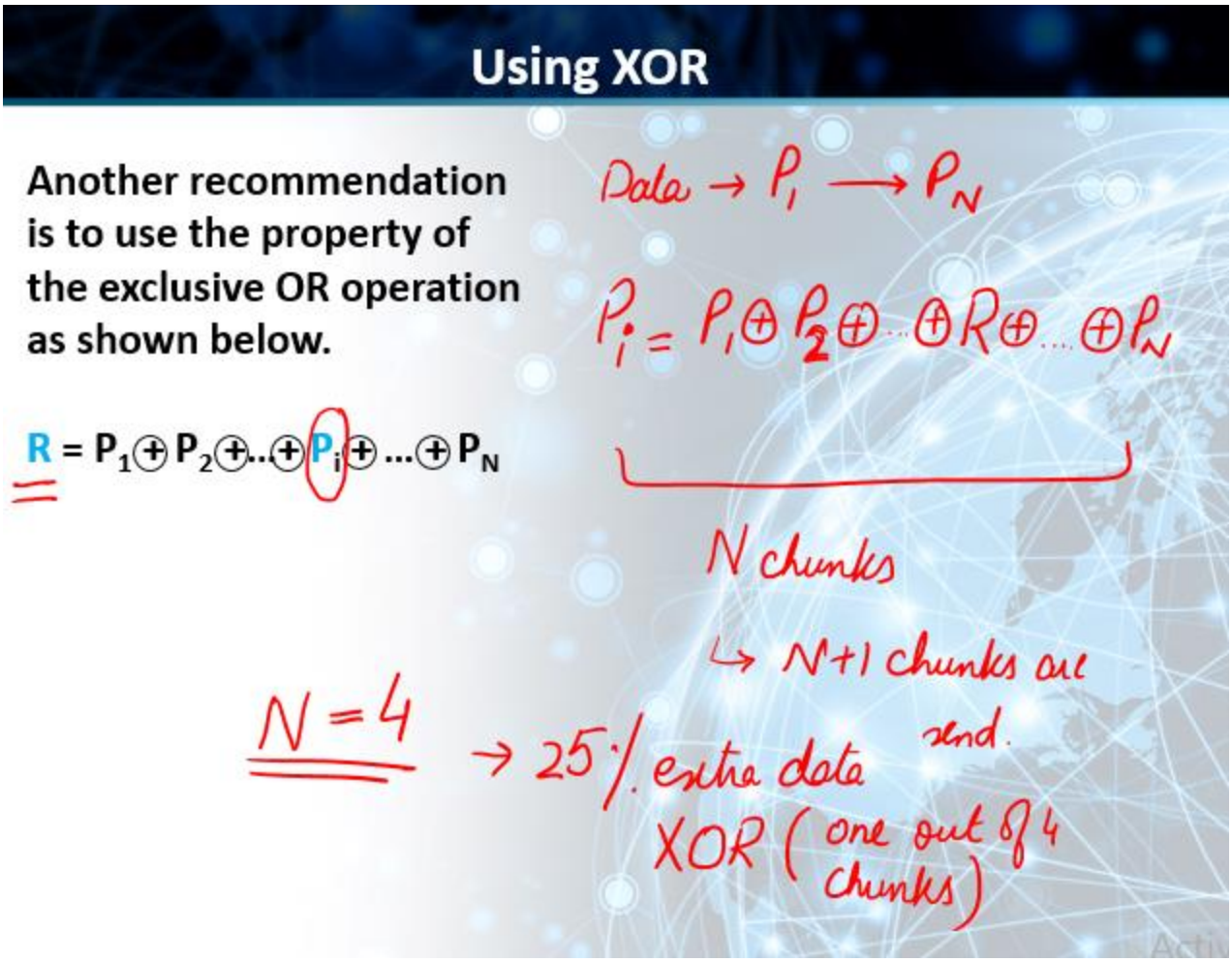
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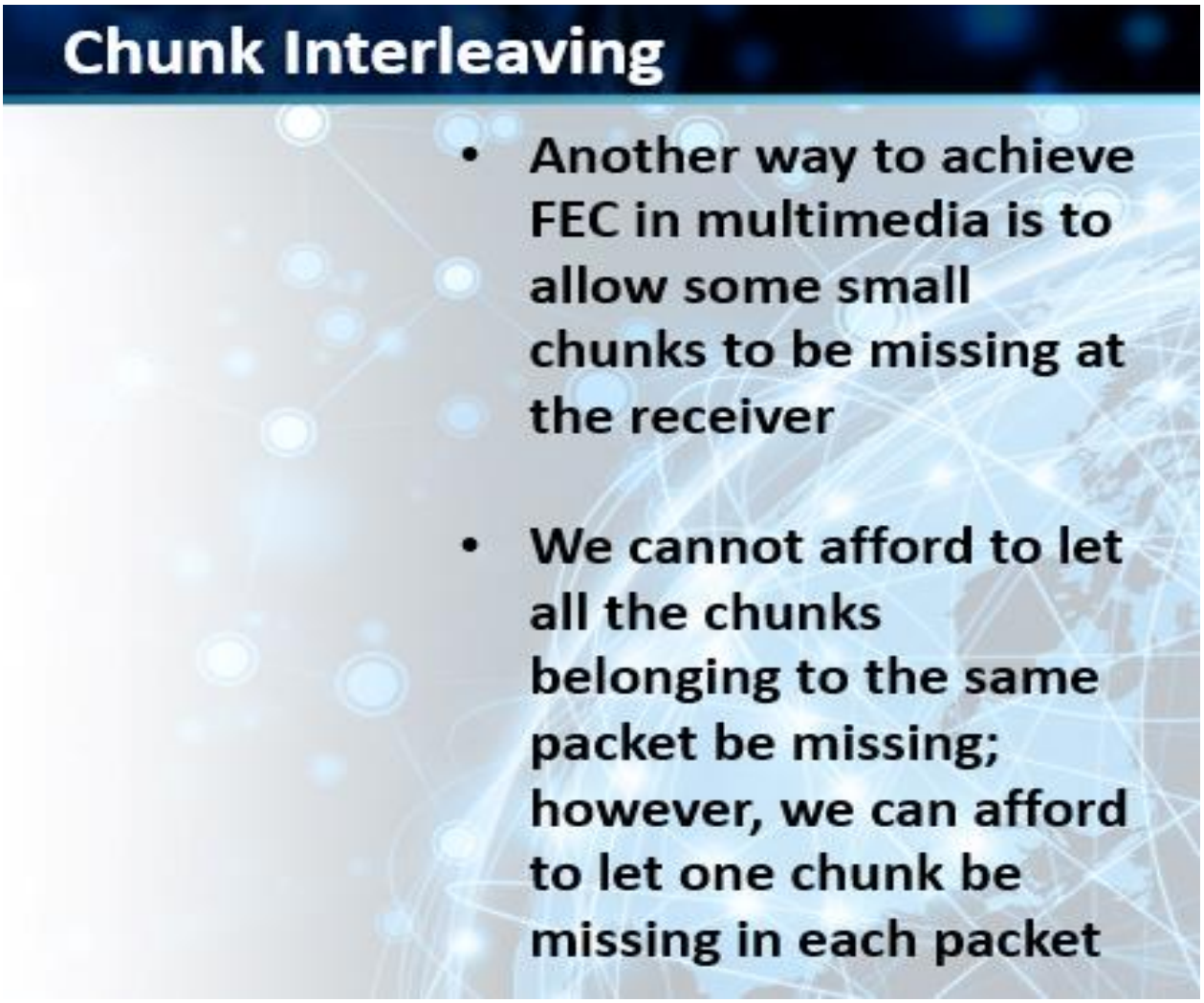
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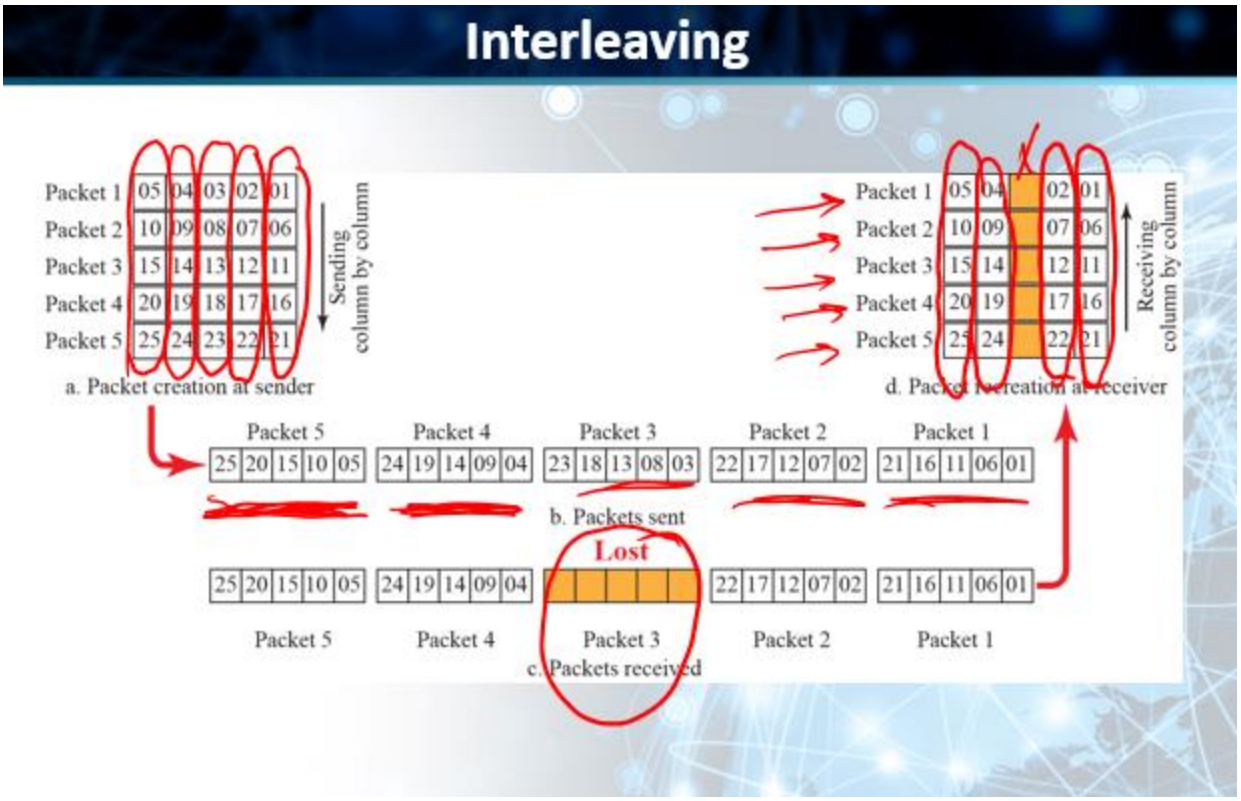
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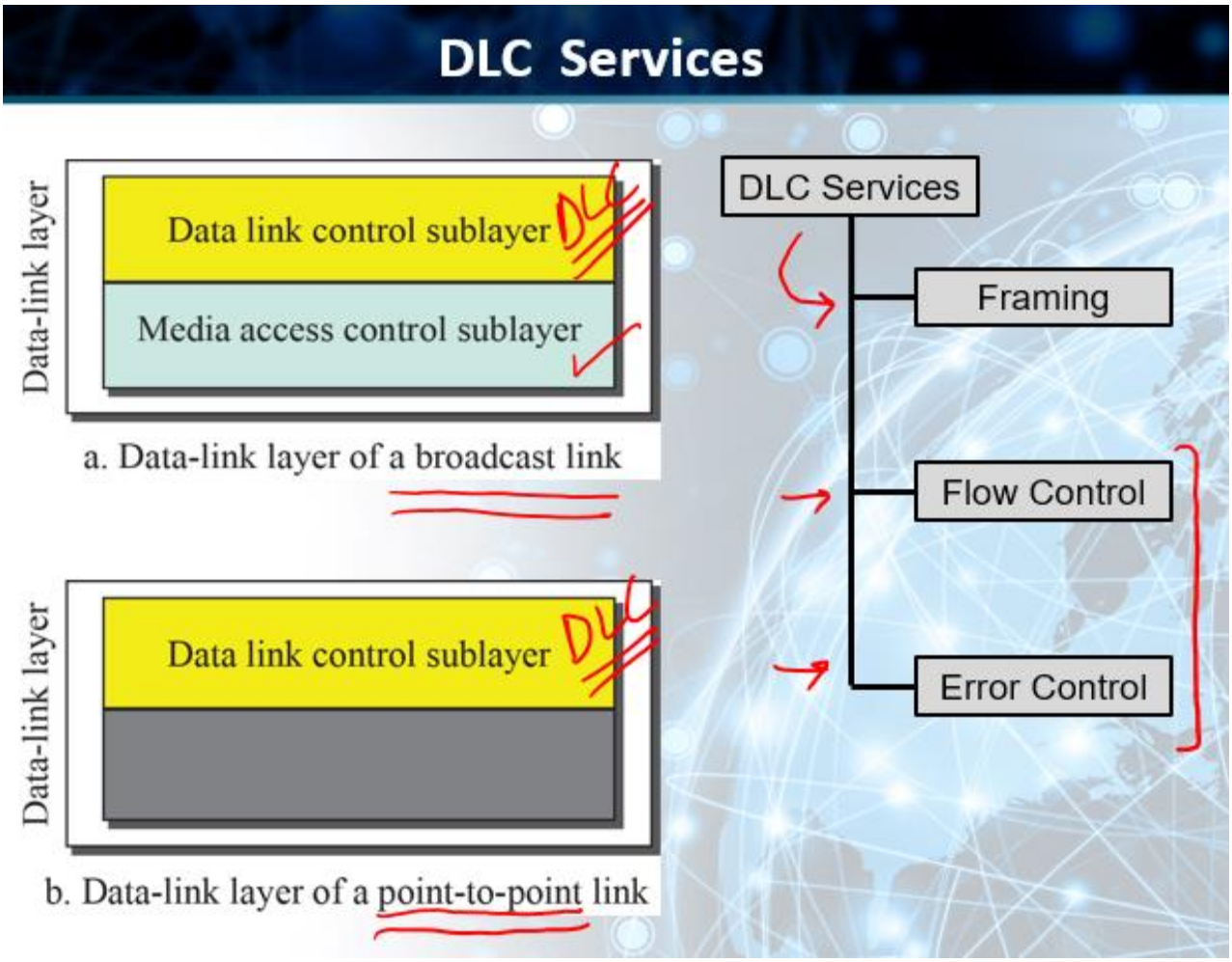
Data Link Control (DLC) Services

- **The data link control (DLC) deals with procedures for communication between two adjacent nodes no matter whether the link is dedicated or broadcast**
- **Data link control functions include framing, flow control and error control**

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Framing

- **Data-Link layer needs to pack bits into frames, so that each frame is distinguishable from another**
- **Our postal system practices a type of framing**
- **Framing separates a message by adding a sender address and a destination address**
- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt

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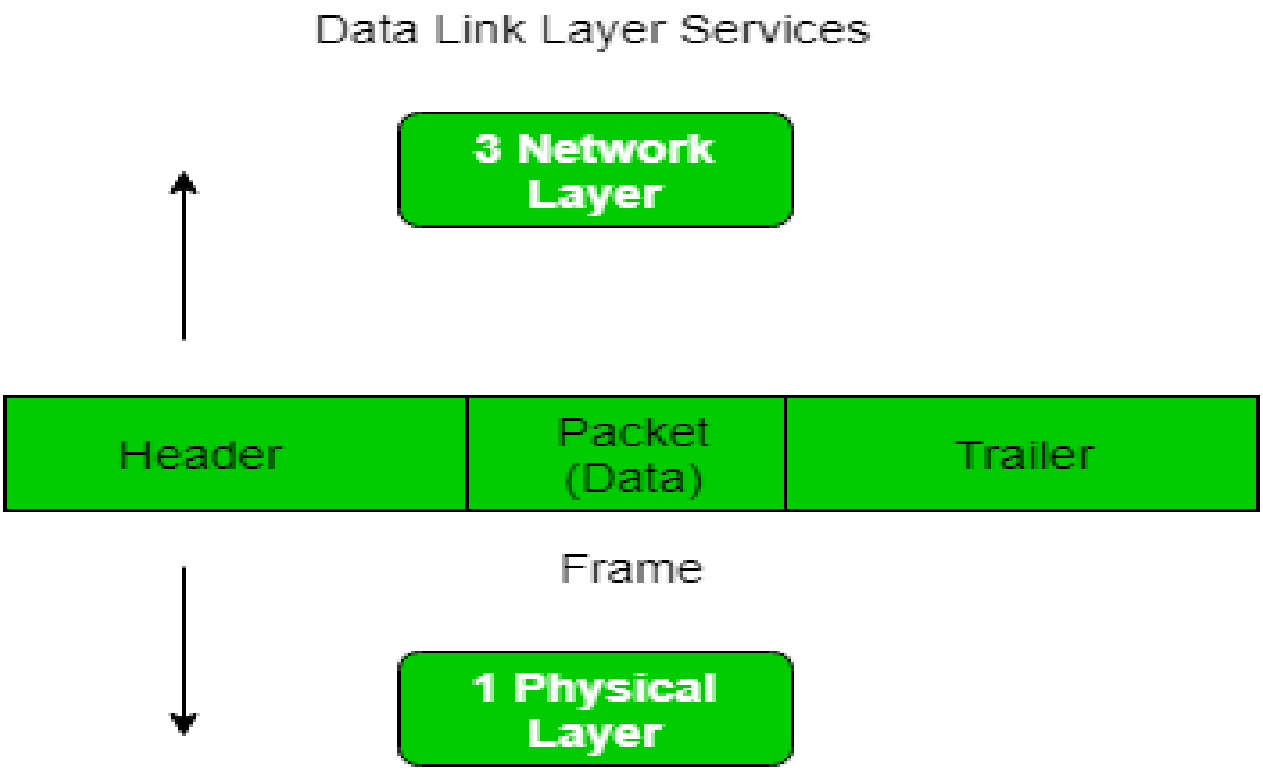
Parts of a Frame

A frame has the following parts −

- Frame Header − It contains the source and the destination addresses of the frame.
- Payload field − It contains the message to be delivered.
- Trailer − It contains the error detection and error correction bits.
- Flag − It marks the beginning and end of the frame.

Delimiter

In computer programming, a delimiter is **a character that identifies the beginning or the end of a character string** (a contiguous sequence of characters). The delimiting character is not part of the character string.



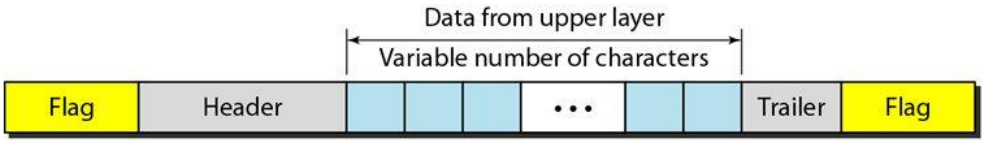
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Framing

- Process of wrapping data with certain info before sending out



- A frame typically consists of
  - Flag: indication for start and end of a frame
  - Header: source/destination addresses, as well as other control information
  - Data from the upper layer
  - Trailer: error detection/correction code

3

Why not one BIG Frame?

Data-link layer takes the packets from the Network Layer and encapsulates them into frames. If the frame size becomes too large, then the packet may be divided into small sized frames. Smaller sized frames make flow control and error control more efficient.

Types of Framing

Framing can be of two types, fixed sized framing and variable sized framing.

Fixed-sized Framing

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Here the size of the frame is fixed and so the frame length acts as delimiter of the frame. Consequently, it does not require additional boundary bits to identify the start and end of the frame.

Example − ATM cells.

Variable – Sized Framing

Here, the size of each frame to be transmitted may be different. So additional mechanisms are kept to mark the end of one frame and the beginning of the next frame.

It is used in local area networks.

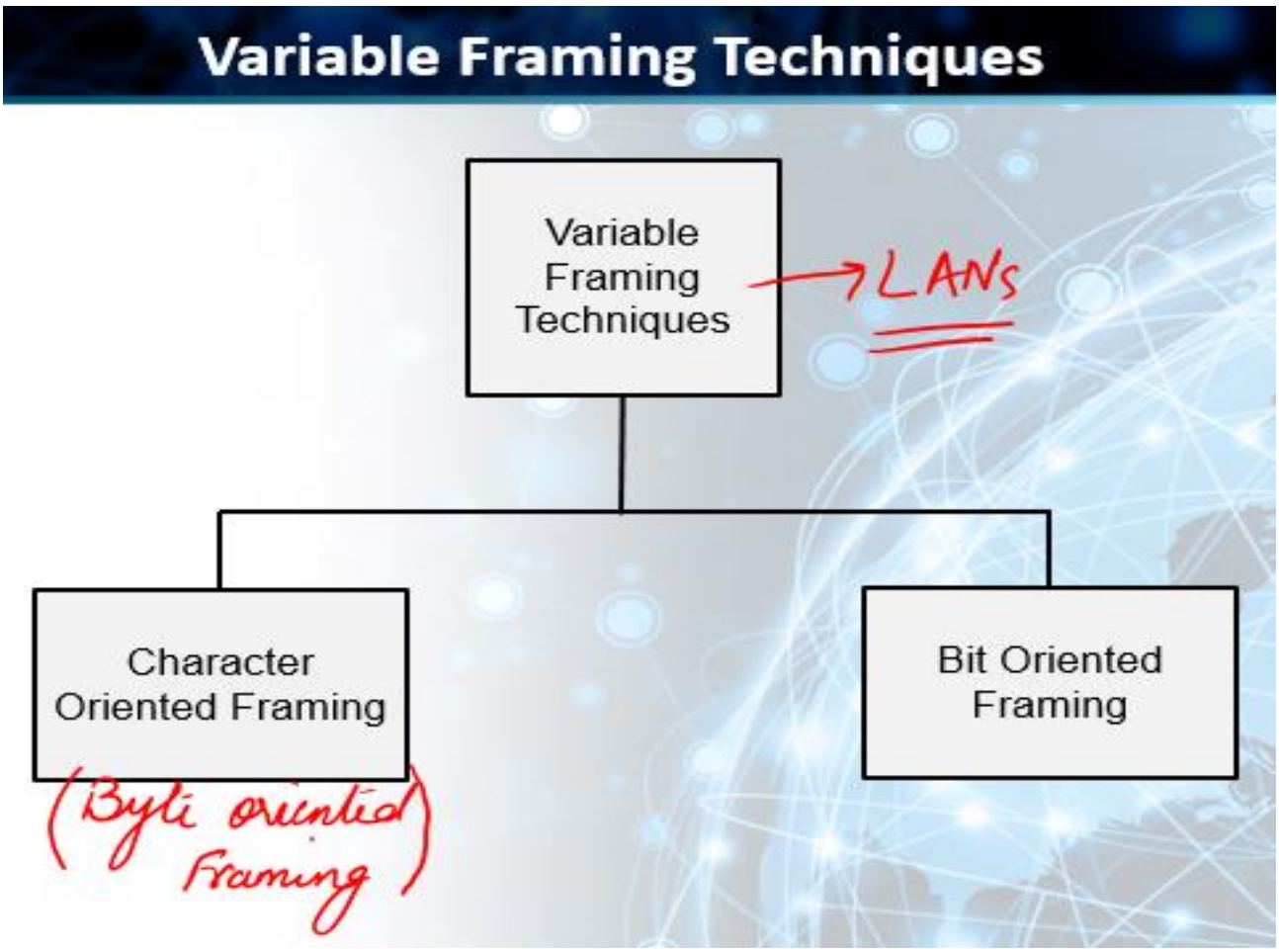
Two ways to define frame delimiters in variable sized framing are −

- **Length Field** − Here, a length field is used that determines the size of the frame. It is used in Ethernet (IEEE 802.3).
- **End Delimiter** − Here, a pattern is used as a delimiter to determine the size of frame. It is used in Token Rings. If the pattern occurs in the message, then two approaches are used to avoid the situation −

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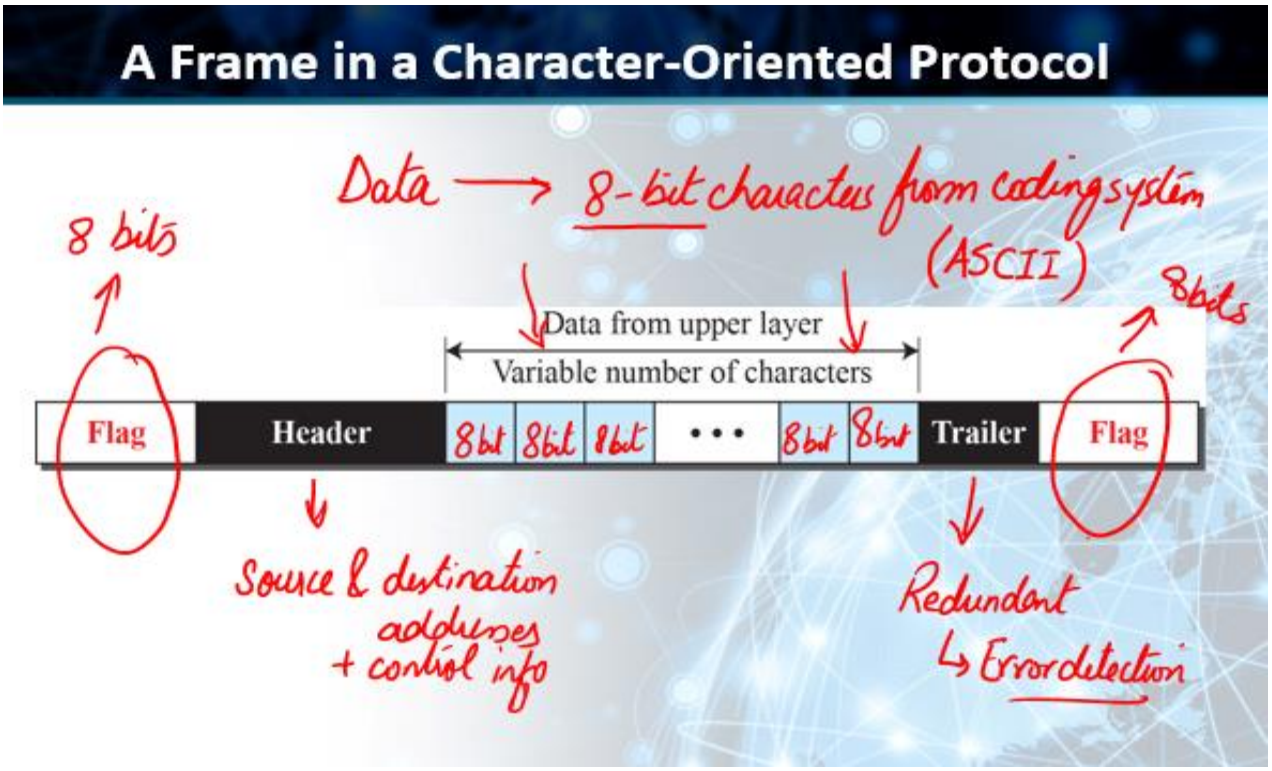
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Connection Oriented Framing

- **Data to be carried are 8-bit characters**

- **Byte – Stuffing** − A byte is stuffed in the message to differentiate from the delimiter. This is also called character-oriented framing.
- **Bit – Stuffing** − A pattern of bits of arbitrary length is stuffed in the message to differentiate from the delimiter. This is also called bit – oriented framing.

Byte Stuffing in Connection-Oriented Framing

- **Connection-oriented Framing used text characters as flags**
- **Nowadays any character used for flag can also be a part of the data**
- **In order to avoid confusing the receiver, we use Byte Stuffing**

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What are escape characters?

An escape sequence contains a backslash (\) symbol followed by one of the escape sequence characters or an octal or hexadecimal number.

...

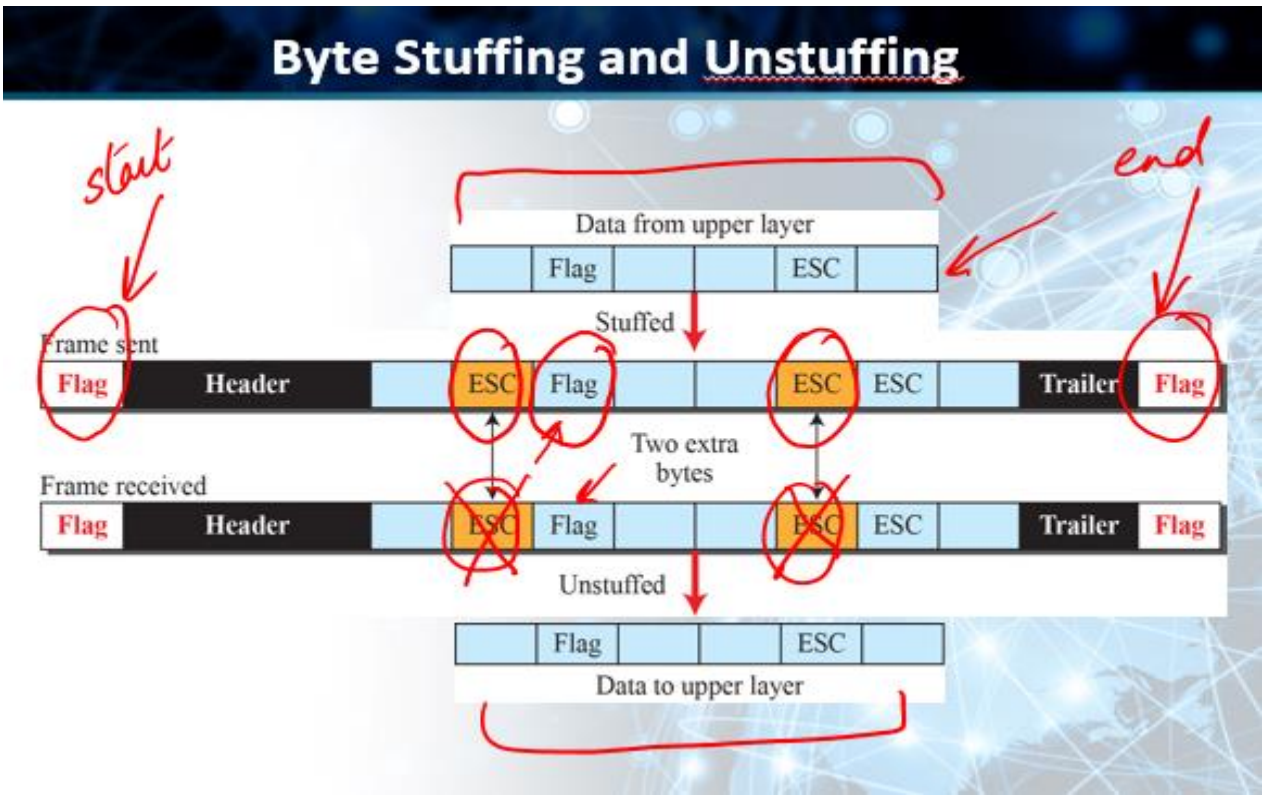
Escape character syntax.

Escape sequence	Character represented
\r	Carriage return
\t	Horizontal tab
\v	Vertical tab
\'	Single quotation mark

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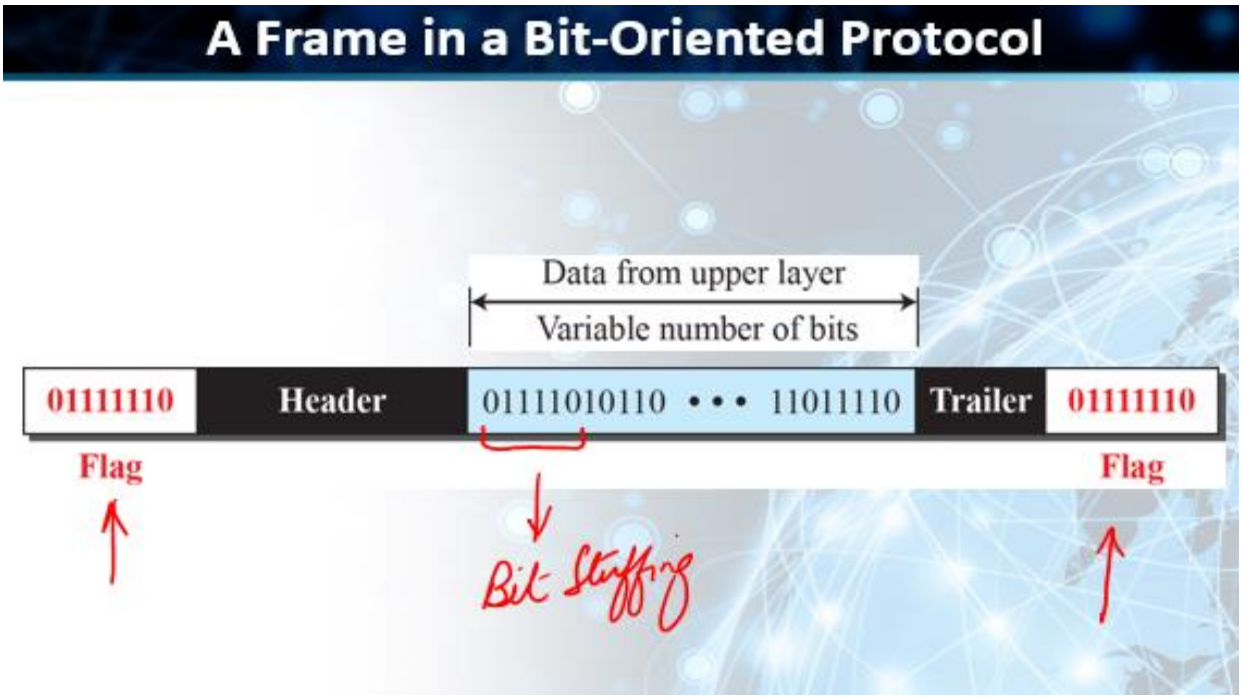
Bit-Oriented Framing

- Data section of frame is a sequence of bits
- We need a delimiter to separate one frame from the other
- A special 8-bit pattern (01111110) to define beginning and end of a frame
- Same issue as Connection-oriented Framing

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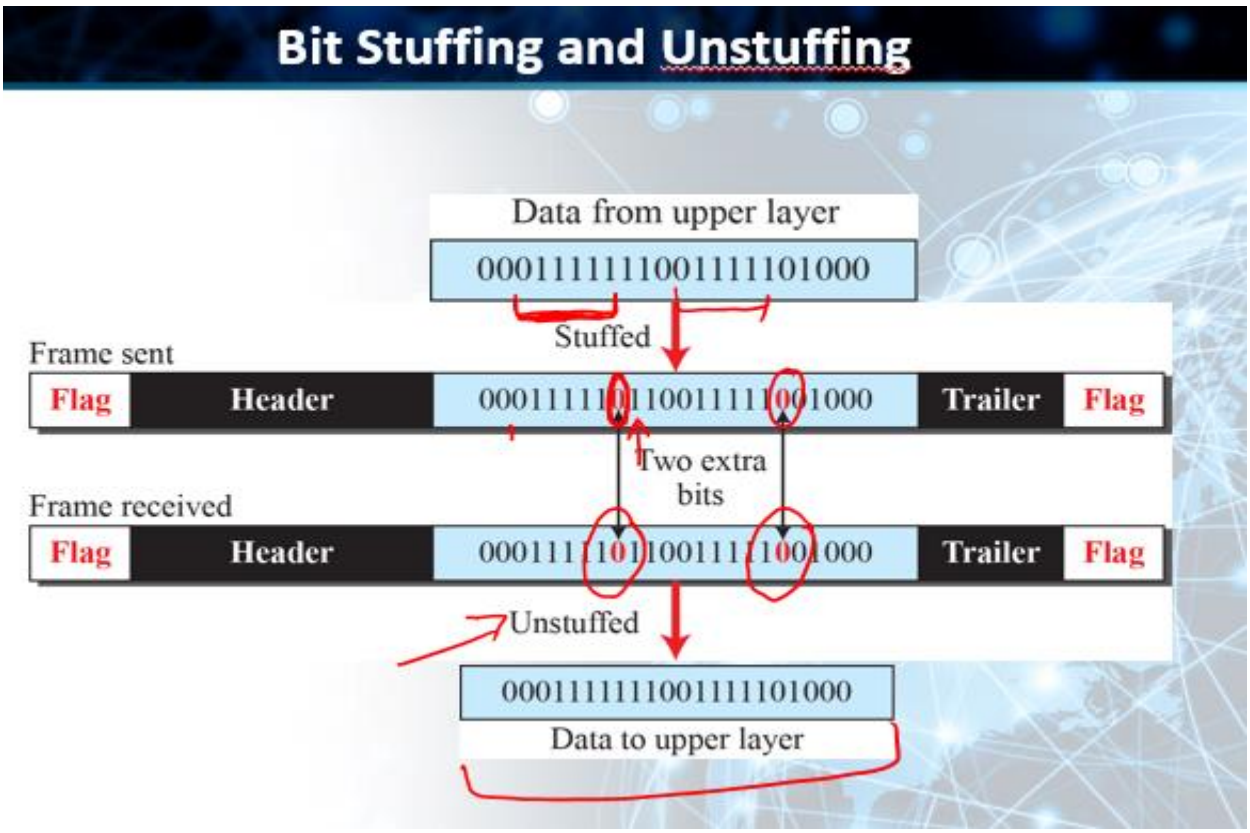
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Flow and Error Control

One of the responsibilities of the data-link control sublayer is flow and error control at the data-link layer

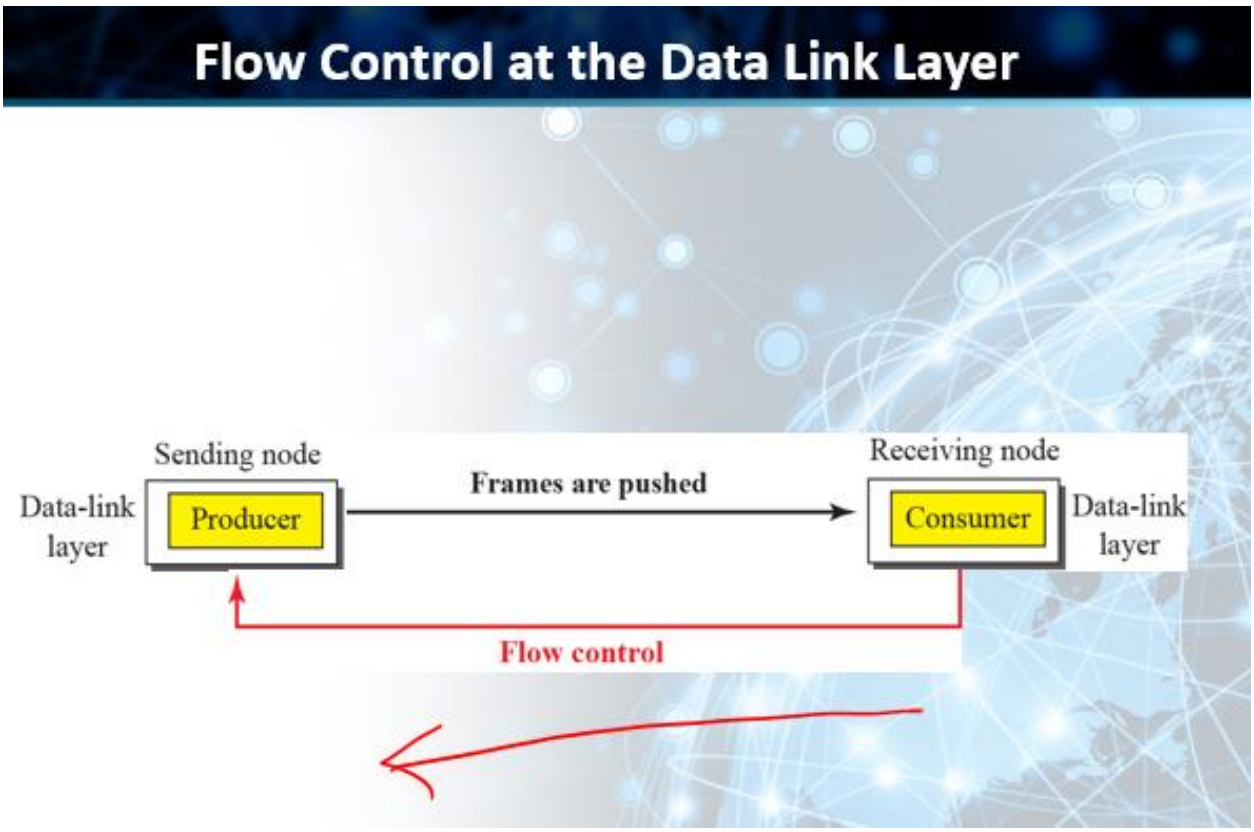
Flow Control

- **Balance between production and consumption rates**
- **If frames are produced faster than they are consumed at the receiving data link layer, the frames will be discarded**
- **Use of buffers; one at sending end and other at receiving end**

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Example

- **Consumers need to communicate with the producers on two occasions:**
  - ✓ **When the buffer is full; &**
  - ✓ **When there are vacancies**
  - ✓ **If the two parties use a buffer with only one slot, the communication can be easier**

Error Control

- Error Control at Data Link layer uses CRC in one of the two ways:
  - ✓ If a frame is corrupted, it is silently discarded and if it is good, it is delivered to network layer

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- ✓ If frame is corrupted, it is silently discarded and if it is good, an acknowledgement is sent to sender

Connectionless and Connection-Oriented

- A DLC protocol can be either connectionless or connection-oriented
- Connectionless: No relationship between the frames
- Connection-Oriented: Frames are numbered and sent in order

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CS601 Lecture No 27 Topic 156 to 160

DATA-LINK LAYER PROTOCOLS

- **Traditionally four protocols have been defined for the data-link layer to deal with flow and error control:**

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- ✓ **Simple Protocol**
- ✓ **Stop-and-Wait Protocol**
- ✓ **Go-Back-N Protocol**
- ✓ **Selective-Repeat Protocol**
- ✓ **Last two protocols have almost disappeared completely**

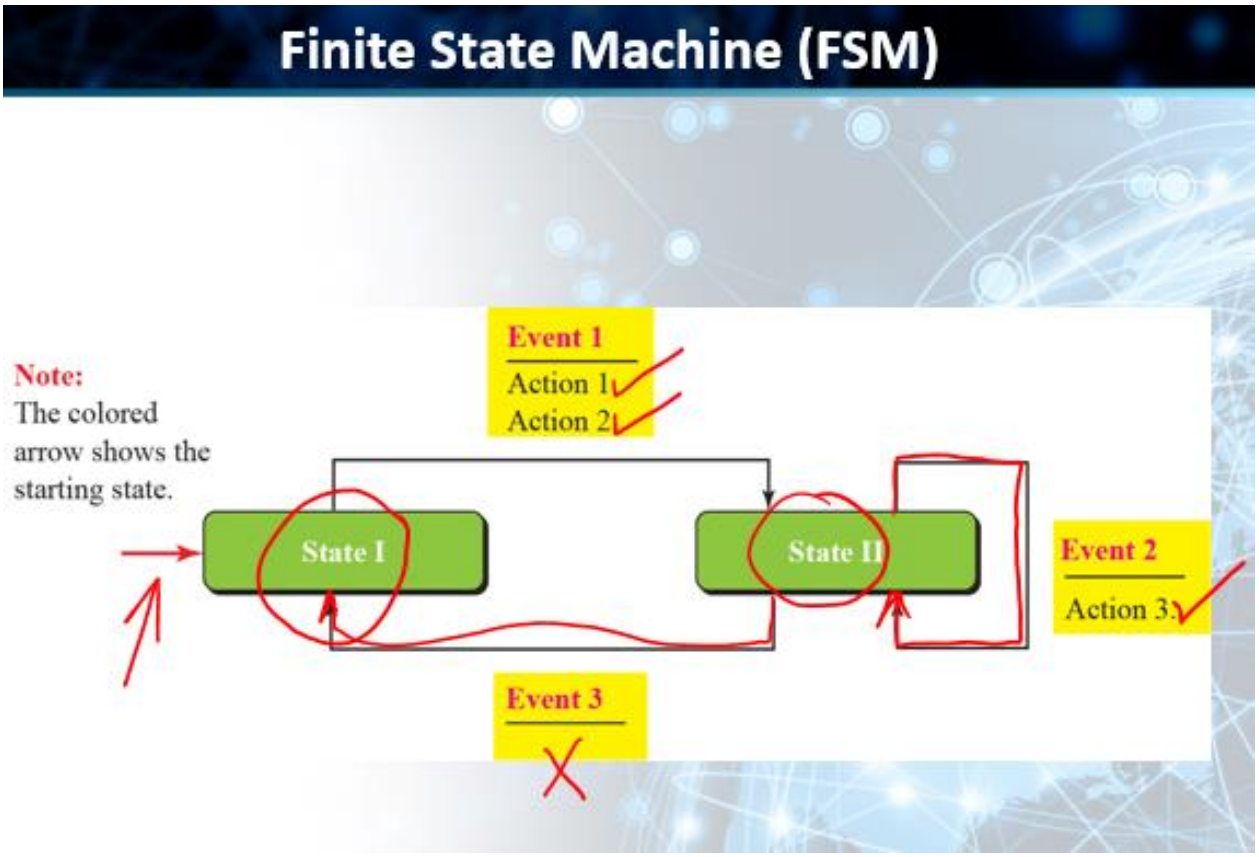
Finite State Machine (FSM)

- **A machine with a finite number of states**
- **Machines stays in one of the states until an event occurs**
- **Each event is associated with 2 reactions:**
  - ✓ **List of actions to be performed**
  - ✓ **Determining the next state**

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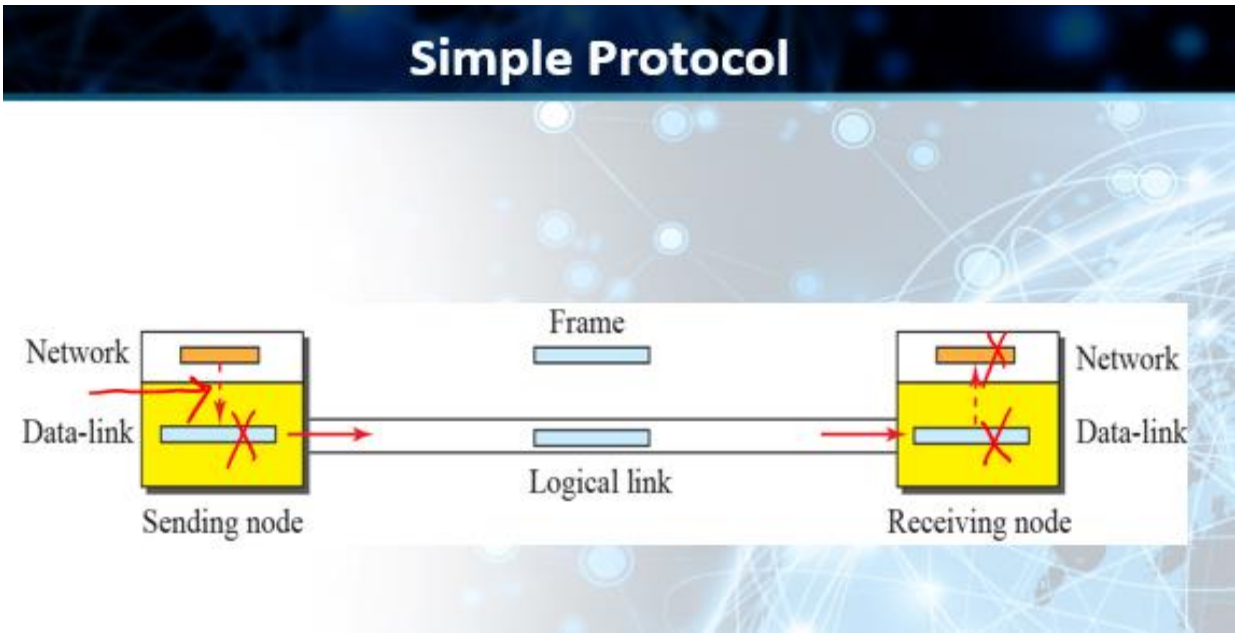
Simple Protocol

- **Simple protocol has neither flow nor error control**
- **Assumption: The receiver can immediately handle any frame it receives**
- **The receiver can never be overwhelmed with incoming frames**

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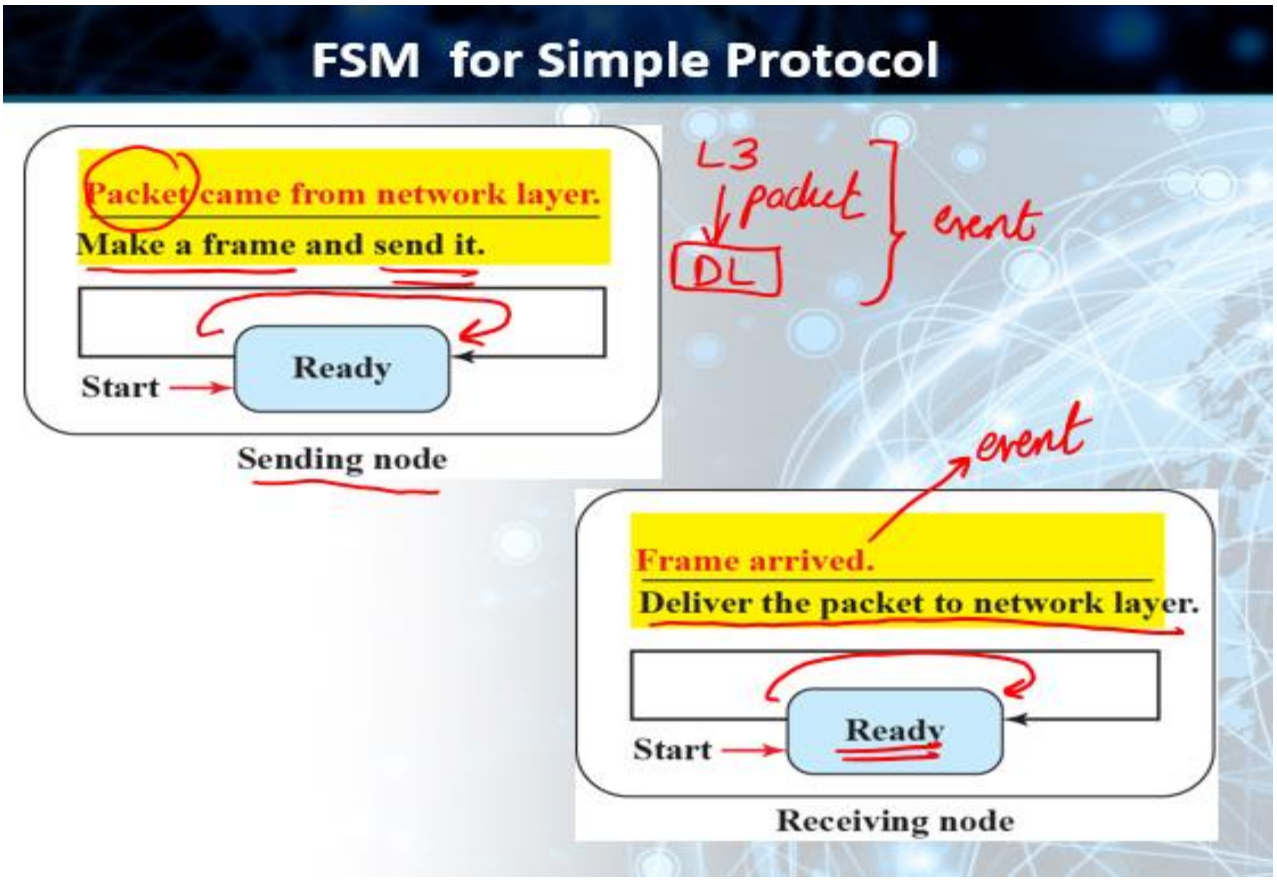
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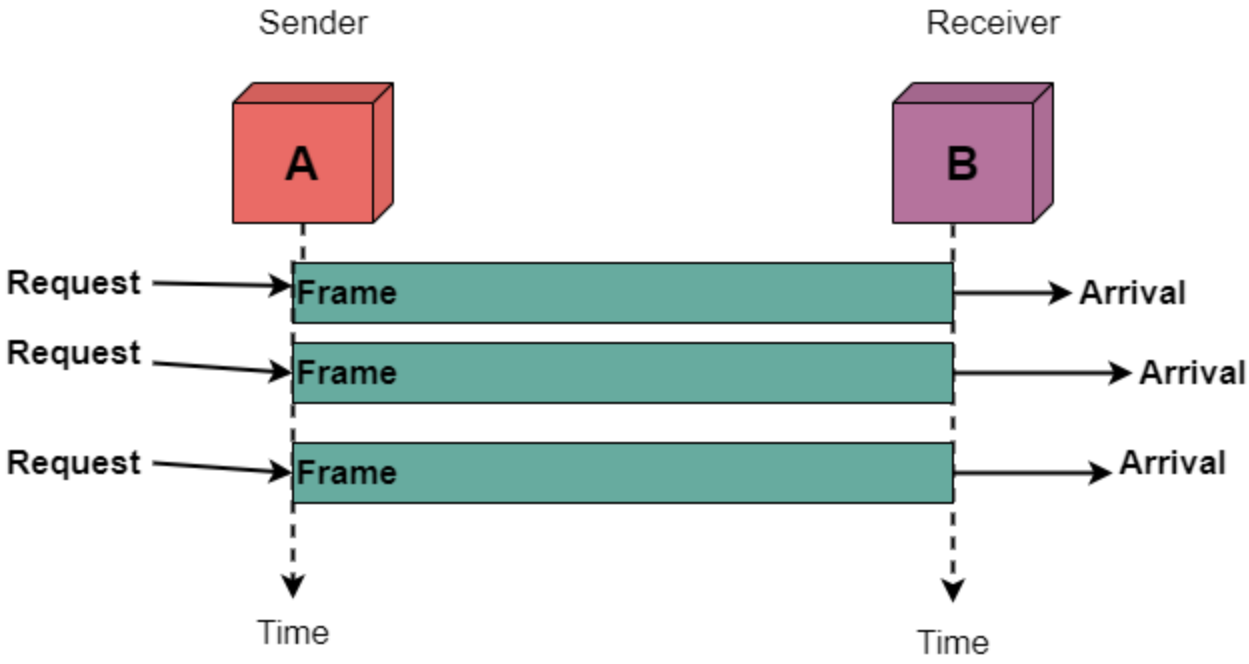
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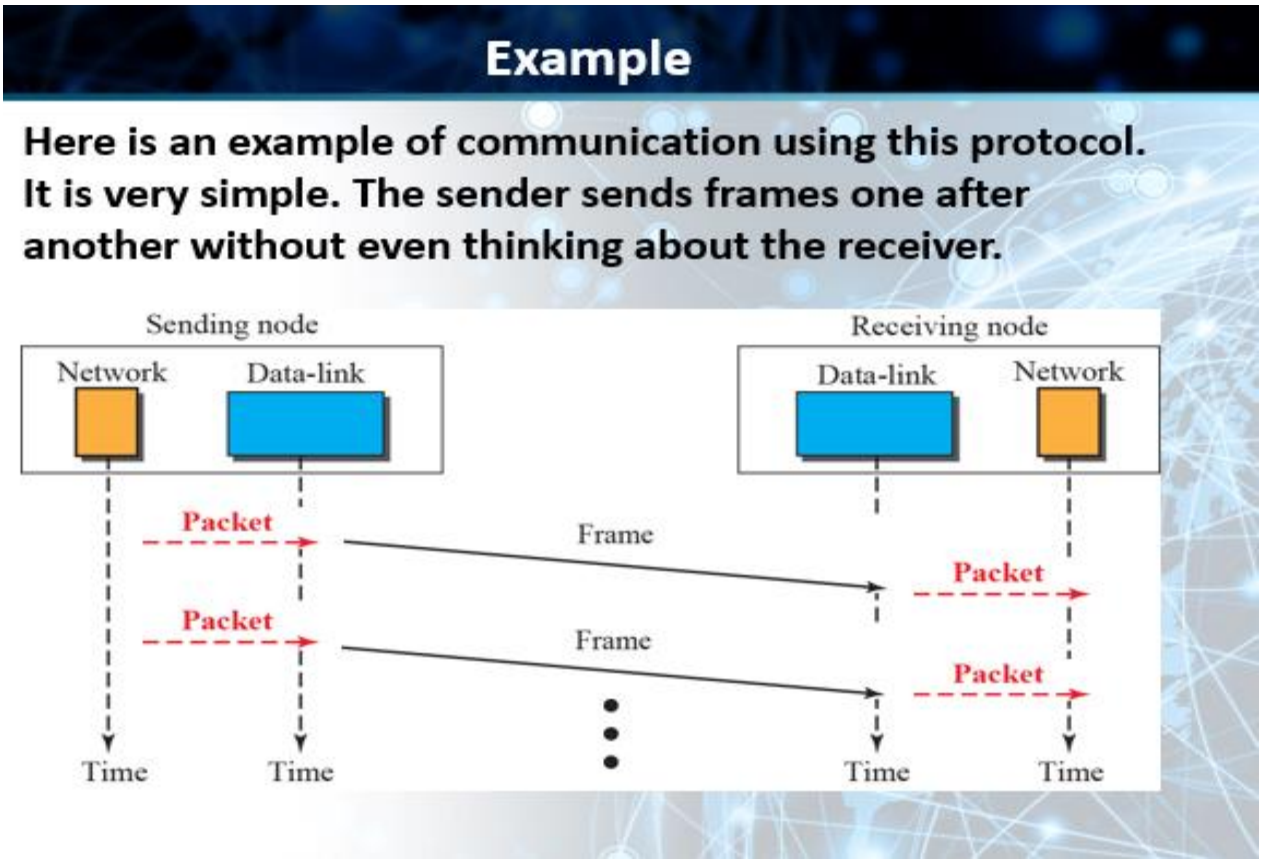
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Stop-and-Wait Protocol

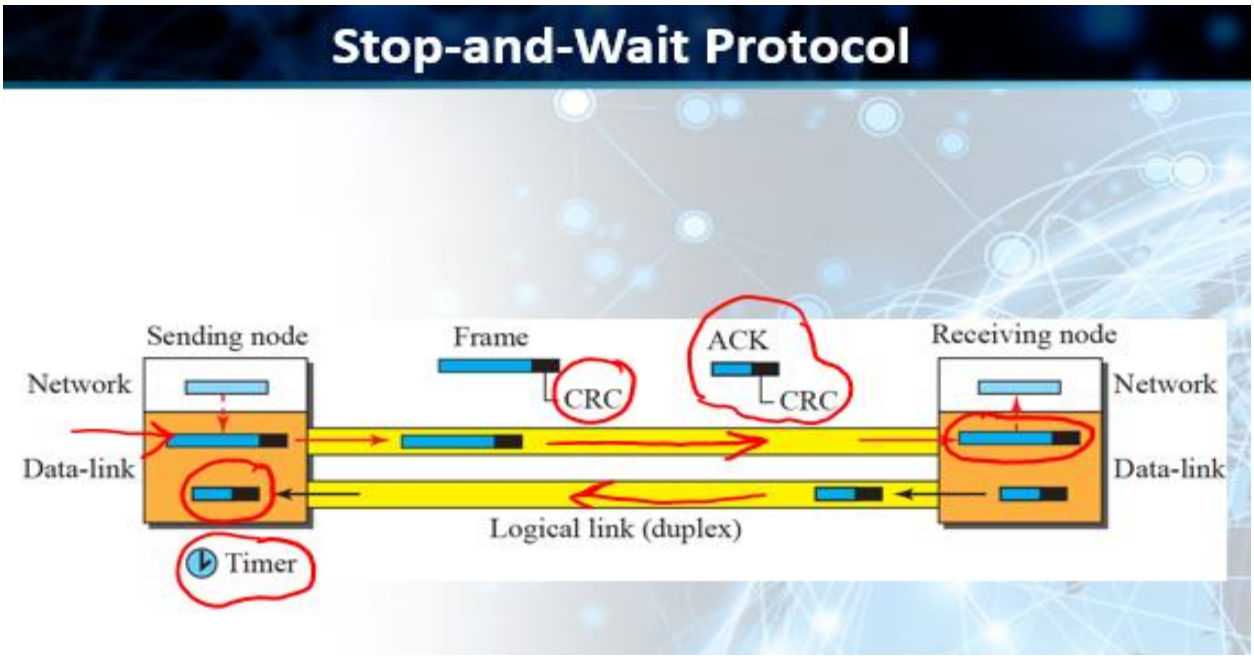
- **Stop-and-Wait protocol uses both flow and error control**
- **The sender sends one frame at a time and waits for an acknowledgment before sending the next one**
- **To detect corrupted frames, we add a CRC code**

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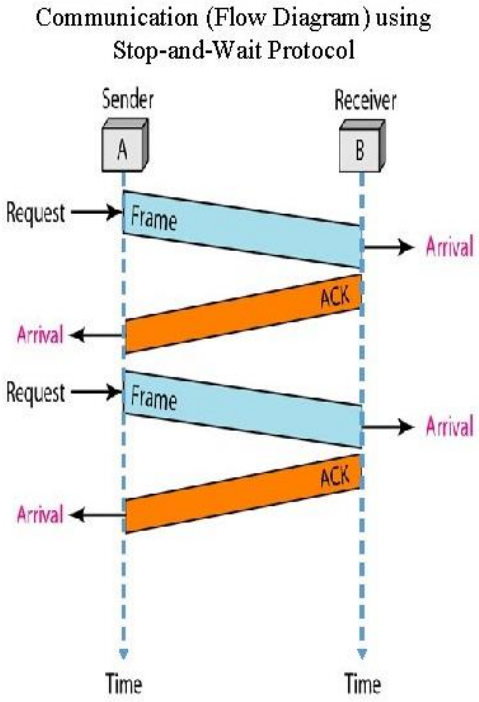
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Stop-and-Wait Protocol

At any time, there is either one ***data frame*** on the forward channel or one ***ACK frame*** on the reverse channel.

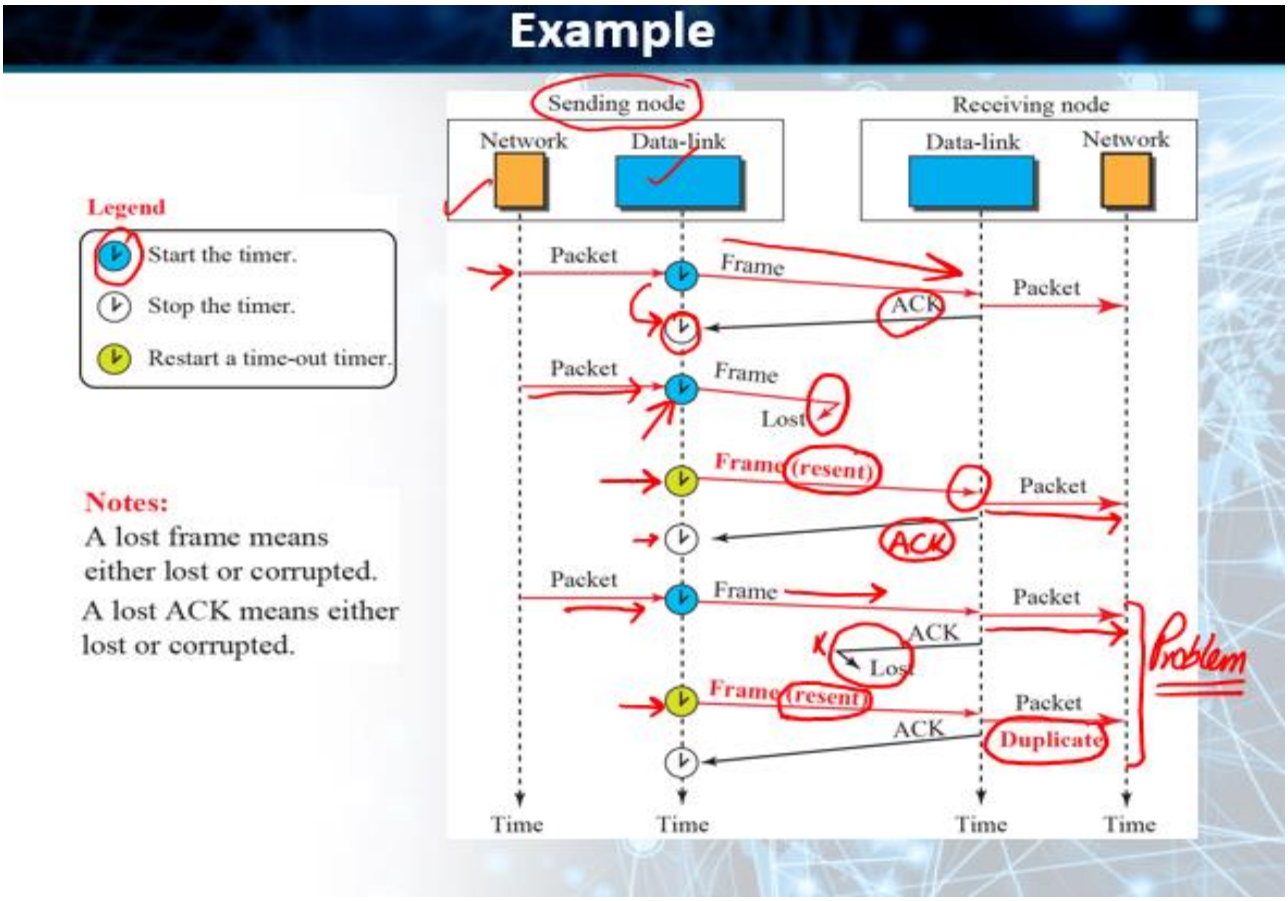
We therefore need a **half-duplex link**.



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Piggybacking

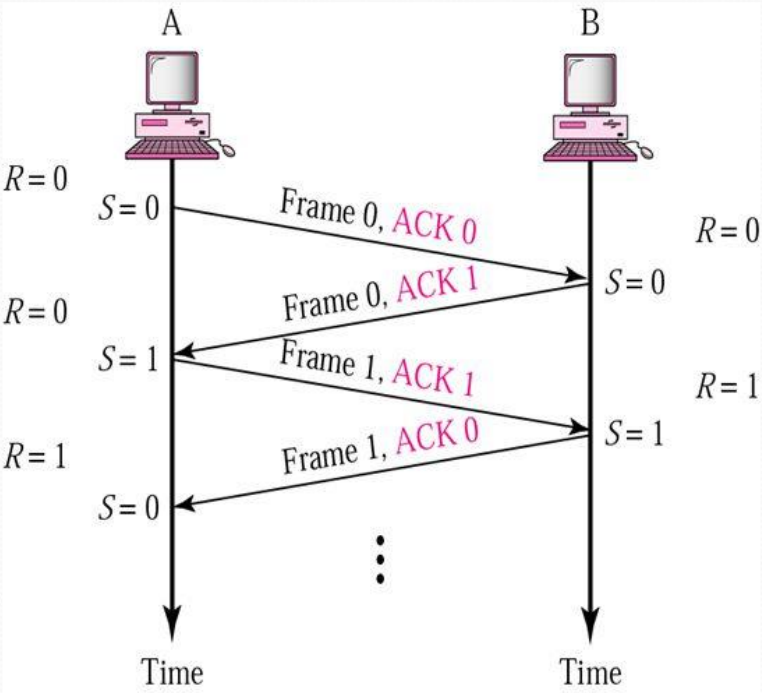
- **Both Simple and Stop-and-wait protocols are designed for unidirectional communication**
- **Data flows in one direction and ACK travels in the other**
- **To make the system efficient, the data in one direction is piggybacked with the acknowledgment in the other direction**

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Piggybacking

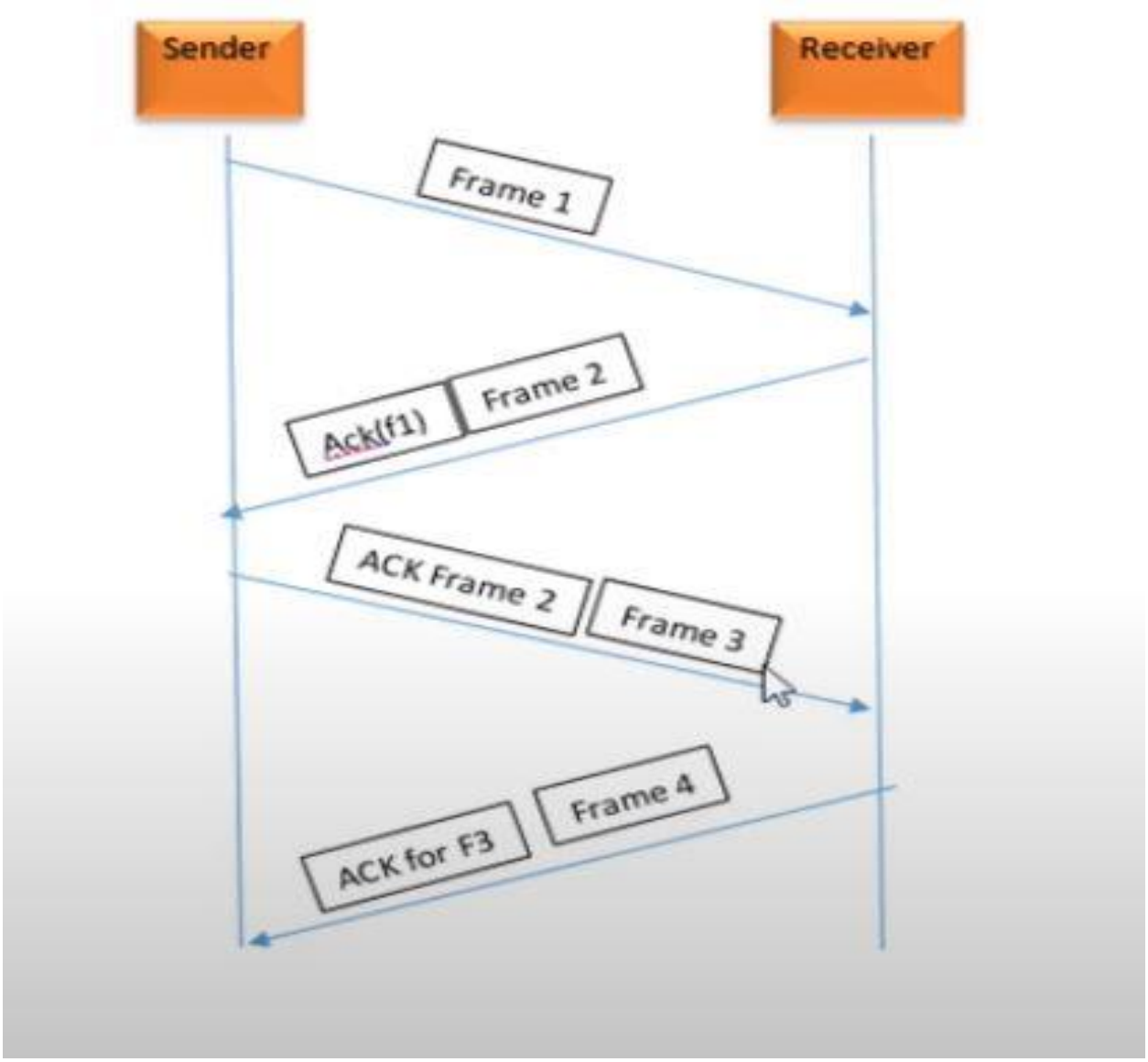


- A method to combine a data frame with ACK.
- Station A and B both have data to send.
- Instead of sending separately, station A sends a data frame that includes an ACK.
- Station B does the same thing.
- Piggybacking saves bandwidth.

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High-level Data Link Control (HDLC)

- **Bit -oriented protocol for communication over point-to-point and multipoint links**
- **It implements Stop-and-Wait protocol**
- **Most of the concepts defined in this protocol is the basis for other protocols such as PPP, Ethernet, or wireless LANs**

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Configurations & Transfer Modes in HDLC

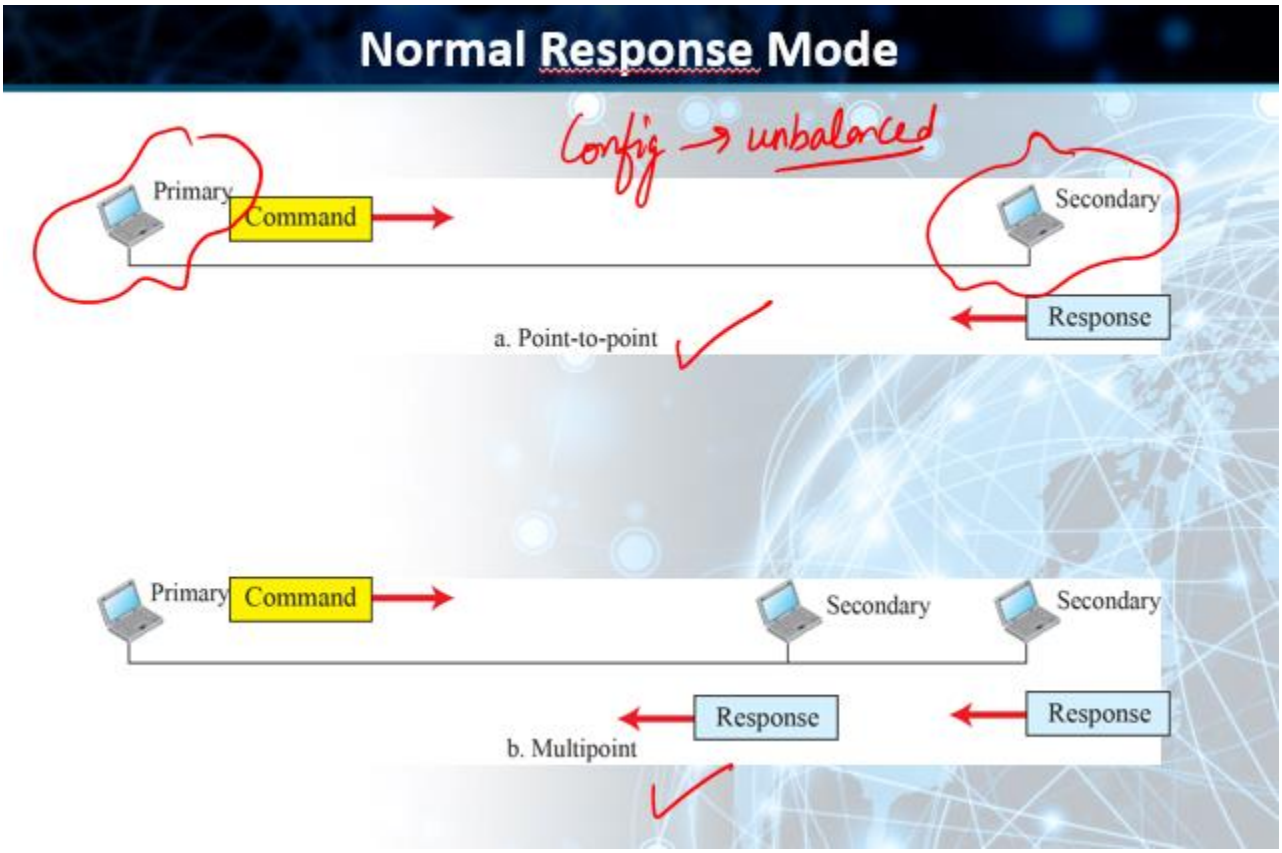
HDLC provides two common transfer modes that can be used in different configurations:

Normal Response Mode (NRM) &

Asynchronous Balanced Mode (ABM

Normal Response Mode (NRM) &

Normal Response Mode (NRM) is **an unbalanced configuration in which only the primary terminal may initiate data transfer**. The secondary terminals transmit data only in response to commands from the primary terminal. The primary terminal polls each secondary terminal to give it an opportunity to transmit any data it has.



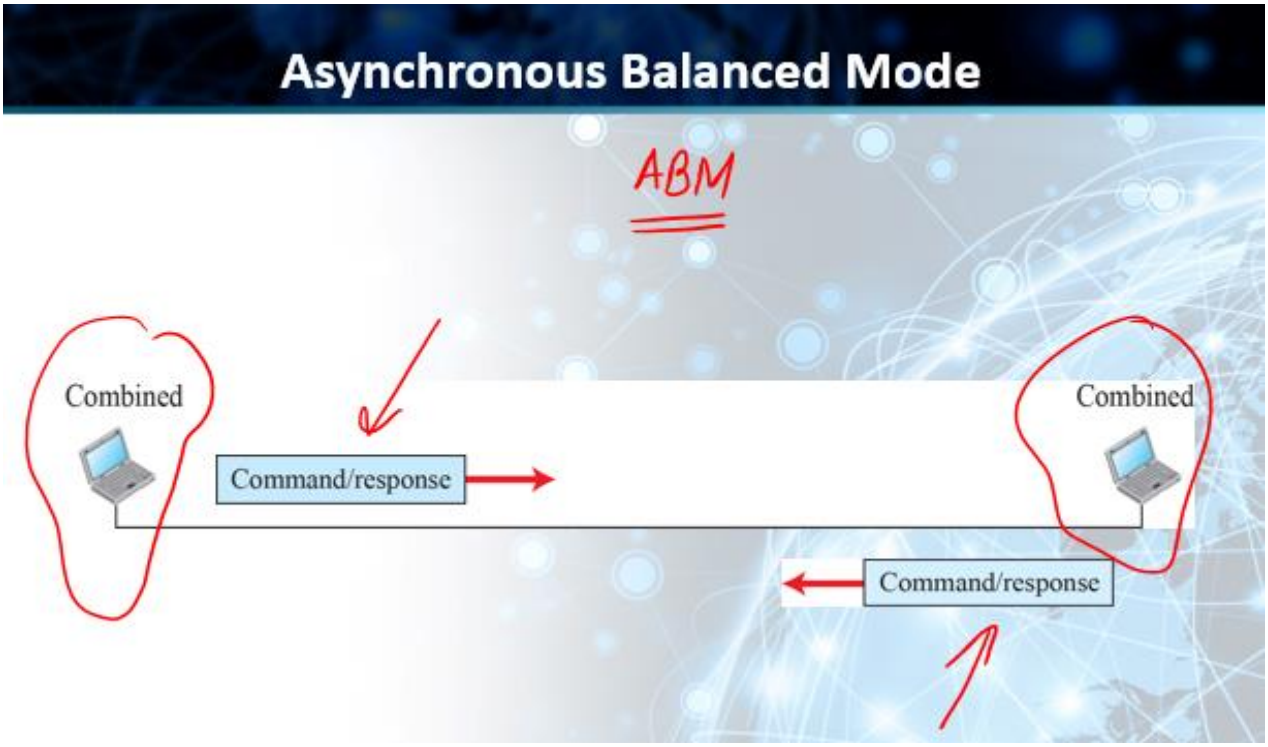
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Asynchronous Balanced Mode (ABM

Asynchronous Balanced Mode (ABM) is a balanced configuration in which either station may initialize, supervise, recover from errors, and send frames at any time. There is no master/slave relationship.



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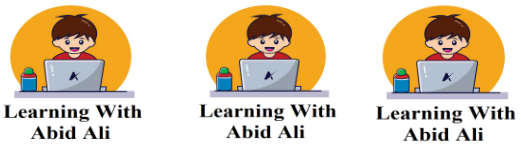
HDLC frames

HDLC defines three **(VIP mcqs)** types of frames

- ✓ **information frames (I-frames)**

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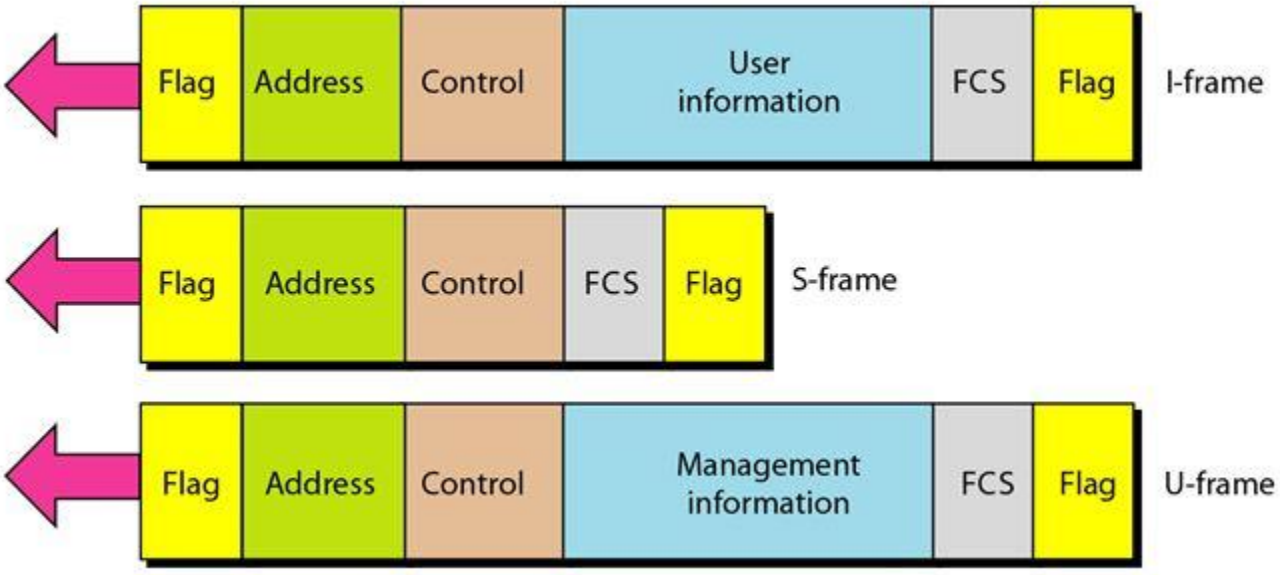
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- ✓ **Supervisory frames (S-frames)**
- ✓ **Unnumbered frames (U-frames)**

- ✓ **I-frame** – I-frames or Information frames carry user data from the network layer. They also include flow and error control information that is piggybacked on user data. The first bit of control field of I-frame is 0.

- ✓ **S-frame** – S-frames or Supervisory frames do not contain information field. They are used for flow and error control when piggybacking is not required. The first two bits of control field of S-frame is 10.

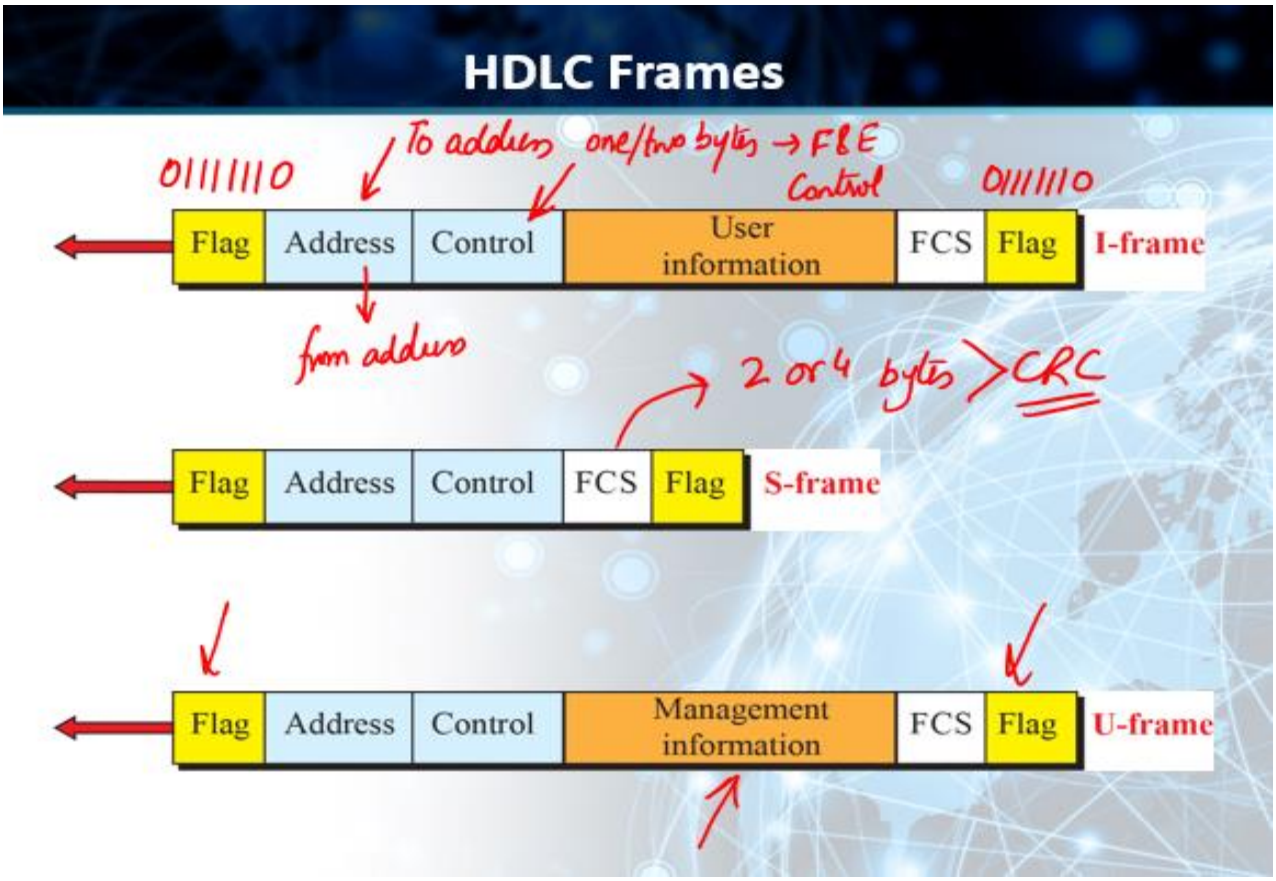
- ✓ **U-frame** – U-frames or Un-numbered frames are used for myriad miscellaneous functions, like link management. It may contain an information field, if required. The first two bits of control field of U-frame is 11.



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Point-to-Point Protocol (PPP)

- **Most common protocol for point-to-point access**
- **Millions of Internet users who need to connect their home computers to the server of an Internet service provider use PPP (VIP mcqs)**
- **To control and manage the transfer of data, there is a need for a PPP(VIP mcqs) at the data-link layer**

Services provided by PPP

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The designers of PPP have included several services to make it suitable for a point-to-point protocol, but have ignored some traditional services to make it simple

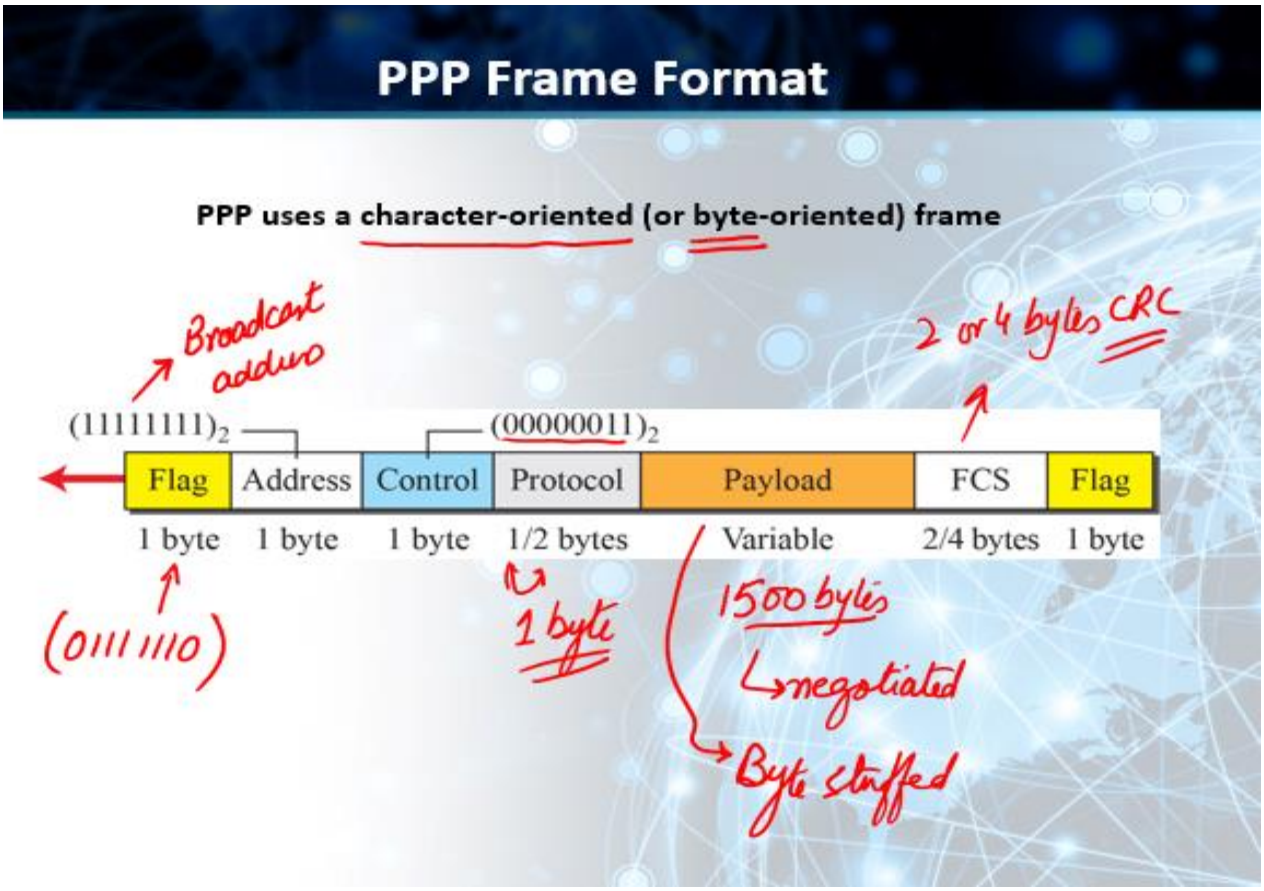
Services Included	Services Not Included
Framing	Flow Control
Link Establishment and Data Exchange	Error Correction (PPP has CRC detection only)
Authentication	No Sequence Numbering
Multilink PPP Address configuration	Absence of sophisticated Addressing Mechanism
Network Address configuration	

PPP Frame Format

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Multiplexing in PPP

- **Although PPP is a link-layer protocol, it uses another set of protocols to establish the link, authenticate and carry the network-layer data**
- **Three sets of protocols are:**
  - **Link Control Protocol (LCP)**
  - **Two Authentication Protocols (APs)**
  - **Several Network Control Protocols (NCPs)**

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Legend

LCP : Link control protocol
AP : Authentication protocol
NCP: Network control protocol

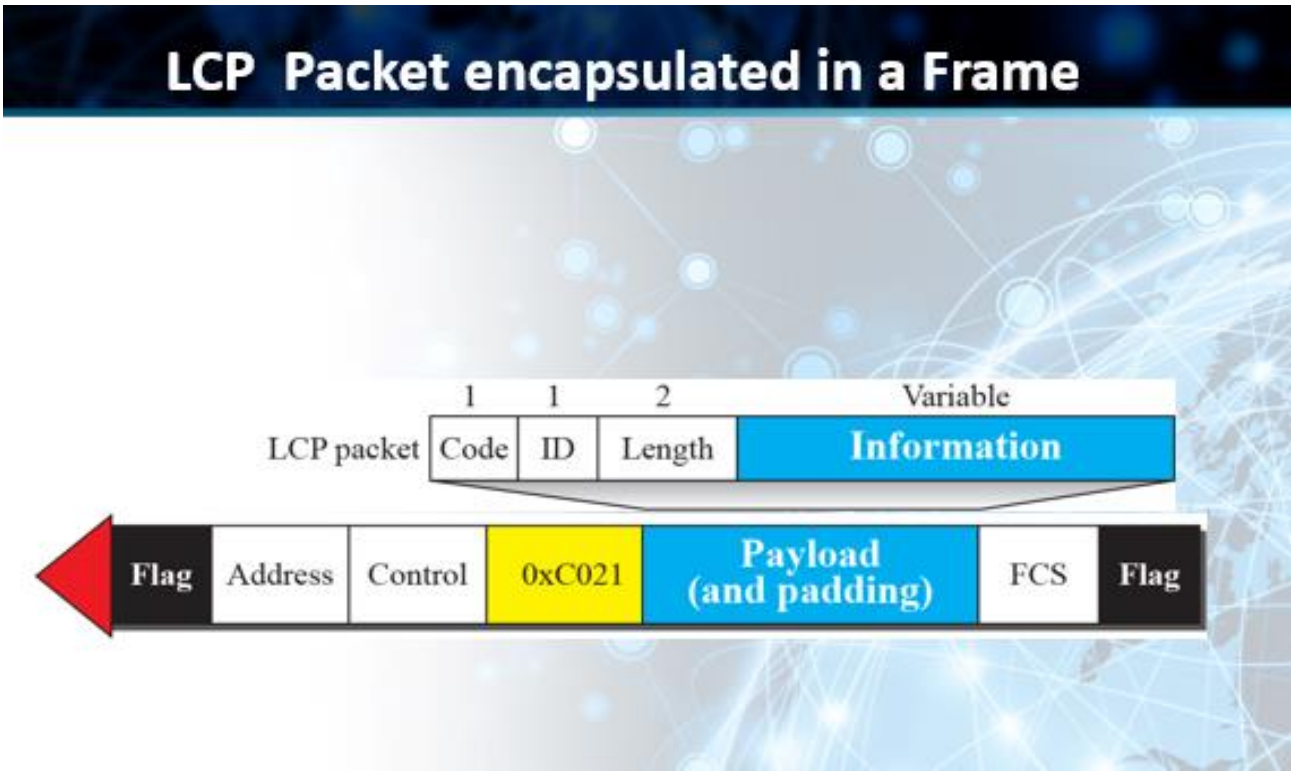
Protocol values:

LCP : 0xC021
AP : 0xC023 and 0xC223
NCP: 0x8021 and ....
Data: 0x0021 and ....

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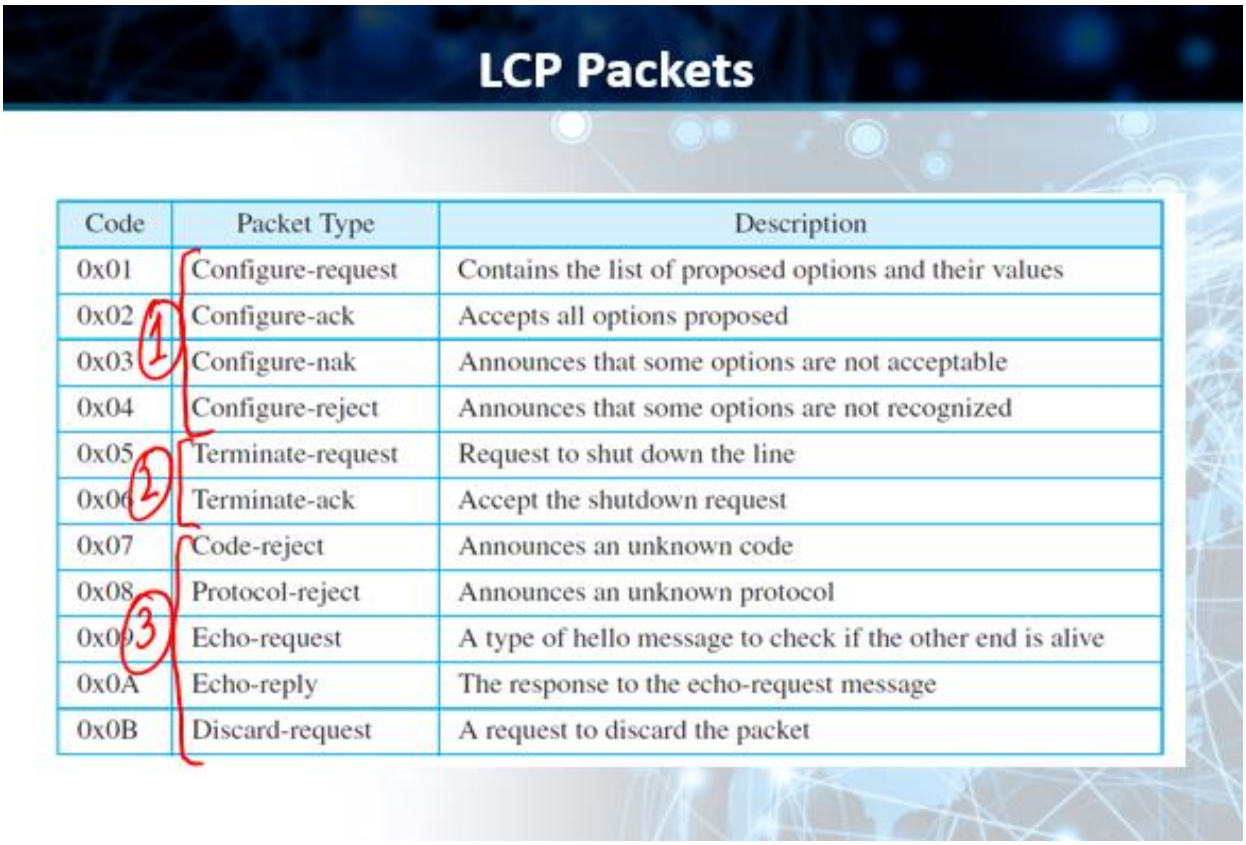
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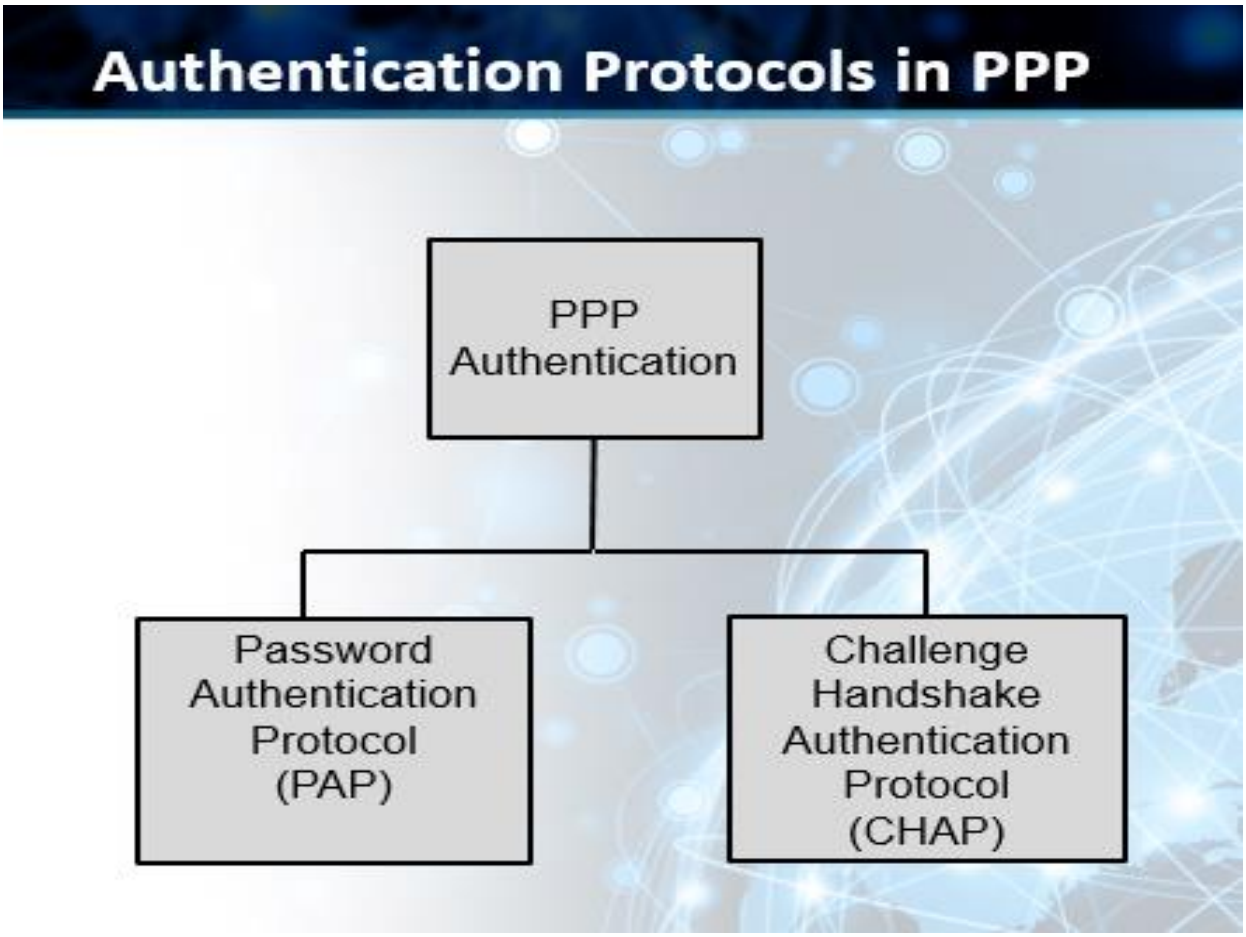
Authentication Protocols in PPP

The PPP authentication protocols are Password Authentication Protocol (PAP) and Challenge-Handshake Authentication Protocol (CHAP). Each protocol uses a secrets database that contains identification information, or security credentials, for each caller that is permitted to link to the local machine.

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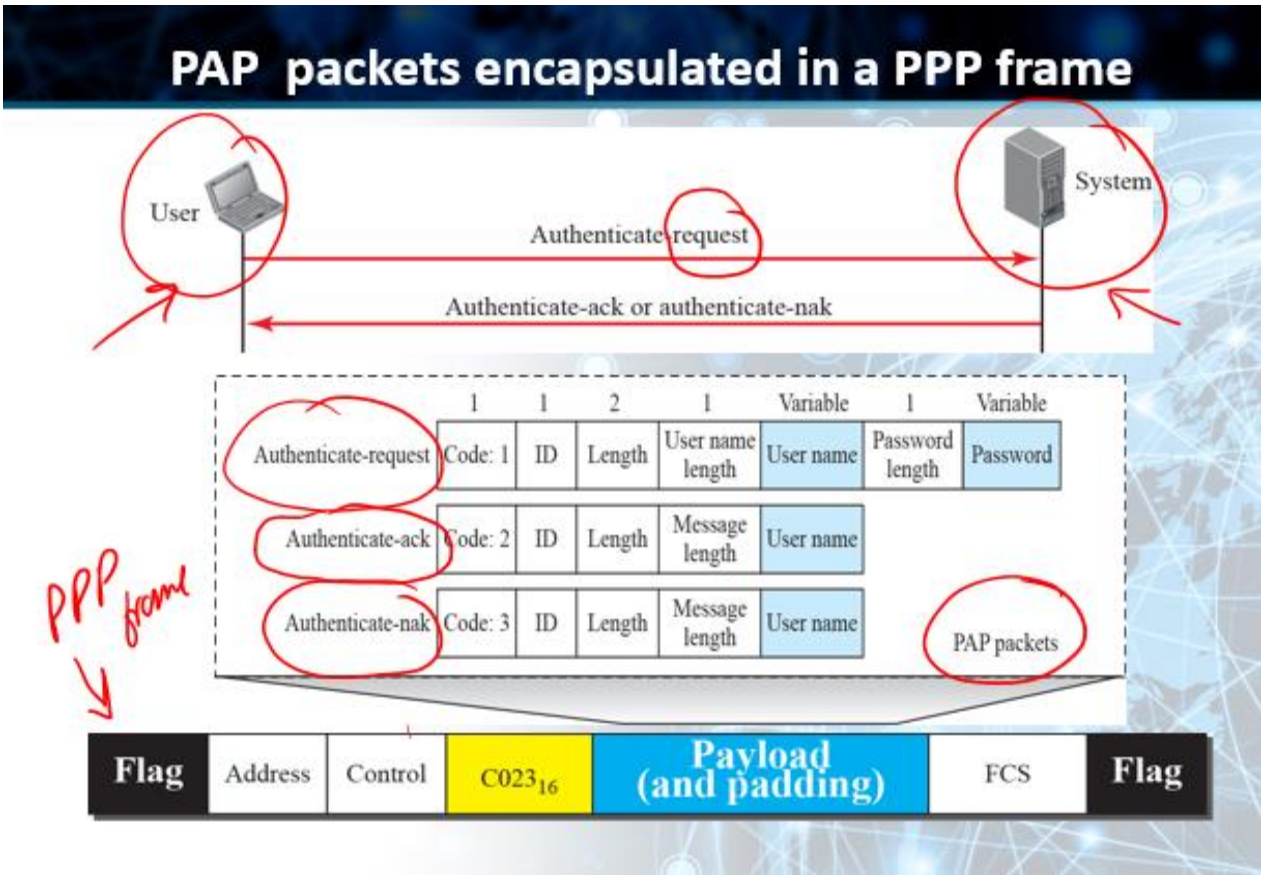
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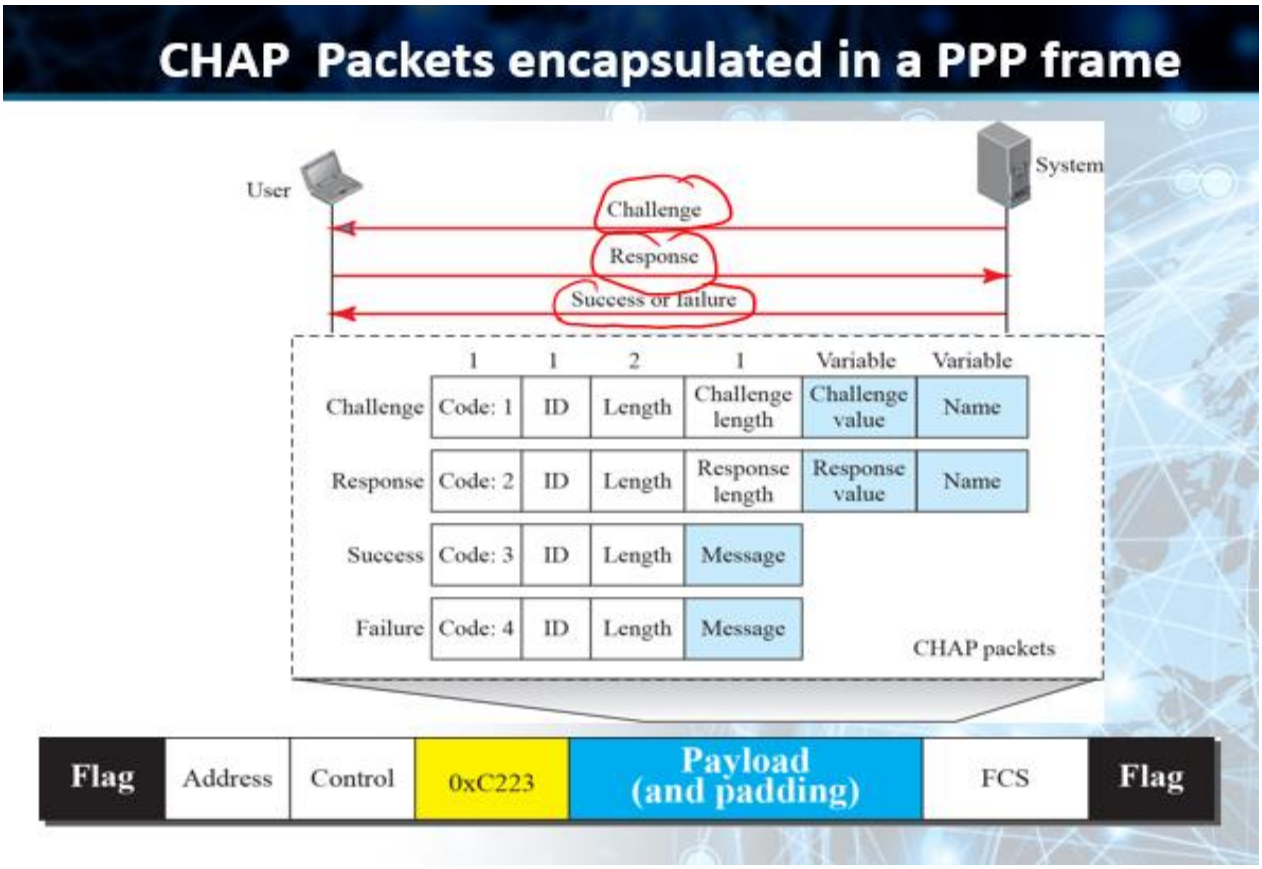
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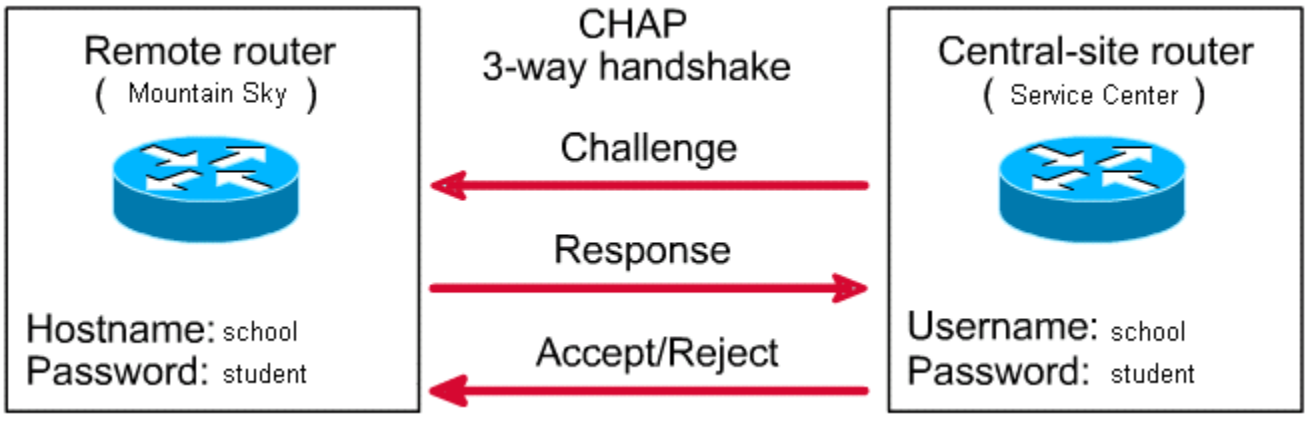






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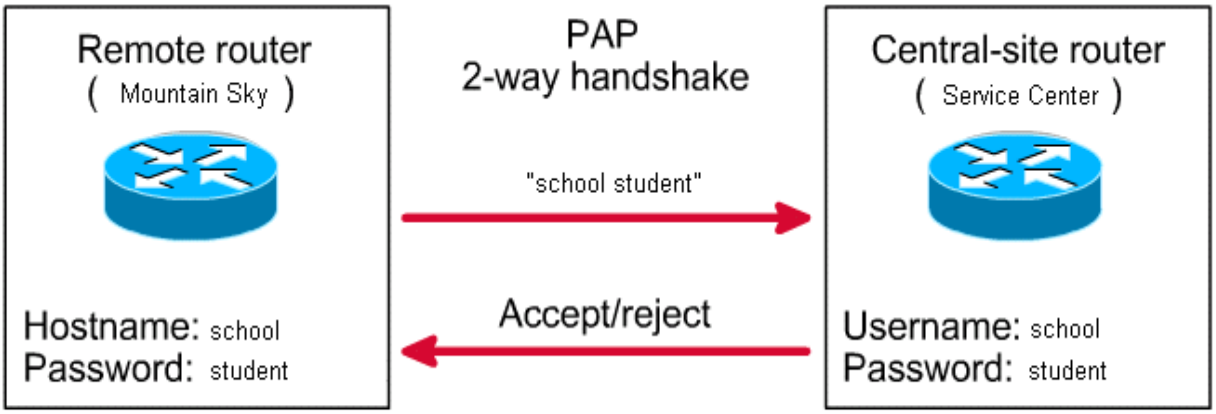


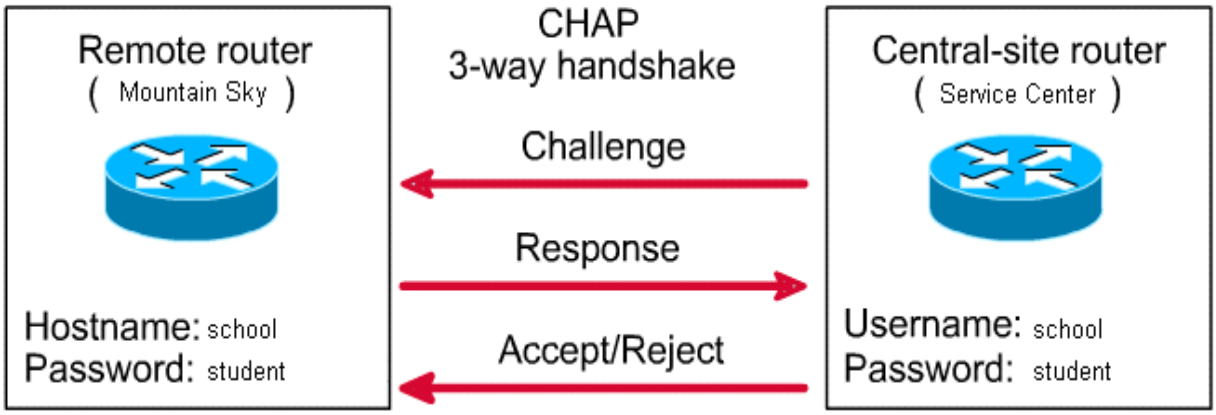


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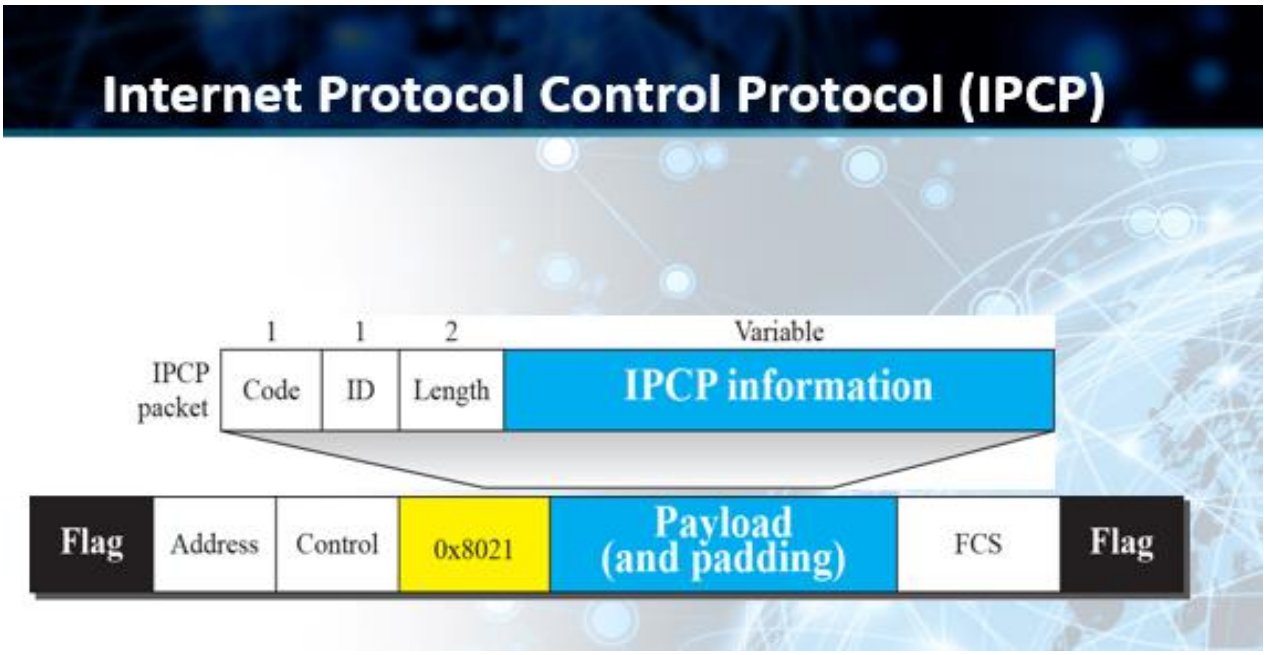




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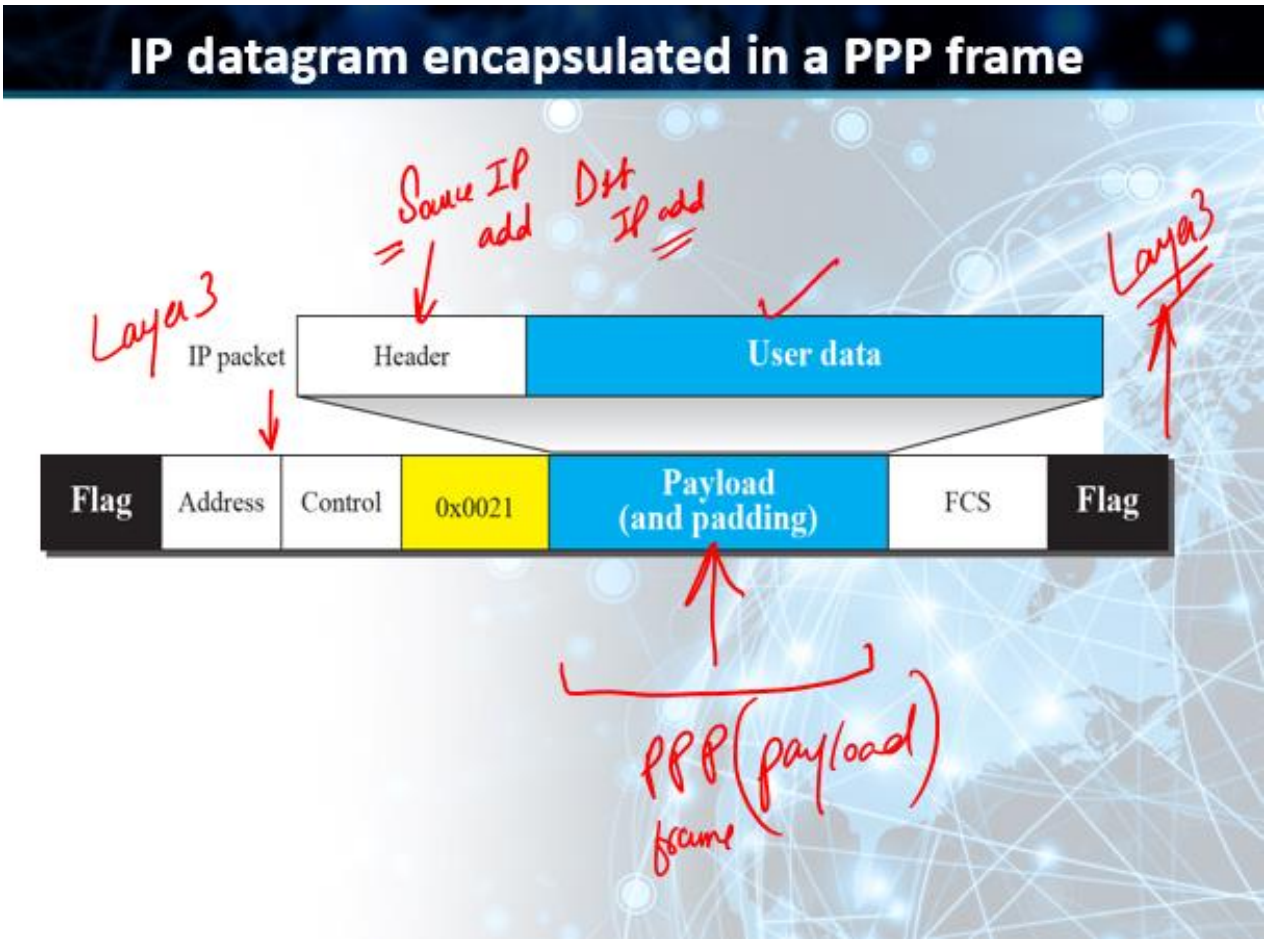
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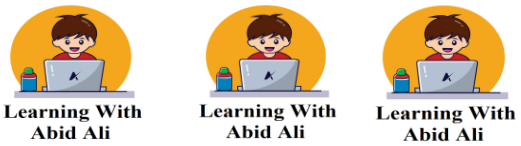
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Code values for IPCP Packets

Code	*IPCP Packet*
0x01	Configure-request
0x02	Configure-ack
0x03	Configure-nak
0x04	Configure-reject
0x05	Terminate-request
0x06	Terminate-ack
0x07	Code-reject

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Lecture Prepared By

Muhammad Abid Ali

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CS601 Lecture No 29 Topic 166 to 170

Media Access Control (MAC) Sub-Layer

- **When nodes use a multipoint or broadcast link, we need a multiple-access protocol to coordinate access to the link**

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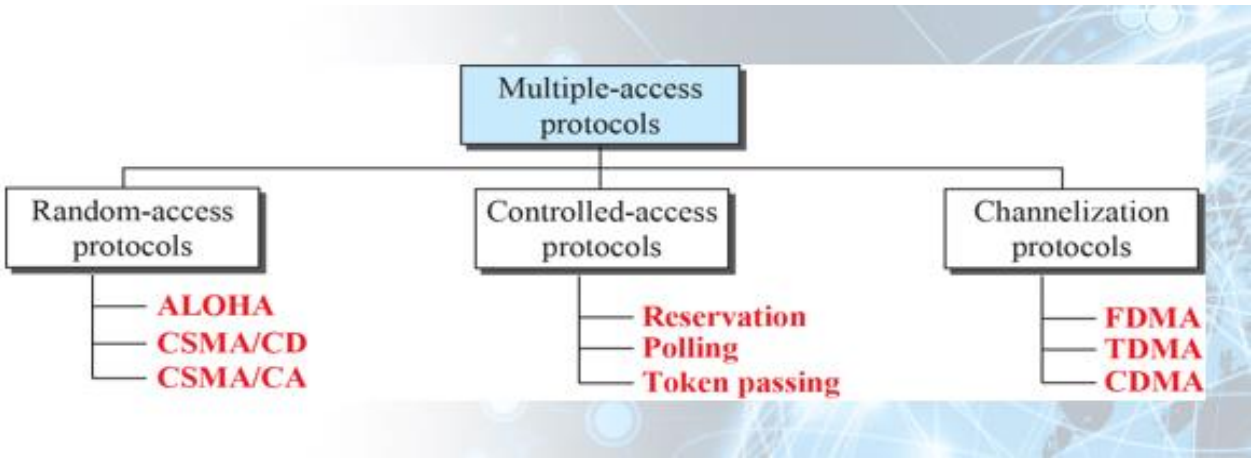
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- **Many protocols have been devised to handle access to a shared link**
- **All of these protocols belong to Media Access Control (MAC) sub-layer**

What is the function of medium access control MAC sub layer?

The medium access control (MAC) is a sublayer of the data link layer of the open system interconnections (OSI) reference model for data transmission. It is responsible for flow control and multiplexing for transmission medium. It controls the transmission of data packets via remotely shared channels.

Taxonomy of Multiple-Access Protocols



Random Access

- **In random-access or contention no station is superior to the other and none is assigned control over the other**
- **Station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send**
- **This decision depends on the state of the medium (idle or busy)**

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ALOHA

- **ALOHA, the earliest random-access method, was developed in early 1970s**
- **Designed for a radio (wireless) LAN, but it can be used on any shared medium**
- **Potential collisions in this arrangement as the medium is shared between the stations**
- **When a station sends data, another station may attempt to do so at the same time**
- **The data from the two stations collide and become garbled**

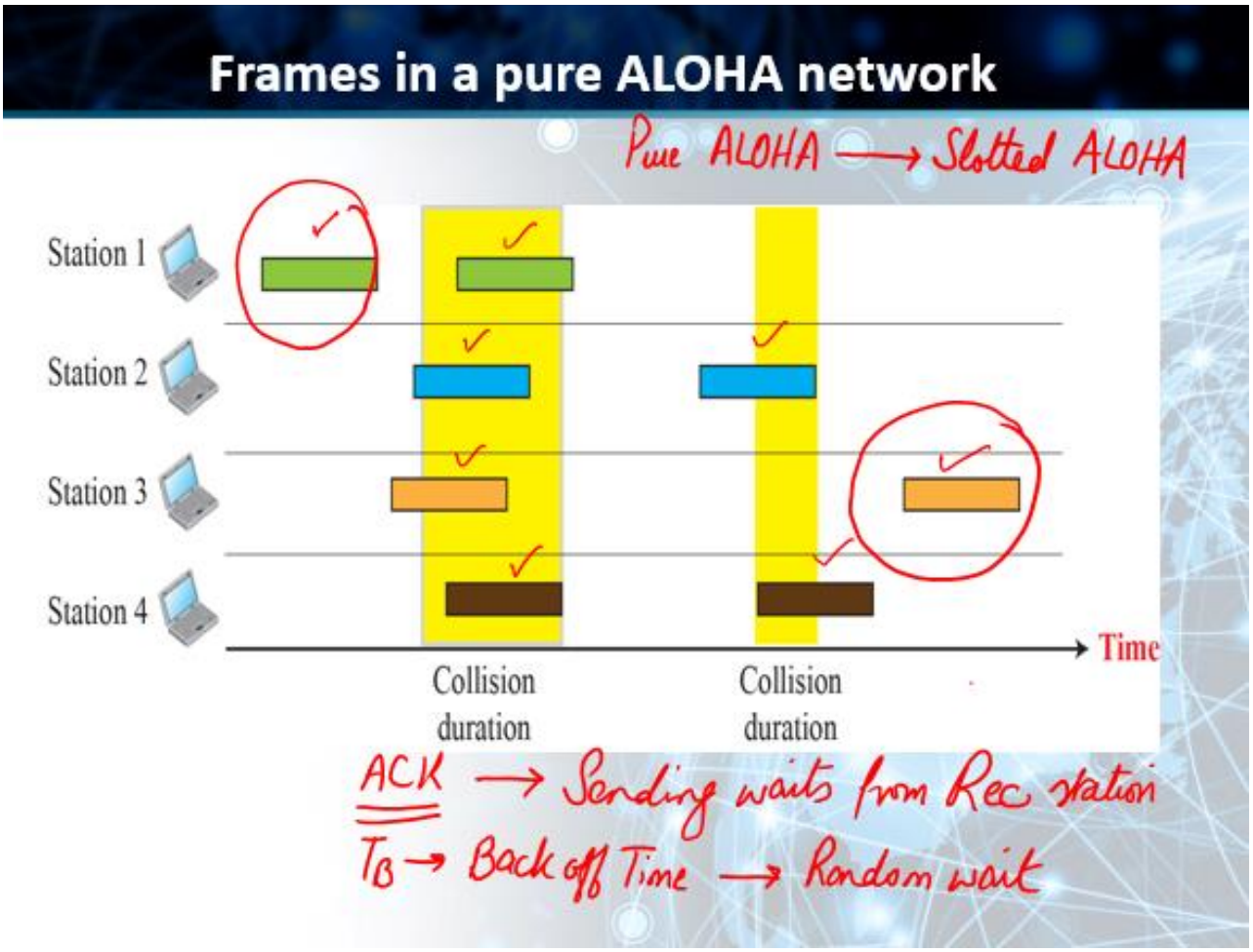
Difference between Pure Aloha and Slotted Aloha

In Slotted Aloha, any station can transmit data only at the beginning of a time slot. In Pure Aloha, time is continuous and is not globally synchronized. In Slotted Aloha, time is discrete and is globally synchronized.

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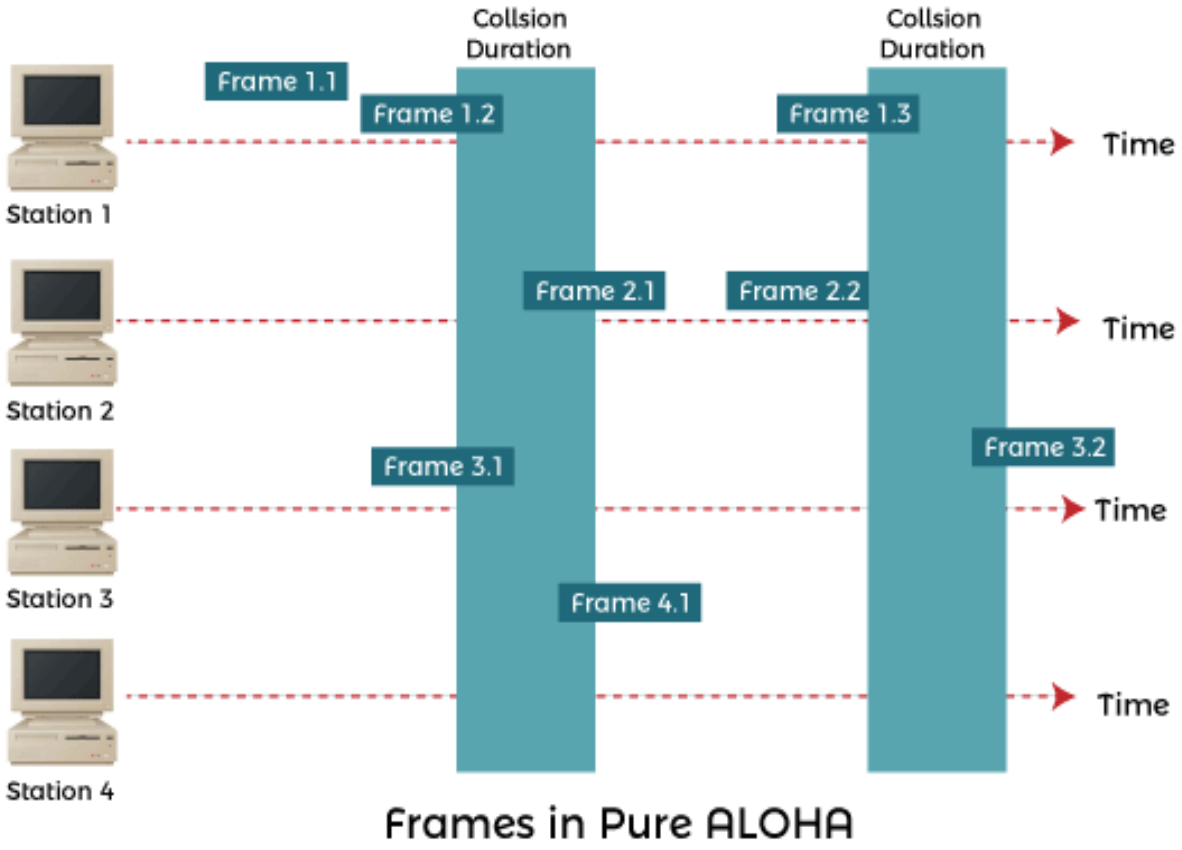
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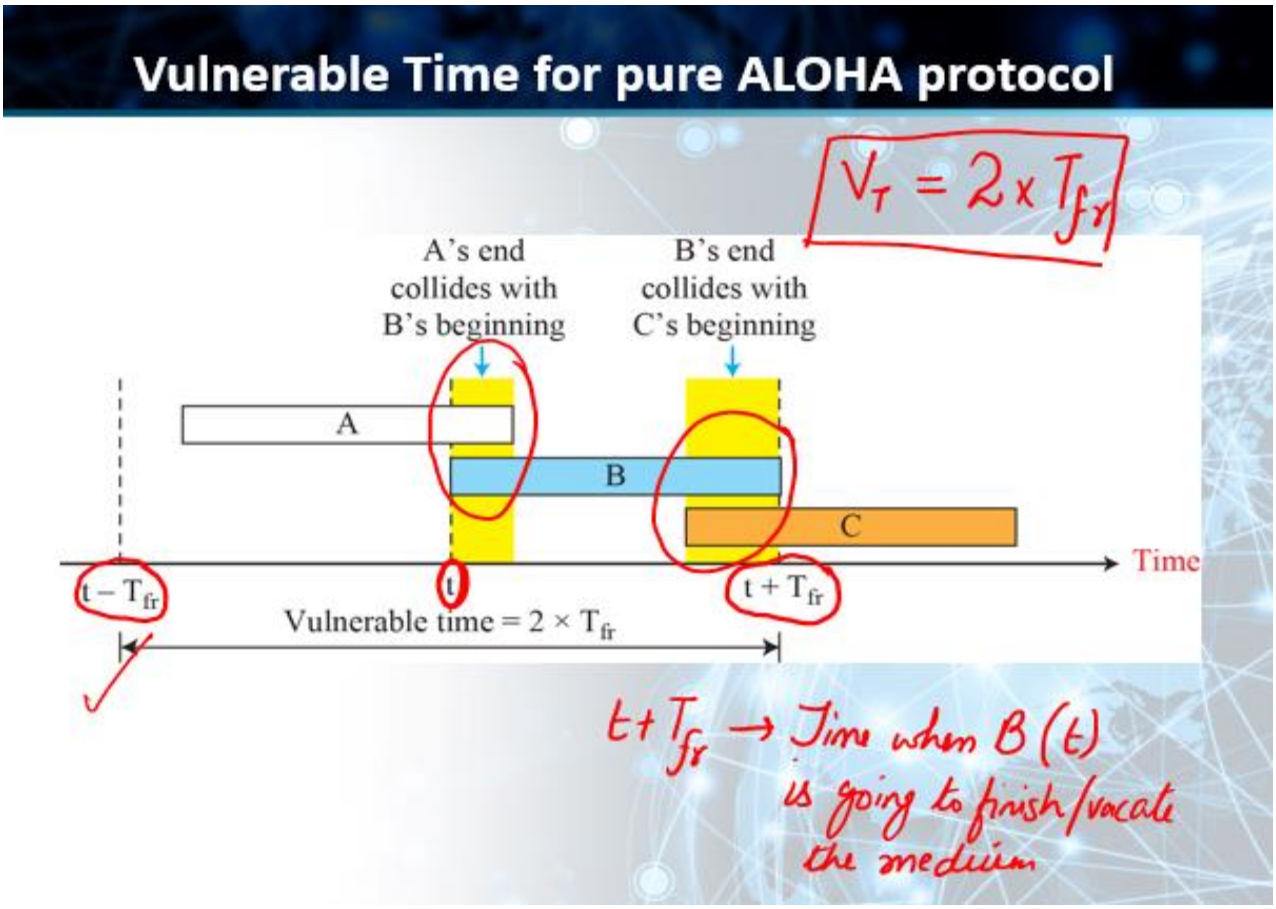


Frames in Pure ALOHA

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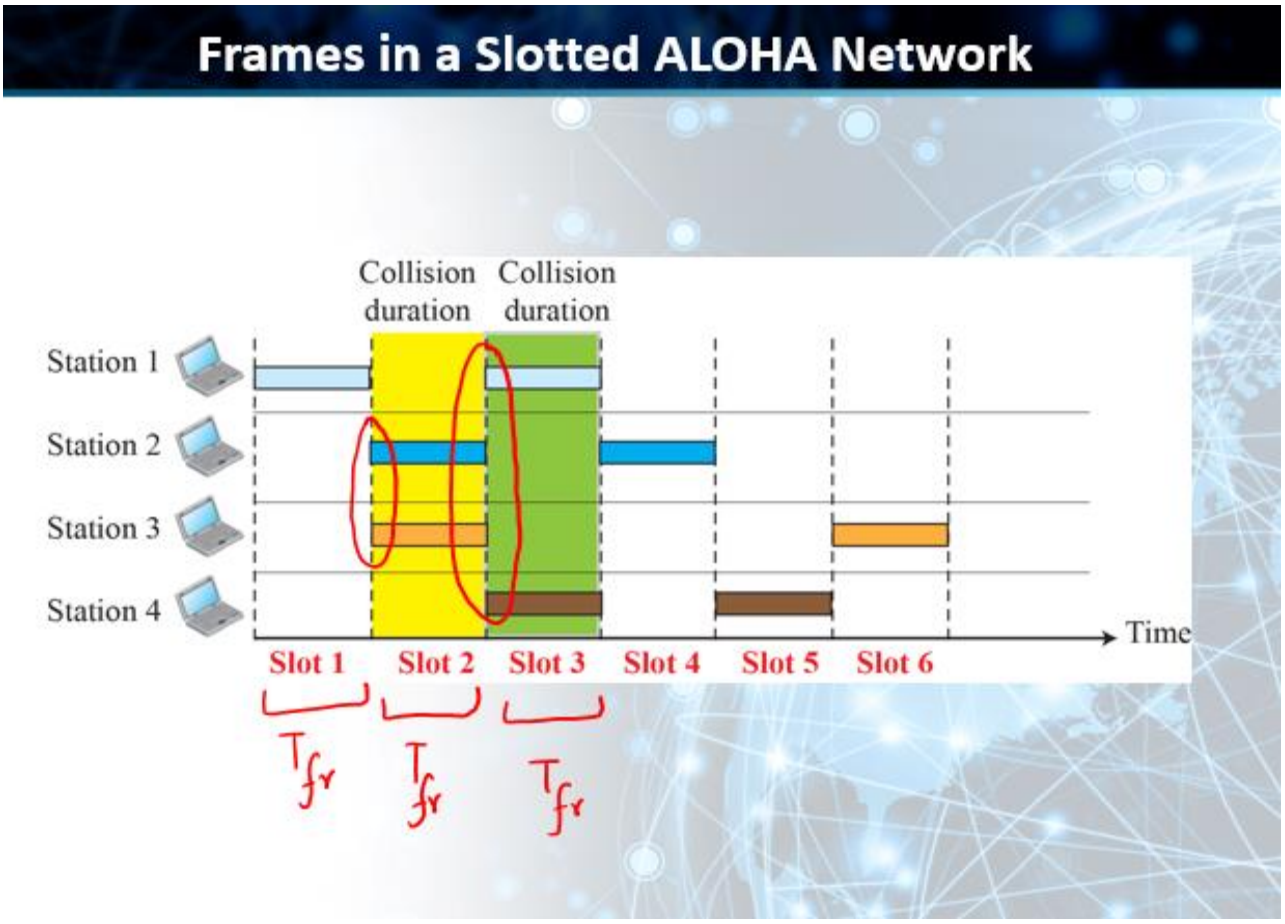
Slotted ALOHA

- **We divide time into slots of Tfr sec and force the station to send only at the beginning of the slot**
- **Invented to improve the efficiency of pure ALOHA**
- **If a station misses the time slot, it must wait until beginning of next time slot reducing vulnerable time to Tfr (vs. 2 x Tfr for pure ALOHA)**

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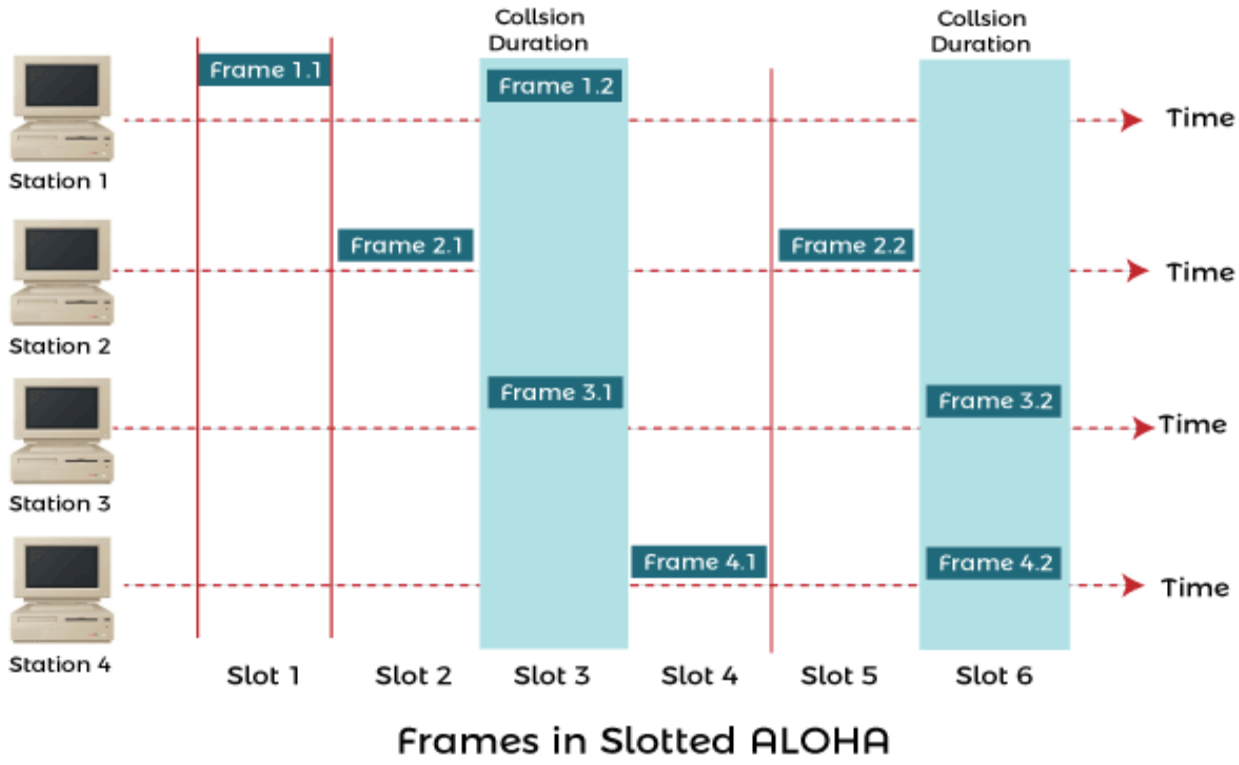
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Frames in Slotted ALOHA

What is aloha?

Aloha is designed for wireless LAN (Local Area Network) but can also be used in a shared medium to transmit data. In aloha, any station can transmit data to a channel at any time. It does not require any carrier sensing.

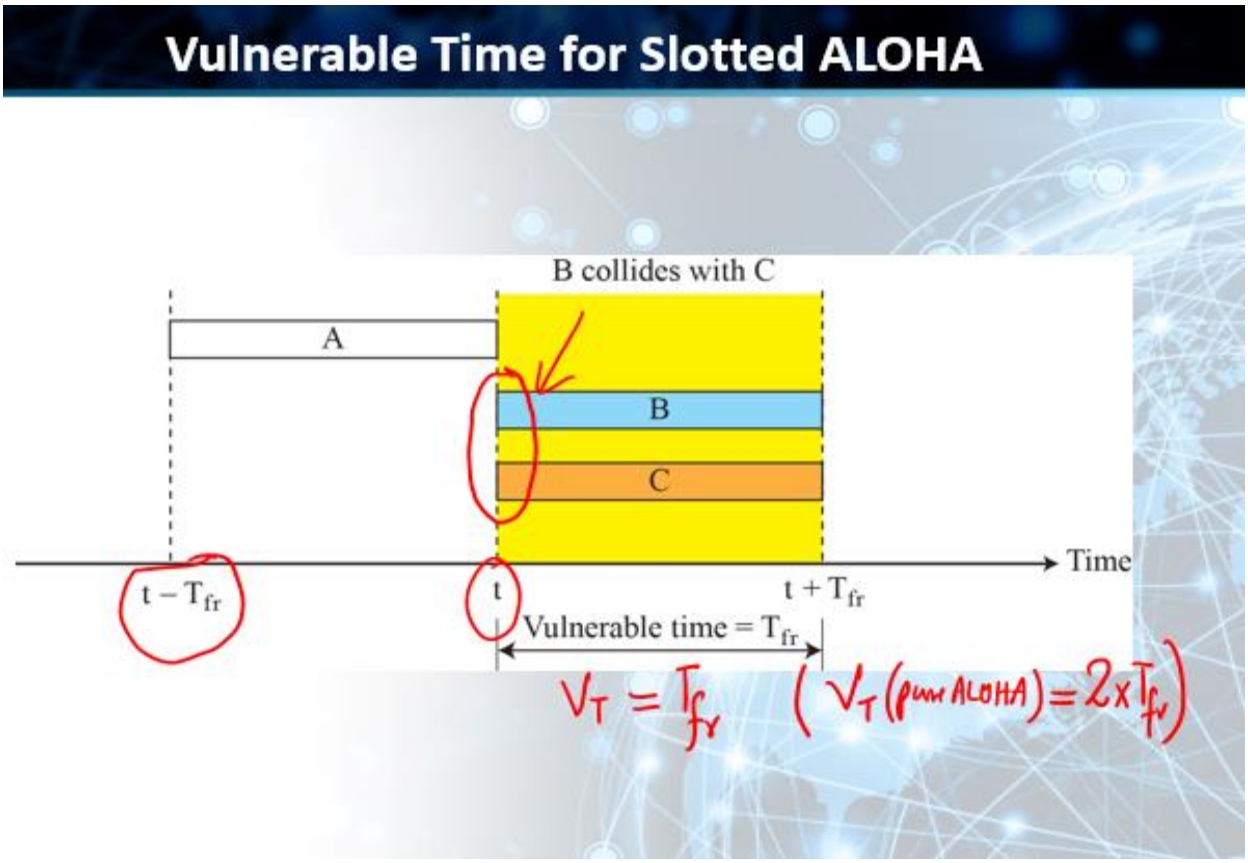
Pure Aloha

Pure aloha is used when data is available for sending over a channel at stations. In pure Aloha, when each station transmits data to a channel without checking whether the channel is idle or not, the chances of collision may occur, and the data frame can be lost.

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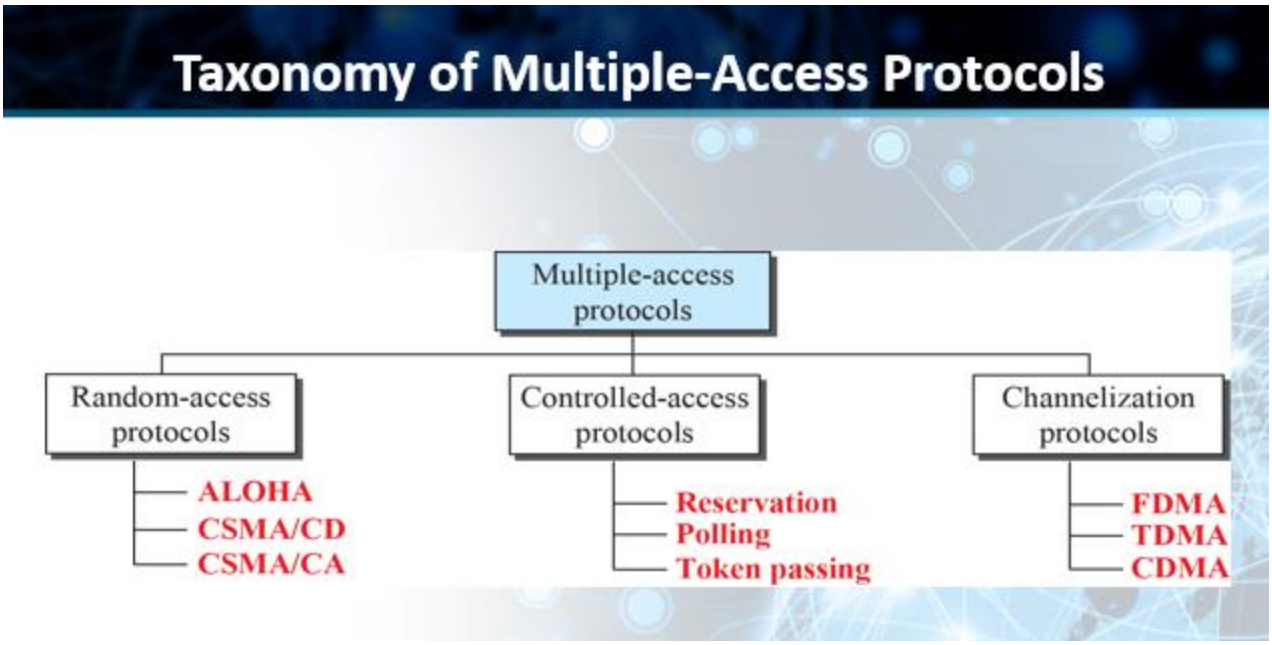
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Carrier Sense Multiple Access (CSMA)

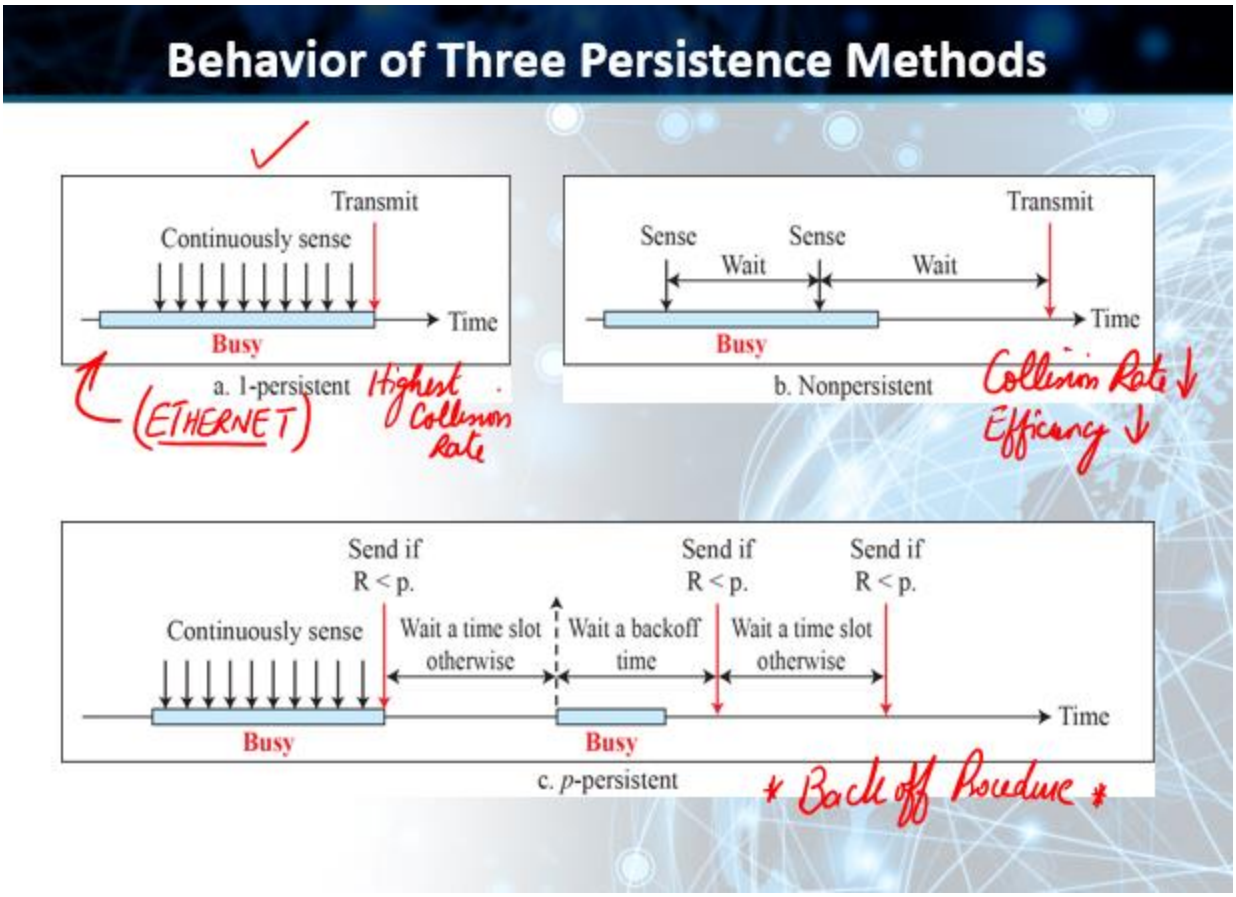
- **To minimize the chance of collision and, therefore, increase the performance, CSMA was developed**
- **The chance of collision is reduced as the station is required to sense/listen to the medium before sending data**
- **'Sense before transmit' or 'listen before talk'**

This method was developed to decrease the chances of collisions when two or more stations start sending their signals over the data link layer. Carrier Sense multiple access requires that each station first check the state of the medium before sending.

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Carrier Sense Multiple Access/Collision Detection

- **CSMA method does not specify the procedure following a collision**
- **CSMA/CD augments the algorithm to handle the collision**
- **The station monitors the medium after it sends a frame to see if the transmission was successful. If there is a collision, the frame is sent again**

In this method, a station monitors the medium after it sends a frame to see if the transmission was successful. If successful, the transmission is finished, if not, the frame is sent again.

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Throughput and Efficiency: The throughput of CSMA/CD is much greater than pure or slotted ALOHA.

Carrier Sense Multiple Access/Collision Avoidance

- **CSMA/CA was invented for Wireless Networks**
- **Collisions are avoided through the use of three strategies:**
  - ✓ **The Interfame Space**
  - ✓ **The Contention Window**
  - ✓ **Acknowledgements**

These are three types of strategies:

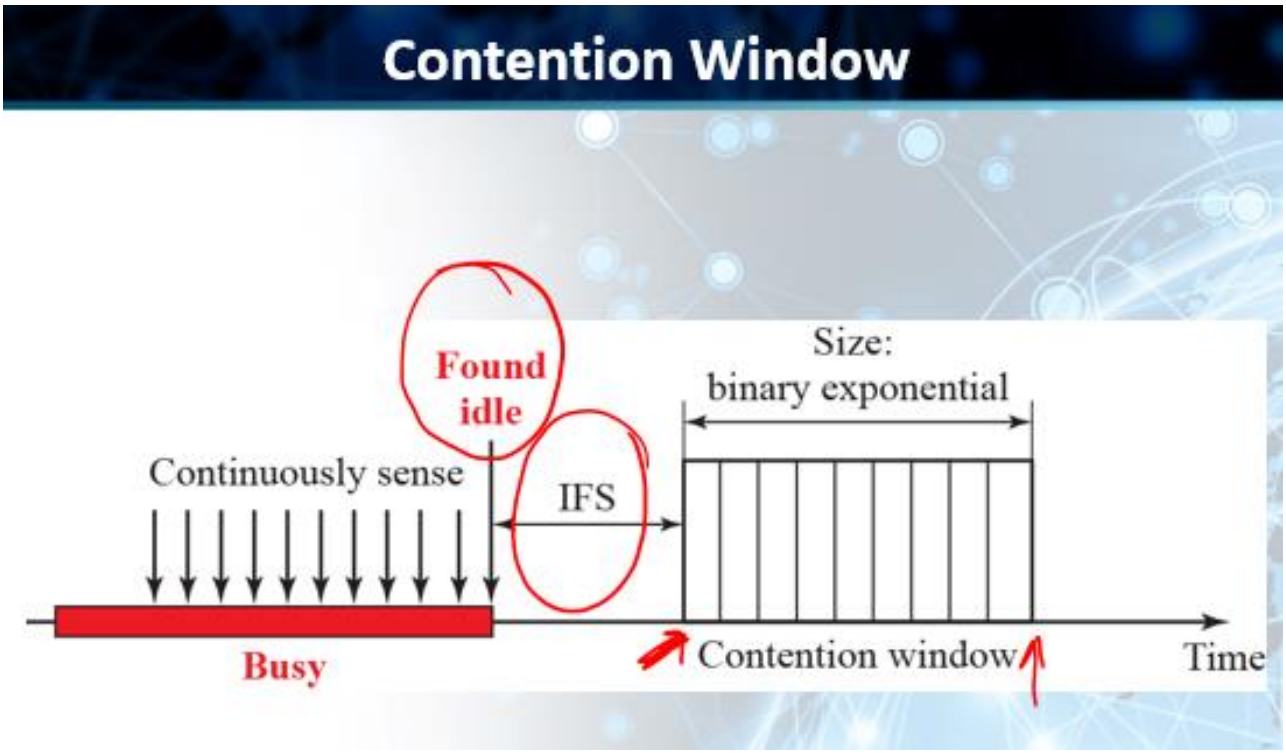
1. **InterFrame Space (IFS):** When a station finds the channel busy it senses the channel again, when the station finds a channel to be idle it waits for a period of time called **IFS time**. IFS can also be used to define the priority of a station or a frame. Higher the IFS lower is the priority.
2. **Contention Window:** It is the amount of time divided into slots. A station that is ready to send frames chooses a random number of slots as **wait time**.
3. **Acknowledgments:** The positive acknowledgments and time-out timer can help guarantee a successful transmission of the frame.

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- **Interframe Space (IFS): Collisions are avoided by deferring transmission even if the channel is idle**
- **Contention Window: Amount of time divided into slots. Station chooses a random number of slots as its wait time (one slot first time and double each time system cannot detect an idle channel)**

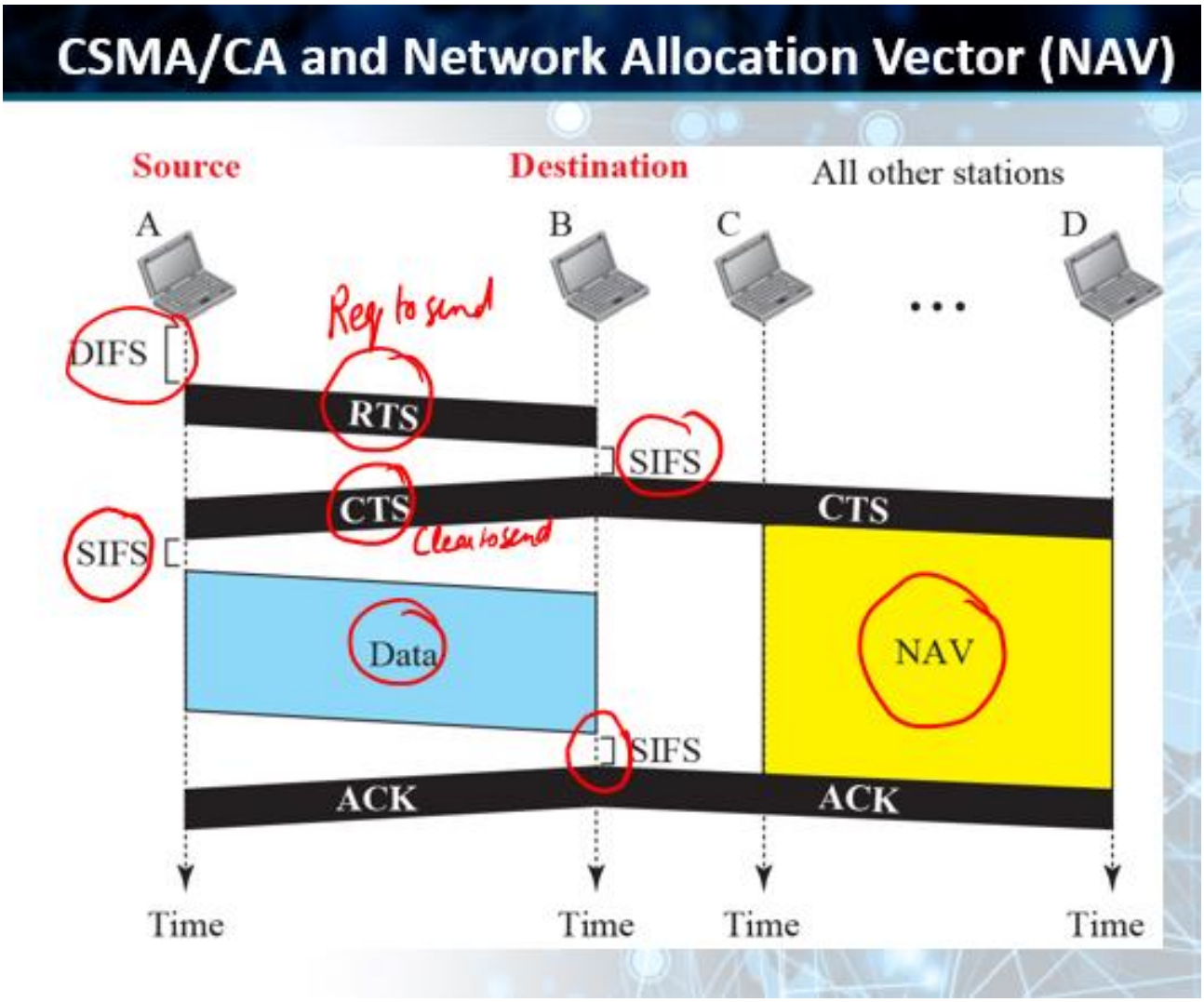


- **Acknowledgement: Positive acknowledgement and time-out timer can help guarantee that the receiver has received the frame**

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CONTROLLED ACCESS

- **The stations consult one another to find which station has the right to send**
- **A station cannot send unless authorized by other stations**
- **We discuss three controlled-access methods:**
  - ✓ **Reservation**

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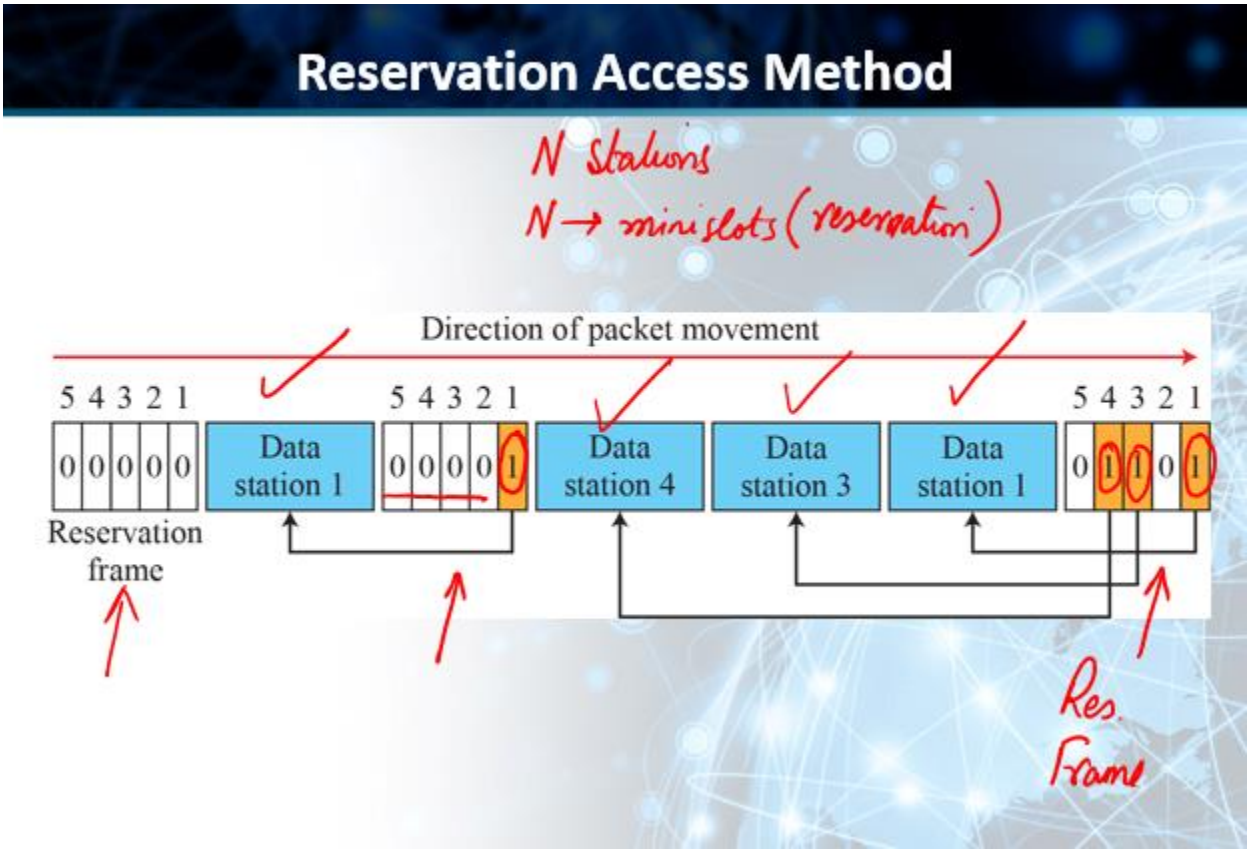
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- ✓ **Polling**
- ✓ **Token Passing**

Reservation

- **In the reservation method, a station needs to make a reservation before sending data**
- **Time is divided into intervals**
- **In each interval, a reservation frame precedes the data frames sent in that interval**



Polling

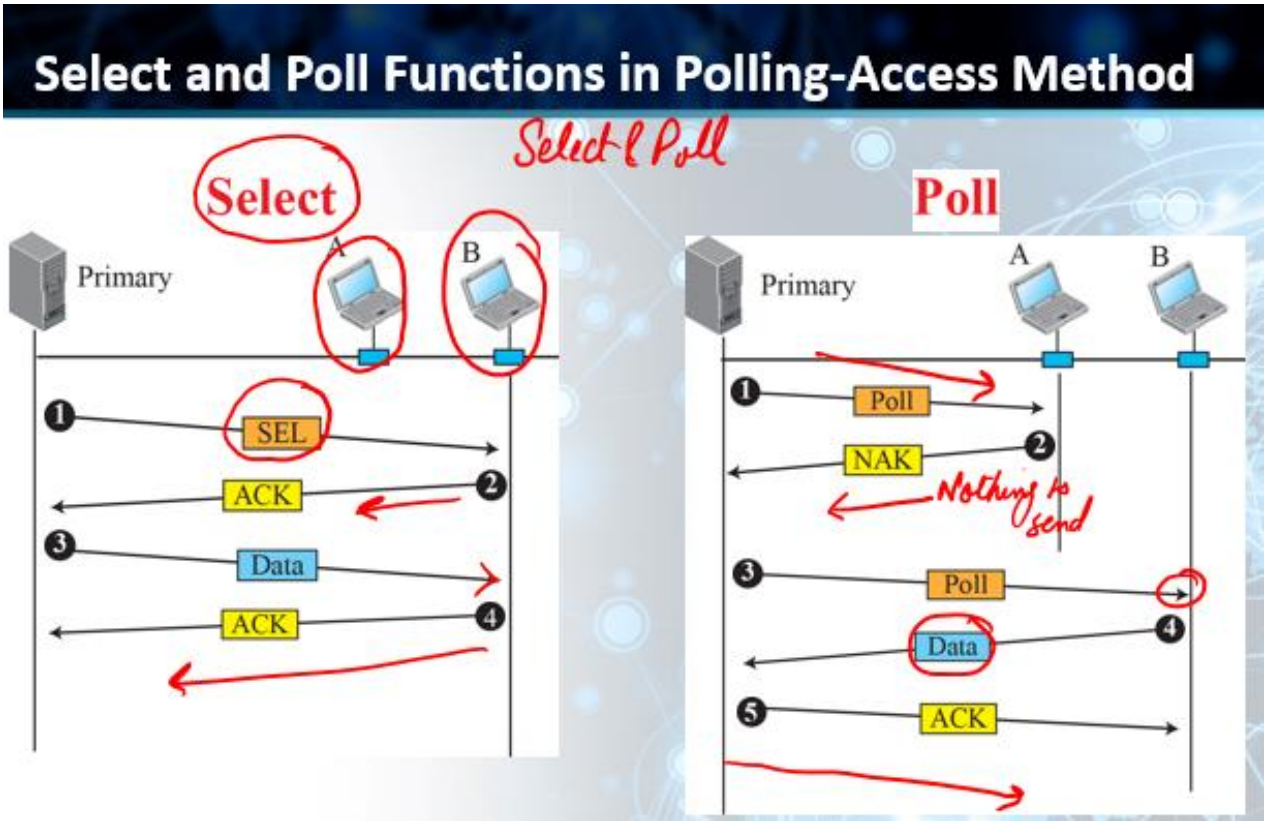
- **Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations**

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- **All data exchanges must be made through primary device even when the ultimate destination is a secondary device**
- **The primary device controls the link; the secondary devices follow its instructions**



Token Passing

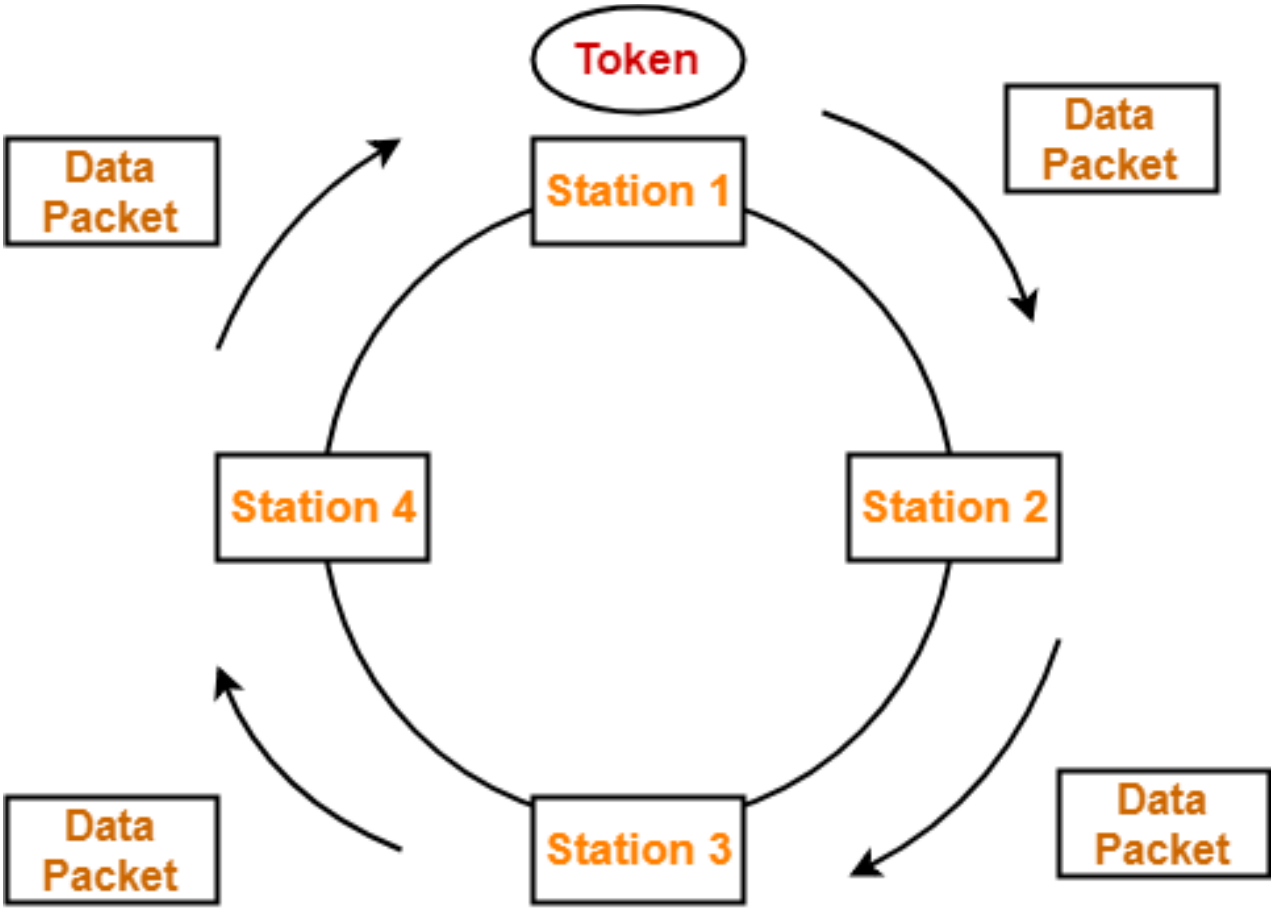
- **In the token-passing method, the stations in a network are organized in a logical ring**
- **For each station, there is a predecessor and a successor**
- **The predecessor is the station which is logically before the station in the ring; the successor is the station which is after the station in the ring**
- **Special packet called TOKEN circulates through the ring**
- **Possession of TOKEN gives the station the right to send the data**

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- **TOKEN Management is required to manage possession time, Token monitoring, priority assignment etc.**



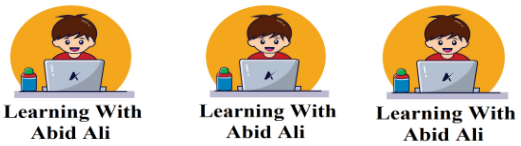
Delayed Token Reinsertion Token Passing

CHANNELIZATION (Channel Partition)

- **The available bandwidth of a link is shared in time, frequency, or through code, among different stations**

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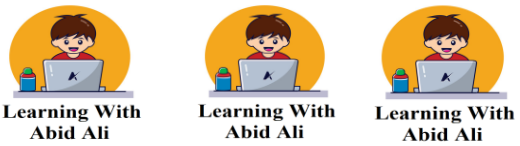
- **We discuss three protocols:**
  - ✓ **Frequency Division Multiple Access (FDMA)**
  - ✓ **Time Division multiple Access (TDMA)**
  - ✓ **Code Division Multiple Access (CDMA)**

Frequency-Division Multiple Access (FDMA)

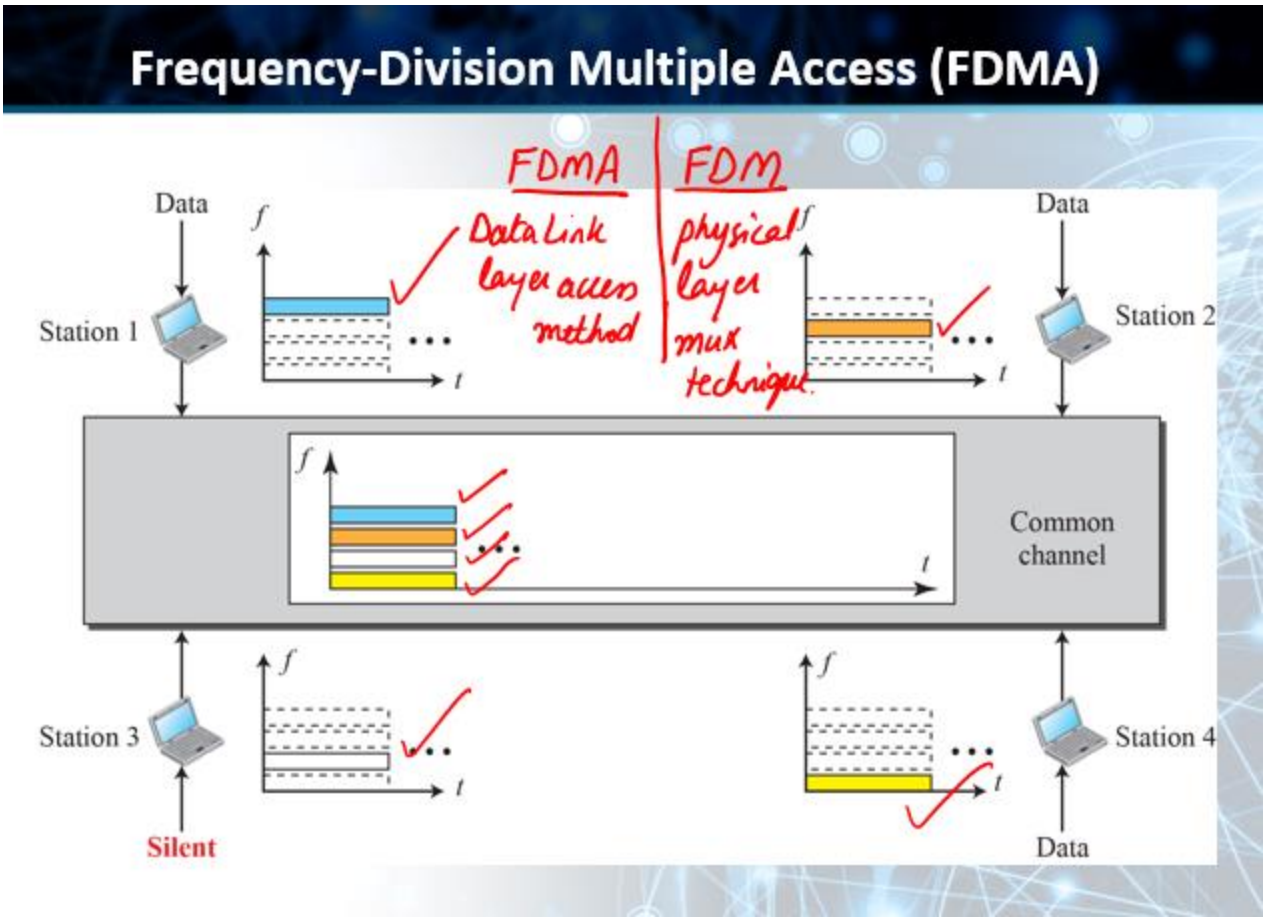
- **In FDMA, the available bandwidth is divided into frequency bands**
- **Each station is allocated a band to send its data i.e. each band is reserved for a specific station, and it belongs to the station all the time**
- **Each station also uses a bandpass filter to confine the transmitter frequencies**

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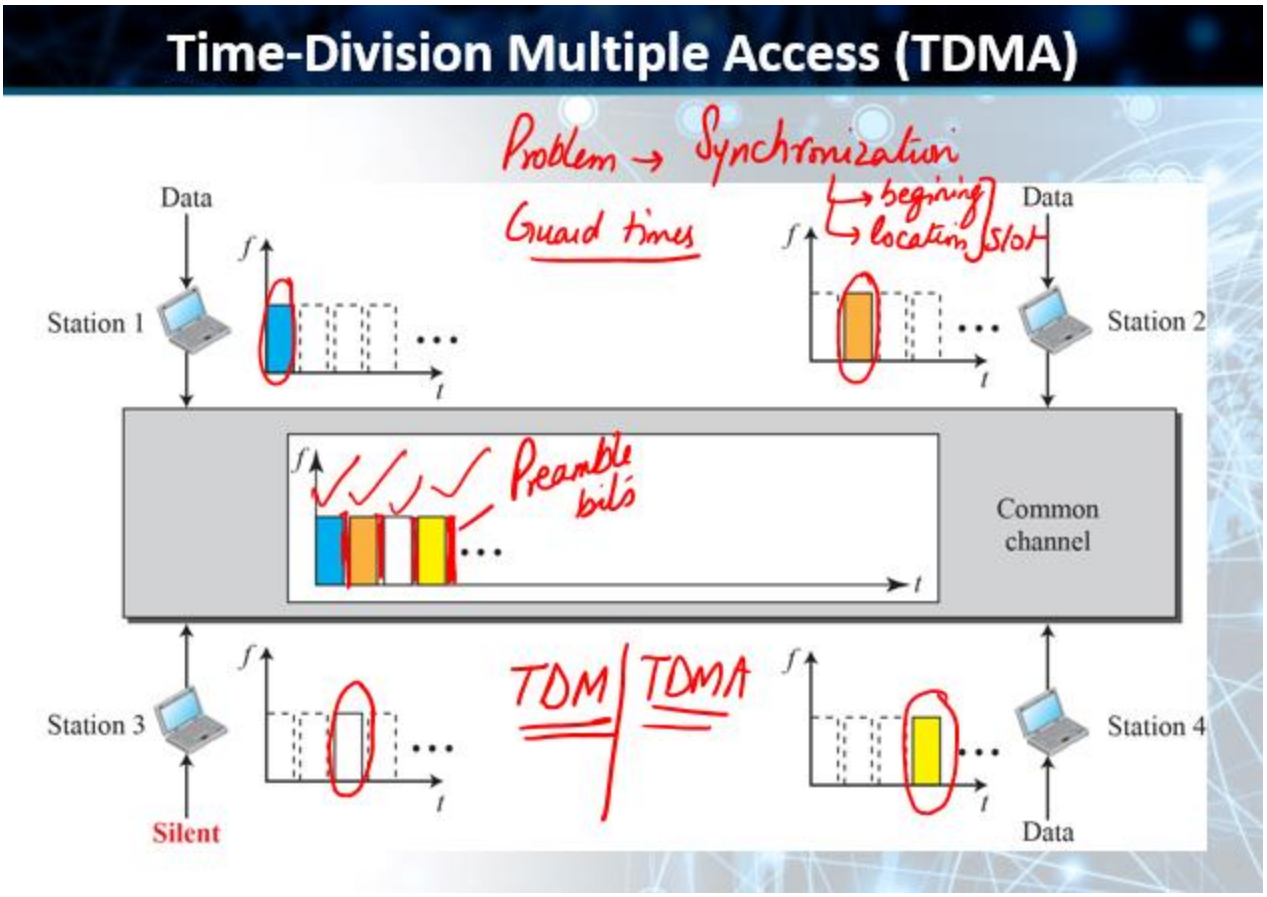
TDMA

- **Stations share the bandwidth of the channel in time**
- **Each station is allocated a time slot during which it can send data**
- **Each station transmits its data in its assigned time slot**

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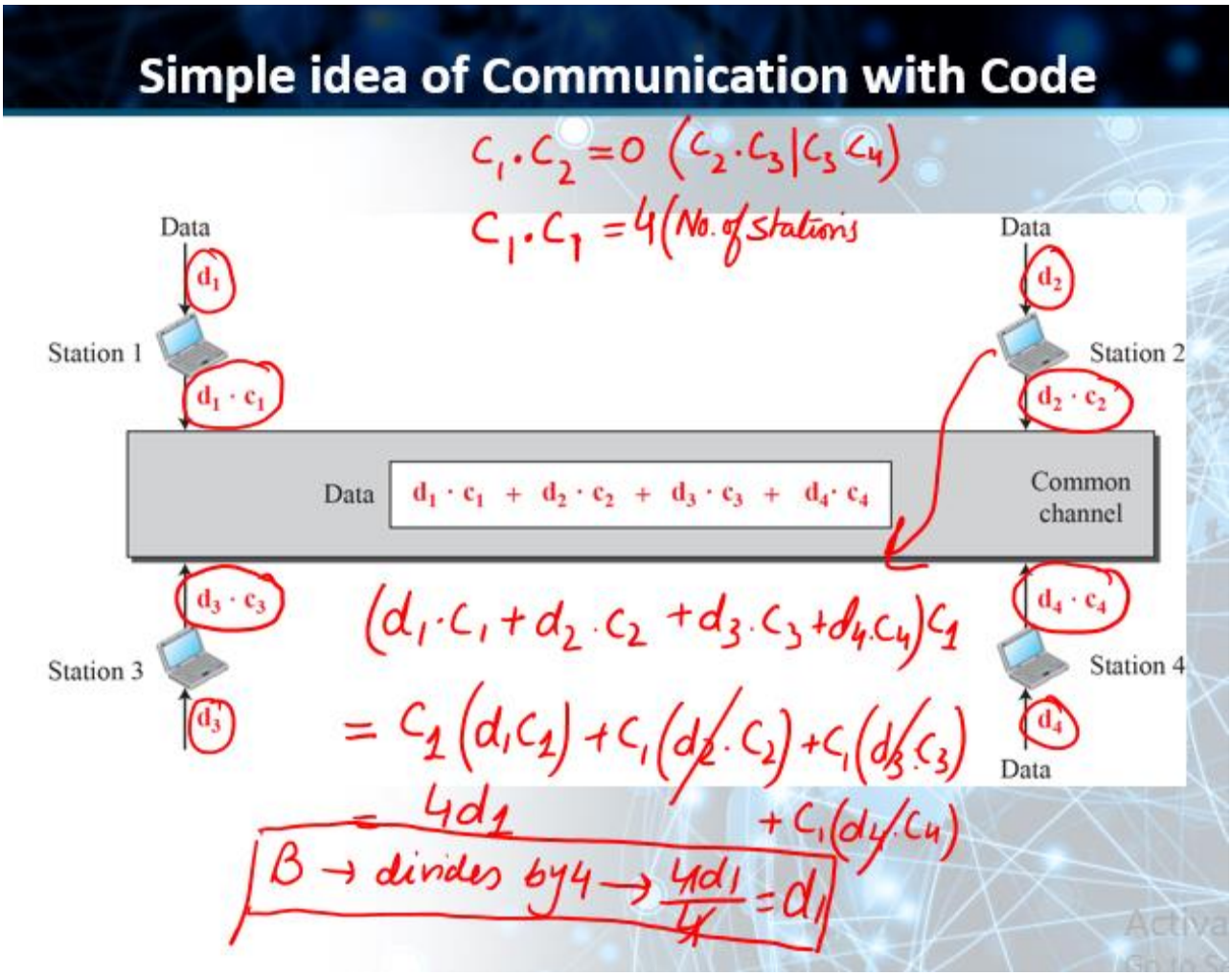
Code Division Multiple Access (CDMA)

- **CDMA differs from FDMA in that only one channel occupies the entire bandwidth of the link**
- **CDMA differs from TDMA in that all stations can send data simultaneously; there is no timesharing**

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Lecture Prepared By

Muhammad Abid Ali

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CS601 Lecture No 31 Topic 180 to 185

Ethernet Protocol

- **Data-link layer and the physical layer are the territory of the local and wide area networks**
- **We can have wired or wireless networks**

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Ethernet protocol is **a typical LAN technology**. Standard Ethernet-based local area networks transmit data at speed up to 10 Mbps. New Ethernet cards known as Fast Ethernet represent high-speed LAN technology as it can provide data transfer rates as high as 100 Mbps.

Where is Ethernet protocol used?



local area networks

Ethernet is a family of wired computer networking technologies commonly used in **local area networks (LAN), metropolitan area networks (MAN) and wide area networks (WAN)**.

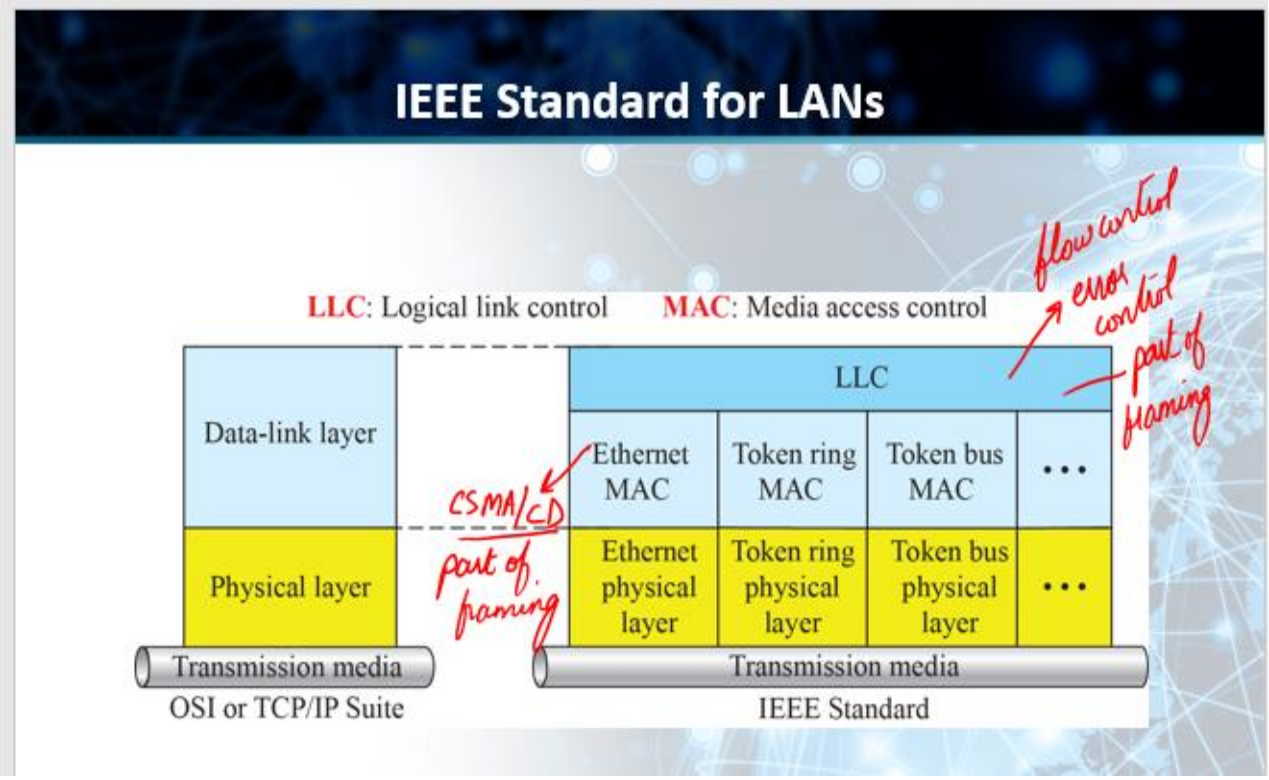
IEEE Project 802 (Institute of Electrical and Electronics Engineers)

- **In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable inter-communication among equipment from a variety of manufacturers**
- **Project 802 did not seek to replace any part of the OSI model or TCP/IP protocol suite**
- **A way of specifying functions of the physical layer and the data-link layer of major LAN protocols**

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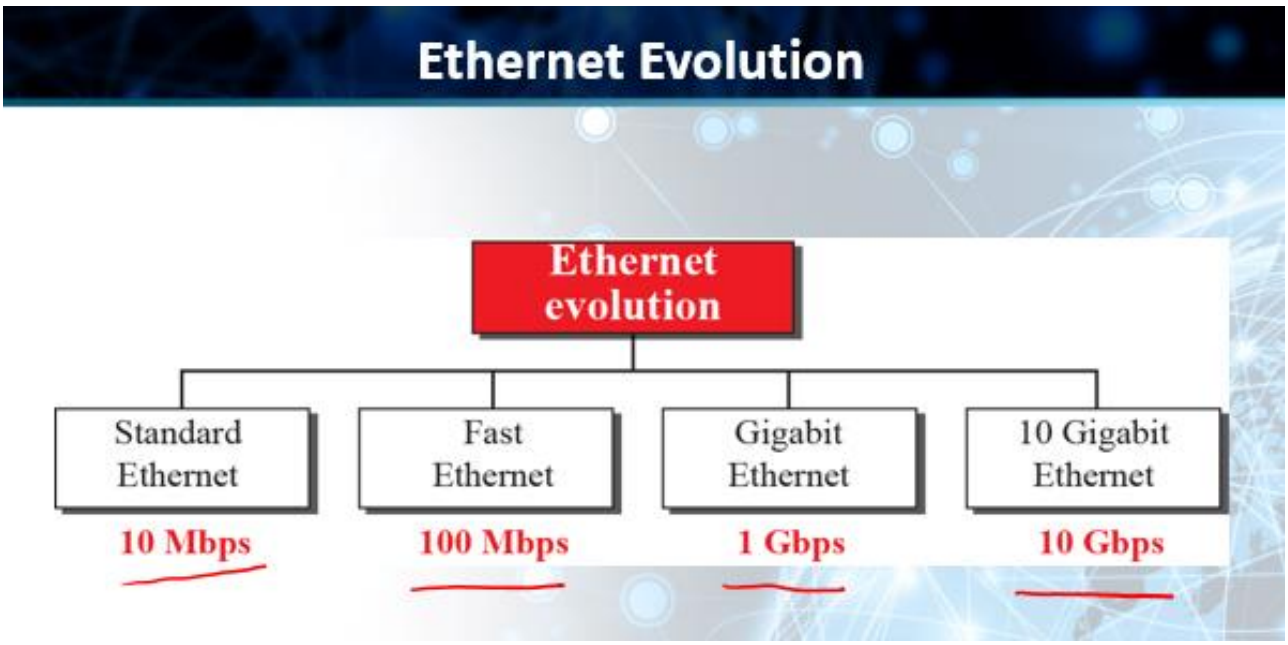
Ethernet Evolution

- **The Ethernet LAN was developed in the 1970s**
- **Since then, it has gone through four generations:**
  - ✓ **Standard Ethernet (10 Mbps)**
  - ✓ **Fast Ethernet (100 Mbps)**
  - ✓ **Gigabit Ethernet (1 Gbps)**
  - ✓ **10 Gigabit Ethernet (10 Gbps**

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Standard Ethernet

- **The original Ethernet technology with the data rate of 10 Mbps is called Standard Ethernet**
- **Most implementations have moved to later evolutions**
- **Still some features of the Standard Ethernet that have not changed during the evolution**

Connectionless & Unreliable Service

- **Each frame is independent of other**
- **No connection establishment or tear down process**
- **The sender may overwhelm receiver with frames and frames are dropped**
- **If frame drops, sender will not know about it unless we are using TCP (Transport)**
- **Ethernet is unreliable like IP and UDP**

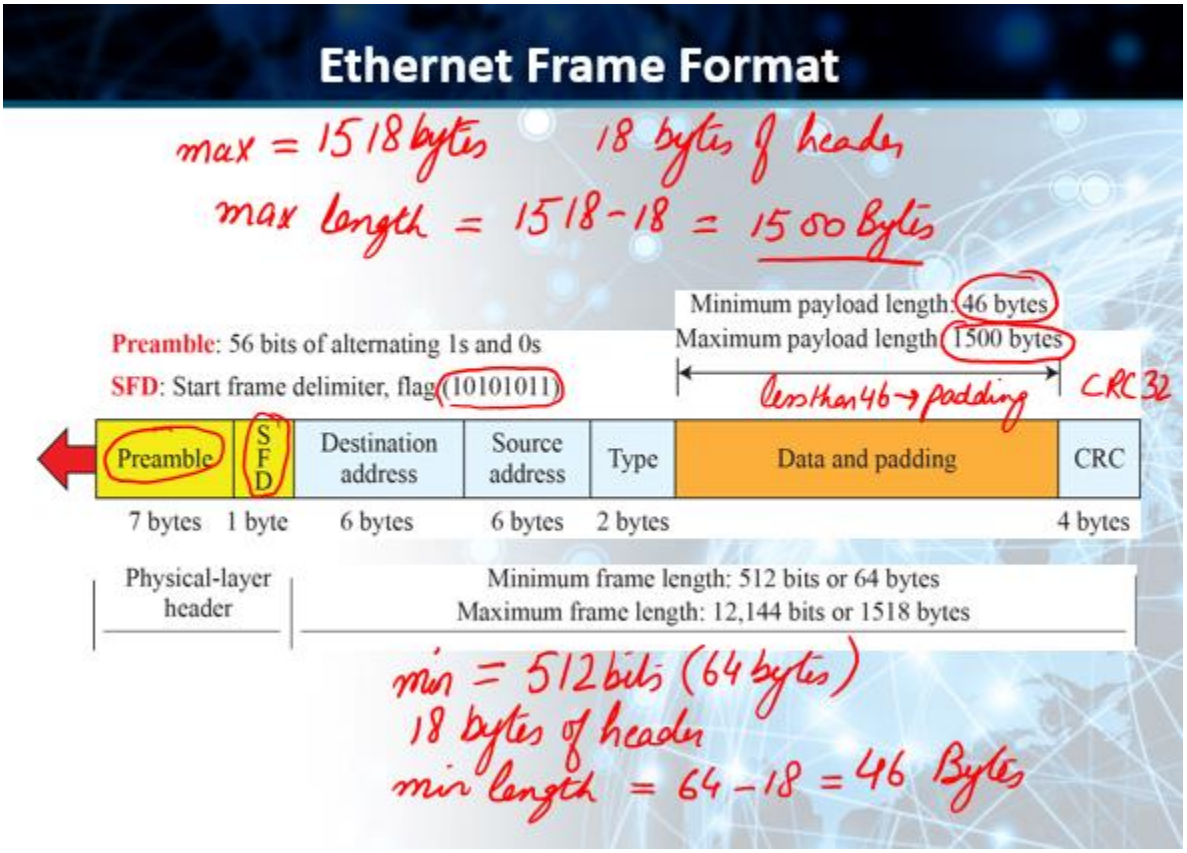
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- **If a frame is corrupted, receiver silently drops it**
- **Left to high level protocols to find out abut it**



Addressing in Standard Ethernet

- **Each station on Ethernet has its own network interface card (NIC)**
- **The NIC fits inside the station and provides the station with a link-layer/physical address**
- **The Ethernet address is 6 bytes (48 bits), normally written in hexadecimal notation, with a colon between the bytes**

Addressing

- **For example, the following shows an Ethernet MAC address:**

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4A:30:10:21:10:1A

Transmission of Address Bits

How the address 47:20:1B:2E:08:EE is sent out online.

The address is sent left to right, byte by byte; for each byte, it is sent right to left, bit by bit, as shown below:

Hexadecimal	47	20	1B	2E	08	EE
Binarys	01000111	00100000	00011011	00101110	00001000	11101110
Transmitted ←	11100010	00000100	11011000	01110100	00010000	01110111

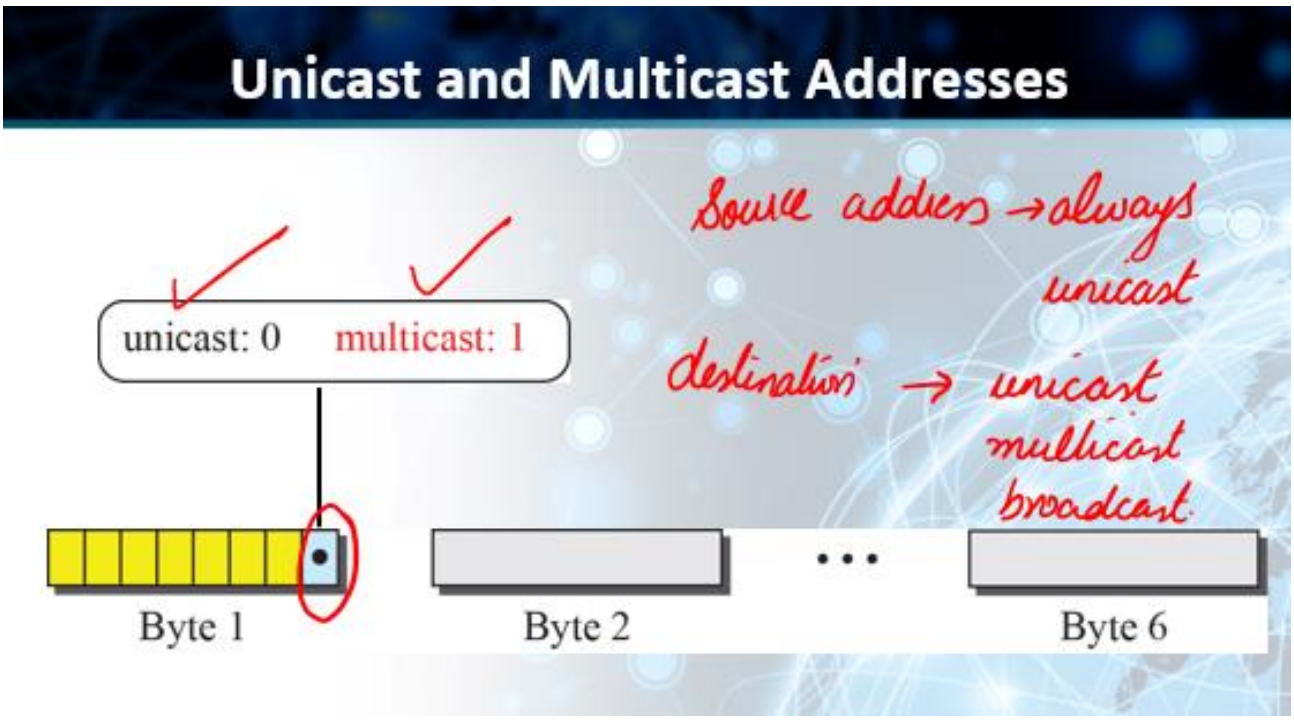
LS MS

→ unicast
→ multicast
→ broadcast

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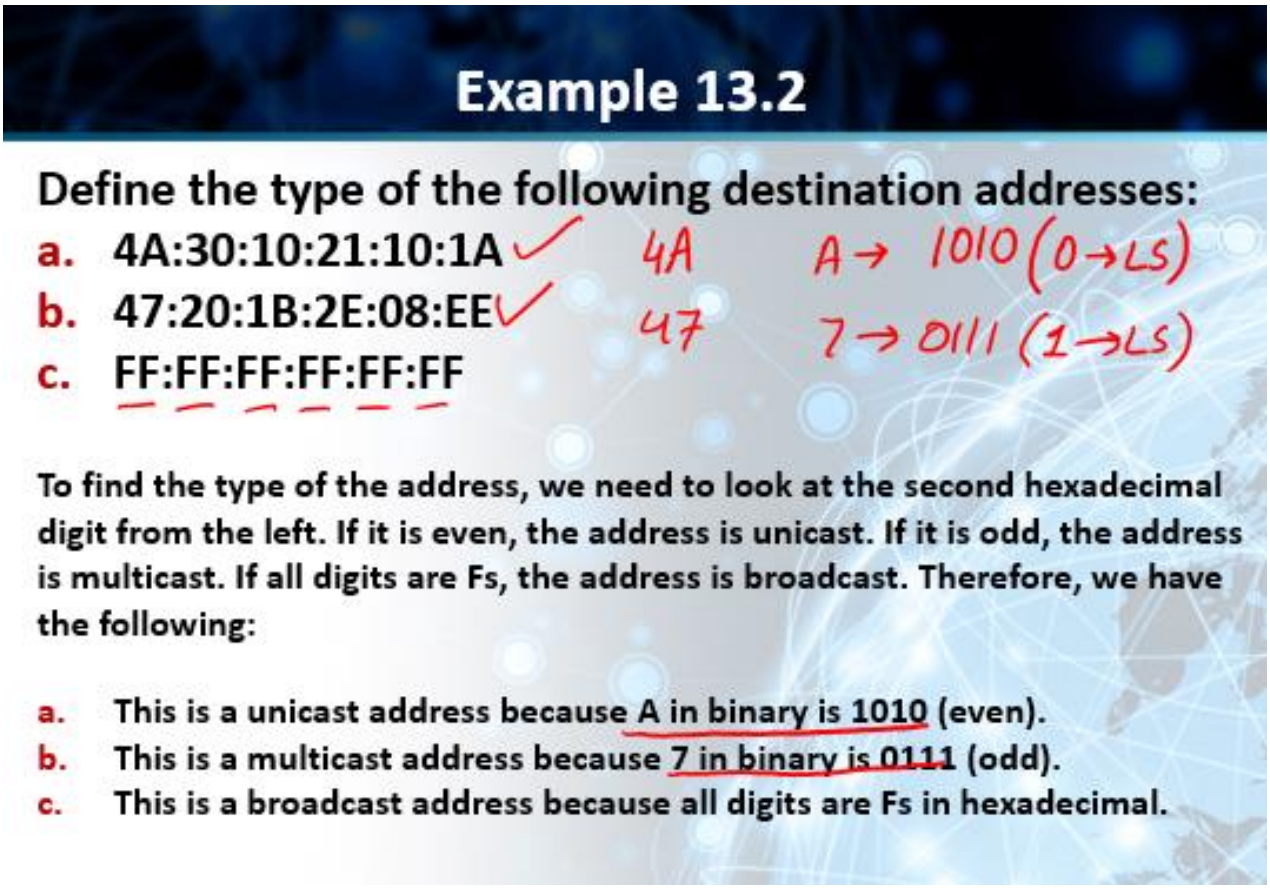
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Access Method in Standard Ethernet

- **Since the network that uses the standard Ethernet protocol is a broadcast network, we need to use an access method to control access to the sharing medium**
- **The standard Ethernet chose CSMA/CD with 1-Persistent Method**

Efficiency of Standard Ethernet

- **The ratio of the time used by a station to send data to the time the medium is occupied by this station**
- **The practical efficiency of standard Ethernet has been measured to be:**

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Efficiency = 1/(1+ 6.4 x a)

where a = number of frames that can fit on a medium

Example

In the Standard Ethernet with the transmission rate of 10 Mbps, we assume that the length of the medium is 2500 m and the size of the frame is 512 bits. The propagation speed of a signal in a cable is normally 2×10^8 m/s.

Propagation Delay = $\frac{2500\ m}{2 \times 10^8\ m/s} = 12.5\ \mu sec$

Transmission Delay = $\frac{512}{10^7} = 51.2\ \mu sec$

IDEAL
a = 0
$\mathcal{E} = 1$

$a = \frac{\text{Prop. Delay}}{\text{Trans Delay}} = \frac{12.5}{51.2} = 0.24$ → 0.24 of a frame occupies medium

$\mathcal{E} = \frac{1}{(1+6.4 \times a)} = 39\ \%$ → medium only 61% of time

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Out Of Topic Example

Problem-07:

Consider a 10 Mbps Ethernet LAN that has stations attached to a 2.5 km long coaxial cable. Given that the transmission speed is 2.3 x 10^8 m/sec, the packet size is 128 bytes out of which 30 bytes are overhead, find the effective transmission rate and maximum rate at which the network can send data.

Solution-

Given-

- Bandwidth = 10 Mbps
- Distance = 2.5 km
- Transmission speed = 2.3 x 10^8 m/sec
- Total packet size = 128 bytes
- Overhead = 30 bytes

Calculating Transmission Delay-

Transmission delay (T_t)

= Packet size / Bandwidth

= 128 bytes / 10 Mbps

= (128 x 8 bits) / (10 x 10^6 bits per sec)

= 1024 / 10^7 sec

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= 102.4 μsec

Calculating Propagation Delay-

Propagation delay (T_p)

= Distance / Speed

= 2.5 km / (2.3 x 10^8 m/sec)

= (2.5 x 10^3 m) / (2.3 x 10^8 m/sec)

= 1.08 x 10^{-5} sec

= 10.8 μsec

Calculating Value of 'a'-

a

= T_p / T_t

= 10.8 μsec / 102.4 μsec

= 0.105

Calculating Efficiency-

Efficiency(η)

= 1 / (1 + 6.44 x a)

= 1 / (1 + 6.44 x 0.105)

= 1 / 1.67

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= 0.59

= 59%

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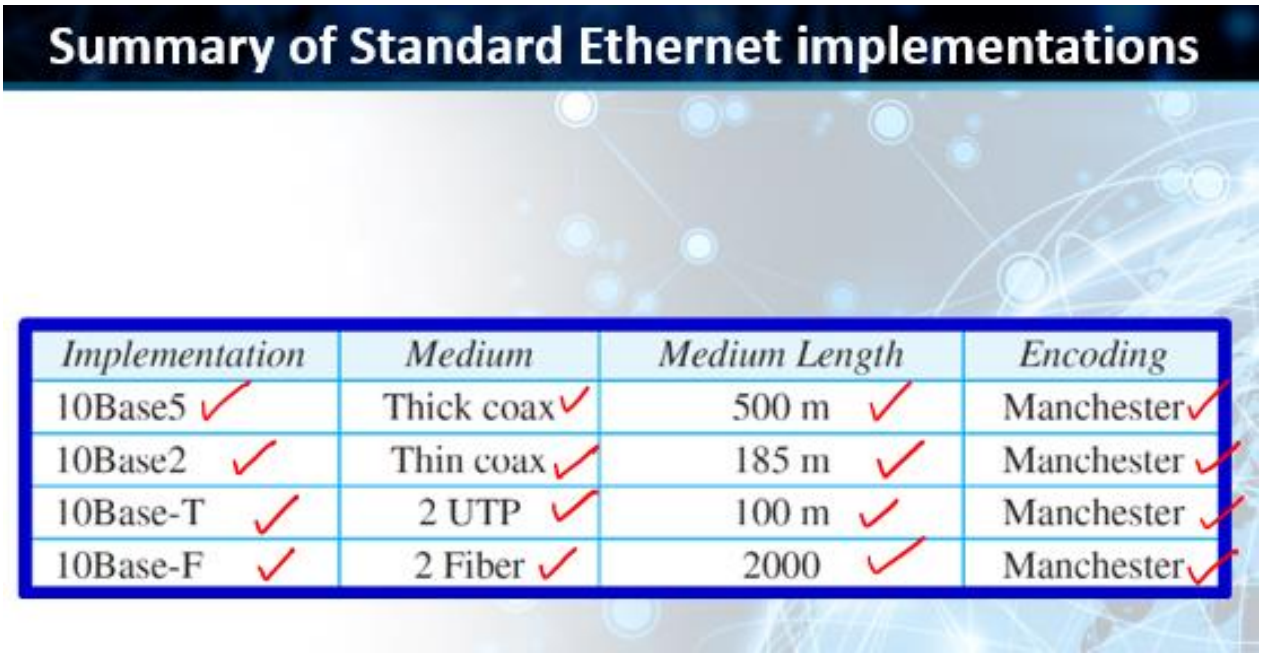
Implementation of Standard Ethernet

- **The Standard Ethernet defined several implementations, but only four of them became popular during the 1980s**

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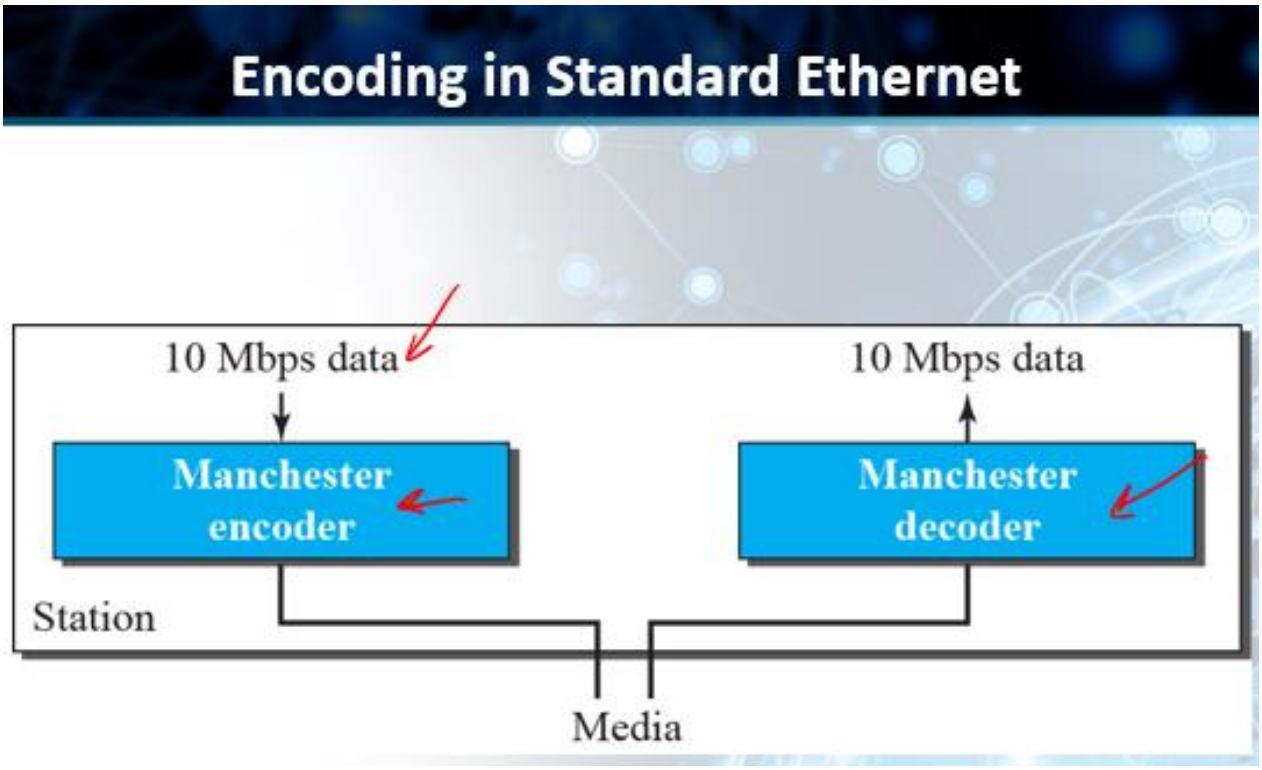


Encoding in Standard Ethernet

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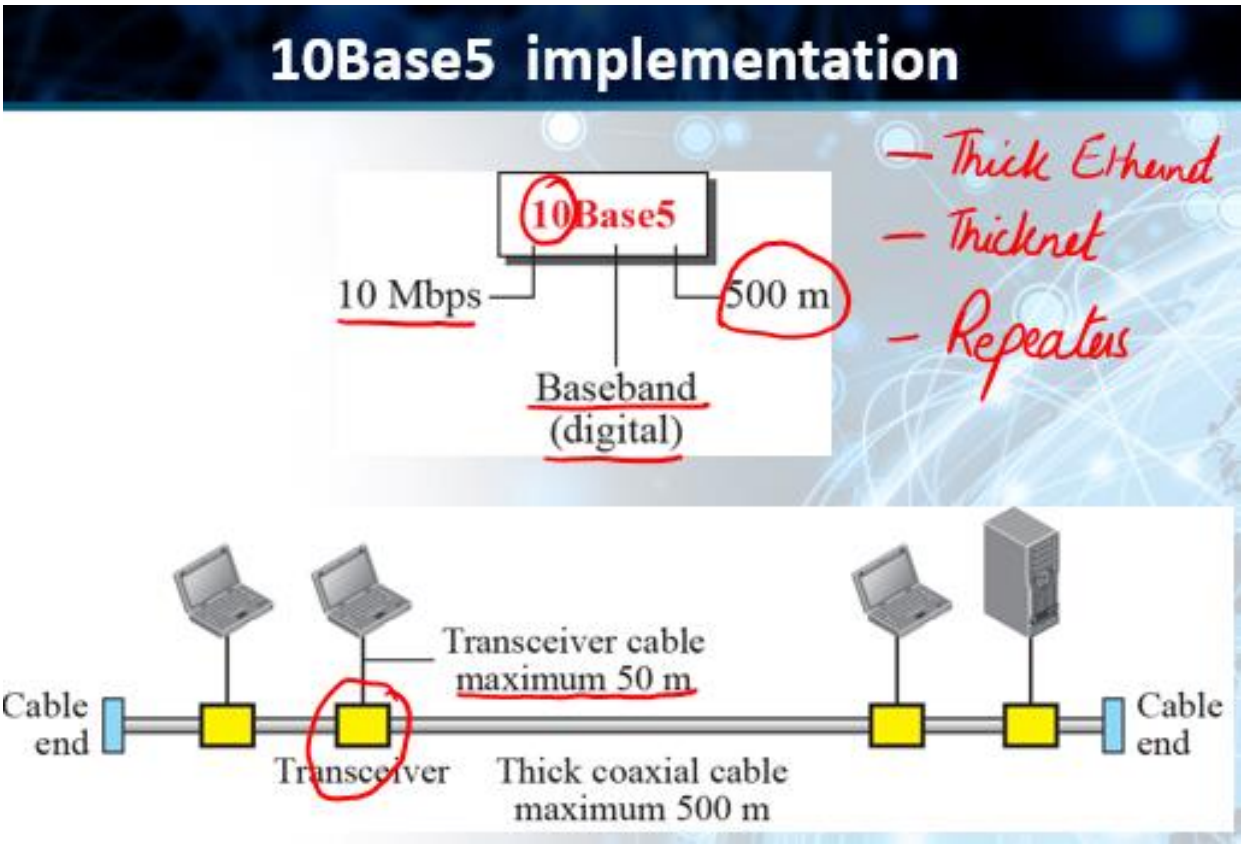
What is meant by Manchester encoding?

In data transmission, Manchester encoding is a form of digital encoding in which data bits are represented by transitions from one logical state to the other. This is different from the more common method of encoding, in which a bit is represented by either a high state such as +5 volts or a low state such as 0 volts.

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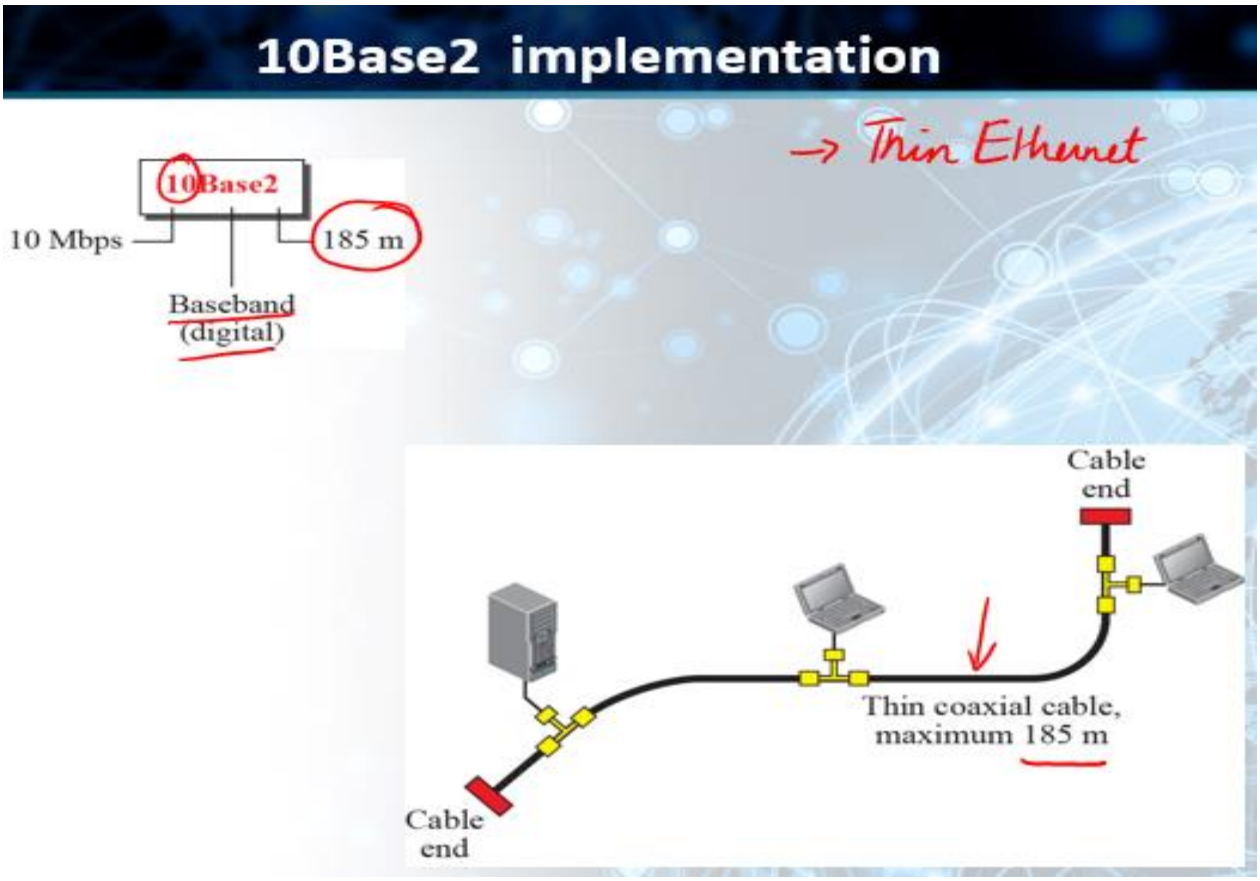
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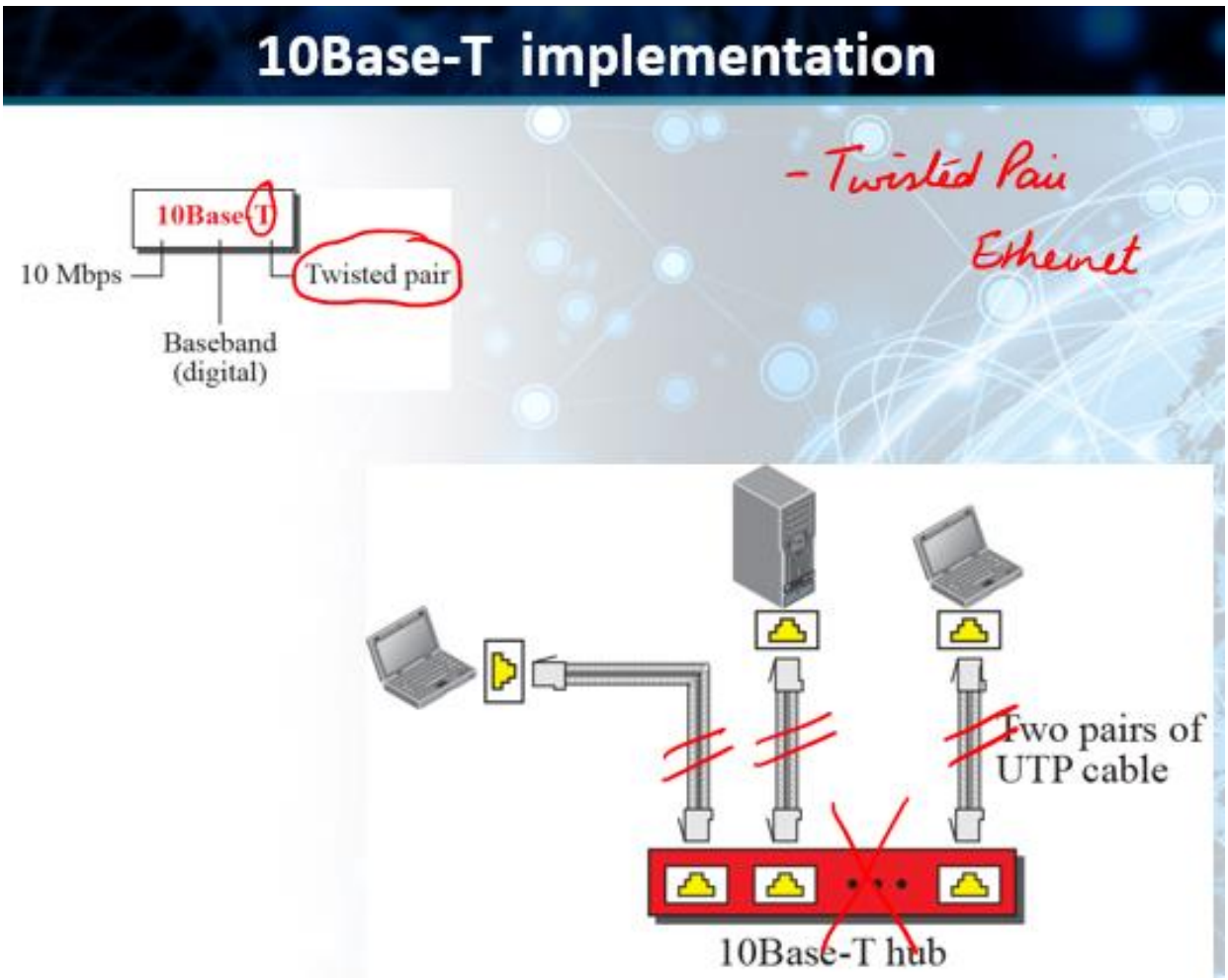
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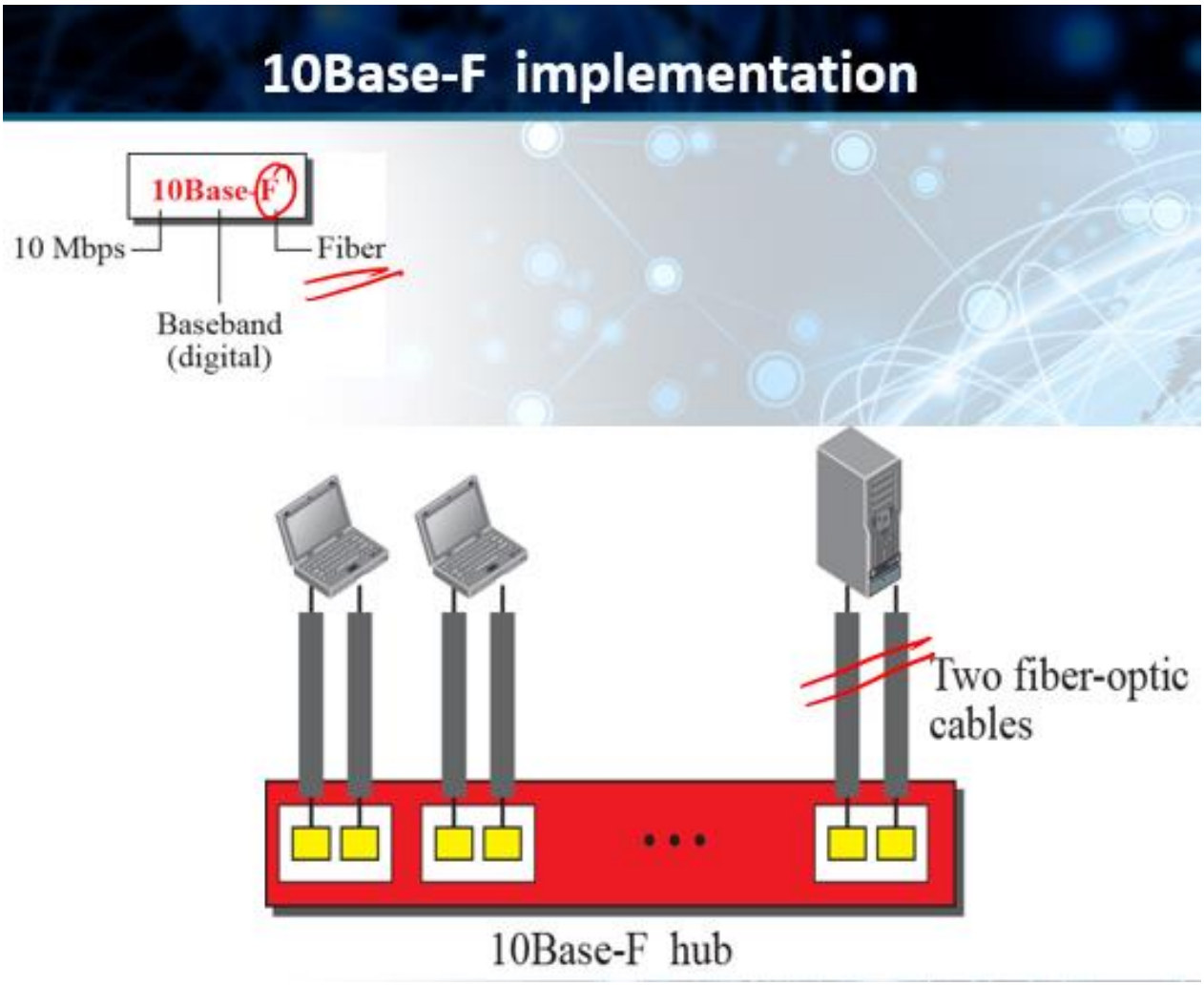
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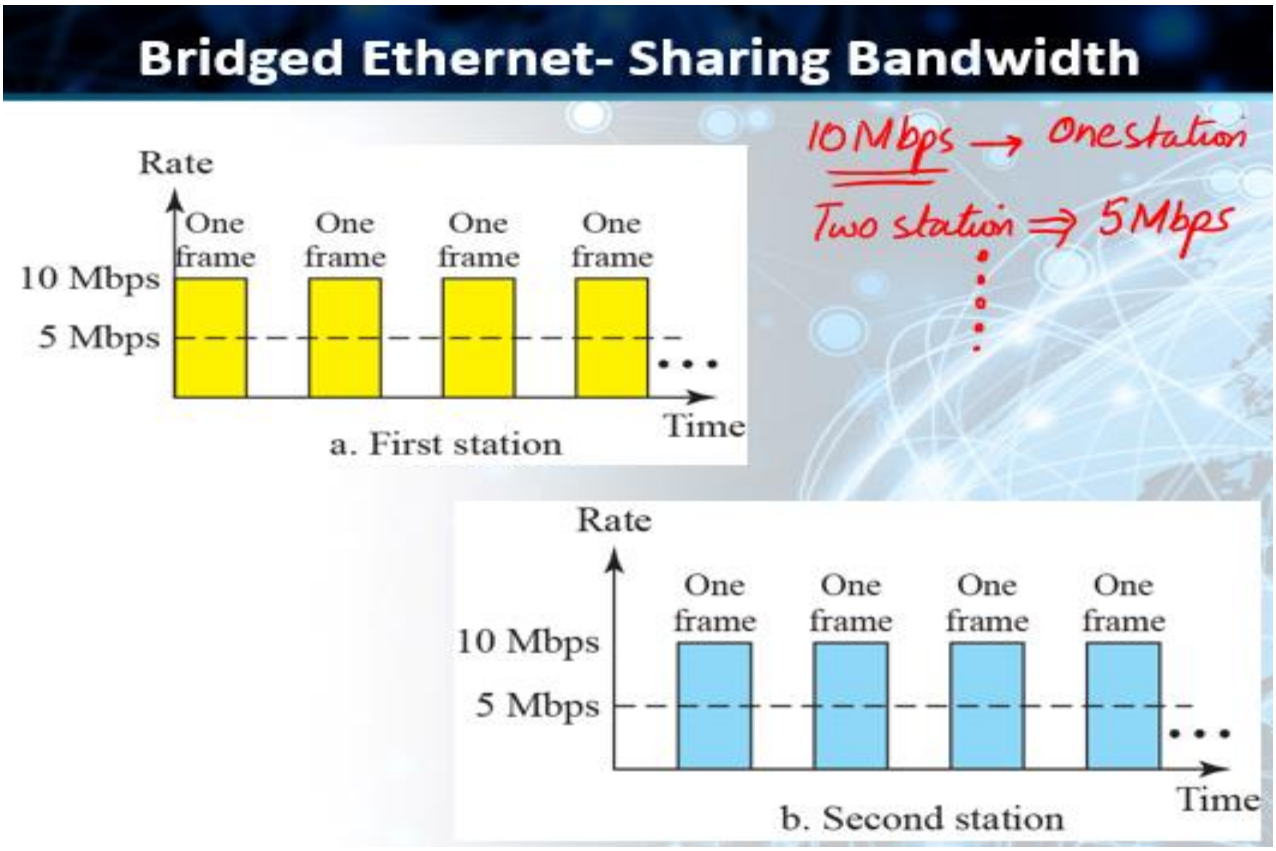
Changes in the Standard

- **The changes that occurred to the 10-Mbps Standard Ethernet opened the road to the evolution of the Ethernet to become compatible with other high-data-rate LANs**
  - ✓ **Bridged Ethernet**
  - ✓ **Switched Ethernet**
  - ✓ **Full-Duplex Ethernet**

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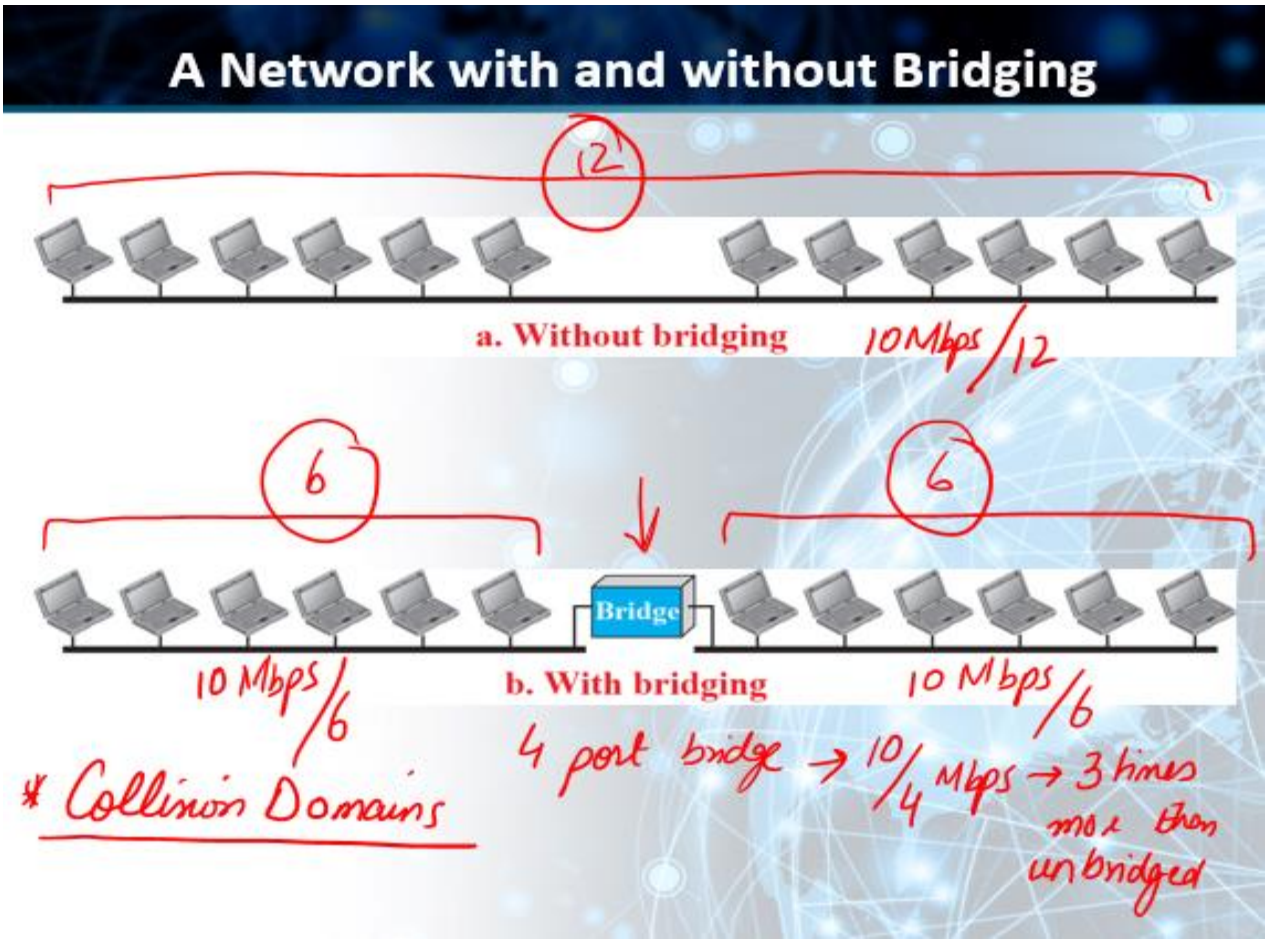
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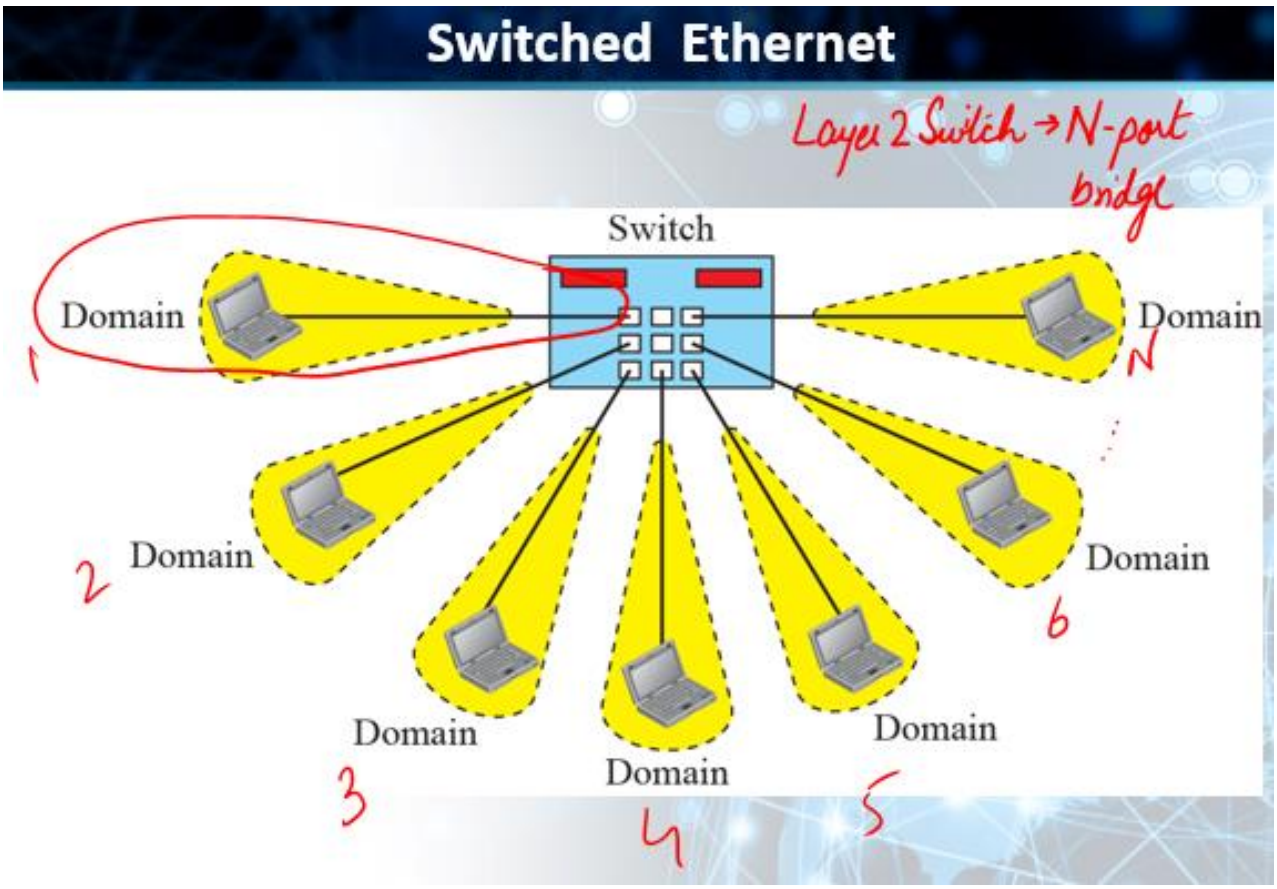
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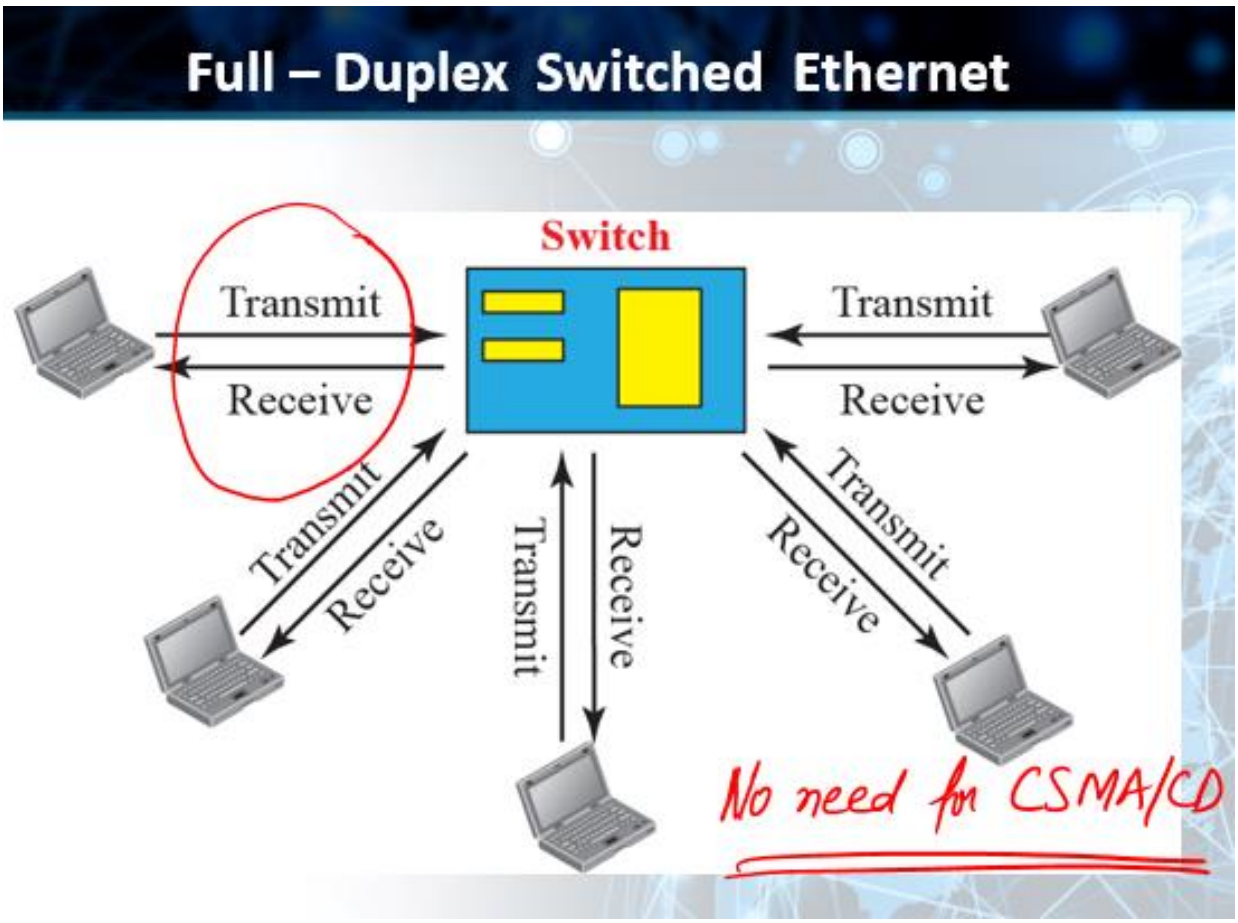
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Fast Ethernet

- **In the 1990s, Ethernet made a big jump by increasing the transmission rate to 100 Mbps, and the new generation was called the Fast Ethernet**
- **To make it compatible with the Standard Ethernet, the MAC sublayer was left unchanged**
- **But the features of the Standard Ethernet that depend on the transmission rate, had to be changed**
- **Goals of Fast Ethernet:**
    - **Upgrade data rate to 100Mbps**

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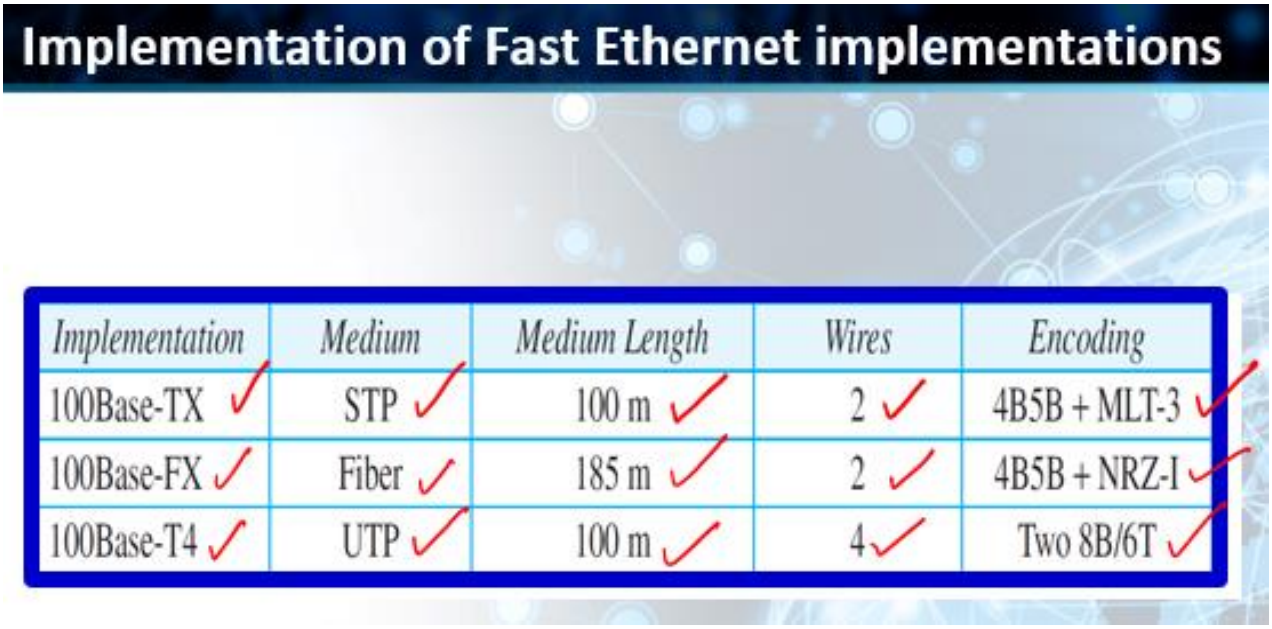
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- **Make it compatible with Standard Ethernet**
- **Keep same 48-bit address**
- **Keep same frame format**

Physical Layer

- **To be able to handle a 100 Mbps data rate, several changes need to be made at the physical layer**

Implementation of Fast Ethernet implementations



Implementation	Medium	Medium Length	Wires	Encoding
100Base-TX	STP	100 m	2	4B5B + MLT-3
100Base-FX	Fiber	185 m	2	4B5B + NRZ-I
100Base-T4	UTP	100 m	4	Two 8B/6T

Gigabit Ethernet

- **Need for an even higher data rate resulted in the design of IEEE Standard 802.3z Gigabit Ethernet Protocol (1000 Mbps)**
- **The goals of the Gigabit Ethernet were:**
  - **Upgrade the data rate to 1 Gbps**

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- **Make it compatible with standard or Fast Ethernet**
- **Use same 48-bit address**
- **Use the same frame format**
- **Keep same minimum and maximum frame lengths**

MAC Sub-layer

- **A main consideration in the evolution of Ethernet was to keep the MAC sublayer untouched**
- **To achieve a data rate of 1 Gbps, this was no longer possible**
- **Gigabit Ethernet has two distinctive approaches for medium access:**
  - ✓ **Half-duplex**
  - ✓ **Full-duplex**

Physical Layer

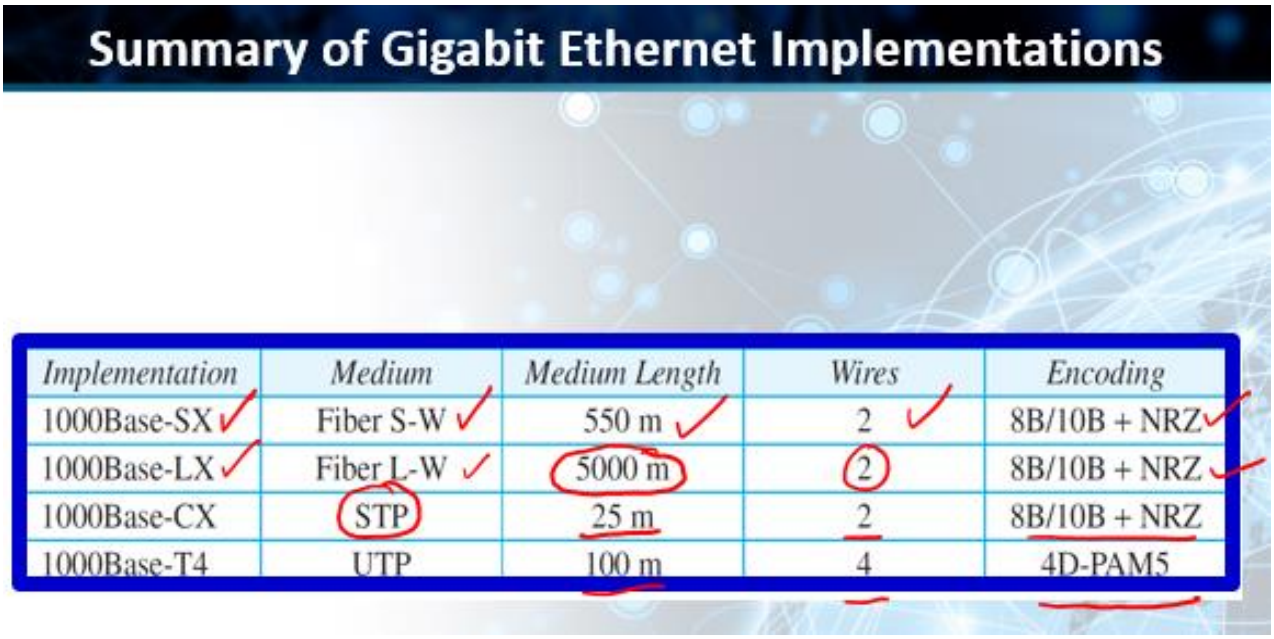
- **The physical layer in Gigabit Ethernet is more complicated than that in Standard or Fast Ethernet**
- **We briefly discuss some features of this layer:**

Summary of Gigabit Ethernet Implementations

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Implementation	Medium	Medium Length	Wires	Encoding
1000Base-SX	Fiber S-W	550 m	2	8B/10B + NRZ
1000Base-LX	Fiber L-W	5000 m	2	8B/10B + NRZ
1000Base-CX	STP	25 m	2	8B/10B + NRZ
1000Base-T4	UTP	100 m	4	4D-PAM5

10-gigabit Ethernet

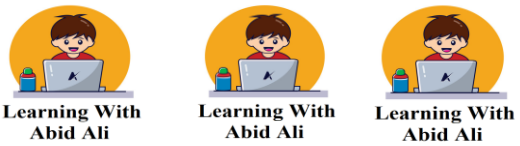
- **The idea is to extend the technology, the data rate, and the coverage distance so that the Ethernet can be used in LANs and MANs (metropolitan area network)**
- **The IEEE committee created 10 Gigabit Ethernet and called it Standard 802.3ae**

Implementation

- **10 Gigabit Ethernet operates only in full-duplex mode, which means there is no need for contention; CSMA/CD is not used in 10 Gigabit Ethernet**
- **Four implementations are most common:**

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Implementation

Implementation	*Medium*	*Medium Length*	*Number of wires*	*Encoding*
10GBase-SR	Fiber 850 nm	300 m	2	64B66B
10GBase-LR	Fiber 1310 nm	10 Km	2	64B66B
10GBase-EW	Fiber 1350 nm	40 Km	2	SONET
10GBase-X4	Fiber 1310 nm	300 m to 10 Km	2	8B10B

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Data Communication CS601 Lecture No 33

Lecture Prepared By

Muhammad Abid Ali

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CS601 Lecture No 33 Topic 191 to 196

Other Wired Networks

- **Access Networks**
  - ✓ **Networks that connect a small LAN to an ISP** (ISP is Internet Service Provider. LAN is Local Area Network)

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- **Wide Area Networks**
  - ✓ **Wired networks used to transfer data over long distances**

Telephone Network

- **The telephone network had its beginnings in the late 1800s**
- **Plain Old Telephone System (POTS) was originally an analog system using analog signals to transmit voice**
- **With the advent of the computer era, the network, in the 1980s, began to carry data in addition to voice**
- **During the last decade, the telephone network has undergone many technical changes and the network is now Digital as well as Analog**

Major Components

The telephone network is made of three major components:

- ✓ **Local Loops**
- ✓ **Trunks**
- ✓ **Switching offices**

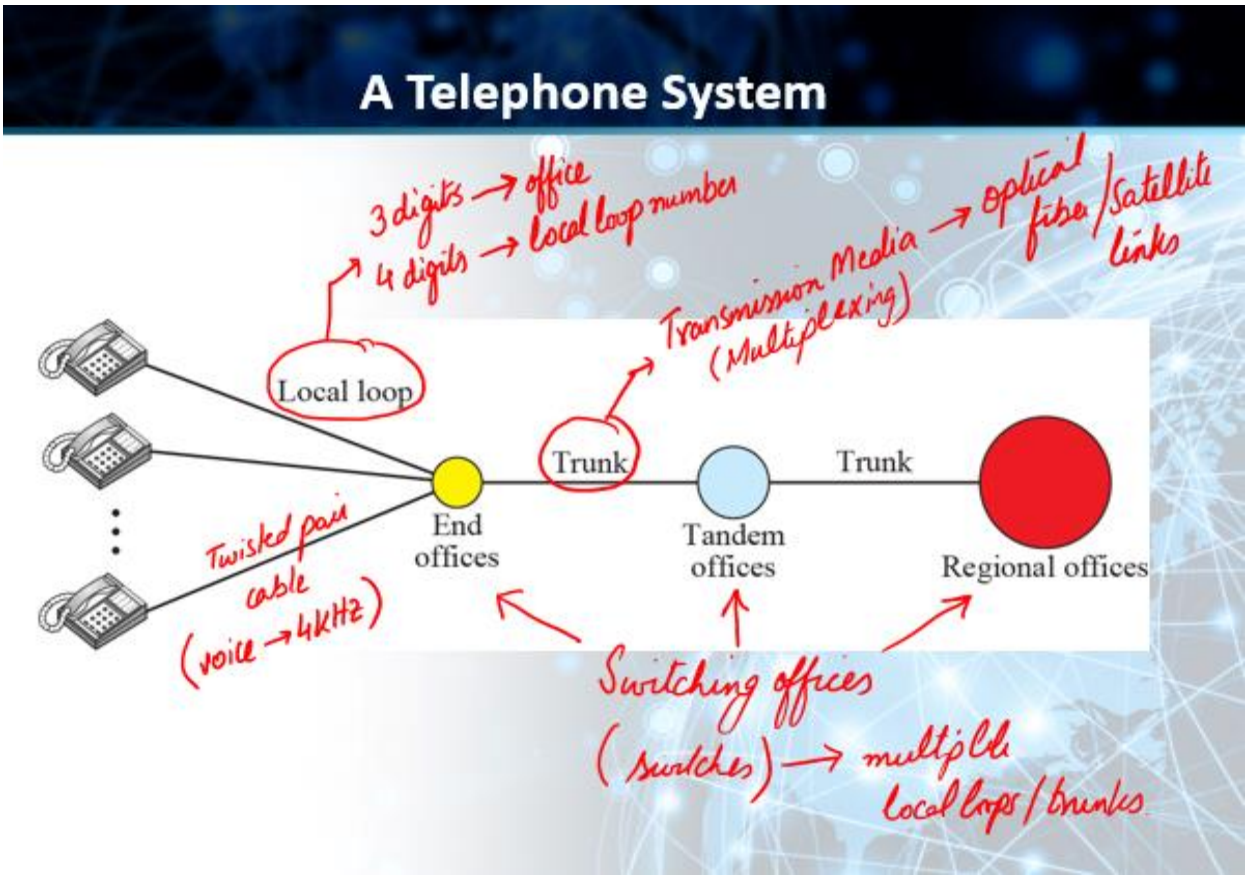
The telephone network has several levels of switching offices:

- ✓ **End offices**
- ✓ **Tandem offices**
- ✓ **Regional offices**

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Local-Access Transport Areas (LATAs)

- **A LATA can be a small or large metropolitan area**
- **A small state may have a single LATA; a large state may have several LATAs**
- **A LATA boundary may overlap with state boundary; part of a LATA can be in one state, part in another state**
- **Services offered by Telephone companies inside a LATA are called Intra-LATA services and between LATAs are called Inter-LATA services**
- **Carrier that handles Intra-LATA are called a Local Exchange Carrier (LEC) and the ones that handle Inter-LATA are called Interexchange Carriers (IXCs)**

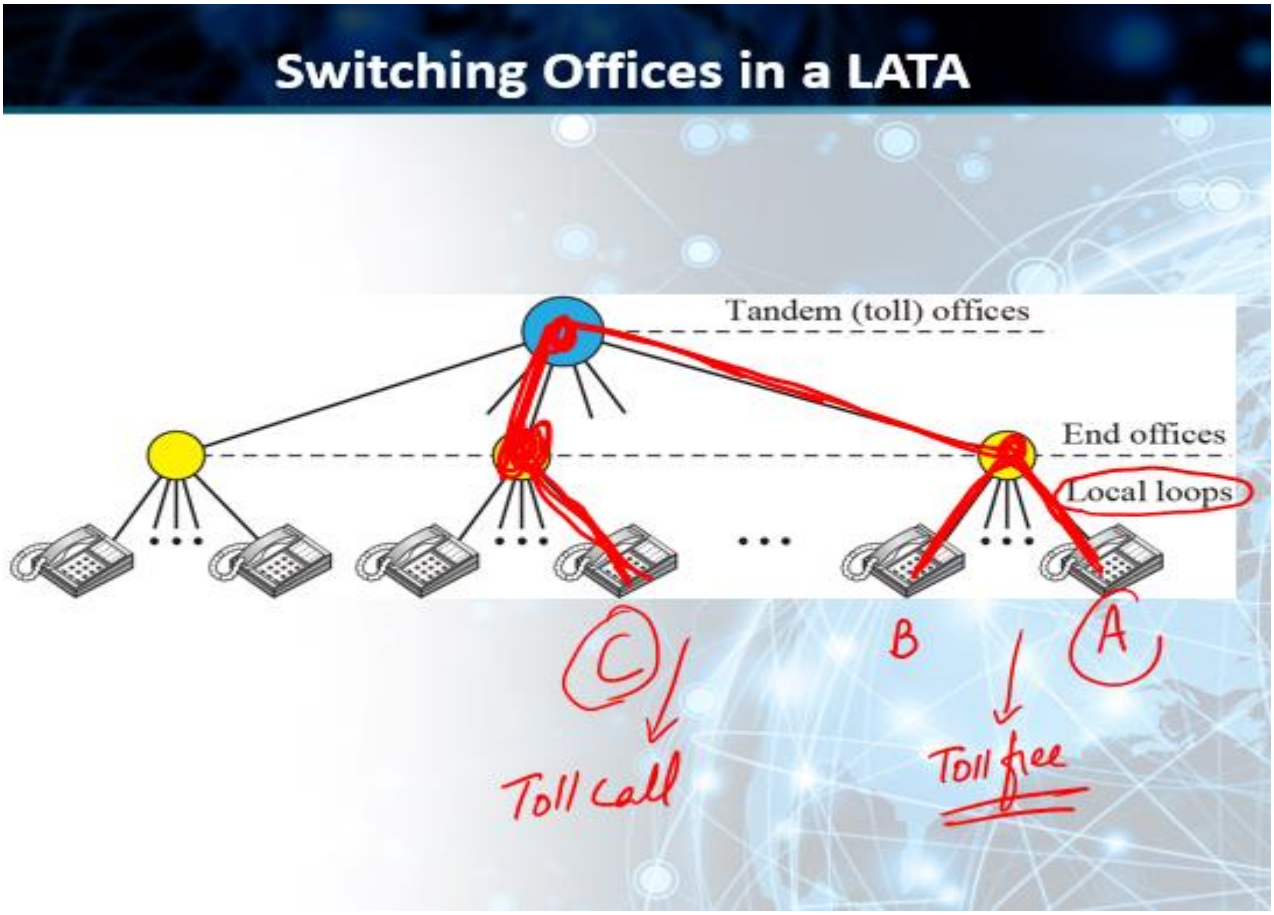
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What is the difference between intra Lata and inter LATA services?

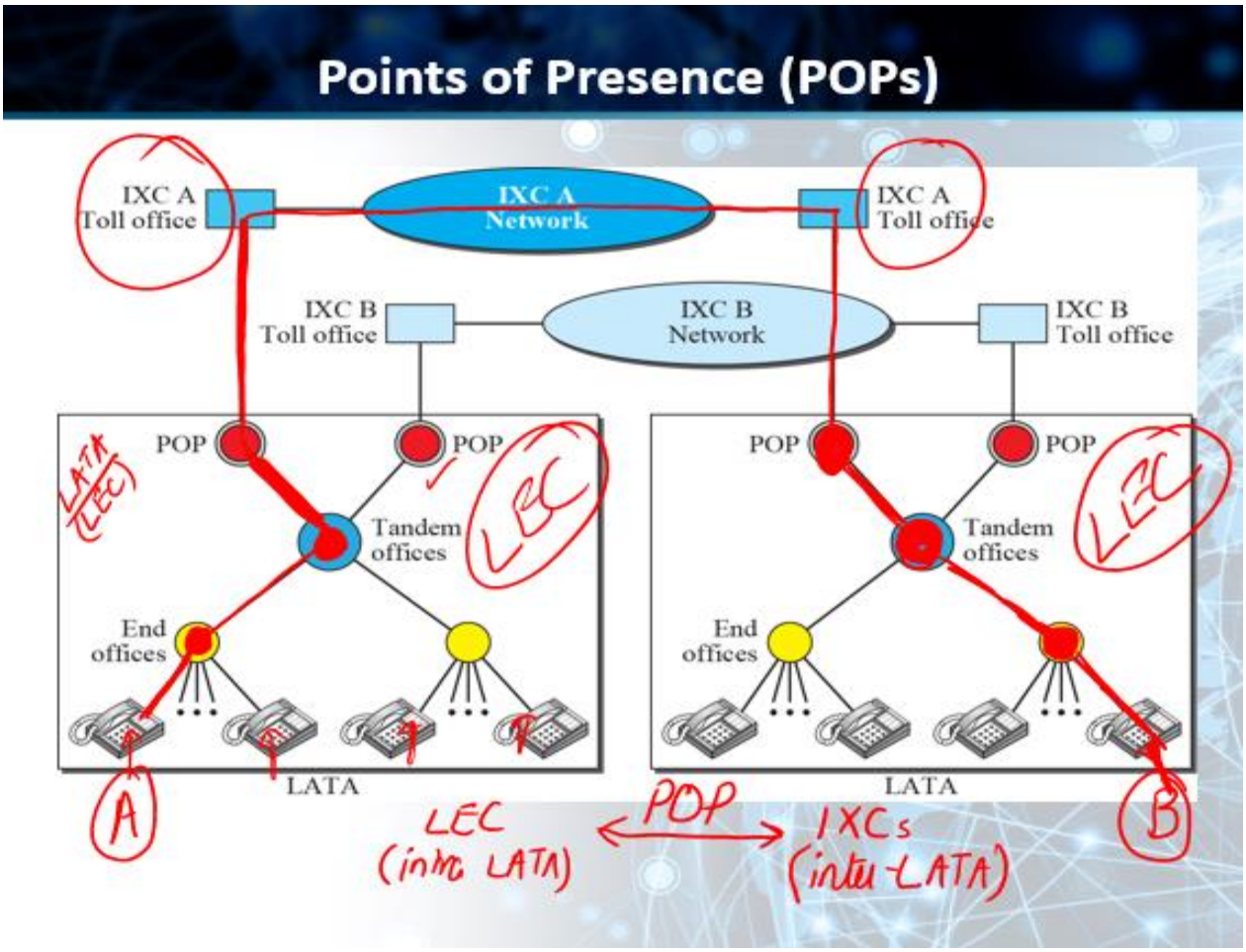
A telephone call within the same LATA (same region). IntraLATA is local telephone service. InterLATA refers to a call from one LATA to another, which can be within a state or from state to state (interstate).



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Signaling

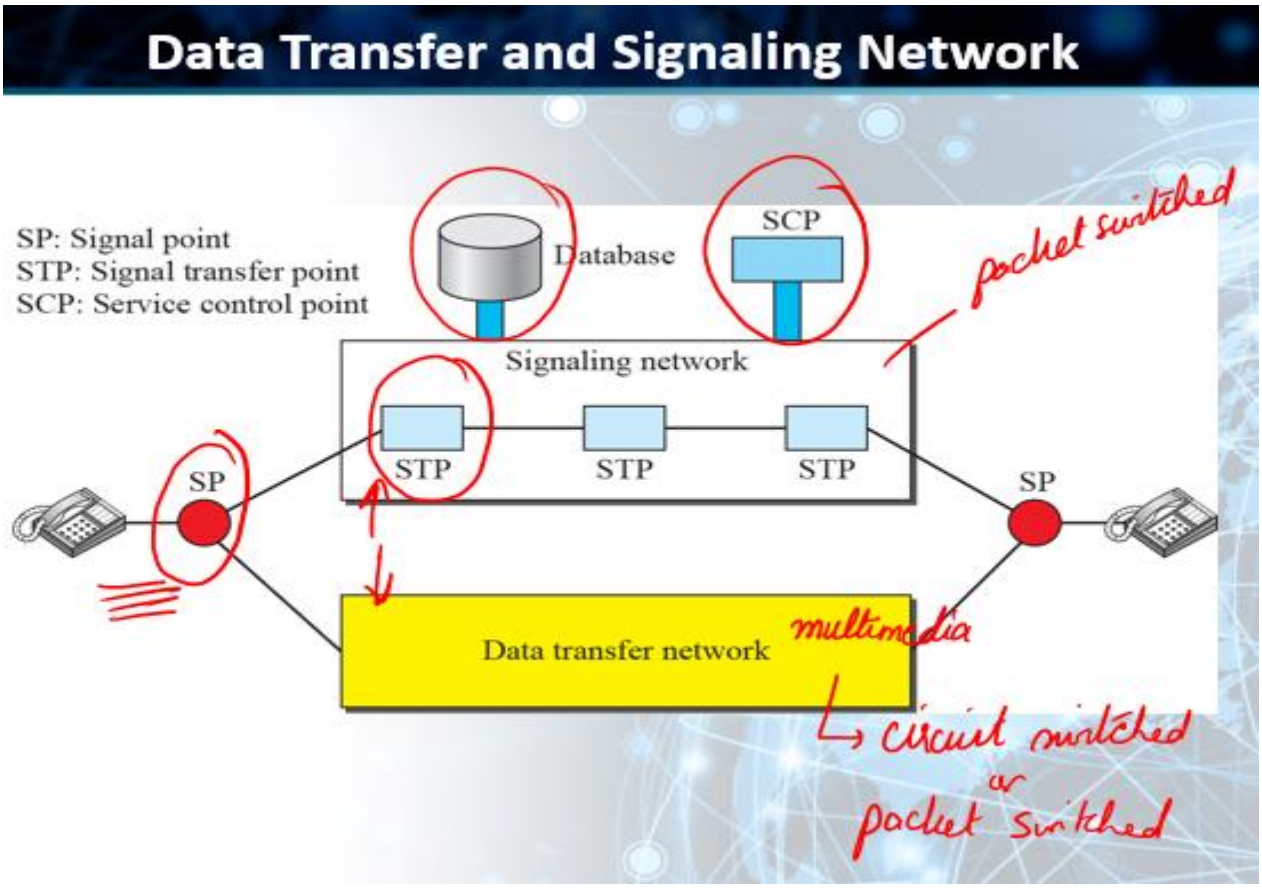
- **The telephone network in the beginning, used a circuit-switched network with dedicated links to transfer voice communication**
- **The operator connected the two parties by using a wire with two plugs inserted into the corresponding two jacks**
- **Later, the signaling system became automatic**
- **Rotary telephones were invented that sent a digital signal defining each digit in a multi-digit telephone number**

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- **As telephone networks evolved into a complex network, the functionality of the signaling system increased**

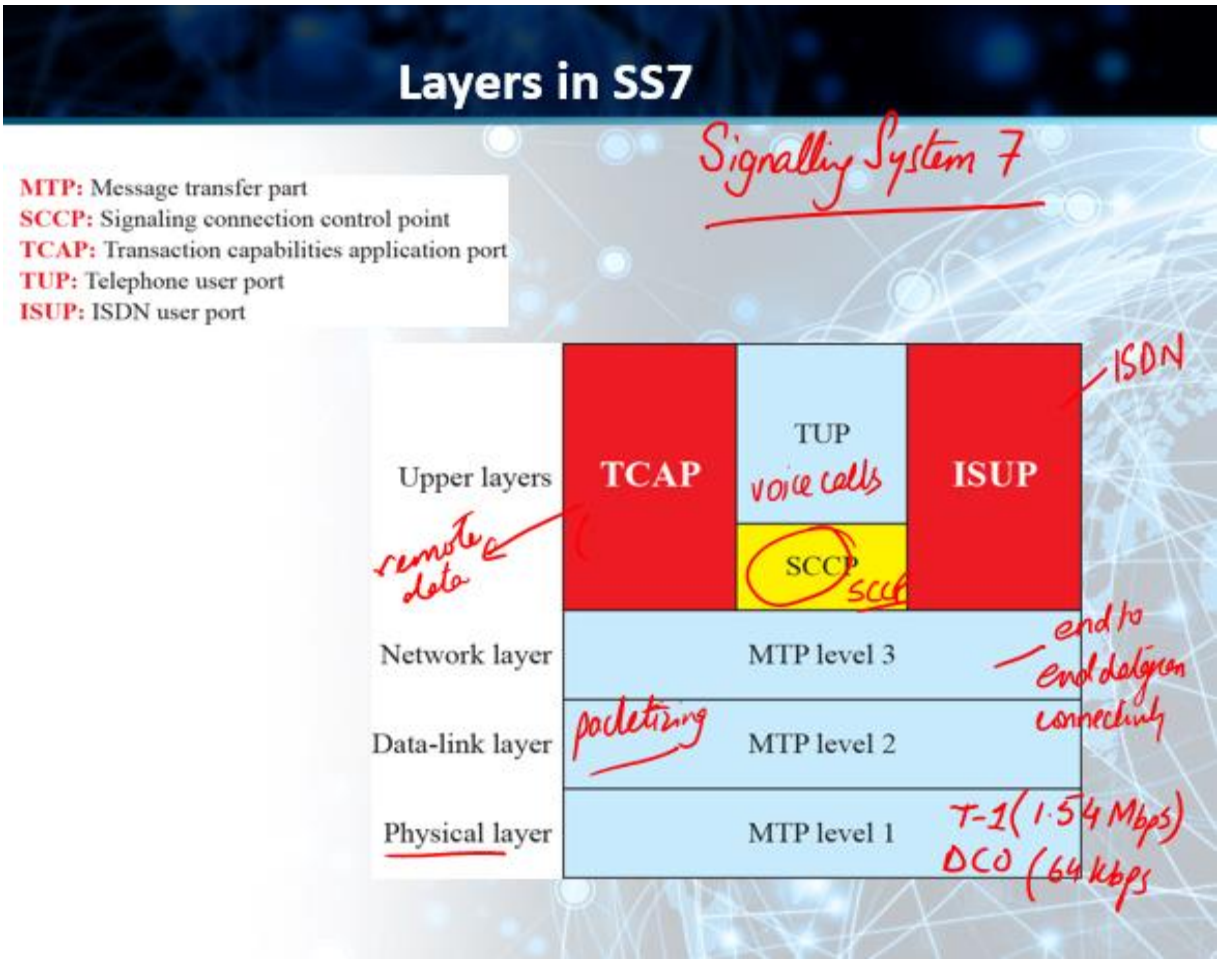


Layers in SS7

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(MTP) Message Transfer Part

Services

- **Telephone companies provide two types of services:**
  - ✓ **Analog Services**
    - ➢ **Analog Switched Services**
    - ➢ **Analog Leased Services**
  - ✓ **Digital Services**

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- **Switched /56 Service**
- **Digital Data Service**

Digital Subscriber Line (DSL)

- **After traditional dial-up modems reached their peak data rate, telephone companies developed another technology, DSL, to provide higher-speed access to the Internet**
- **DSL supports high-speed digital communication over the existing telephone**
- **DSL technology is a set of technologies, each differing in the first letter (ADSL, VDSL, HDSL, and SDSL)**

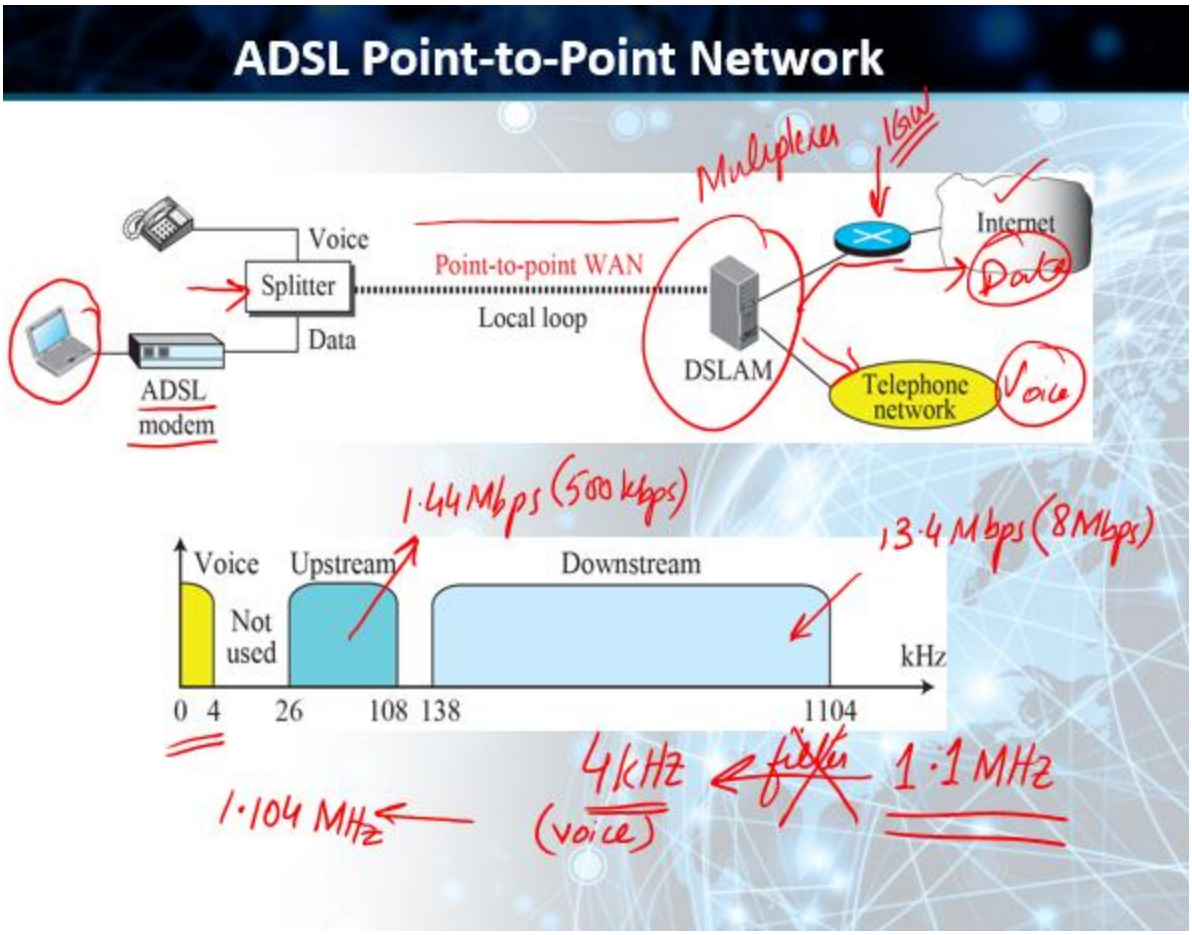
What is DSL ADSL SDSL VDSL?

Common types of DSL are Symmetric Digital Subscriber Line (SDSL, with matching upload and download speeds), Asymmetric Digital Subscriber Line (ADSL, featuring faster download speeds than upload), and Very High-Rate Digital Subscriber Line (VDSL, featuring much faster asymmetric speeds).

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Cable Network

- **The Cable TV networks were initially created to provide remote subscribers access to TV programs**
- **Cable networks enabled access to remote broadcasting stations via microwave connections**
- **Cable TV also found a good ISP market by using some of the channels originally designed for video**

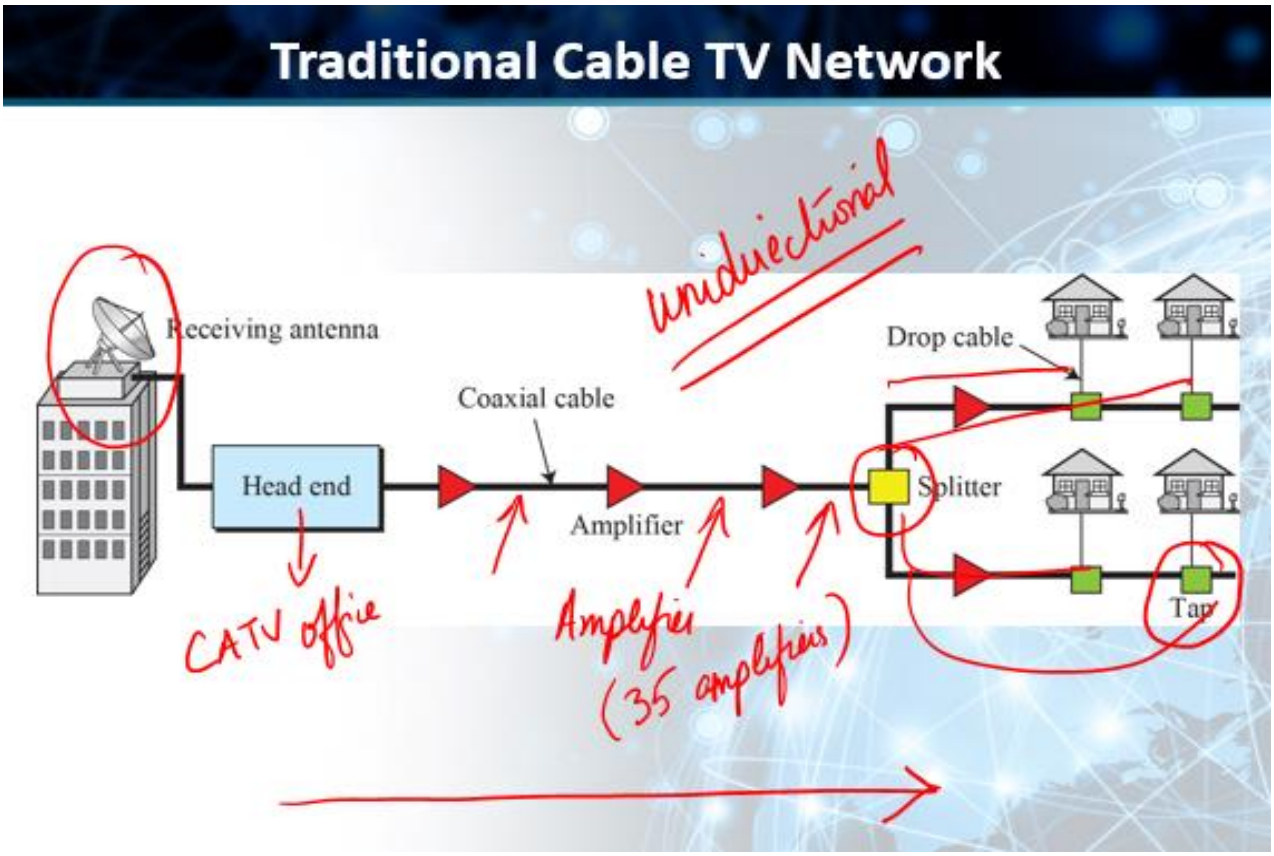
Traditional Cable Networks

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- **Cable TV started to distribute broadcast video signals to locations with poor or no reception in the late 1940s**
- **It was called community antenna television (CATV) because an antenna at the top of a tall hill or building received the signals from the TV stations**



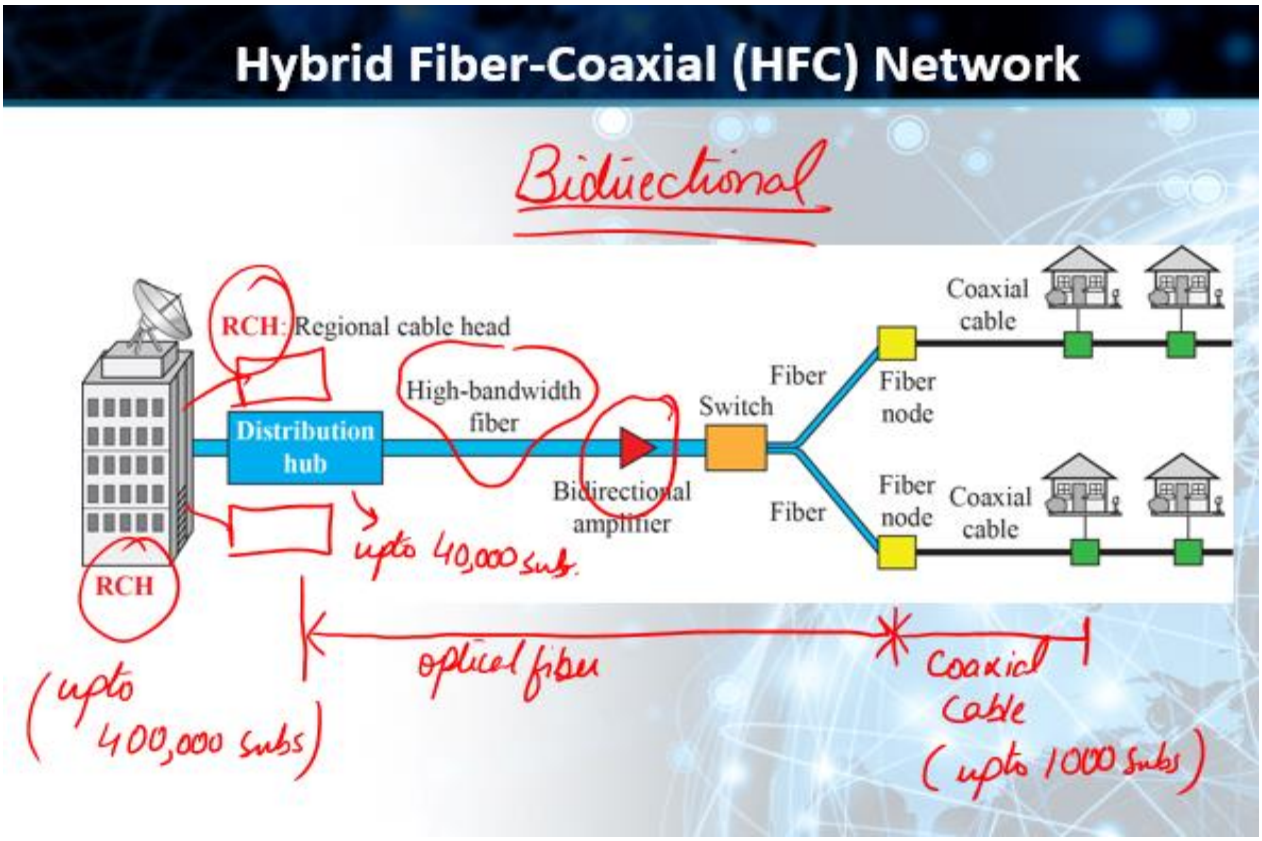
Hybrid Fiber Coaxial (HFC) Network

- **Second generation of cable network is called a Hybrid Fiber-Coaxial (HFC) network**
- **The network uses a combination of fiber-optic and coaxial cable**

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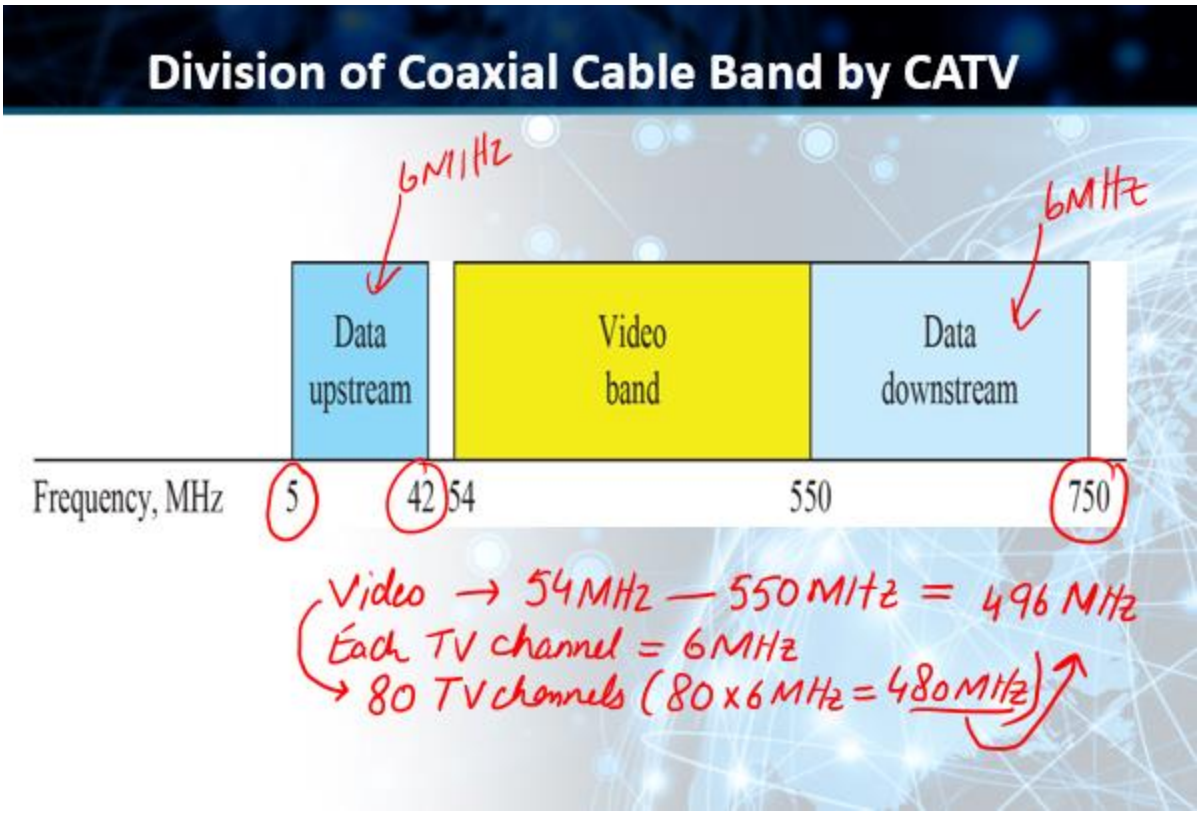
Cable TV for Data Transfer

- **Cable companies are now competing with telephone companies for the residential customer who wants high-speed data transfer**
- **DSL technology provides high-data-rate connections for residential subscribers over the local loop BUT UTP is susceptible to Interfence**
- **This imposes an upper limit on the data rate. A solution is the use of the cable TV network**

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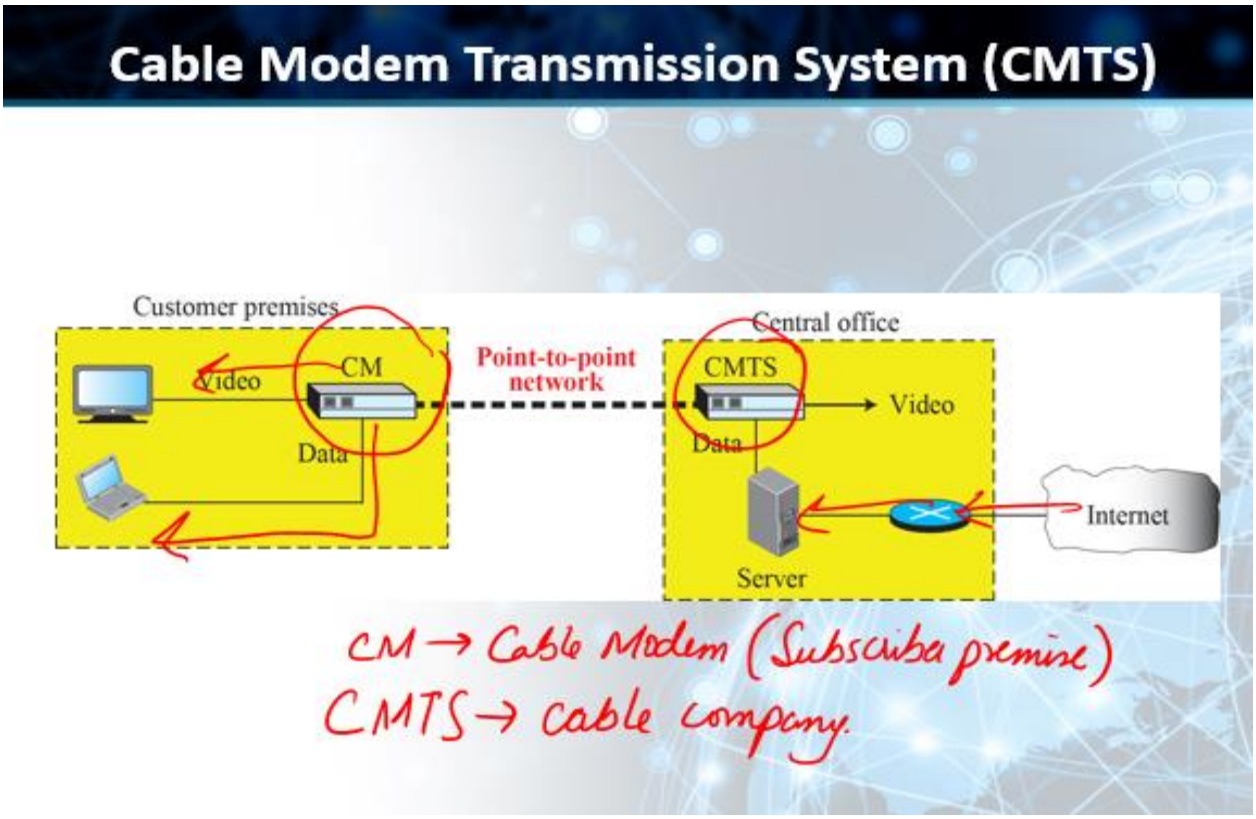
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Lecture Prepared By

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CS601 Lecture No 34 Topic 197 to 204

Synchronous Optical Network (SONET)

- **A wide area network (WAN) that is used as a transport network to carry loads from other WANs**
- **ITU–T standard called Synchronous Digital Hierarchy (SDH)**

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- **Architecture of a SONET system consists of signals, devices, and connections**

SONET Architecture

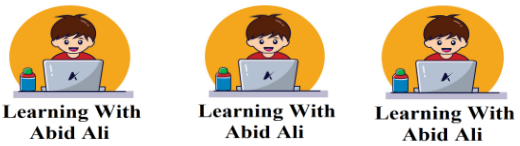
- **Signals**
  - ✓ Synchronous Transport Signals (STS)
  - ✓ Optical Carriers (OCs)
  - ✓ Synchronous Transport Module (STM)

- **SONET Devices**
  - ✓ STS Mux/Demux
  - ✓ Regenerators
  - ✓ Add-Drop Multiplexer and Terminals

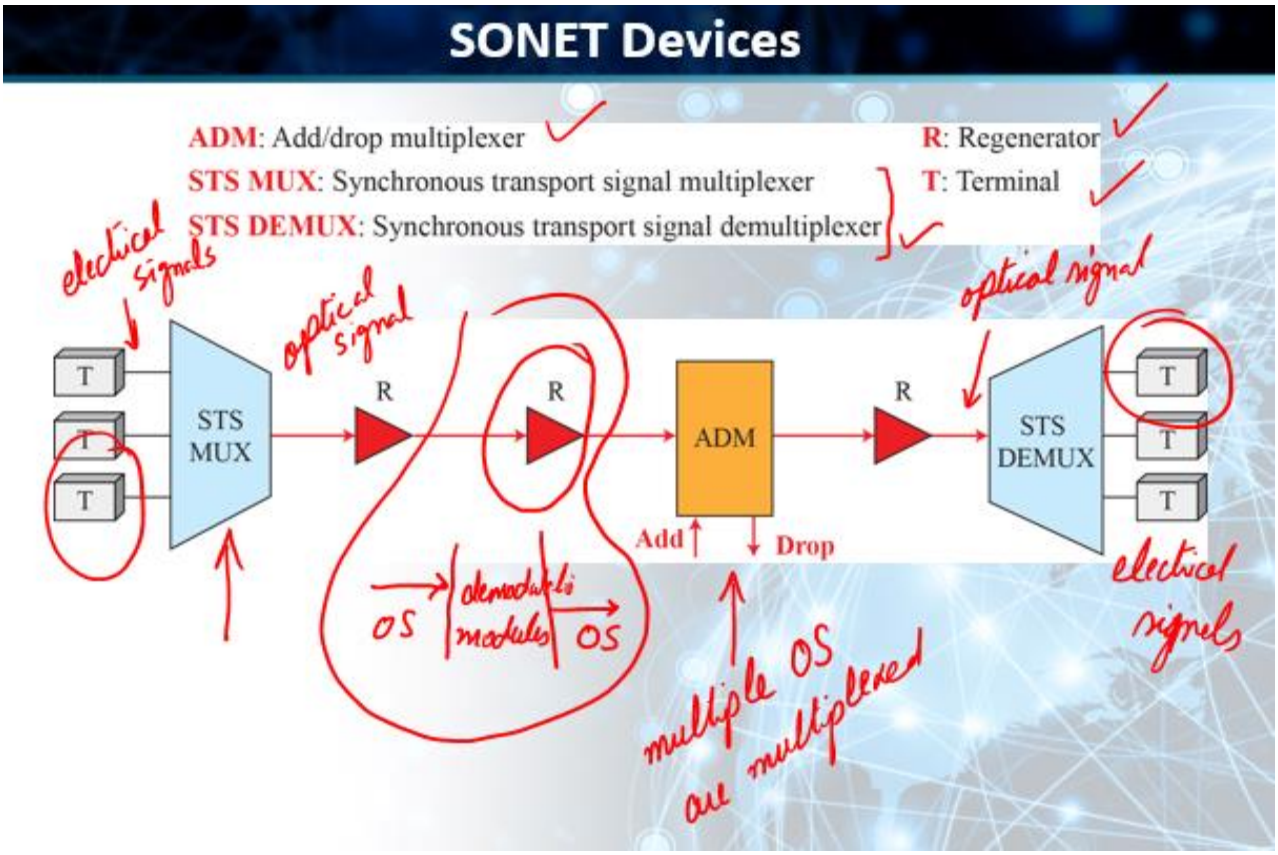
- **Connections**
  - ✓ Section
  - ✓ Line
  - ✓ Path

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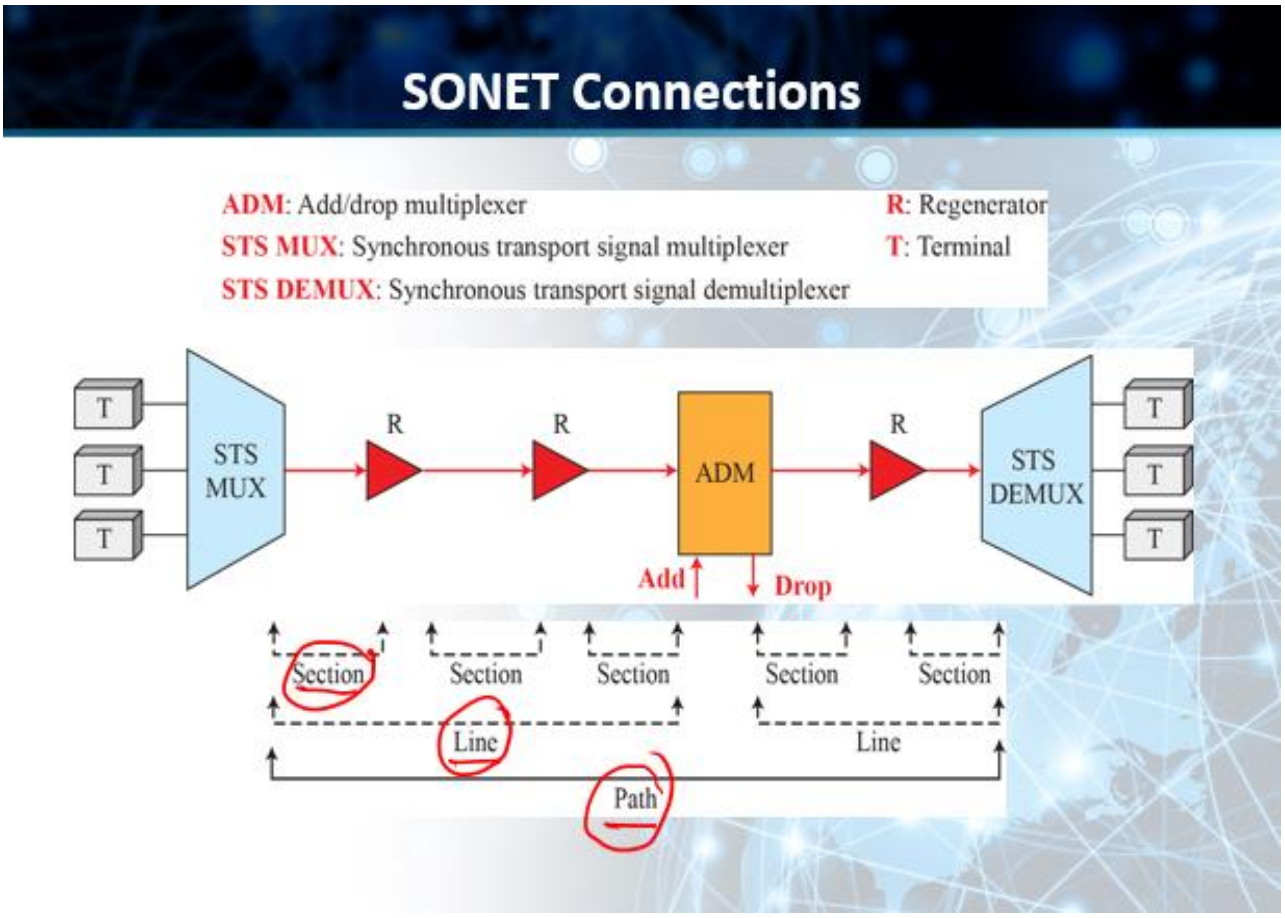
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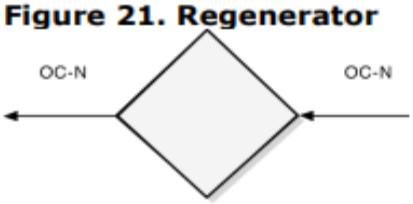
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Regenerator

A regenerator is needed when, due to the long distance between multiplexers, the signal level in the fiber becomes too low.

The regenerator clocks itself off of the received signal and replaces the section overhead bytes before retransmitting the signal. The line overhead, payload, and POH are not altered (see *Figure 21*).

Figure 21. Regenerator



SONET Layers

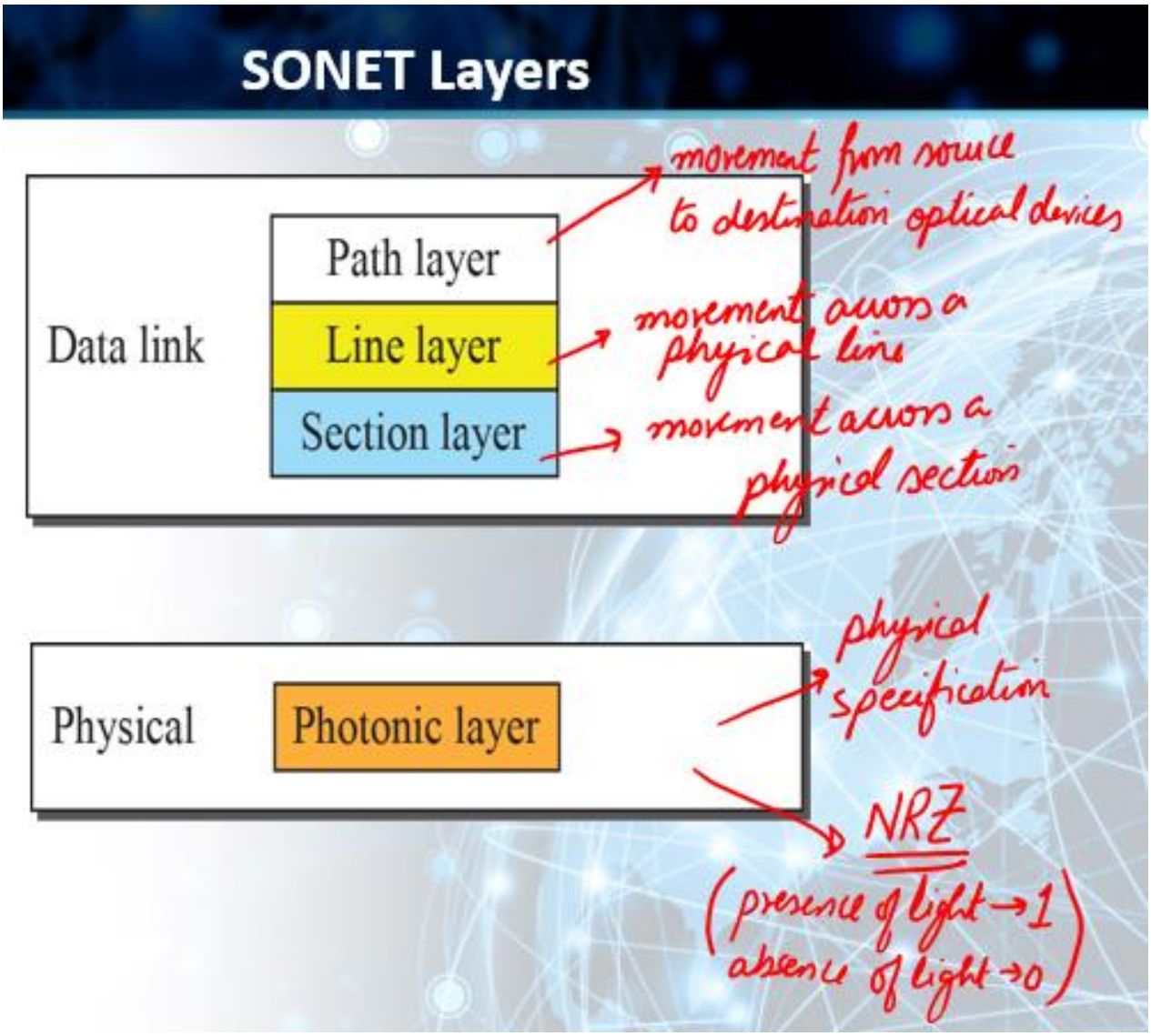
- **The SONET standard includes four functional layers:**
  - ✓ **The Path Layer**
  - ✓ **The Line Layer**
  - ✓ **The Section Layer**
  - ✓ **The Photonic Layer**

- **The layers correspond to both the physical and the data-link layers**

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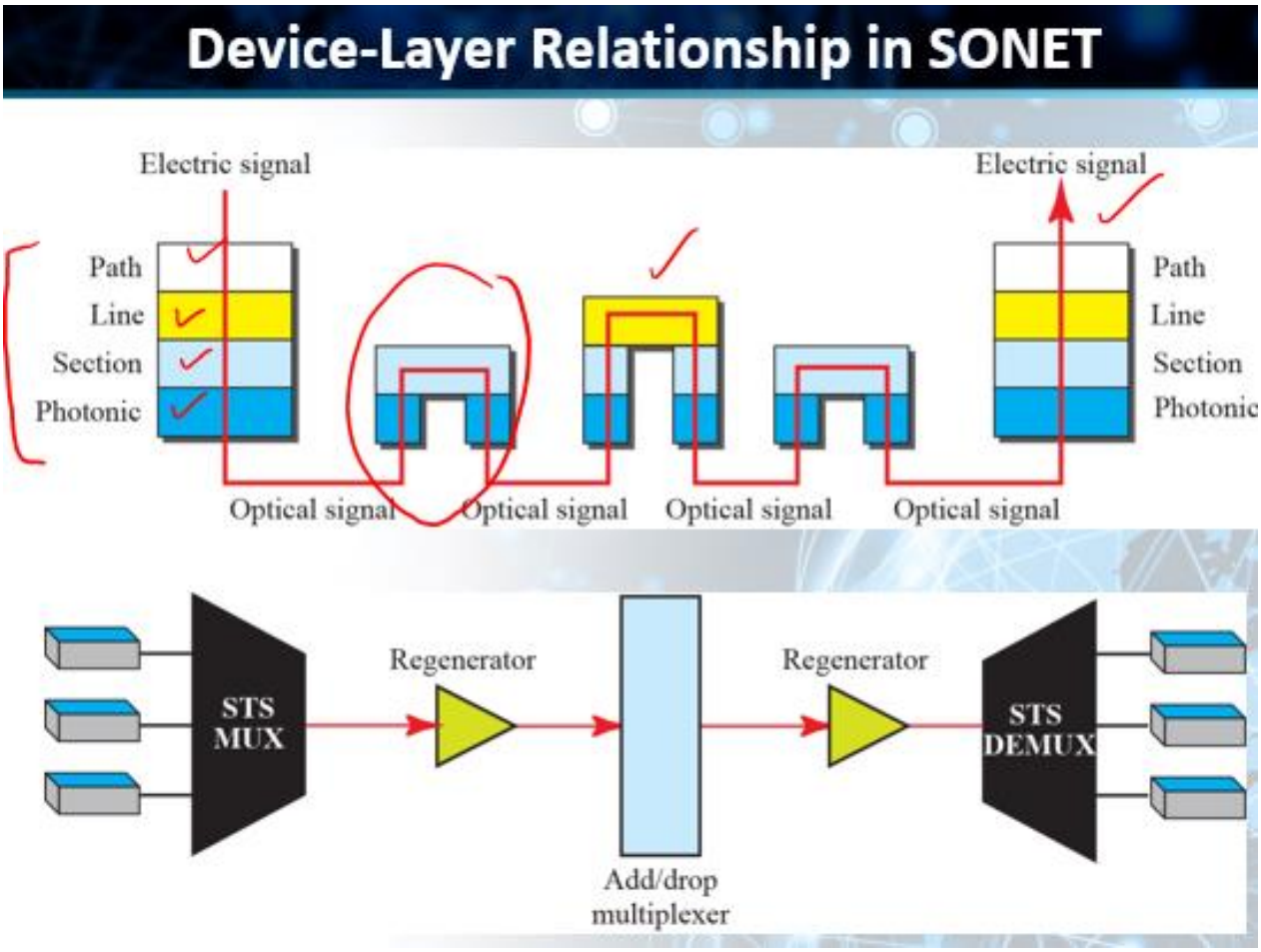
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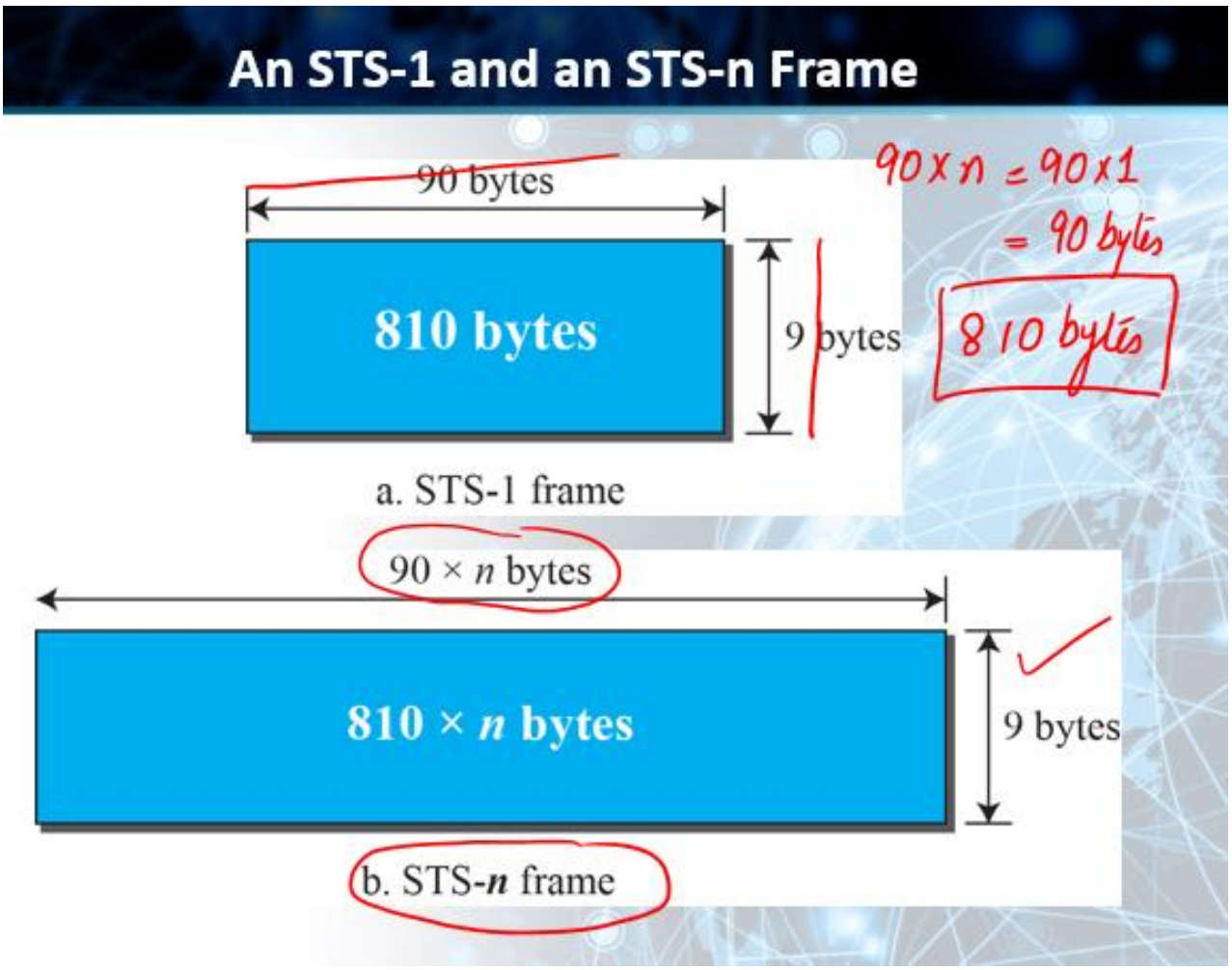
SONET Frames

- **Each synchronous transport signal STS-n is composed of 8000 frames**
- **Each frame is a two-dimensional matrix of bytes with 9 rows by 90×n columns**
- **STS-1 frame is 9 rows by 90 columns (810 bytes), and an STS-3 is 9 rows by 270 columns (2430 bytes)**

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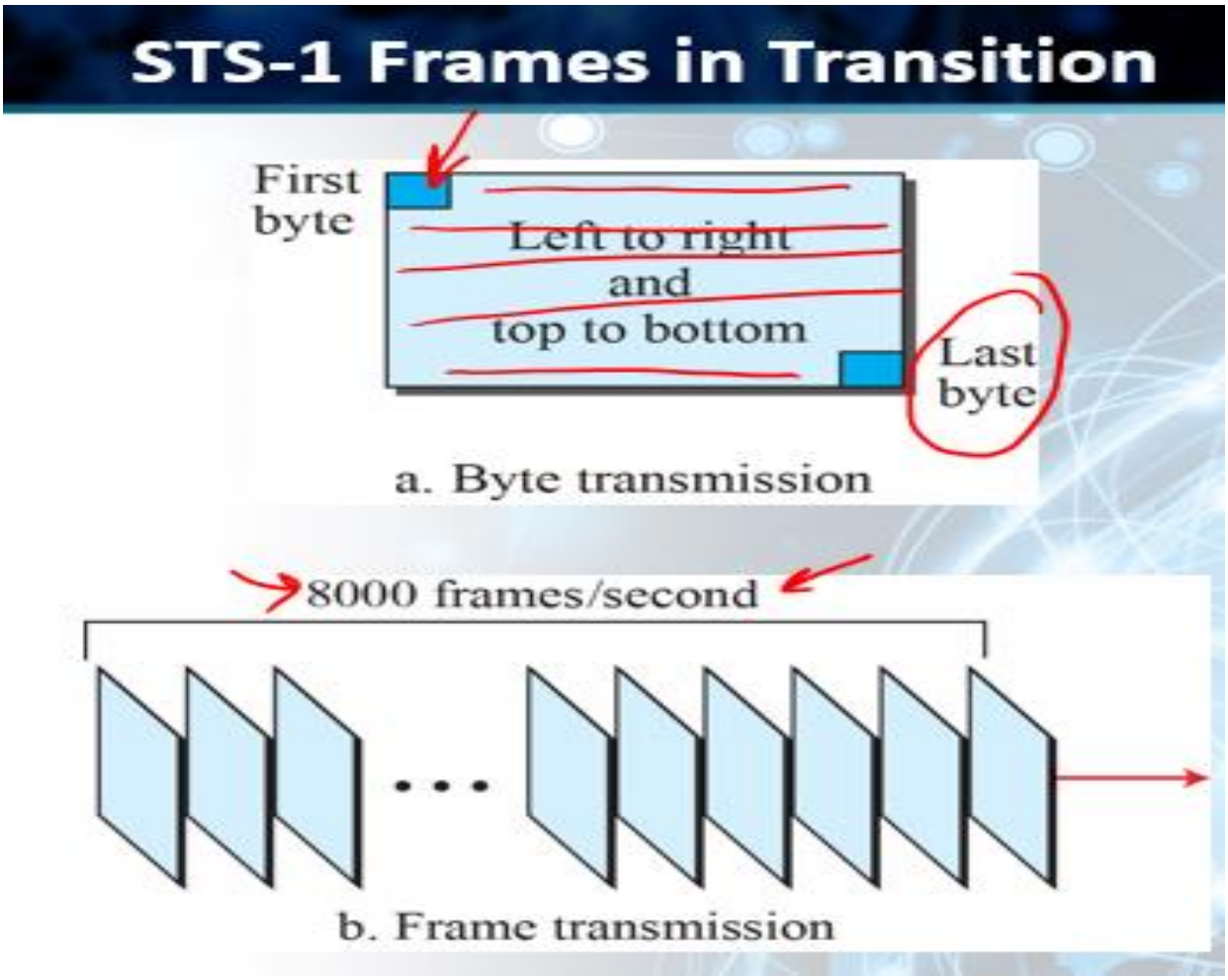
SONET Signal	Bit Rate (Mbps)
STS–1, OC–1	51.840
STS–3, OC–3	155.520
STS–12, OC–12	622.080
STS–48, OC–48	2,488.320

STS-1 Frames in Transition

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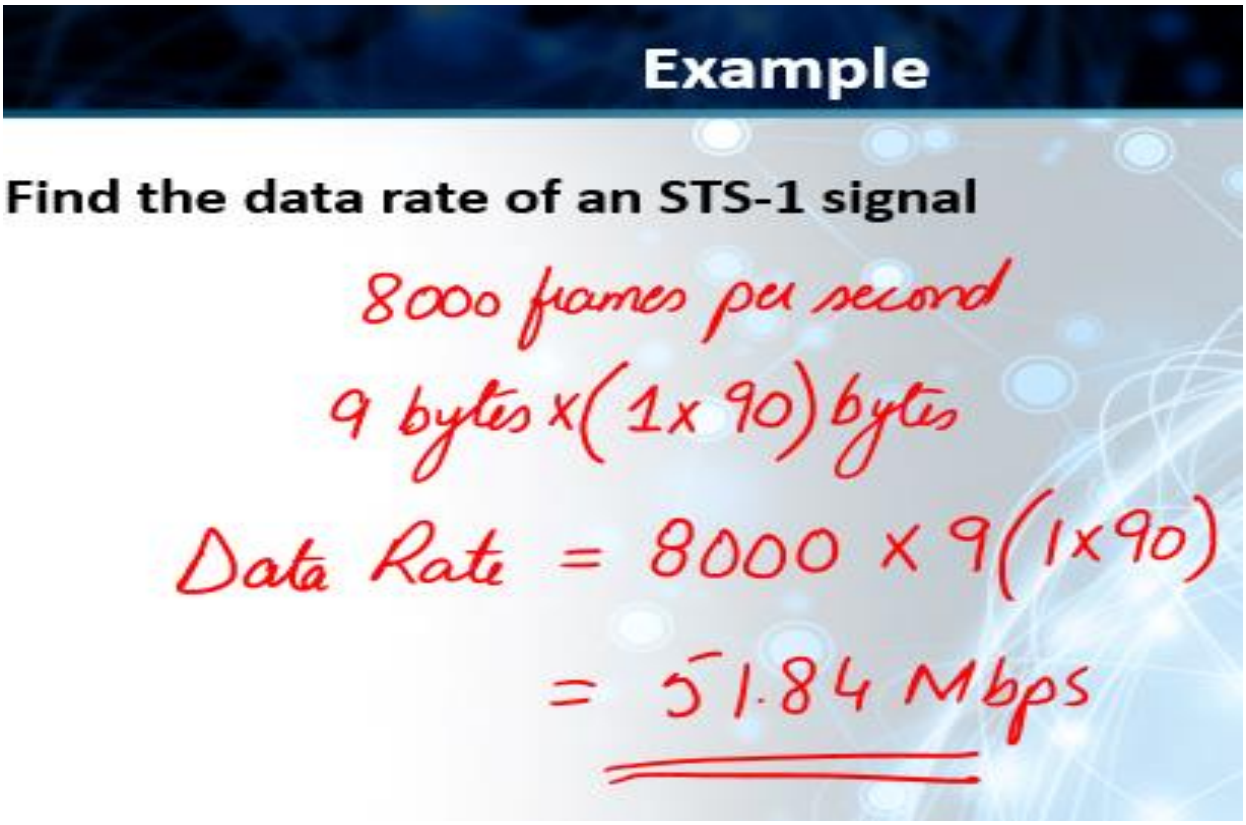
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STS–1 is a specific sequence of 810 bytes (6,480 bits), which includes various overhead bytes and an envelope capacity for transporting payloads. It can be depicted as a 90-column by 9-row structure. With a frame length of 125 µs (8,000 frames per second), STS–1 has a bit rate of **51.840 Mbps**.

(9) x (90 bytes/frame) x (8 bits/byte) x (8,000 frames/s) = 51,840,000 bps = 51.840 Mbps

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Example 17.1

Find the data rate of an STS-1 signal. SONET sends 8000 frames per second.

Solution

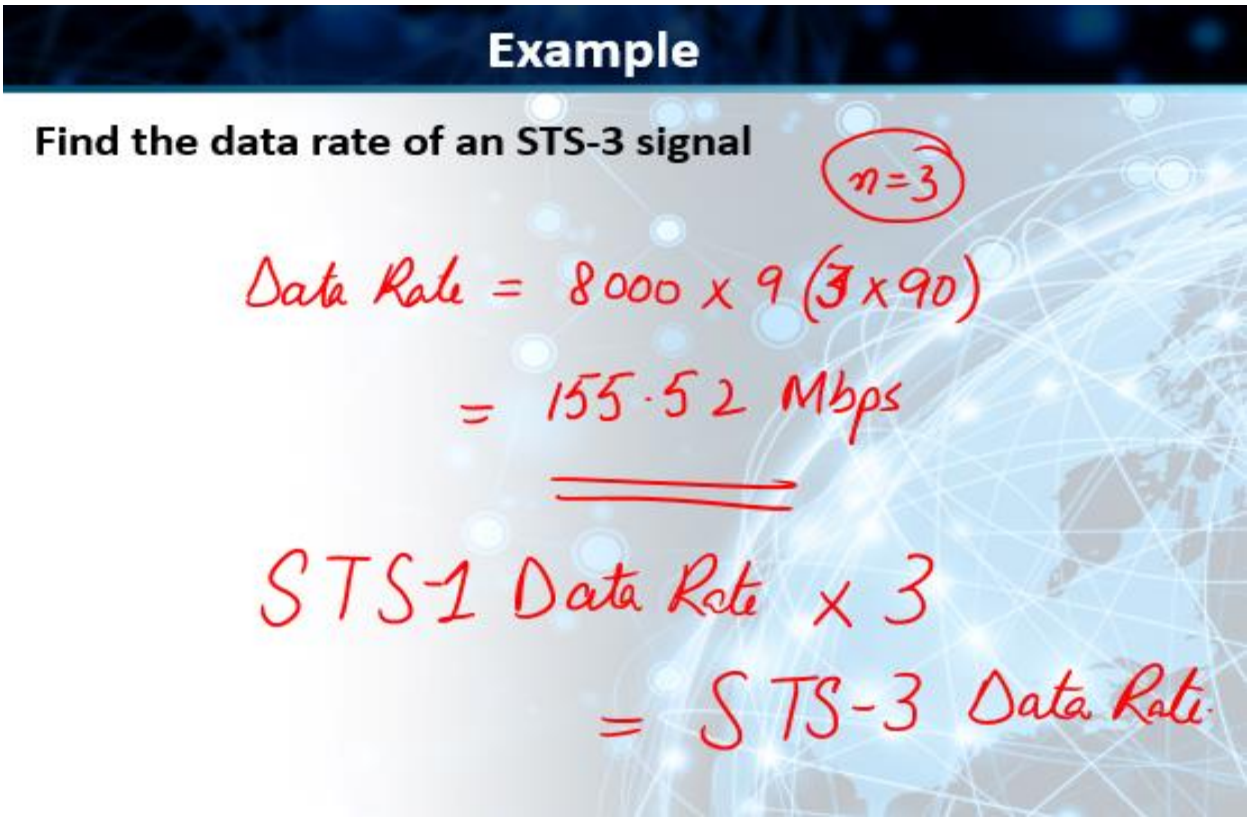
STS-1, like other STS signals, sends 8000 frames per second. Each STS-1 frame is made of 9 by (1 × 90) bytes. Each byte is made of 8 bits. The data rate is

$$\text{STS-1 data rate} = 8000 \times 9 \times (1 \times 90) \times 8 = 51.840 \text{ Mbps}$$

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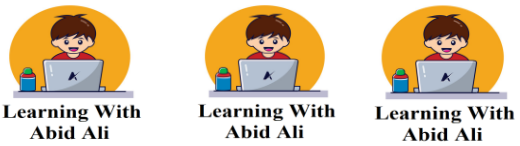
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Example 17.2

Find the data rate of an STS-3 signal.

Solution

STS-3, like other STS signals, sends 8000 frames per second. Each STS-3 frame is made of 9 by (3 × 90) bytes. Each byte is made of 8 bits. The data rate is

$$\text{STS-3 data rate} = 8000 \times 9 \times (3 \times 90) \times 8 = 155.52 \text{ Mbps}$$

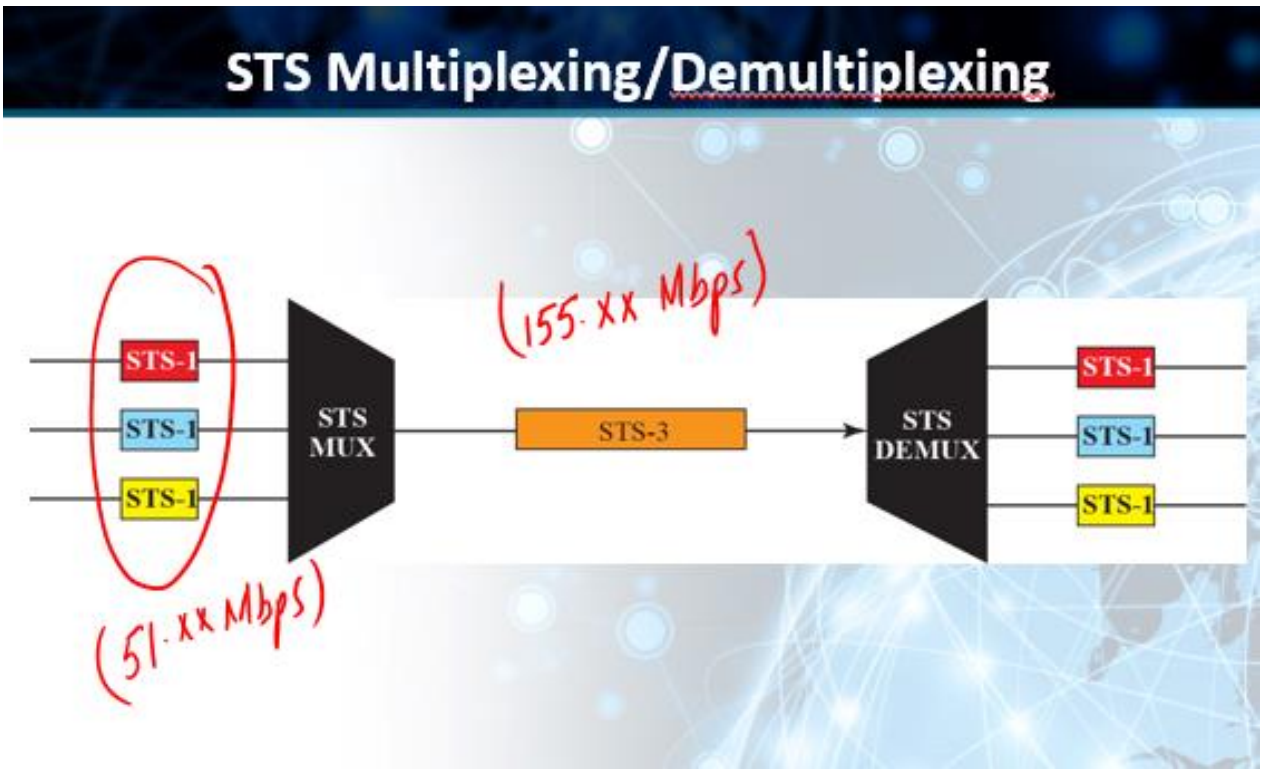
STS Multiplexing

- **In SONET, frames of lower rate can be synchronously time-division multiplexed into a higher-rate frame**
- **For example, three STS-1 signals (channels) can be combined into one STS-3 signal (channel), four STS-3s can be multiplexed into one STS-12, and so on**

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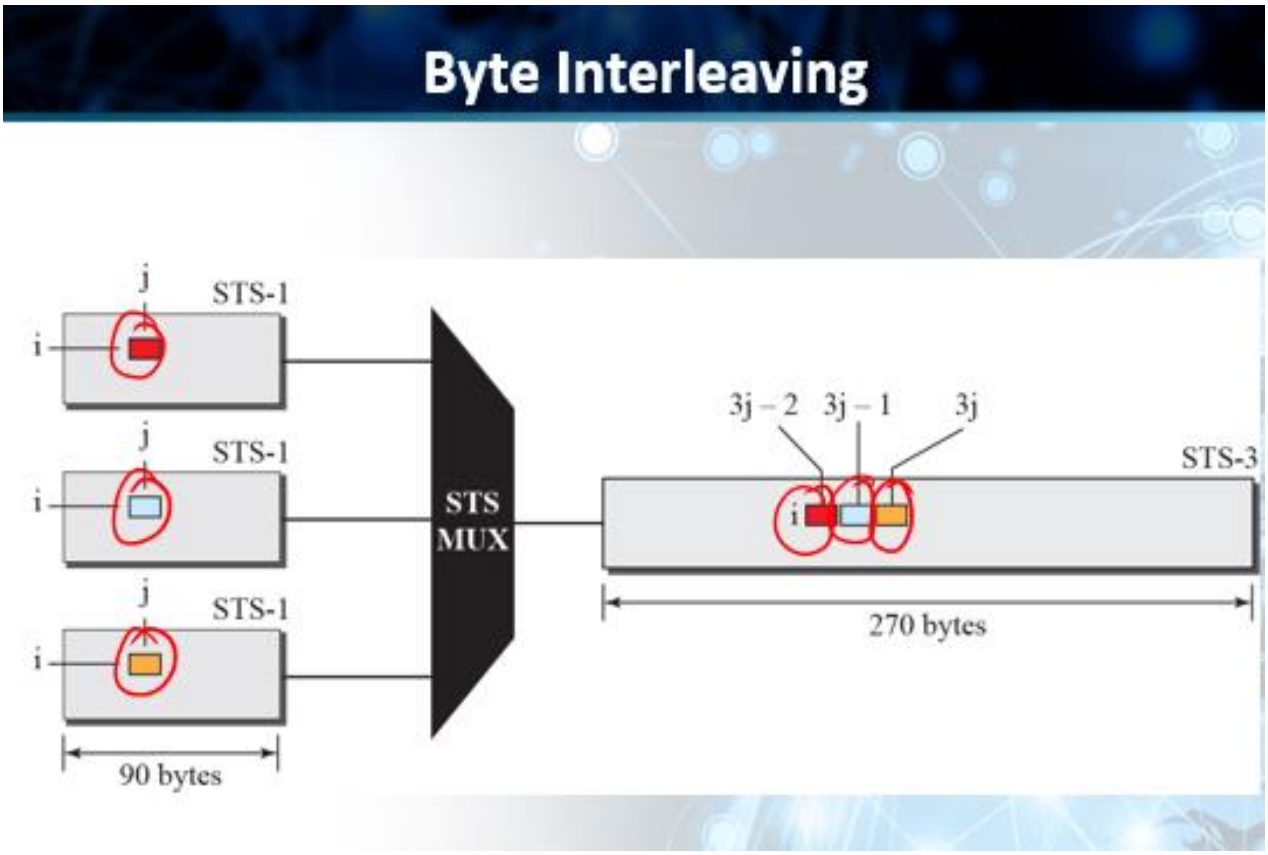
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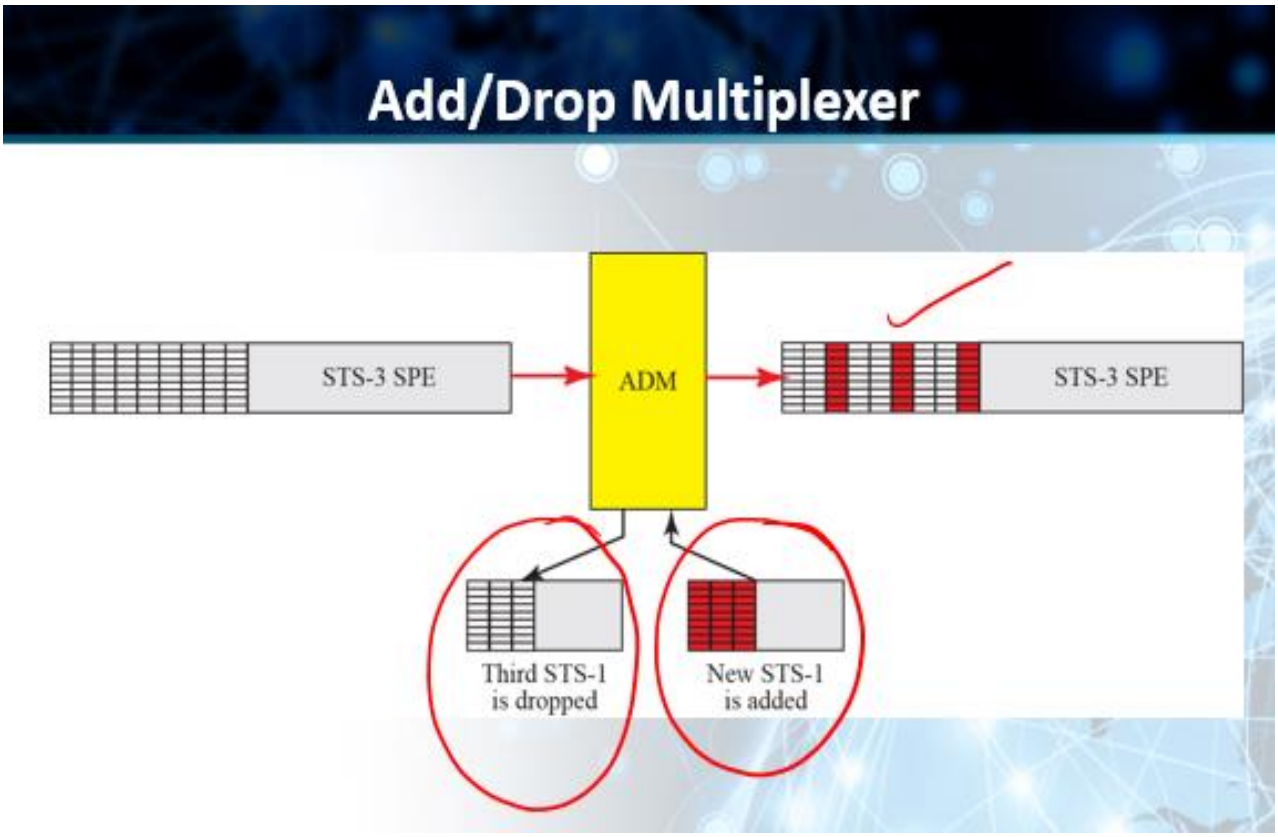
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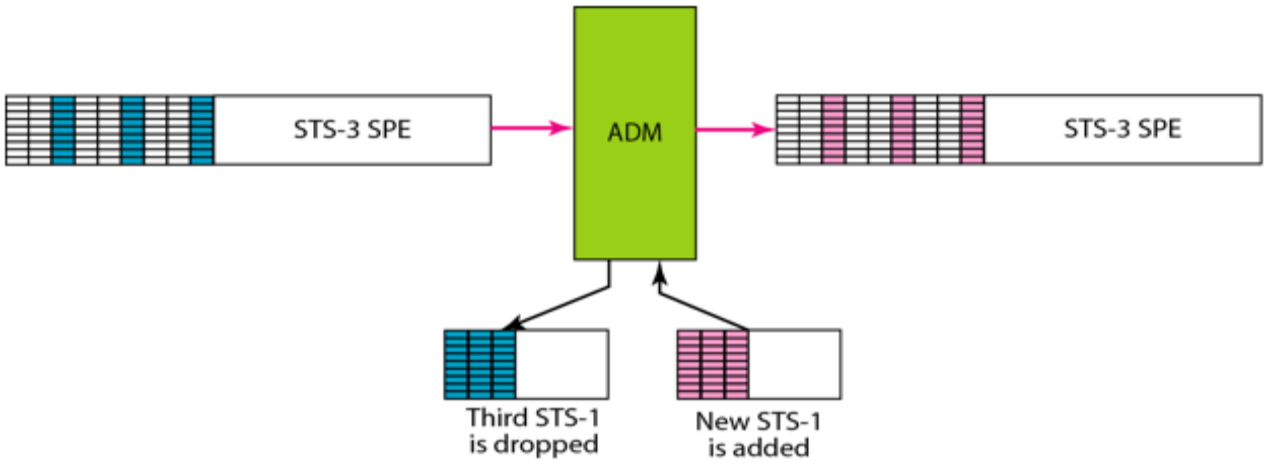


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Figure 17.16 *Dropping and adding STS-1 frames in an add/drop multiplexer*



Notice that ADM only operates a physical layer – it drops one STS-1 and adds a new one in its place.

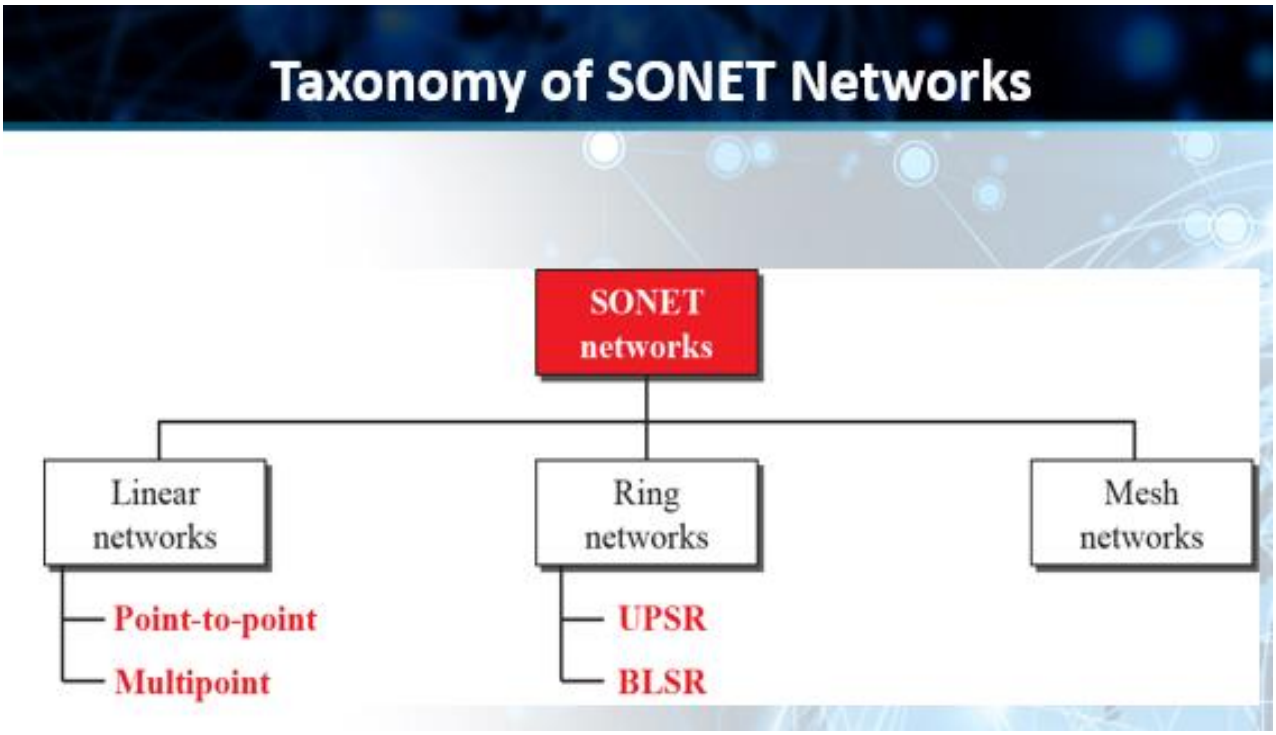
SONET Networks

- **SONET network can be used as a high-speed backbone carrying loads from other networks such as ATM or IP**
- **We can roughly divide SONET networks into three categories:**
  - ✓ **Linear Networks**
  - ✓ **Ring Networks**
  - ✓ **Mesh networks**

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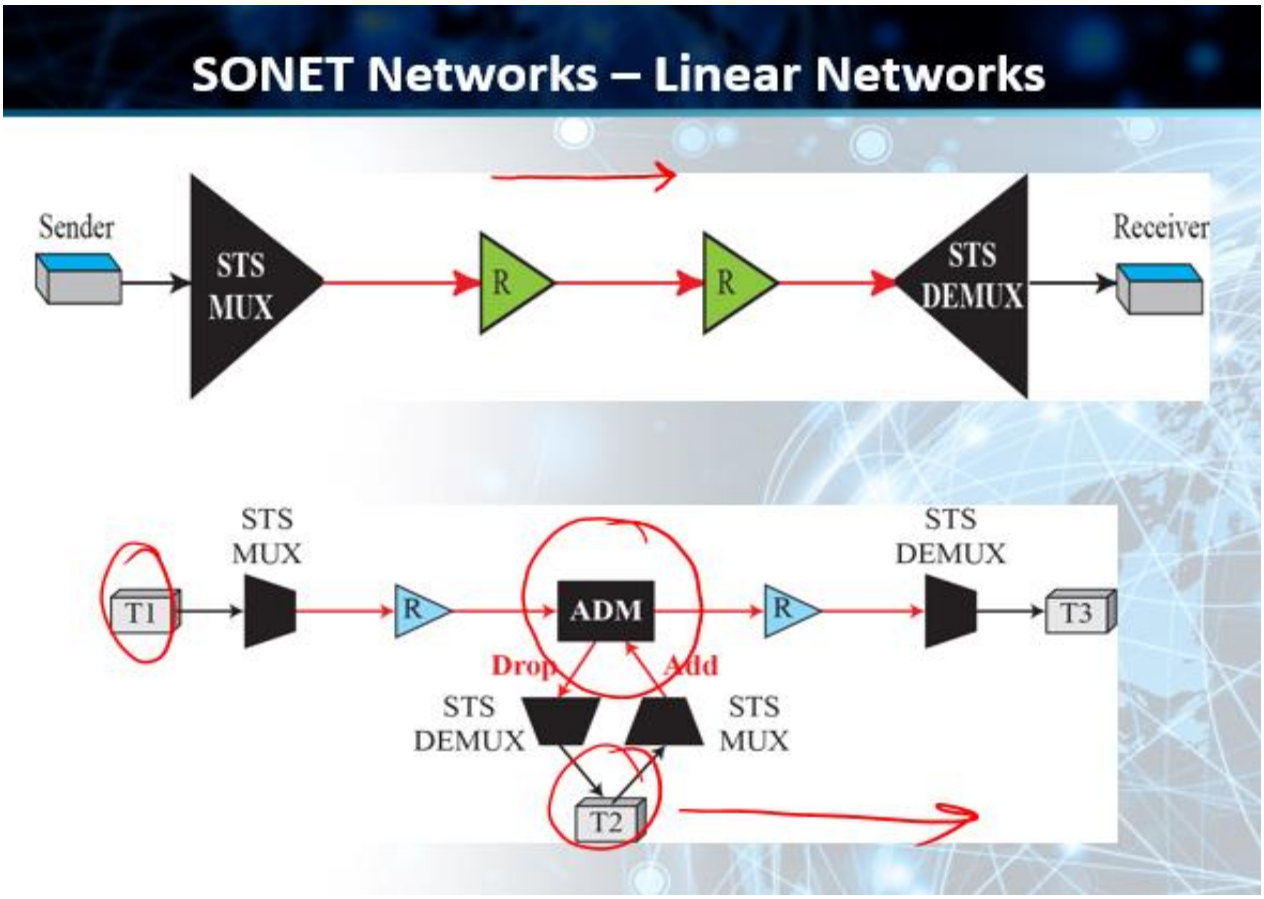
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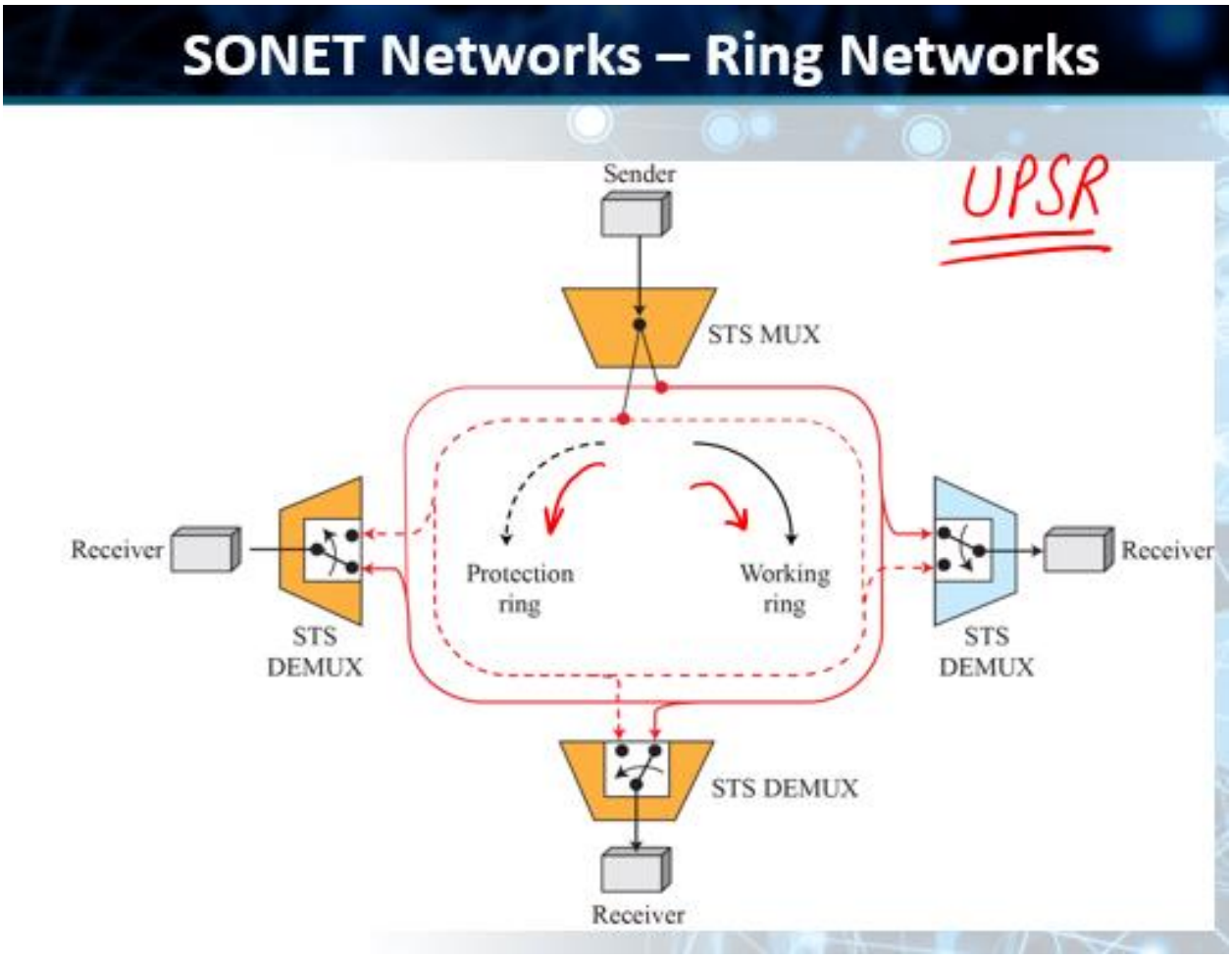
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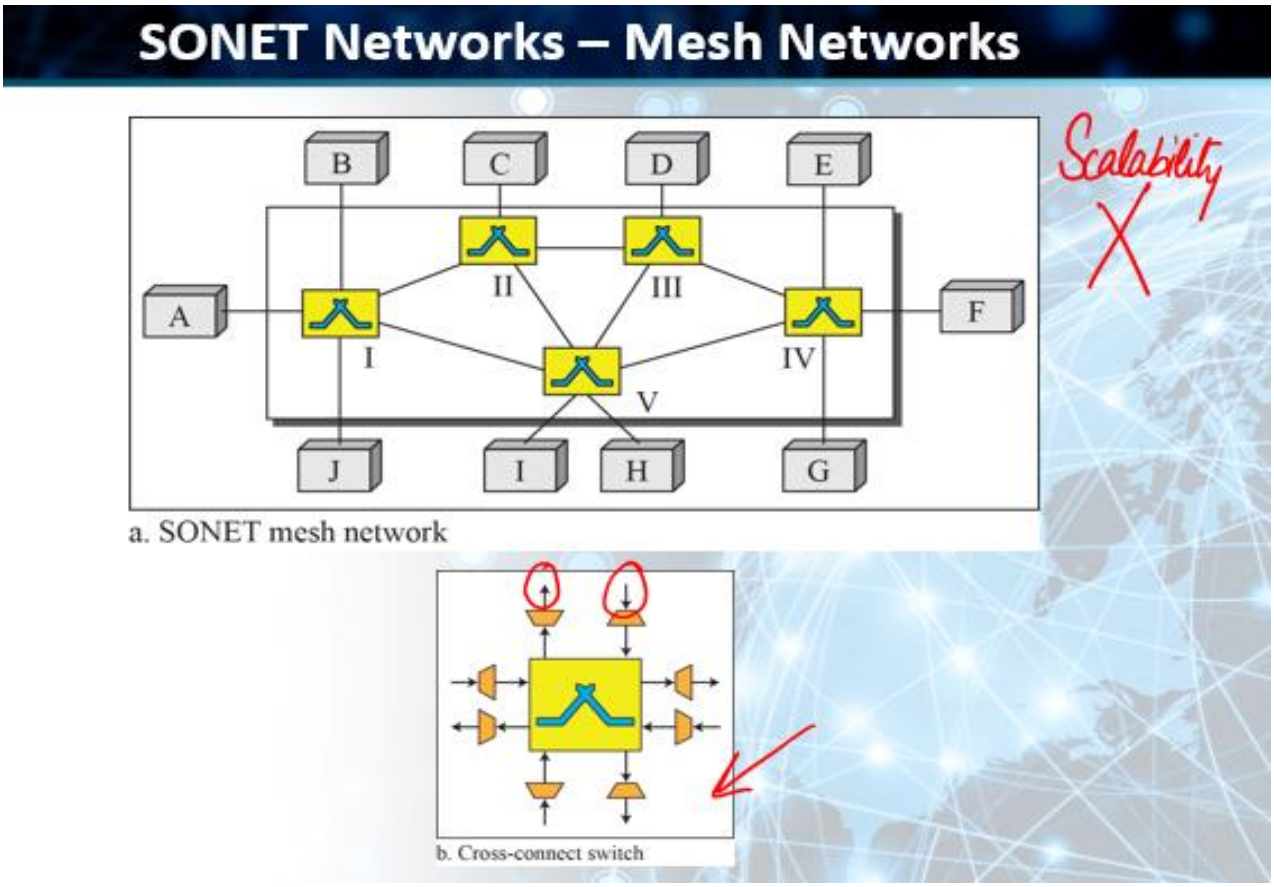
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ATM

- **Asynchronous Transfer Mode (ATM) is a switched wide area network based on the cell relay protocol designed by the ATM forum**
- **The combination of ATM and SONET will allow high-speed interconnection of networks**

Problems

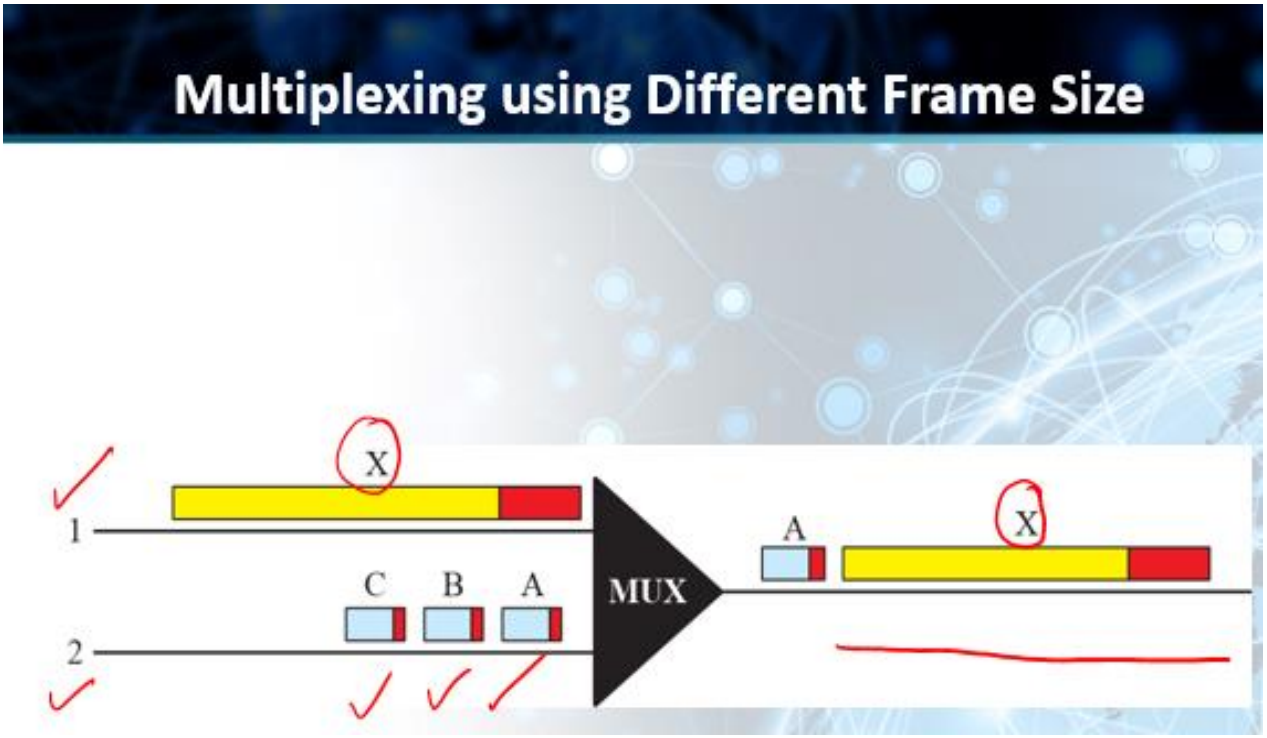
- **Some of the problems associated with existing systems are:**
  - ✓ **Frame Networks**
  - ✓ **Mixed Network Traffic**
- **Solution**

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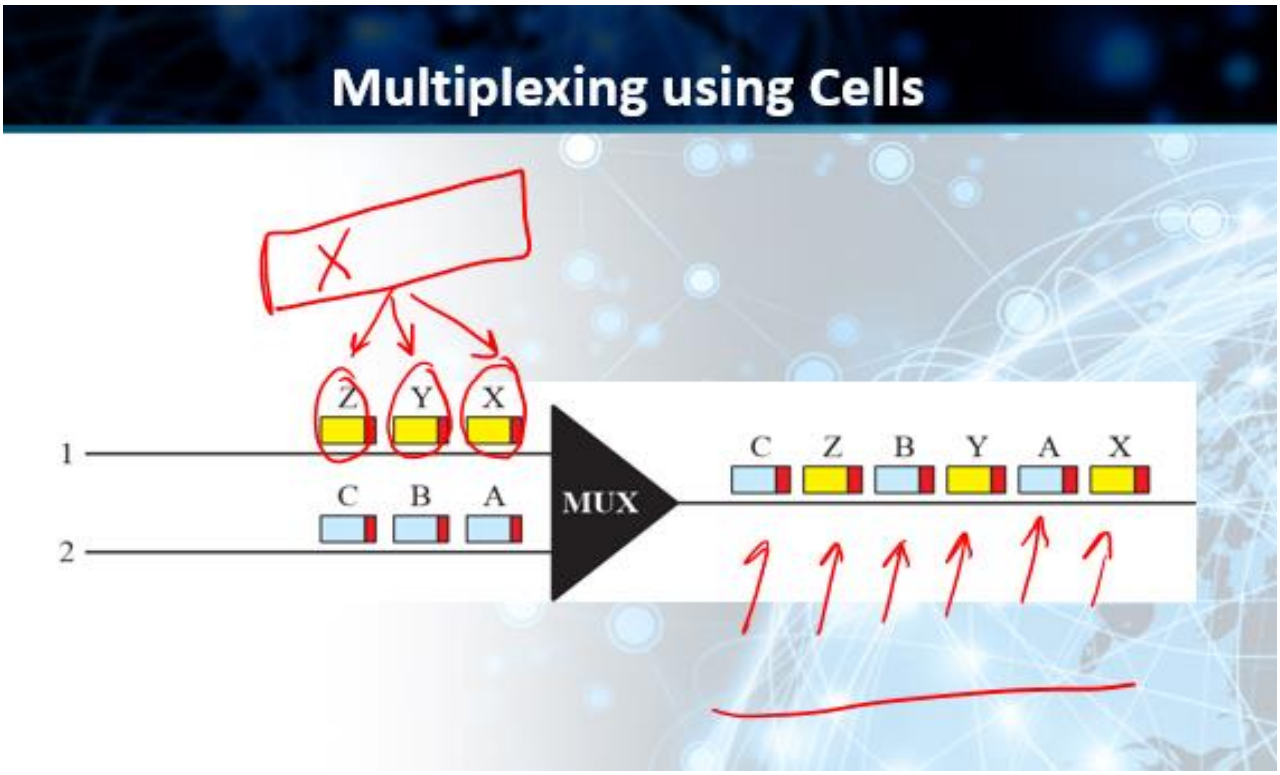
- ✓ **Cell Networks**
- ✓ **Asynchronous TDM**



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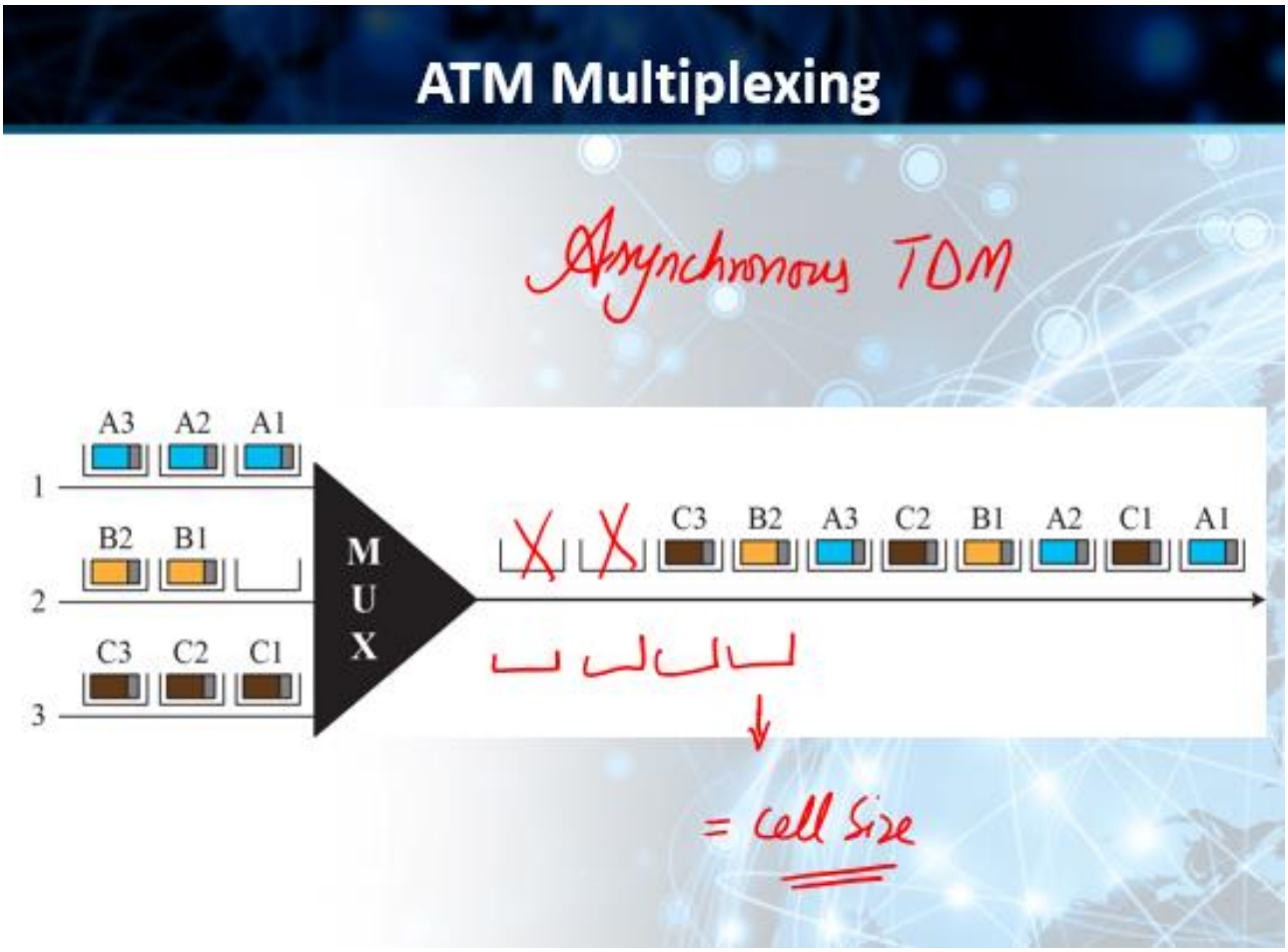
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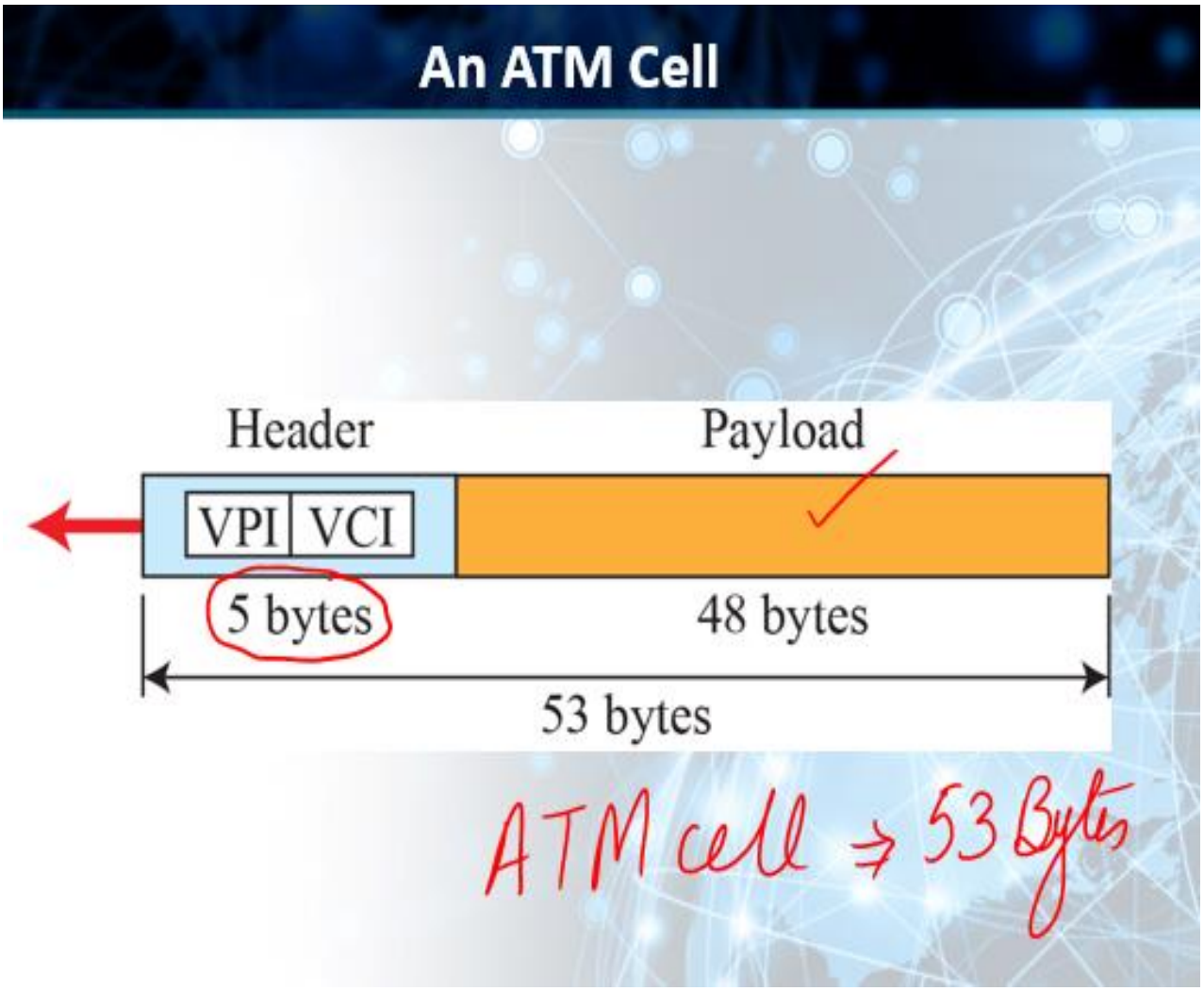
Architecture

- **ATM is a cell-switched network**
- **The user access devices, called the endpoints, are connected through a user-to-network interface (UNI) to the switches inside the network**
- **The switches are connected through network-to-network interfaces (NNIs)**

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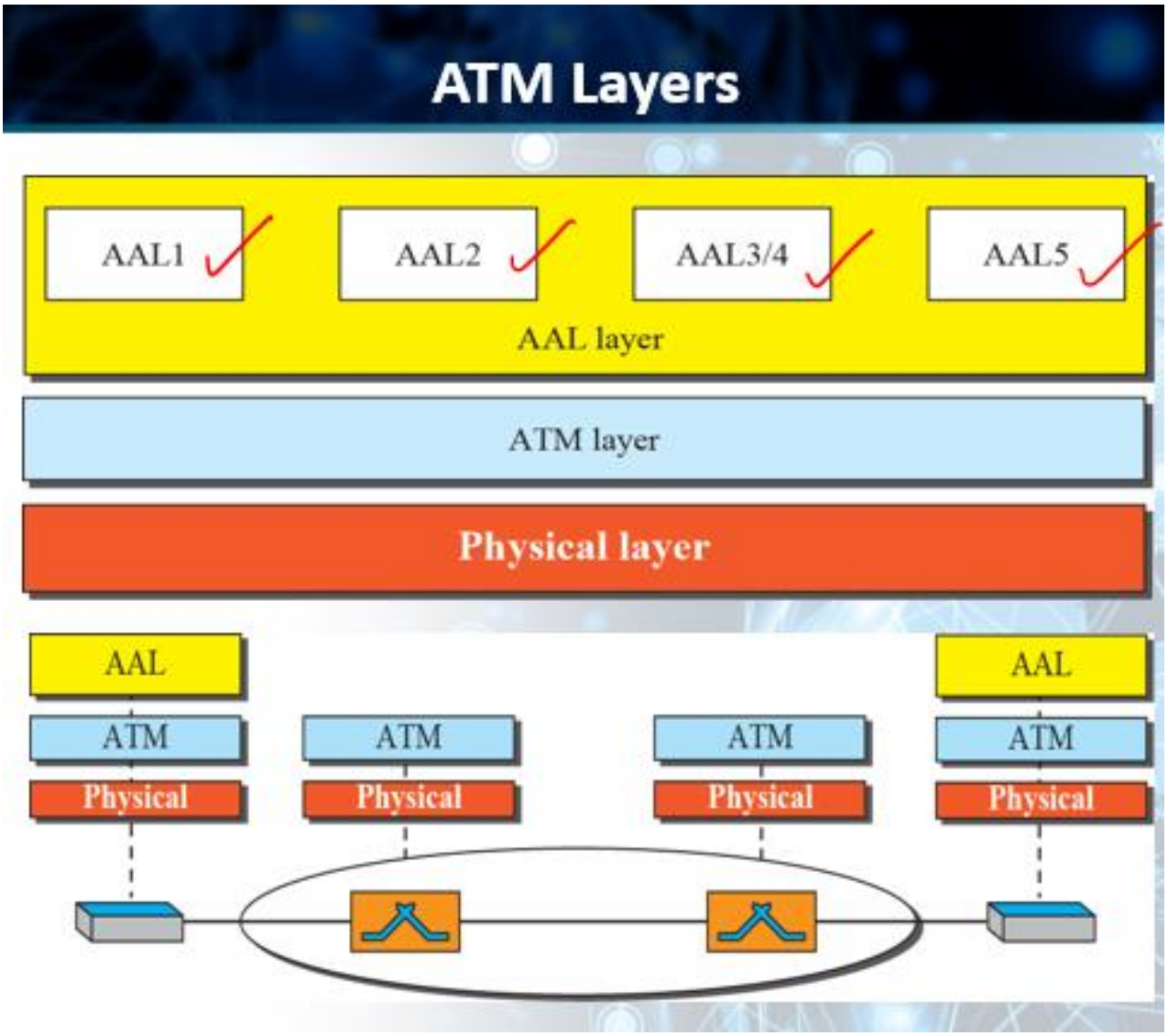
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Lecture Prepared By

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CS601 Lecture No 35 Topic 205 to 212

Wireless communication

- **Wireless communication is one of the fastest-growing technologies**
- **The demand for connecting devices without the use of cables is increasing everywhere**

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- **Wireless LANs can be found on college campuses, in office buildings, and in many public areas**

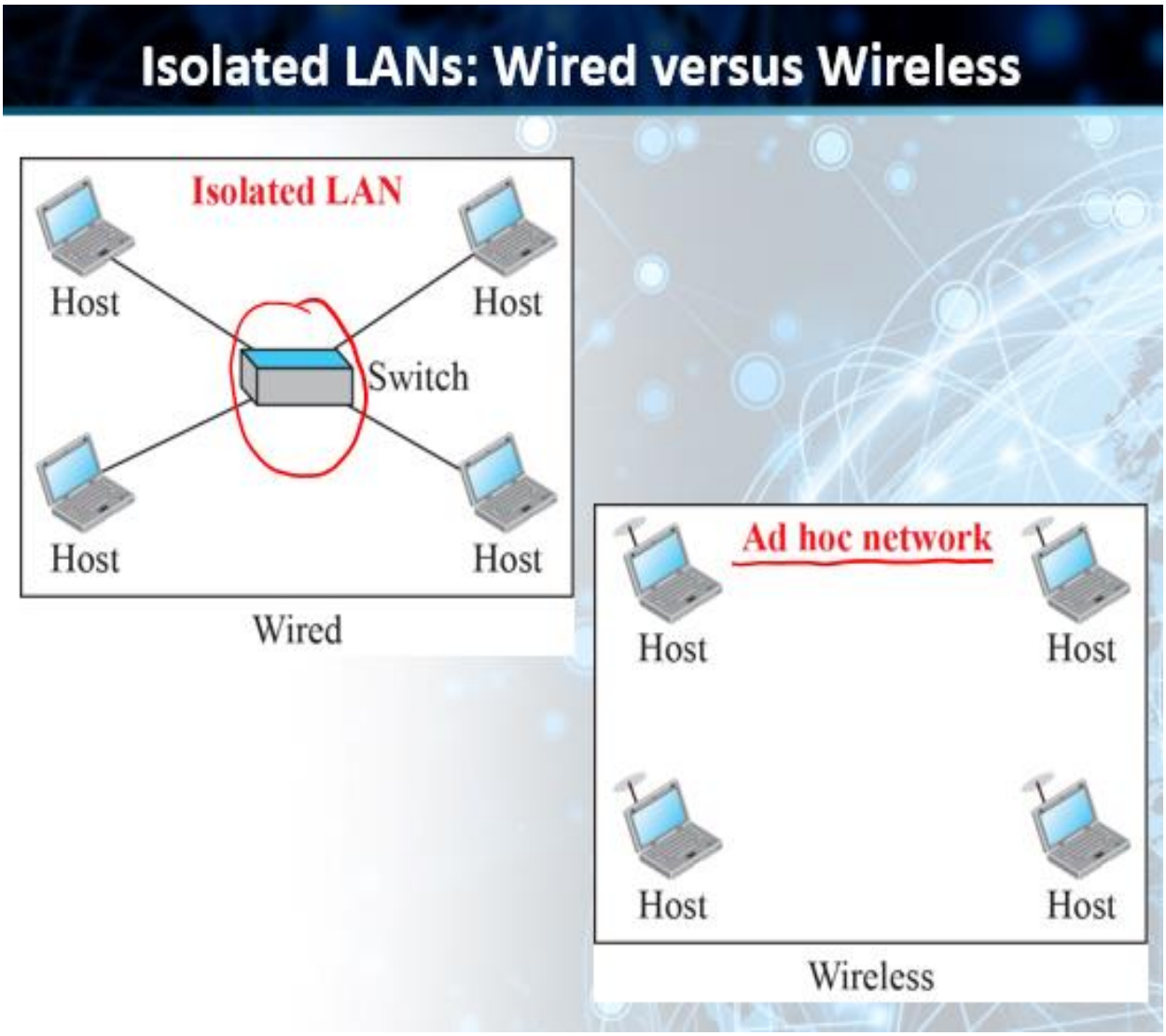
Architectural Comparison

- **Architecture comparison of wired and wireless LANs**
  - ✓ **Medium**
  - ✓ **Hosts**
  - ✓ **Isolated LANs**
  - ✓ **Connection to other Networks**
  - ✓ **Moving between Environments**

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Characteristics of a Wireless LAN

- **Several characteristics of wireless LANs either do not apply to wired LANs or the existence of these is negligible and can be ignored**
  - ✓ **Attenuation**
  - ✓ **Interference**
  - ✓ **Multipath Propagation**

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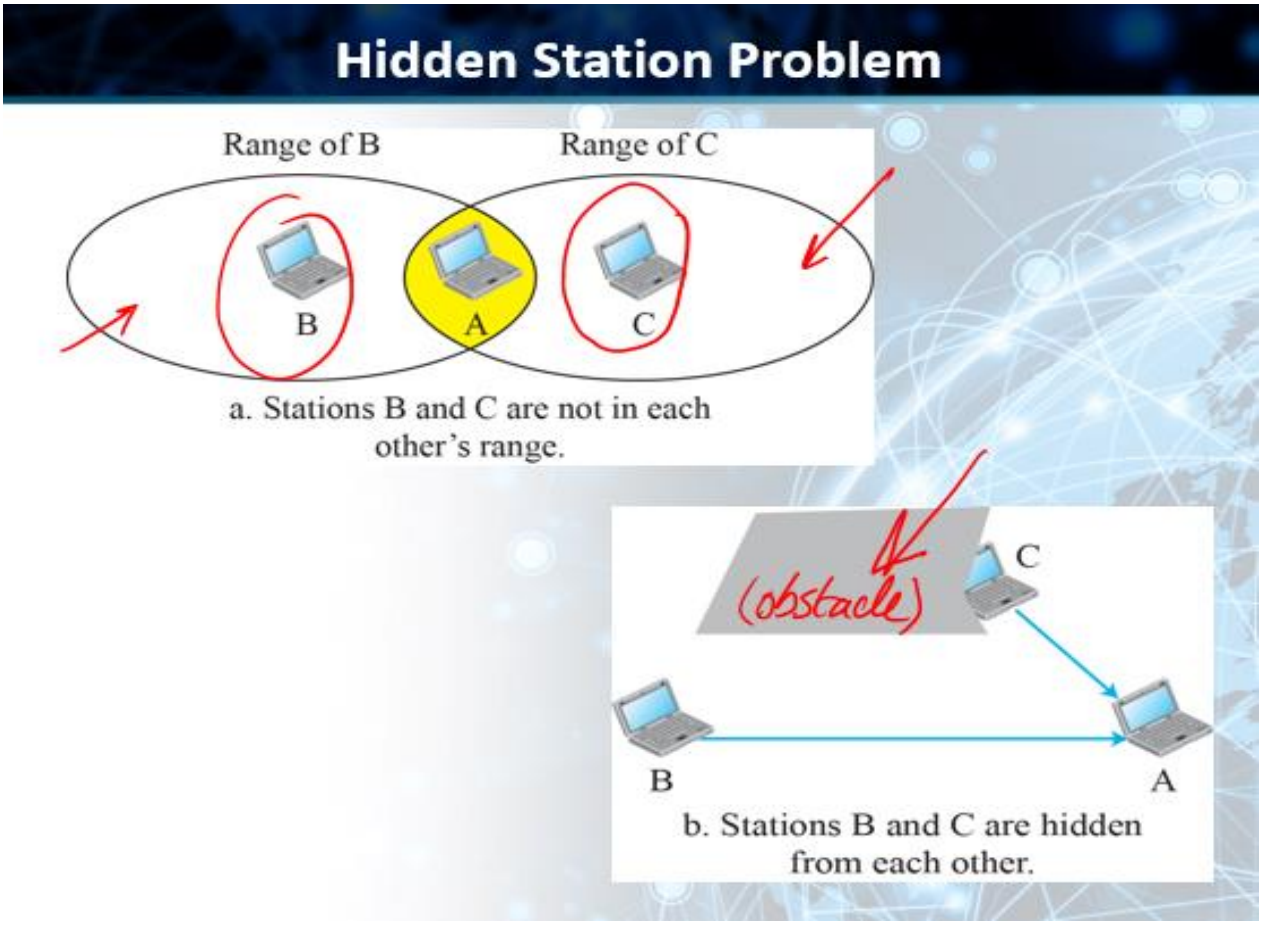
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- ✓ **Error**

Access Control

- **Most important issue in a wireless LAN is how a wireless host can get access to the shared medium (air)**
- **CSMA/CD does not work in wireless LANs for three reasons:**
    1. **Wireless hosts don't have power to send and receive at the same time**
- **The hidden station problem prevents collision detection**
- **The distance between stations can be large**



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IEEE 802.11 PROJECT

- **IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data-link layers**
- **It is sometimes called Wireless Ethernet**
- **The term WiFi (short for wireless fidelity) as a synonym for wireless LAN (certified by WiFi alliance)**

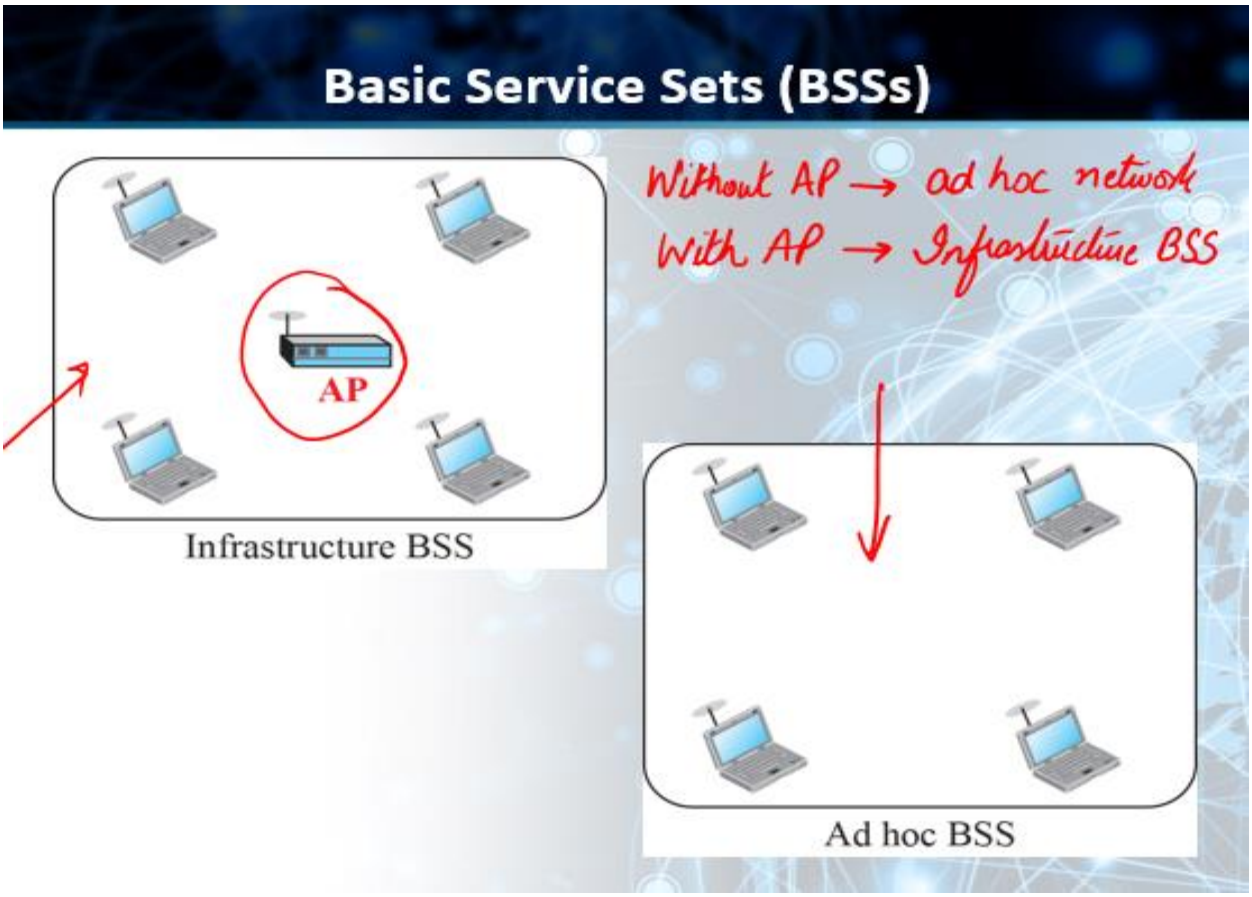
Architecture

- **The standard defines two kinds of services:**
  - ✓ **The basic service set (BSS); and**
  - ✓ **The Extended service set (ESS)**

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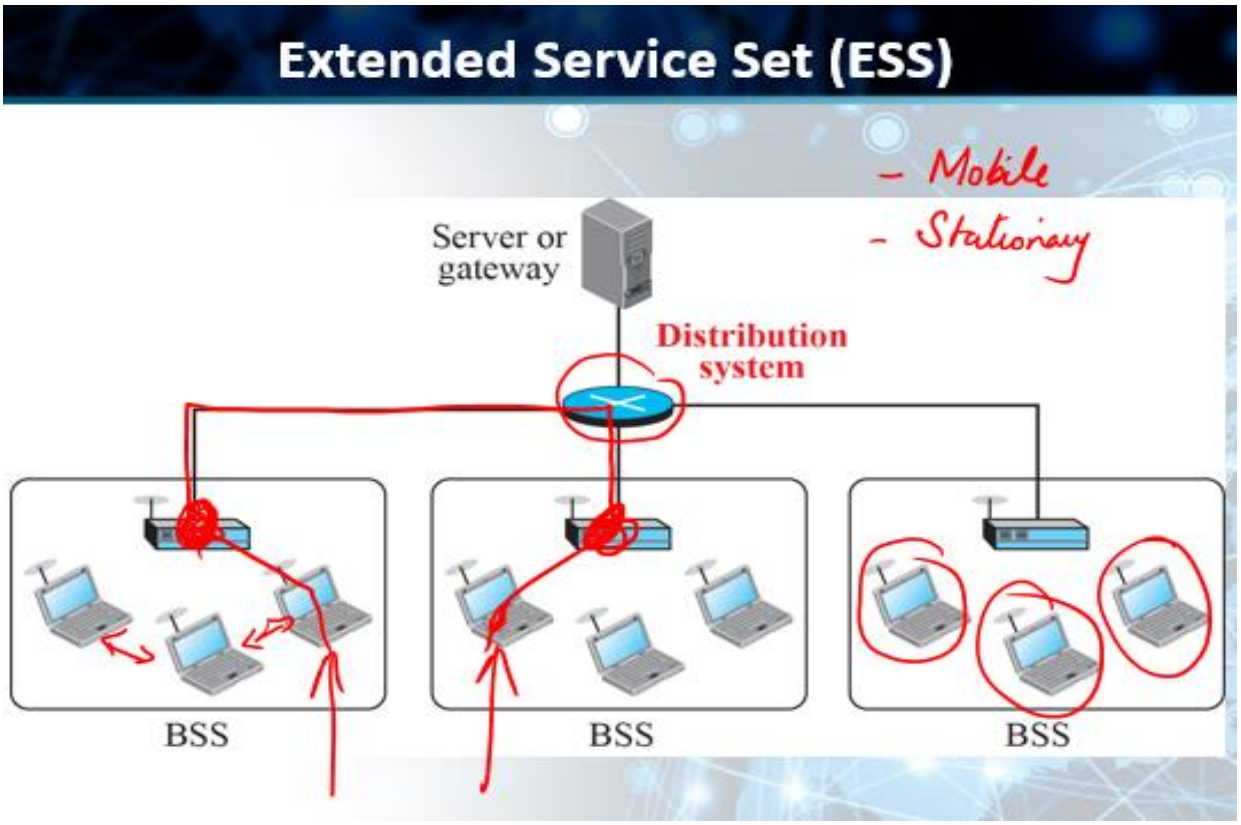
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Types of Stations

- **No-Transition Mobility**
- **BSS-Transition Mobility**
- **ESS-Transition Mobility**

MAC Sub-layer Wireless communication

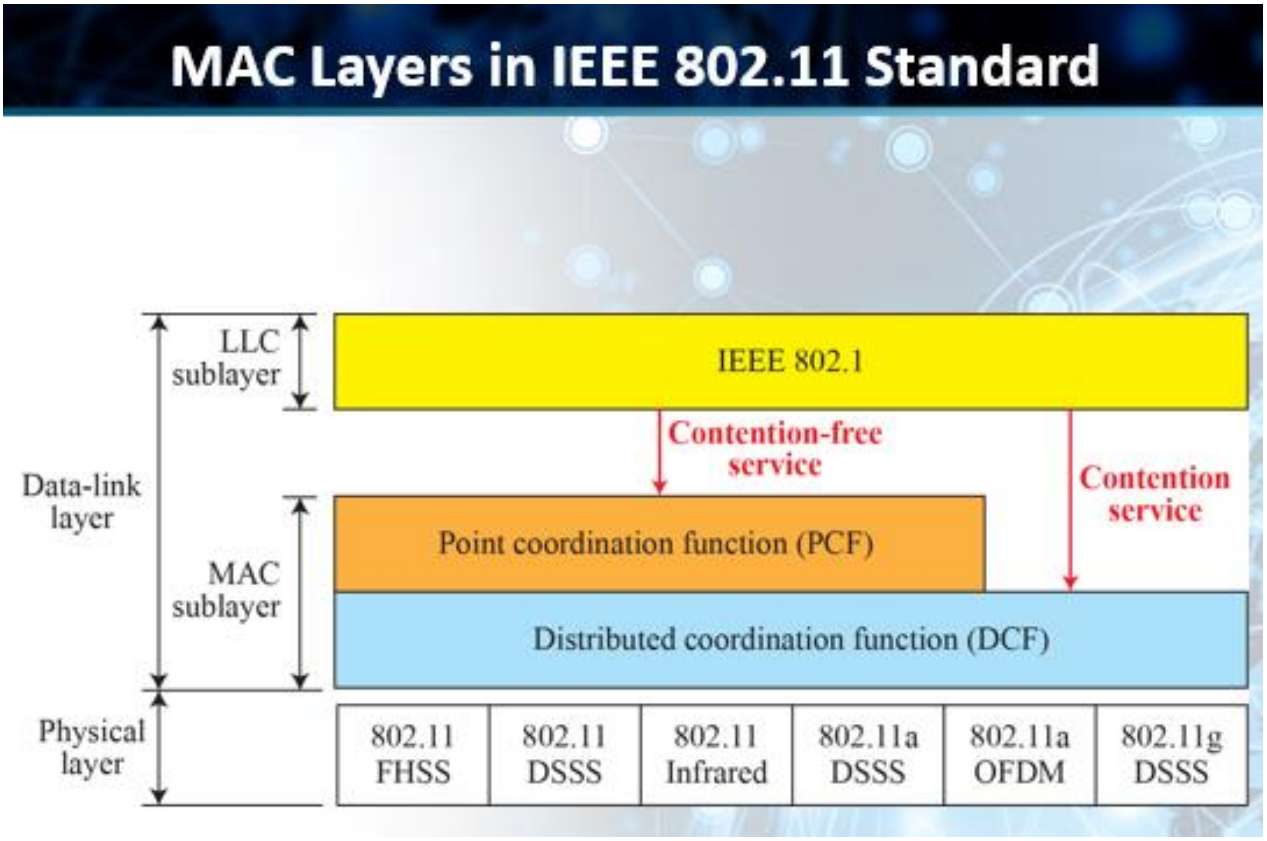
- **IEEE 802.11 defines two MAC sub-layers:**
  - ✓ **The Distributed Coordination Function (DCF) ; and**

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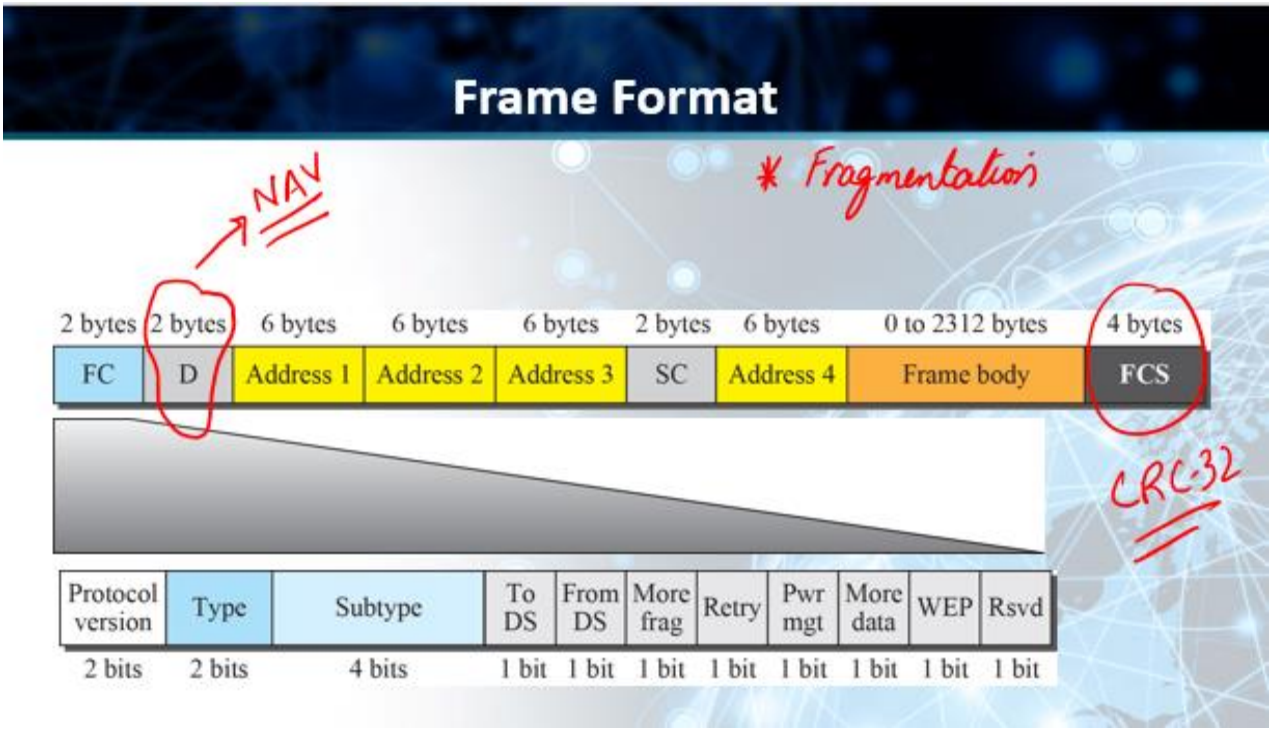
- ✓ **The Point Coordination Function (PCF)**



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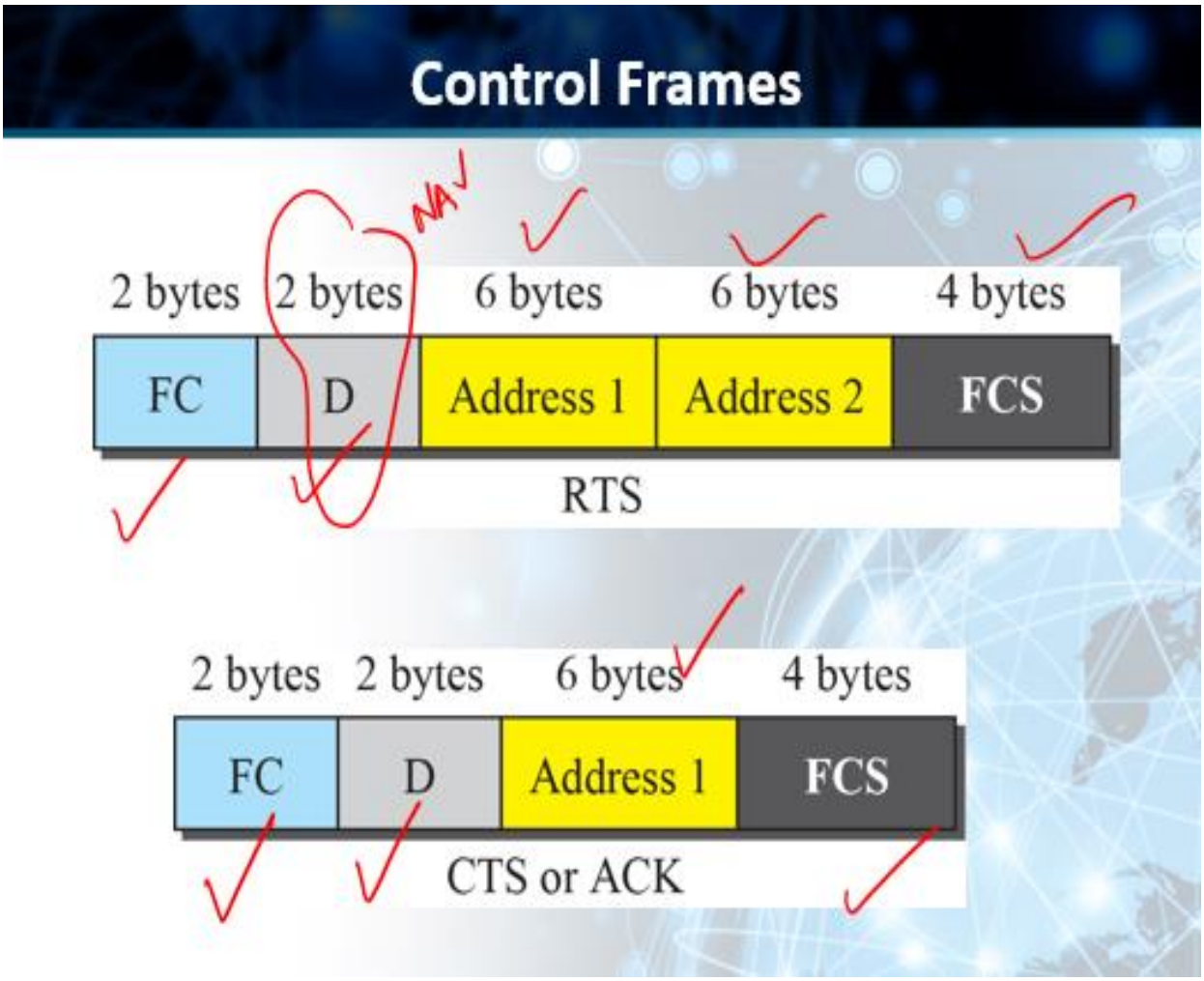
Frame Types

- **Management Frames**
- **Control Frames**
- **Data Frames**

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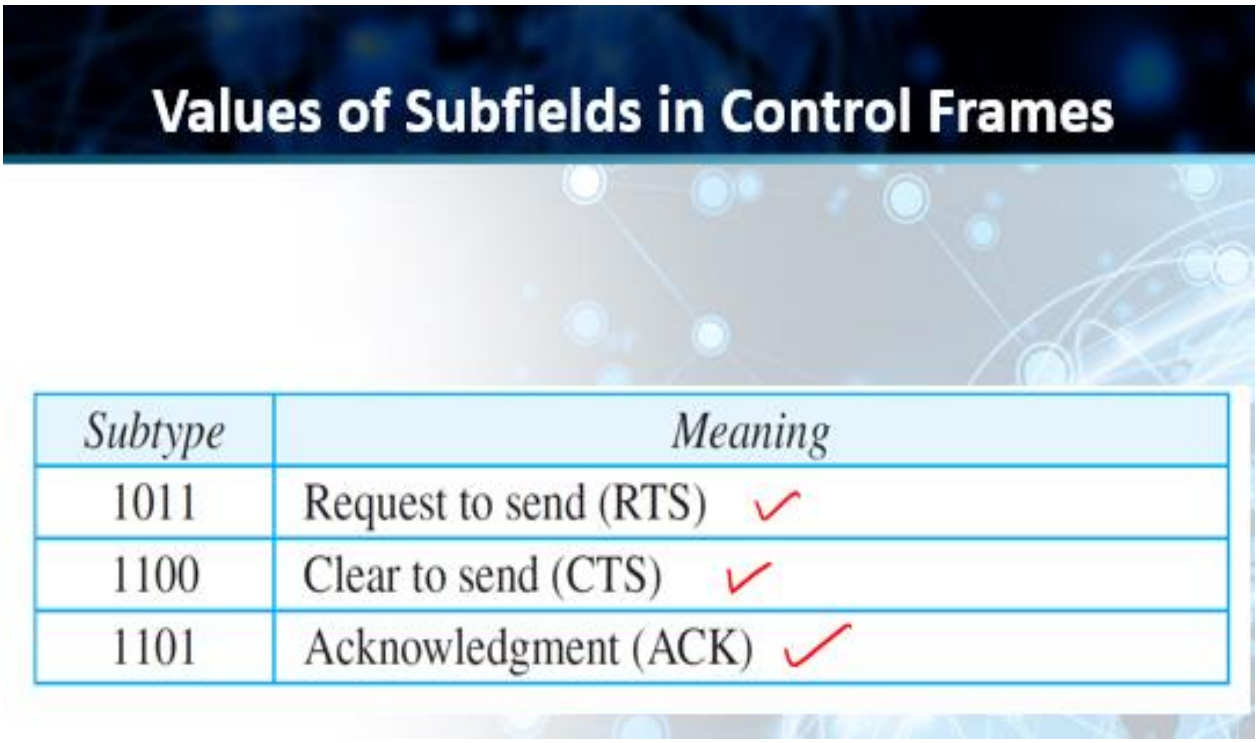
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Subtype	Meaning
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

Physical Layer

- **All physical implementations, except the infrared, operate in the industrial, scientific, and medical (ISM) band, which defines 3 unlicensed bands in 3 ranges:**
  - ✓ **902–928 MHz**
  - ✓ **2.400–4.835 GHz**
  - ✓ **5.725–5.850 GHz**

BLUETOOTH

- **Bluetooth is a wireless LAN technology designed to connect devices of different functions when they are at a short distance from each other**
- **A Bluetooth LAN is an ad hoc network**

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- **The devices, sometimes called gadgets, find each other and make a network called a Piconet**
- **Bluetooth is a wireless LAN technology designed to connect devices of different functions when they are at a short distance from each other**
- **A Bluetooth LAN is an ad hoc network**
- **The devices, sometimes called gadgets, find each other and make a network called a Piconet**

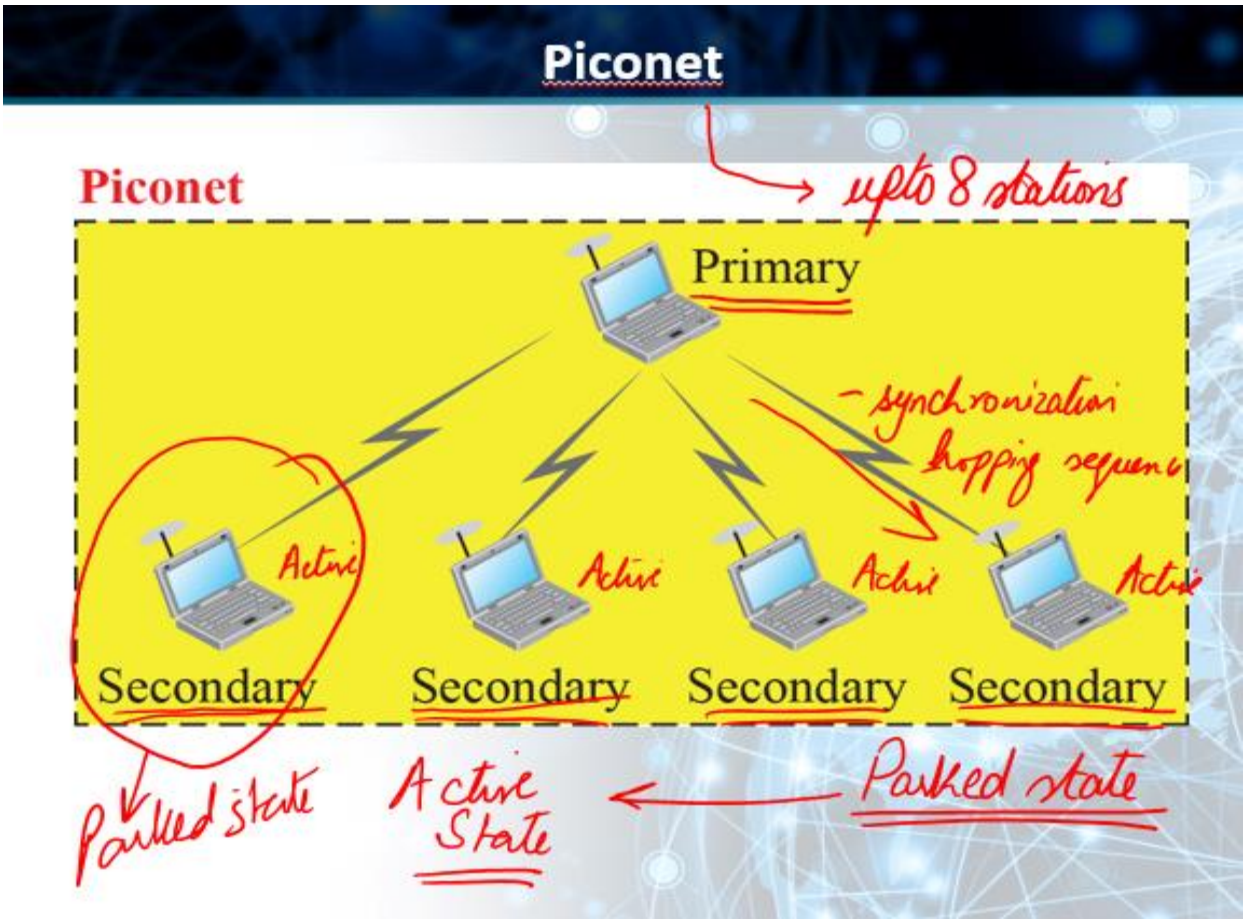
Architecture

- **Bluetooth defines two types of networks:**
  - ✓ **Piconet**
  - ✓ **Scatternet**

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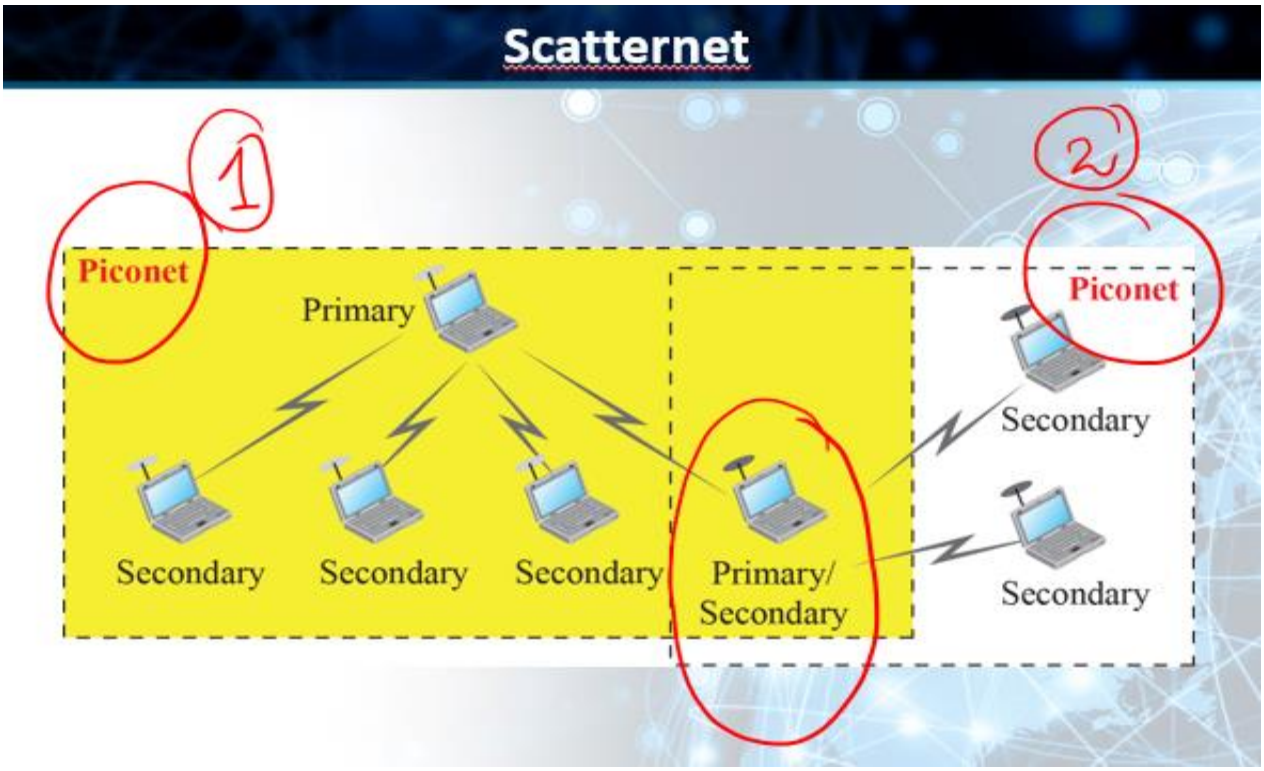
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Bluetooth Devices

- **A Bluetooth device has a built-in short-range radio transmitter**
- **The current data rate is 1 Mbps with a 2.4-GHz bandwidth**
- **This means that there is a possibility of interference between the IEEE 802.11b wireless LANs and Bluetooth LANs**

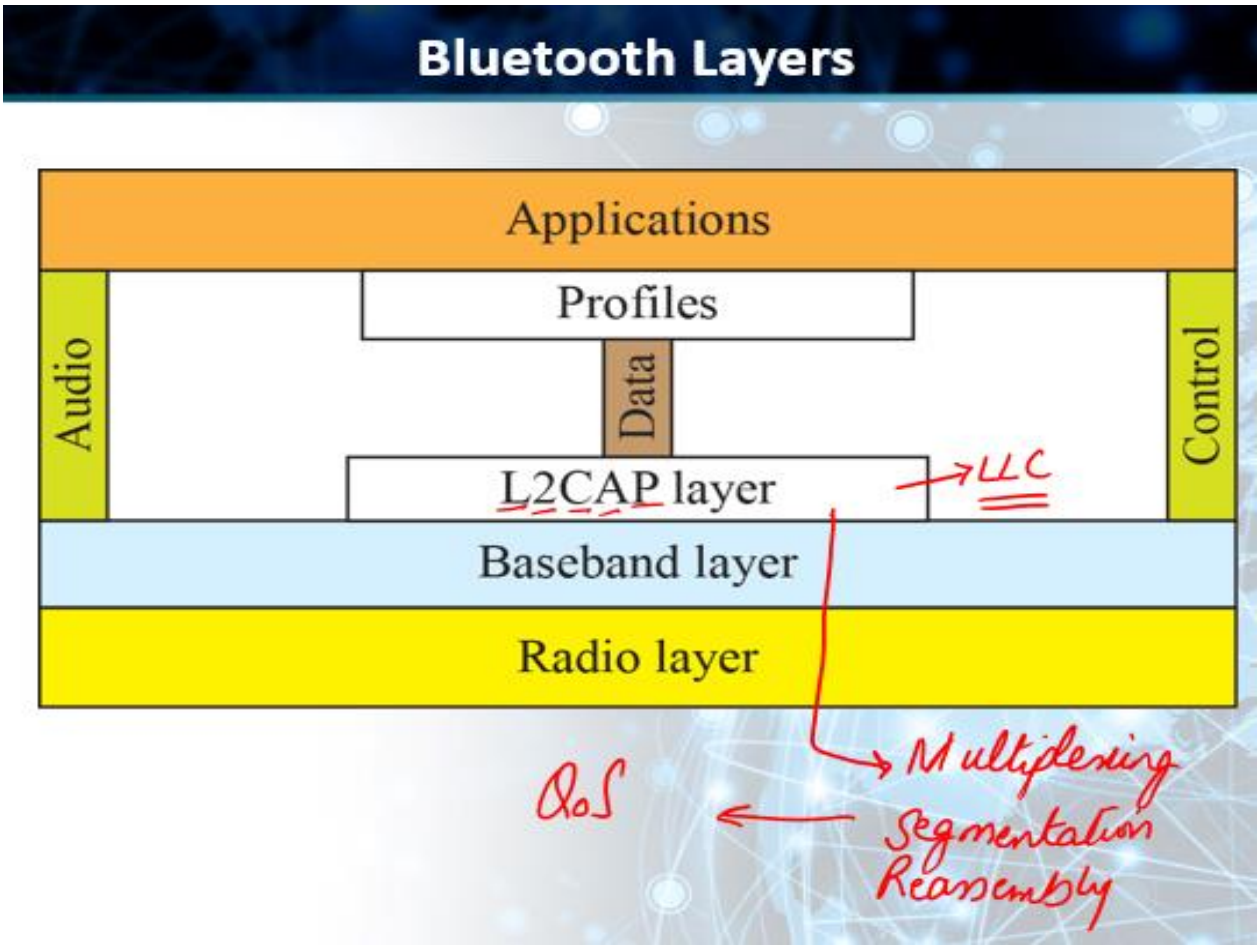
Bluetooth Layers

- **Bluetooth uses several layers that do not exactly match those of the Internet model we have defined already**

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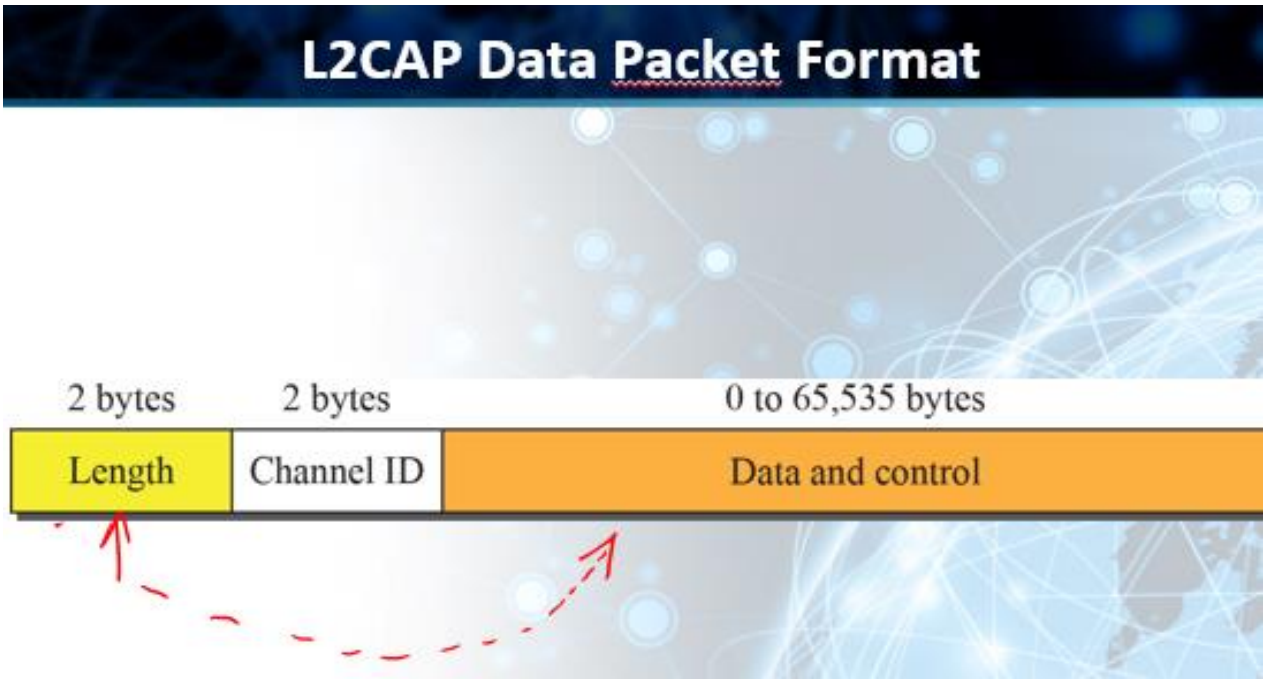
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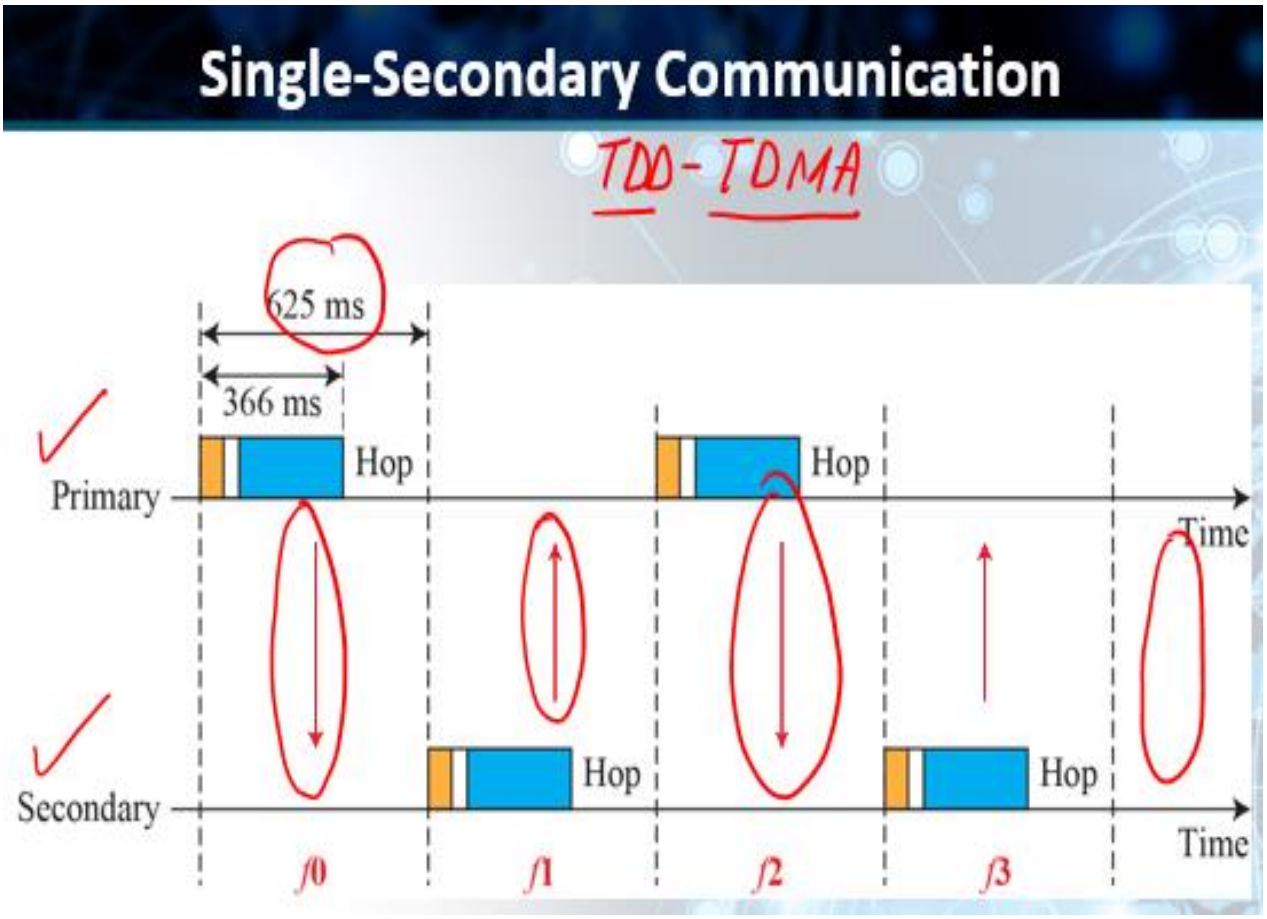
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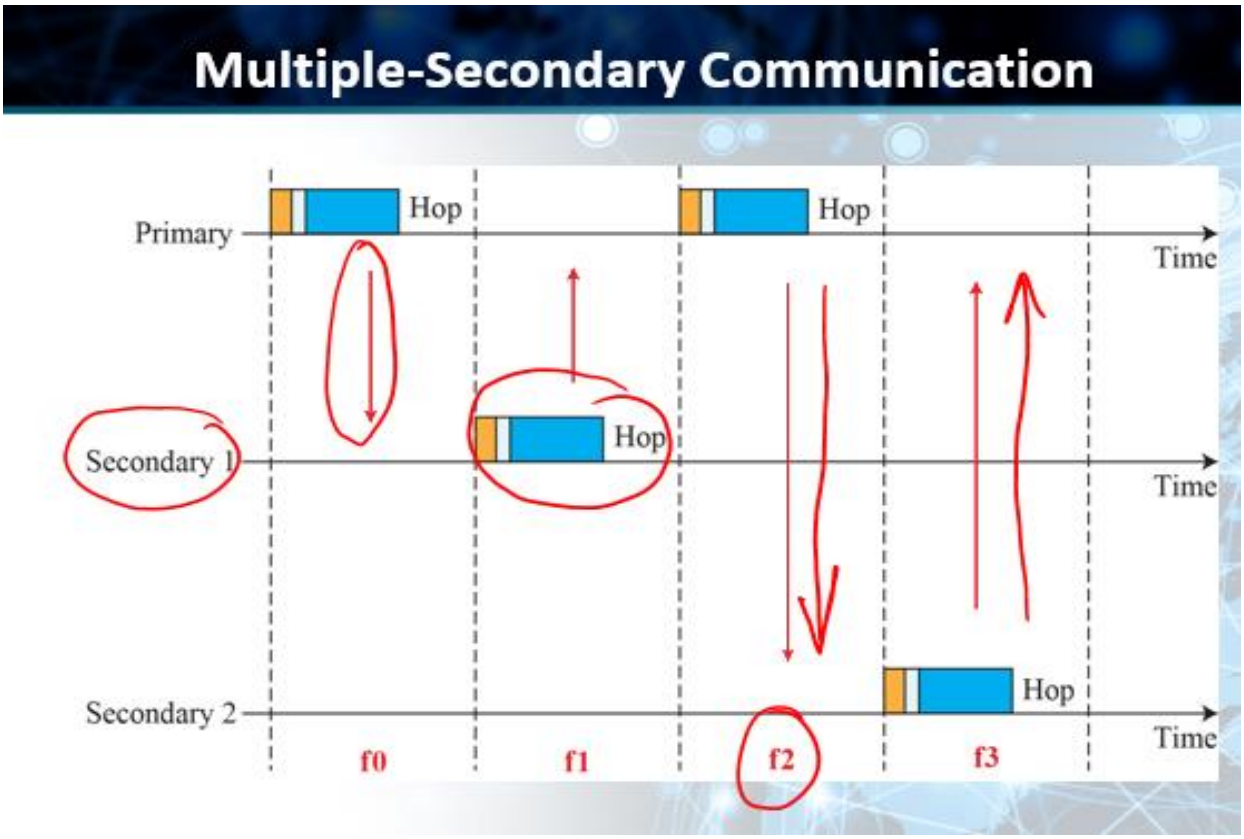
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CS601 Lecture No 36 Topic 213 to 220

Connecting Devices

- **Hosts and networks do not normally operate in isolation**
- **Connecting devices connect hosts together to make a network or connect networks together to make an internet**

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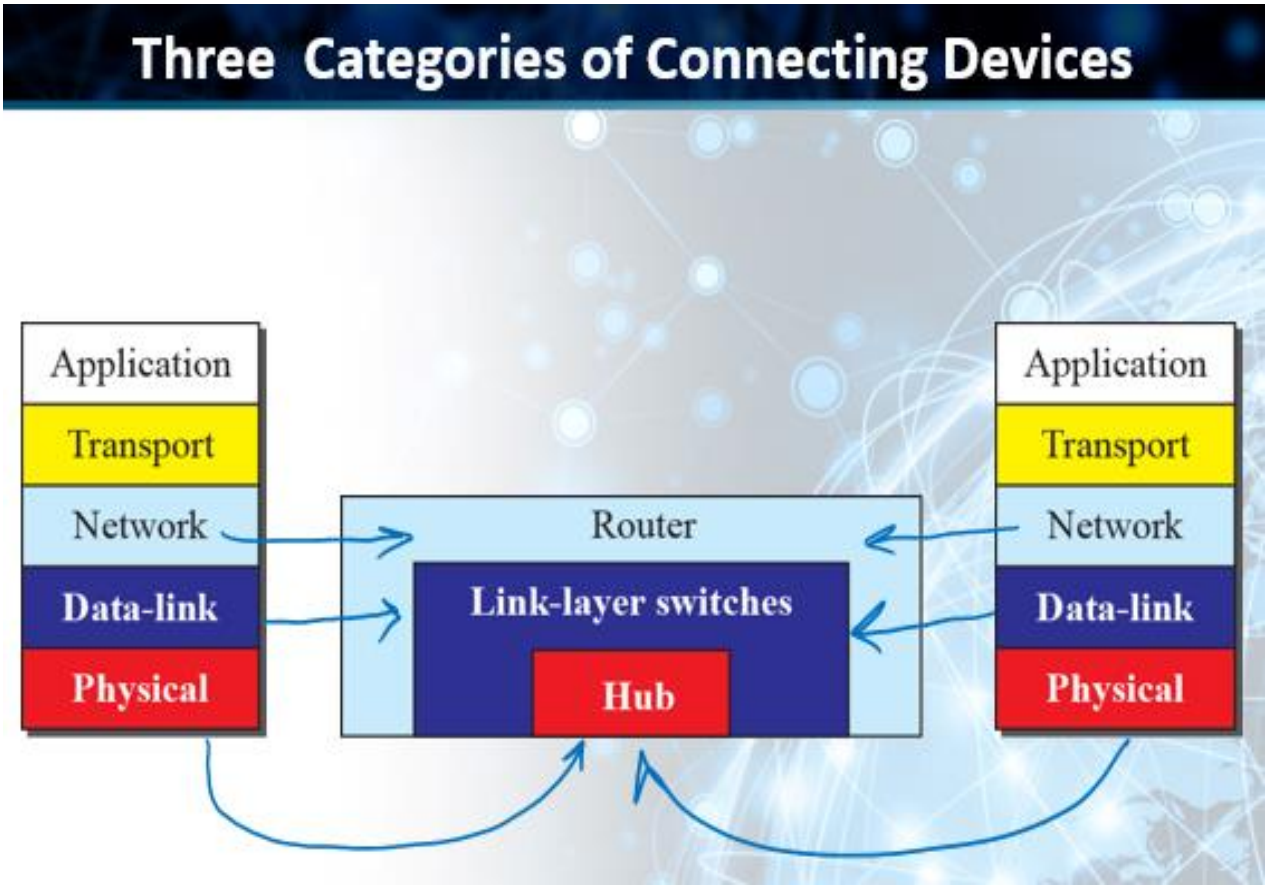
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- **Connecting devices can operate in different layers of the Internet model**

Three kinds of connecting devices:

- ✓ **Hubs**
- ✓ **Link-layer switches**
- ✓ **Routers**

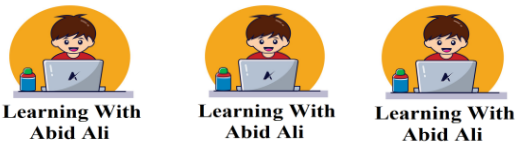


Hubs

- **Hub is a device that operates only in the physical layer**

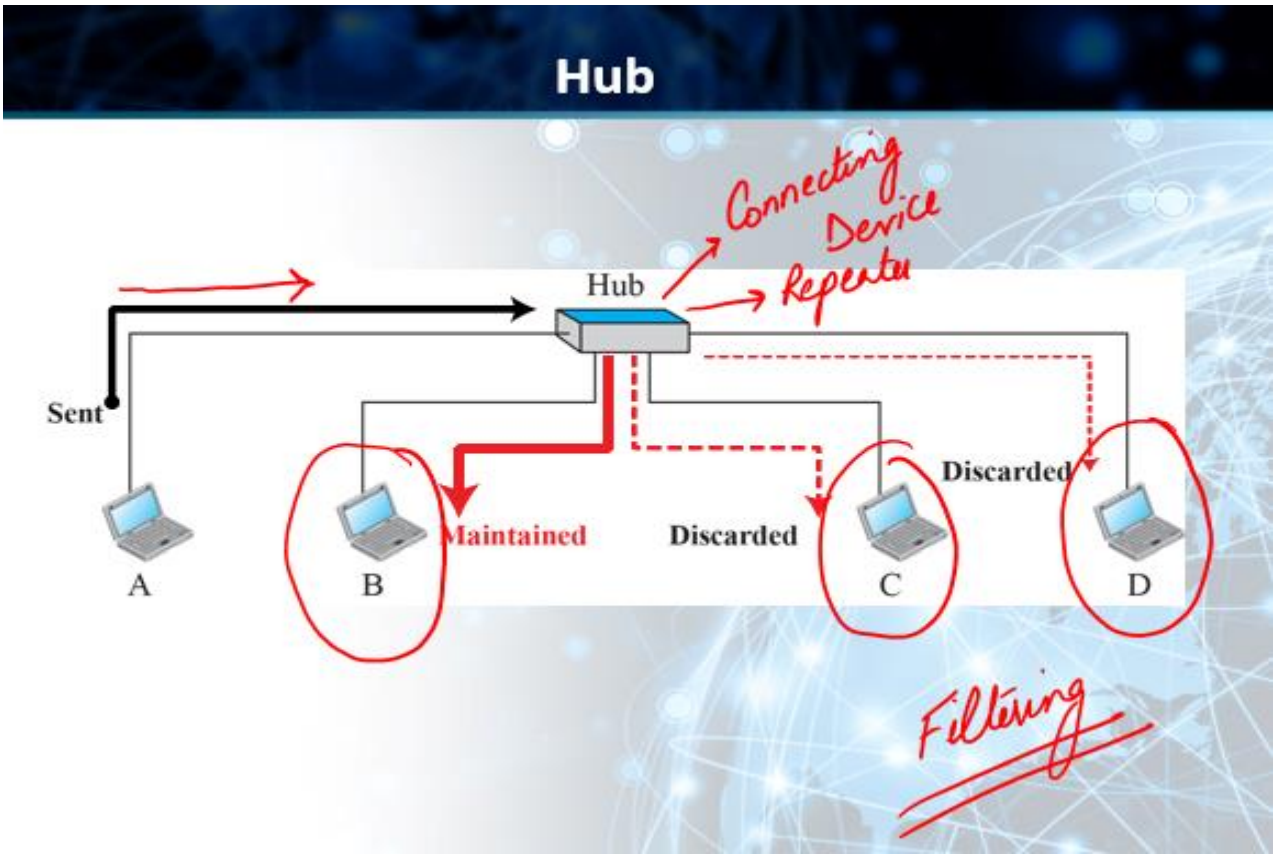
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- **Signals that carry information within a network can travel a fixed distance before attenuation impacts the data**
- **A hub (repeater) receives a signal and, before it becomes too weak or corrupted, regenerates it**



Link-Layer Switches

- **A link-layer switch (or switch) operates in both the physical and the data-link layers**
- **As a physical-layer device, it regenerates the signal it receives**
- **As a link-layer device, the link-layer switch can check the MAC addresses (source and destination) contained in the frame**

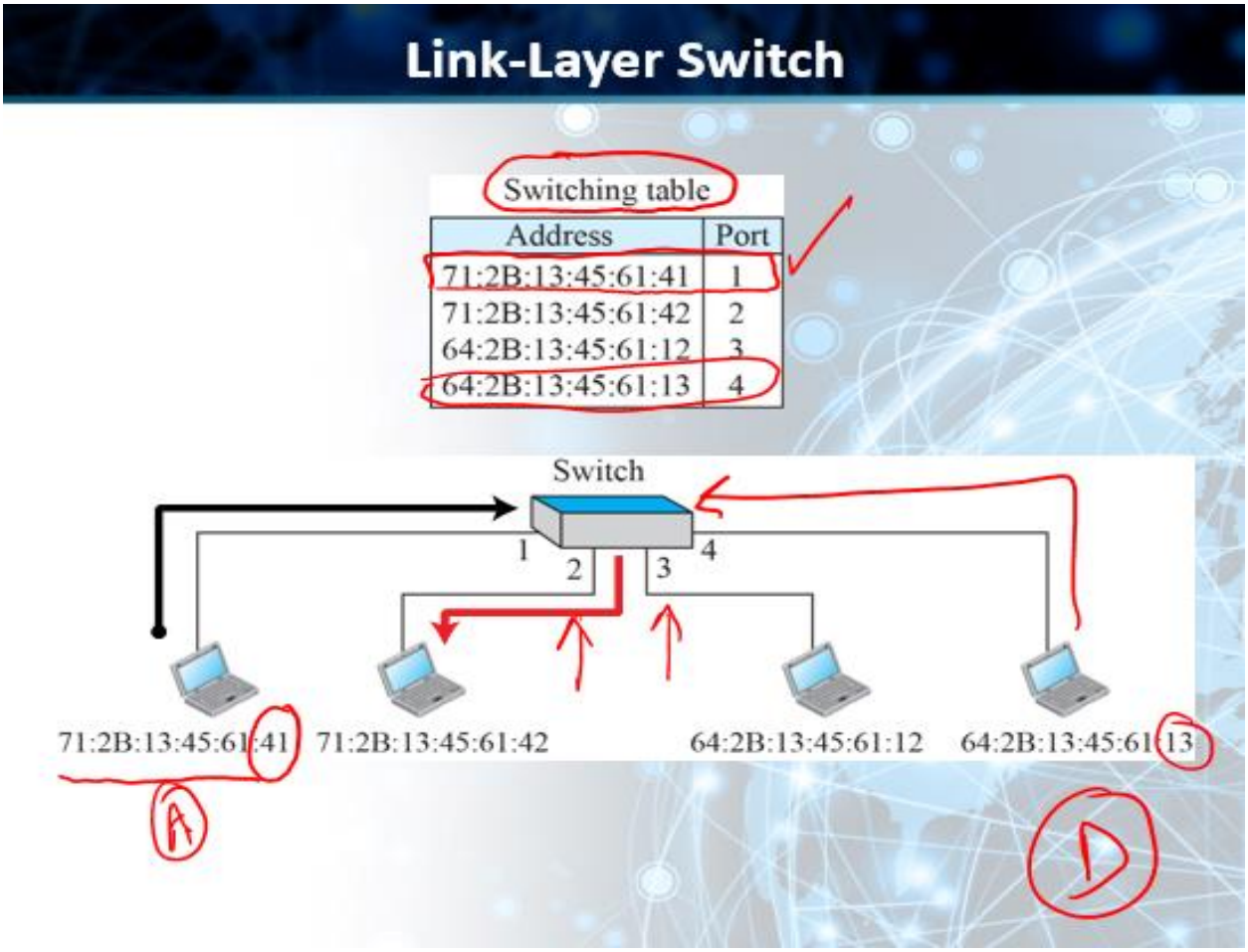
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Switch versus Hub

- **Switch has the 'Filtering' capability**
- **Unlike hub, a switch can check the destination address of a frame and decide on outgoing port**
- **Switch eliminates collisions and does not require carrier sensing**
- **Switches connect heterogeneous devices**



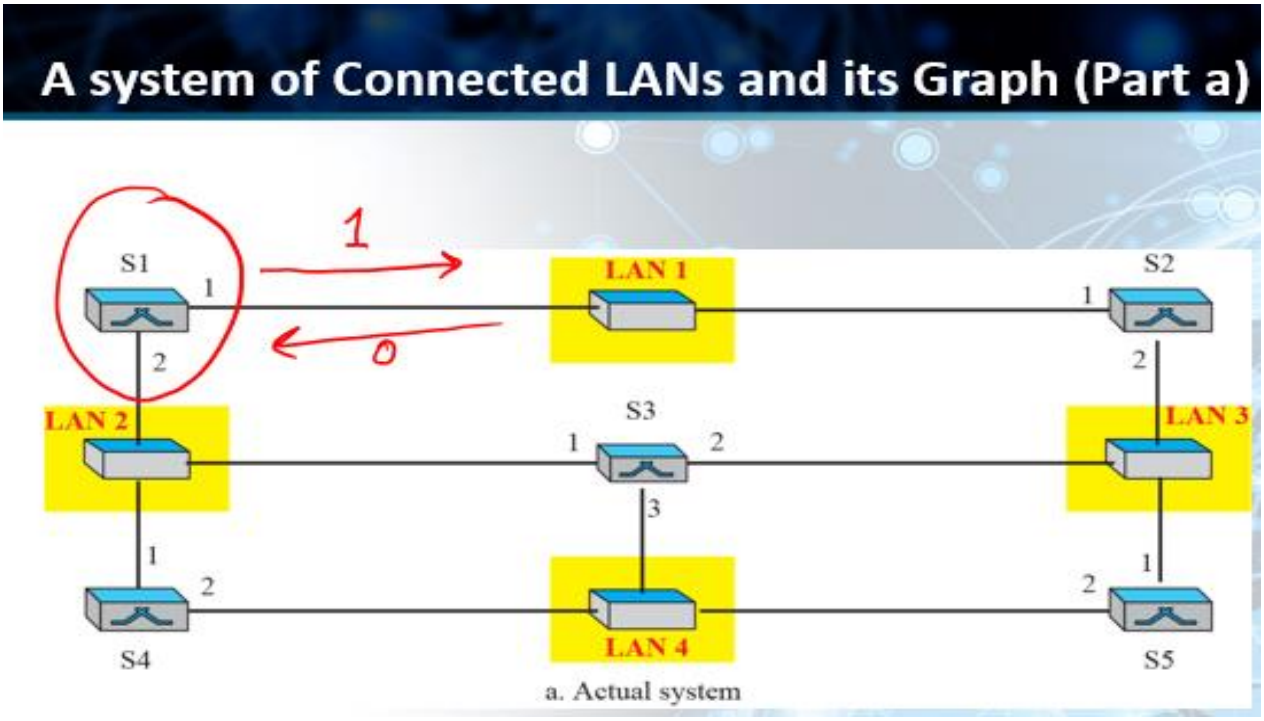
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Spanning Tree Algorithm

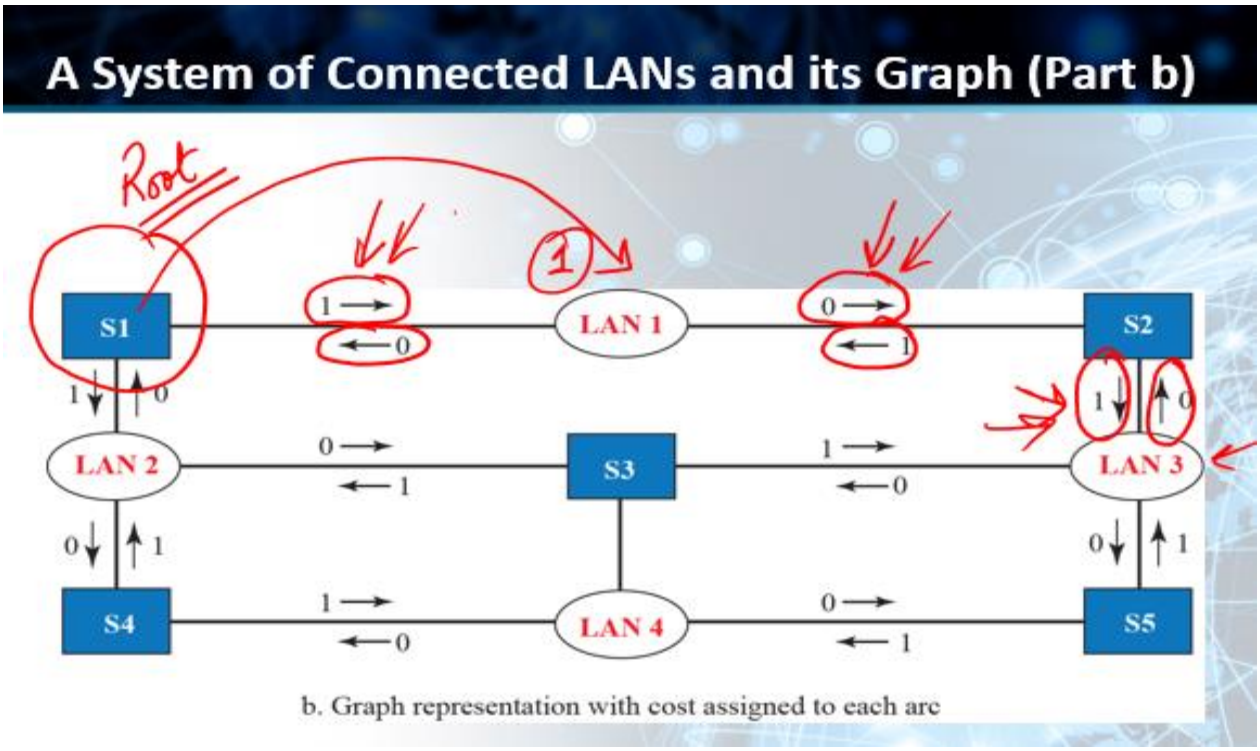
- **In graph theory, Spanning Tree is a graph in which there is no loop**
- **In a switched LAN, this means creating a topology in which each LAN can be reached from any other LAN through one path only (no loop)**
- **To find the spanning tree, we assign a cost (metric) to each LAN link**



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Routers

- **We compare routers to two-layer switch and a hub**
- **A router is a three-layer device; it operates in the physical, data-link, and network layers**

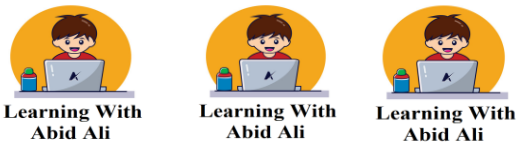
Router vs. Switch

Three differences between a router and a repeater or a switch:

1. A router has a physical and logical (IP) address for each of its interfaces.
2. A router acts only on those packets in which the link-layer destination address matches the address of the interface at which the packet arrives.

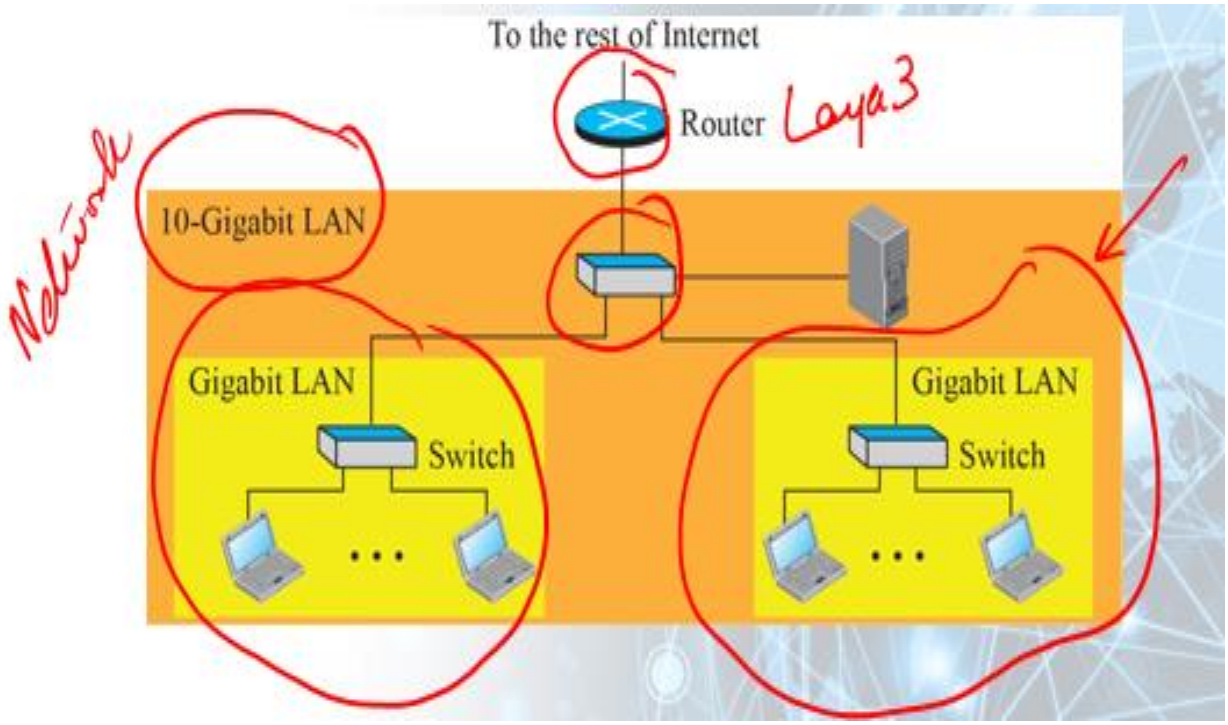
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3. A router changes the link-layer address of the packet (both source and destination) when it forwards the packet.

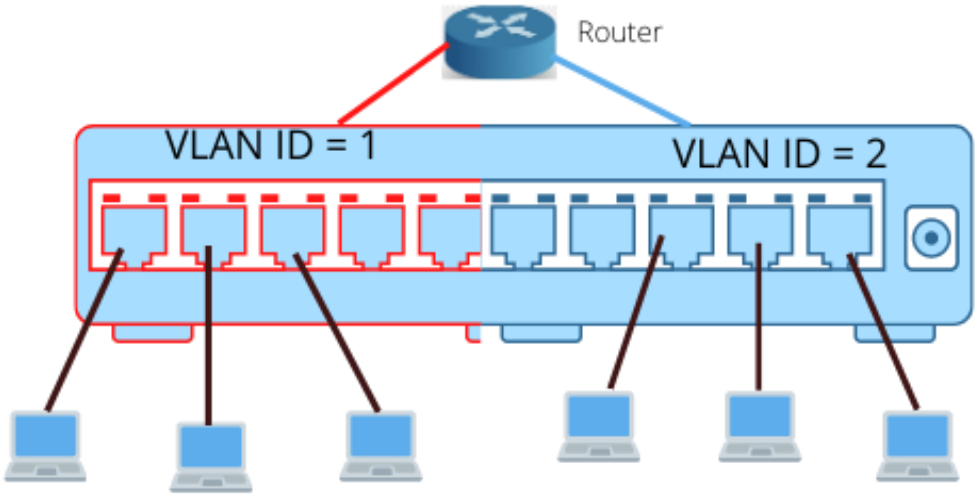


VIRTUAL LANS (VLAN)

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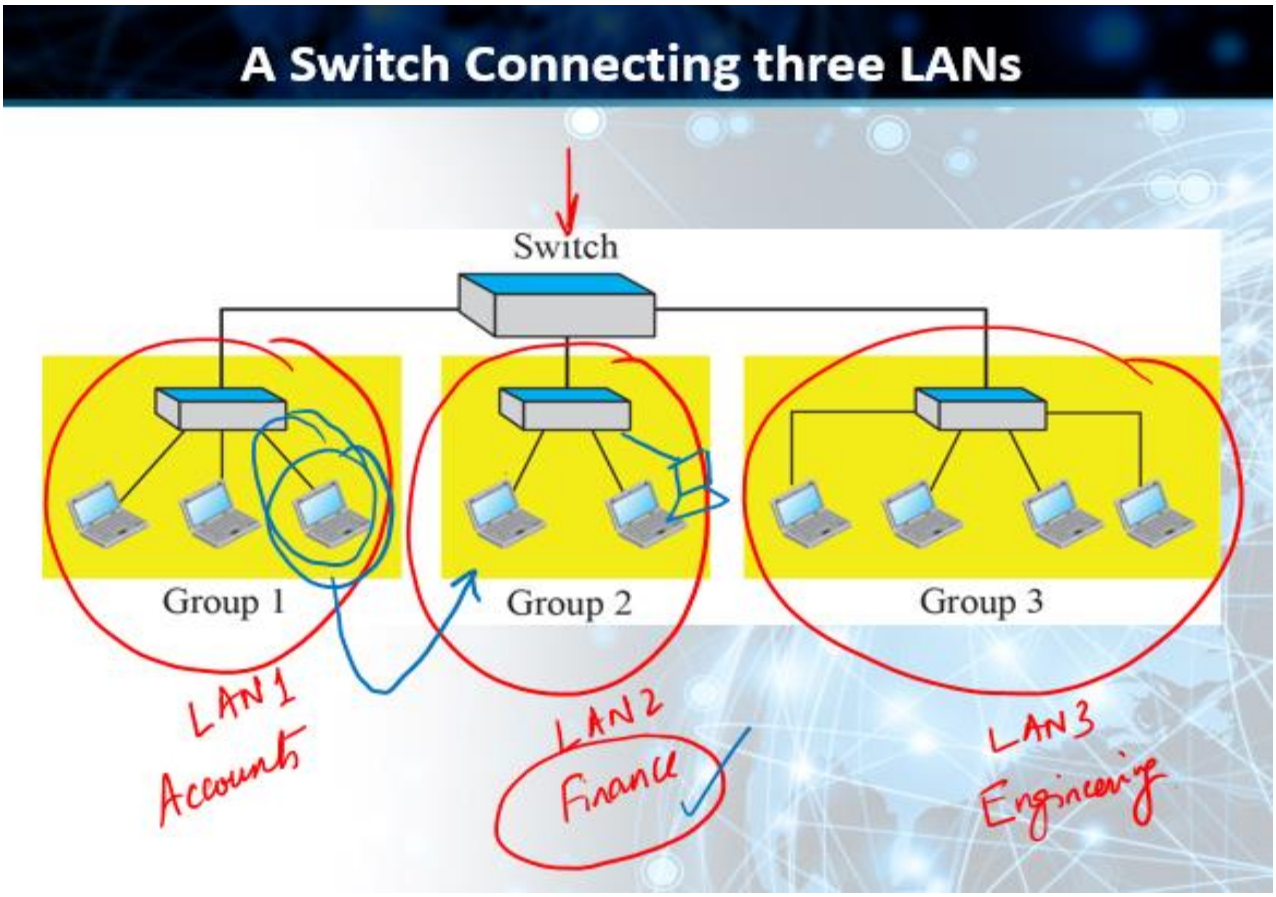
LAN With VLAN

- **A VLAN is a LAN configured by software, not by physical wiring**
- **A station is considered part of a LAN if it physically belongs to that LAN i.e. The criterion of membership is geographic**
- **Provides a virtual connection between two stations belonging to two different physical LANs**

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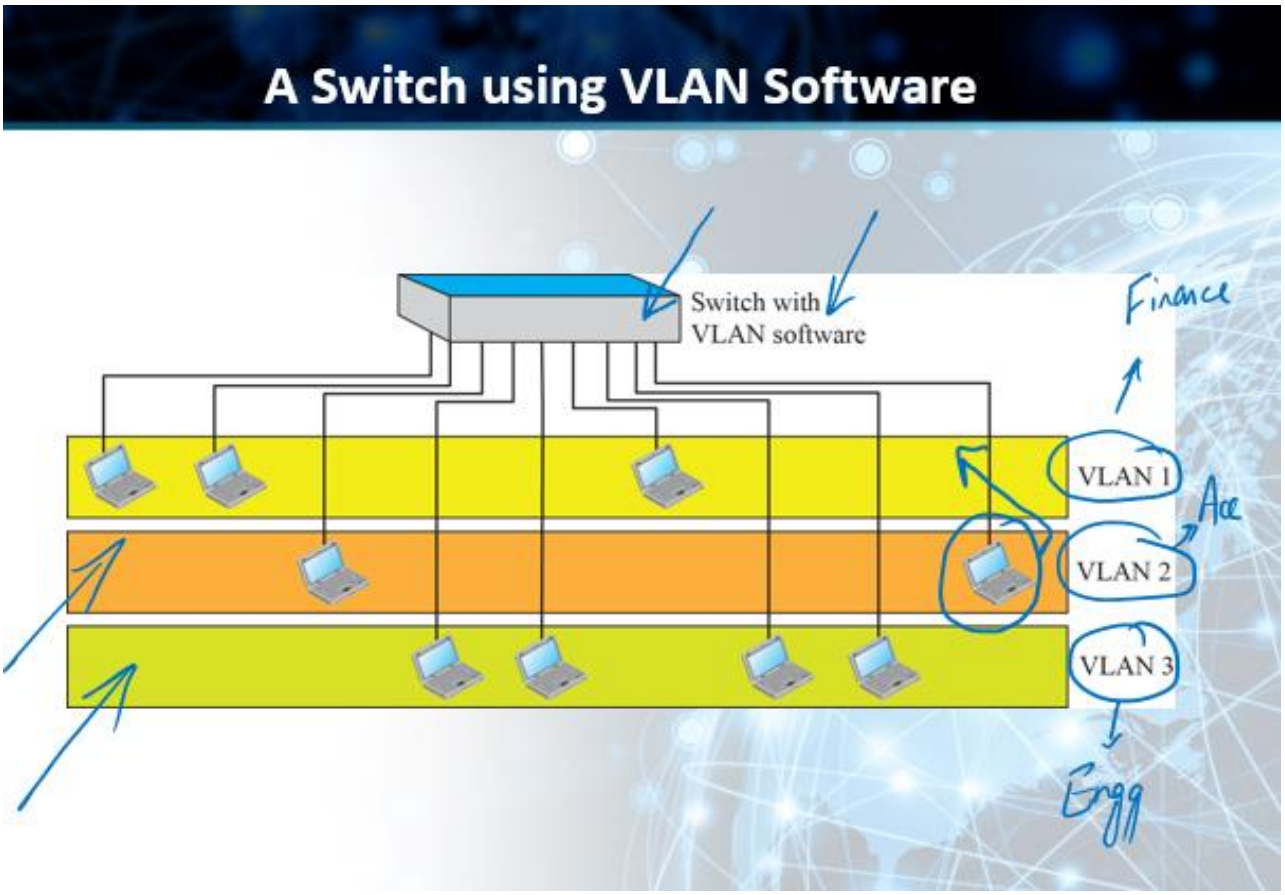
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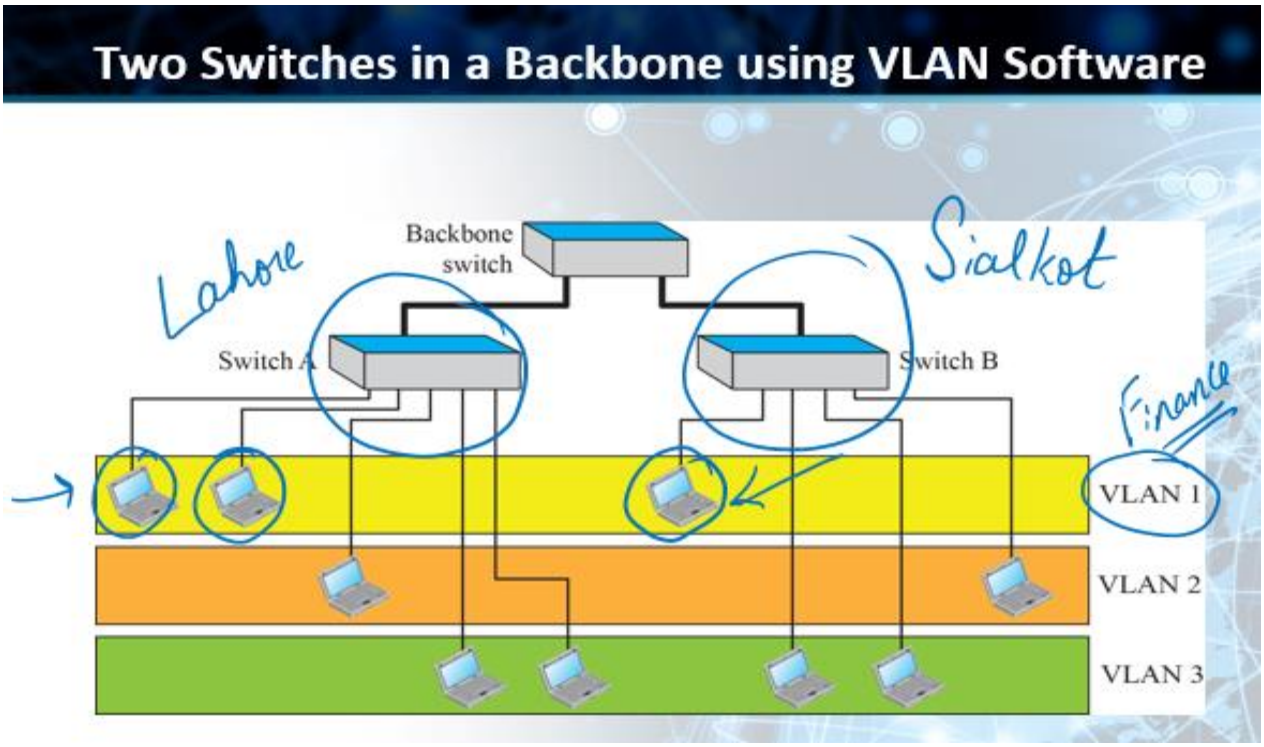
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Membership of a VLAN

- **What characteristic can be used to group stations in a VLAN?**
- **Vendors use different characteristics such as interface numbers, port numbers, MAC addresses, IP addresses, IP multicast addresses, or a combination of two or more of these**

Communication between Switches

In a multi-switched backbone, each switch must know:

Which station belongs to which VLAN; and

The membership of stations connected to other switches

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Advantages of using VLANs

- **Cost and Time Reduction**
- **Creating virtual Workgroups**
- **Security**

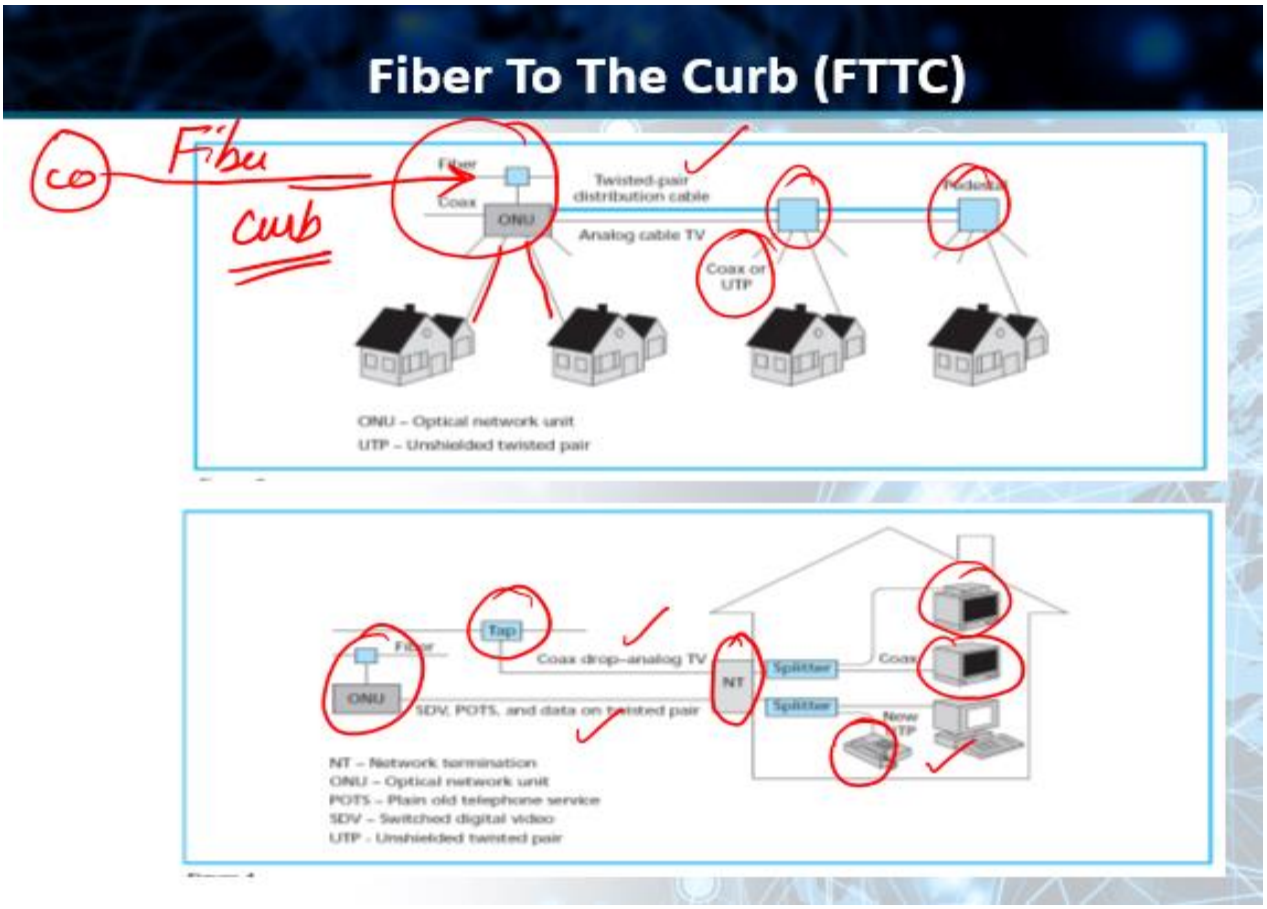
Fiber To the Curb (FTTC)

- **An access network in which fiber is used for part, but not the entire link from the provider to the end-user**
- **An optical to electrical (O/E) conversion takes place somewhere near the end-user**
- **The terminal network segment of a FTTC network is usually twisted pair or coaxial cable**
- **The final optical receiver in a FTTC network typically serves several customers**

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Fiber To the Home (FTTH)

- **Need: High-speed data, reliable voice and high-quality video**
- **Problems:**
  - ✓ **How to get high speed lines out to each customer?**
  - ✓ **How to future-proof the architecture?**
- **Solution: <u>FTTH</u>**
- **Fiber-to-the-home (FTTH) is the installation of optical fiber from a telephone switch directly into the subscriber's home**
- **It is one of the latest access technologies**

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- **FTTH is also referred to as Fiber-to-the-Building (FTTB)**